GREEN CLIMATE FUND

**Meeting of the Board**
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

**GCF/B.37/02/Add.10**

2 October 2023

# Consideration of funding proposals – Addendum X

## Funding proposal package for FP219


**Summary**

This addendum contains the following seven parts:

a) A funding proposal titled "Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat’s assessment;
e) Independent Technical Advisory Panel’s assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

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# Funding Proposal

| | |
|---|---|
| Programme title: | Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains |
| Countries: | Togo, Senegal and Guinea |
| Accredited Entity: | African Development Bank |
| Date of first submission: | *[2018/10/29]* |
| Date of current submission | *[2023/08/11]* |
| Version number | *[V.013]* |

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# Contents



***Note to Accredited Entities on the use of the funding proposal template***

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

**Please submit the completed proposal to:**

fundingproposal@gcfund.org

**Please use the following name convention for the file name:**

“FP[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

| Abbreviation | Definition |
|---|---|
| ACSs | Agricultural Cooperative Societies |
| ADAZZ | Authority for the Development and Administration of Special Economic and Industrial Zones |
| AE | Accredited Entity |
| AEZ | Agro-Ecological Zone |
| AFAPMs | Administrative, Financial and Accounting Procedures Manuals |
| AFAWA | Affirmative Finance Action for Women in Africa |
| AfDB | African Development Bank |
| AIPs | Agro-Industrial Parks |
| ANGE | Togo National Environmental Management Agency |
| APHs | Agro-Processing Hubs |
| APRODAT | Agency for Promotion and Development of Agropoles in Togo |
| APRs | Annual Performance Reports |
| ATCs | Agricultural Transformation Centres |
| AVCD | Agricultural Value Chain Development |
| AWM | Small-scale Agricultural Water Management Infrastructure |
| AWP | Annual Work Program |
| AWPB | Annual Work Program and Budget |
| BoA | Bureau of Agriculture |
| BOAD | West African Development Bank |
| BoFED | Bureau of Finance and Economic Development |
| BPP | Procurement Methods and Procedures |
| BPS | Borrower Procurement Systems |
| CBA | Cost-Benefit Analysis |
| CBAPs | Capacity Building Action Plans |
| CCLM | Climate Limited-area Modelling Community |
| CDD | Maximum Number of Consecutive Dry Days |
| CIF | Climate Investment Fund |
| CORDEX | Coordinated Regional Climate Downscaling Experiment |
| CRA | Climate Resilient Agriculture |
| CRP | Comprehensive Resettlement Plan |
| CRU | Climate Research Unit |
| CSOs | Civil Society Organizations |
| CSP | Country Strategy Papers |
| CWR | Crop Water Requirement |
| DCOM | Design, Construct, Operate and Maintain |
| DSCR | Debt Service Coverage Ratio |
| DEEG | Directorate of Equity and Gender Equality |
| DIDS | Drip Irrigation Distribution Systems |
| DIT | Drip Irrigation Technology |
| DP | Direct Procurement |
| DPM | Direct Payment Method |
| ECOWAS | Economic Community of West African States |
| EE | Executing Entity |
| EIA | Environmental Impact Assessment |
| ERR | Economic Rate of Return |
| ESCOs | Energy Service Companies |
| ESIA | Environmental and Social Impact Assessments |
| ESMP | Environmental and Social Management Plan |
| ESRP | Environmentally and Socially Responsive Procurement |
| FAO | Food and Agriculture Organization |
| FBAs | Farm-based Associations |
| FEMs | Financial and Economic Models |

| | |
|---|---|
| FM | Financial Management |
| FMA | Financial Management and Assessment |
| FWUC | Farmer Water User Community |
| FWUG | Farmer Water User Group |
| GAFSP | Global Agriculture and Food Security Program |
| GAP | Gender Action Plan |
| GCF | Green Climate Fund |
| GDP | Gross Domestic Product |
| GHG | Greenhouse Gas |
| RIAIPDC | Regional Integrated Agro-Industrial Park Development Corporation |
| IAIPs | Integrated Agro-Industrial Parks |
| IERs | Interim Evaluation Reports |
| IFAD | International Fund for Agriculture Development |
| IMF | International Monetary Fund |
| IMG | Inter-Ministerial Group |
| INDCs | Intended Nationally Determined Contributions |
| ISS | Integrated Safeguards System |
| JSF | Joint Steering Committee |
| LCB | Limited Competitive Bidding |
| LCOEs | Levelized Cost of Energy |
| LGP | Length of the Growing Period |
| LGC | Long Growing Cycle |
| LoC | Letter of Credit |
| LUCF | Land Use Change and Forestry |
| MDAs | Ministries, Departments and Agencies |
| MDBECG | Multilateral Development Banks’ Evaluation Cooperation Group (ECG). |
| MDBs | Multilateral Development Banks |
| MDIPMI | Ministry of Industrial Development and Small and Medium Industries |
| MFIs | Multilateral Financial Institutions |
| MOA | Ministry of Agriculture |
| MOAI | Ministry of Agriculture and Industry |
| MOI | Ministry in Charge of Industry |
| MoSAPWL | Ministry of Social Action, the Promotion of Women and Literacy |
| MOTI | Ministry of Trade and Industry |
| MSMEs | Micro, Small and Medium Enterprises |
| MTCO2e | Metric Tons of Carbon dioxide Equivalent |
| NCS | Non-consulting Services |
| NDCs | Nationally Determined Contributions |
| NGOs | Non-Governmental Organizations |
| NPV | Net Present Value |
| O&M | Operation & Maintenance |
| OCP | Open Competitive Bidding |
| OECD-DAC | Organization for Economic Cooperation, and Development Assistance Committee |
| OFAG | Office of the Federal Auditor General |
| OS | Operational Safeguard |
| PAPs | Project Affected Persons |
| PARs | Project Appraisal Reports |
| PFS | Programme Feasibility Studies |
| PIU | Project Implementing Unit |
| PNEEG | National Policy on Equity and Gender Equality |
| POM | Programme Operational Manual |
| PPCR | Pilot Program on Climate Resilience |
| PPMPs | Pest and Pesticide Management Plans |
| PPP | Public Private Partnerships |

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| | |
|---|---|
| PRCA | Procurement Risks and Capacity Assessment |
| PSC | Project Steering Committee |
| PSMC | Project Steering and Monitoring Committee |
| RAP | Resettlement Action Plan |
| RCM | Regional Climate Model |
| RCP4.5 | Representative Concentration Pathway 4.5 |
| RCP8.5 | Representative Concentration Pathway 8.5 |
| REOIs | Requests for Expressions of Interest |
| RG | Reimbursement Guarantee |
| RMCs | Regional Member Countries |
| RRR | Resources Recovery Reuse |
| SAM | Special Account Method |
| SBDs | Standard Bidding Documents |
| SCPZs | Staple Crops Processing Zones |
| SESA | Strategic Environmental and Social Assessment |
| SHFHs | Smallholder Farm Households |
| SIGCFs | Computerized Accounting and Financial Management Systems |
| SMEs | Small and Micro Enterprises |
| SNEEG | National Strategy for Equity and Gender Equality |
| SPEI | Standardized Precipitation and Evapotranspiration Index |
| SPI | Standardized Precipitation Index |
| SSS | Single-Source Selection |
| TA | Technical Assistance |
| ToC | Theory of Change |
| ToTs | 'Training of Trainers' |
| TRP | Technical Responsible Party |
| TF | Technical Task Force |
| UA | Units of Account |
| UNFCCC | United Nations Framework Convention on Climate Change |
| VCD | Value Chain Development |
| WABEs | Women-led Agribusiness Enterprises |

## ECT/PROGRAMME SUMMARY

| | | | |
|---|---|---|---|
| **A.1. Project or programme** | Programme | **A.2. Public or private sector** | Public |
| **A.3. Request for Proposals (RFP)** | *If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.*<br><br>Not applicable | | |

**A.4. Result area(s)**

*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*

| | **GCF contribution** | **Co-financers' contribution[1]** |
|---|---|---|
| **Mitigation total** | 44 % | Enter number % |
| ☒ Energy generation and access | 25 % | 0 % |
| ☐ Low-emission transport | Enter number % | Enter number % |
| ☐ Buildings, cities, industries and appliances | Enter number % | Enter number % |
| ☒ Forestry and land use | 19 % | 0 % |
| **Adaptation total** | 56 % | 100 % |
| ☒ Most vulnerable people and communities | 23 % | 26 % |
| ☒ Health and well-being, and food and water security | 33 % | 74 % |
| ☐ Infrastructure and built environment | Enter number % | Enter number % |
| ☐ Ecosystems and ecosystem services | Enter number % | Enter number % |

| | | | | |
|---|---|---|---|---|
| **A.5. Expected mitigation outcome**<br><br>*(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)* | *16.9million tCO2eq (programme lifetime)* | **A.6. Expected adaptation outcome**<br><br>*(Core indicator 2: direct and indirect beneficiaries reached)* | *Indicate total number of direct and indirect beneficiaries: 6.72Million* | |
| | | | *Direct beneficiaries: 1.104 million (54% are females)[2]* | *Indirect beneficiaries: 5.612 million (54% are females)[3]* |
| | | | *Indicate % of direct beneficiaries vis-à-vis total population: 3.2%* | *Indicate % of indirect beneficiaries vis-à-vis total population: 16.4%* |

| | | | |
|---|---|---|---|
| **A.7. Total financing (GCF + co-finance[4])** | 271.70 Million USD | **A.9. Project size** | Large (Over USD 250 million) |
| **A.8. Total GCF funding requested** | 102.79 Million USD<br>*For multi-country proposals, please fill out annex 17.* | | |

[1] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] The direct beneficiaries (1,104,728) breakdown is as follows: Togo is 428,853 (197,272 males & 231,580 females); Senegal is 233,215 (107,278 males & 125,936 females) and; Guinea is 442,660 (203,623 males & 239,036 females)
[3] The indirect beneficiaries (6,717,144) breakdown is as follows: Togo is 747,077 (343,655 males & 403,422 females); Senegal is 3,198,020 (1,471,089 males & 1,726,931 females) and; Guinea is 1,667,318 (766,966 males & 900,351 females).
[4] Refer to the Policy of Co-financing of the GCF.

| | | | |
|---|---|---|---|
| **A.10. Financial instrument(s) requested for the GCF funding** | *Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.*<br>☒ Grant 75.79 Million USD<br>☒ Loan 26.99 Million USD<br>☐ Guarantee Enter number<br>☐ Equity Enter number<br>☐ Results-based payment Enter number | | |
| **A.11. Implementation period** | 5(60) years(months) | **A.12. Total lifespan** | 25(300) years(months) |
| **A.13. Expected date of AE internal approval** | *This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.*<br>*11/24/2023* | **A.14. ESS category** | *Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.*<br>I-1 |
| **A.15. Has this FP been submitted as a CN before?** | Yes ☒ No ☐ | **A.16. Has Readiness or PPF support been used to prepare this FP?** | Yes ☐ No ☒ |
| **A.17. Is this FP included in the entity work programme?** | Yes ☒ No ☐ | **A.18. Is this FP included in the country programme?** | Yes ☒ No ☐ |
| **A.19. Complementarity and coherence** | *Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.*<br>Yes ☐ No ☒ | | |
| **A.20. Executing Entity information** | **Togo:** Government of Togo acting through Agency for Promotion and Development of Agropoles in Togo (APRODAT)<br>**Senegal:** Government of Senegal acting through Ministry of Industrial Development and Small and Medium Industries (MDIPMI).<br>**Guinea:** Government of Guinea acting through Authority for the Development and Administration of Special Economic and Industrial Zones (ADAZZ) | | |
| **A.21. Executive summary (max. 750 words, approximately 1.5 pages)** | | | |

Climate change threatens the lives and livelihoods of over millions in extreme poverty, especially in the sub-Saharan region of Africa. Togo, Senegal and Guinea, are all located in the major climate 'hotspots' in the region. They are already suffering from the effects of climate change as a result of historical warming and highly variable precipitation patterns alongside increased frequency and intensity of extreme events. The high dependence on rain-fed agriculture and natural resources, as well as their relatively low adaptive capacity makes these countries highly vulnerable to climate change impacts and variability. For instance, climate change and variability has resulted in harmful consequences on the economy due to the large dependence on the climate-sensitive agricultural sector in Togo. Consequently, leading to food and nutritional insecurity, degradation of forest resources, difficult access to energy, water resources and quality health care. The country recorded temperature increases of 1.2°C in 2020, which is an increase of 20% compared to 2012[5], and precipitation decrease of amplitudes ranging from 15 mm to 98 mm. These climate conditions translate into physical risks manifested by floods, drought, high temperatures, shifts in rainfall seasons, strong winds, variable rainfall distribution, soil erosion and coastal erosion that influence all development sectors. In Senegal, an average temperature increases of about 1.06°C since 1960 is observed with minimum and maximum values of 0.58°C in Dakar and 1.88°C in Ziguinchor[6], respectively. The southern parts of the country receive more rainfall compared to the northern region. The observed climate trends, particularly, the rise in temperatures and the drop in rainfall, have negative effects on the development of the country's priority economic sectors, such as, biodiversity, agriculture, livestock, water resources, fishing, and coastal zone. In Guinea, increase in average temperatures and slight increase in average annual rainfall is observed with increased inter-annual and intra-annual variability. These climate conditions have huge implications on the country's agriculture, livestock and forestry sectors.

The **Staple Crops Processing Zone (SCPZ)** program is specifically designed to help rationalize interventions in the agriculture sector of the program countries towards activities that contribute to emission reduction from agricultural activities and improve resilience of agroecosystems, agricultural assets, and beneficiaries. The program aims to reduce climate change vulnerability and greenhouse gas (GHG) emissions within the agricultural value chains in four regions from the three highly indebted poor countries in Africa. The host regions include Kara in Togo, Southern agropole in Senegal, and Boké and Kankan in Guinea. This will help stimulate productivity and value addition, competitiveness, generate employment and increase incomes of the most vulnerable people and communities in these countries. Since the program targets smallholder farmers that widely dominate the agricultural value chains in Africa, there is very little scope for these smallholder farmers to pay for the investments other than operation and maintenance costs. There is a need to accelerate investment in the region to address these issues, especially since the onset of the Covid-19 pandemic, which has had severe impacts on the region's economic development.

The GCF funds, along with the African Development Bank (AfDB) funds, other program financiers and the respective countries contributions, will be used to overcome critical infrastructure deficits, financial, capacity and institutional barriers that constrain the effective implementation of an integrated approach to climate adaptation and mitigation. Specifically, the funds will be invested in: (i) strengthening the resilience of critical SCPZs value chain infrastructure; (ii) building the resilience of SCPZs agricultural Products Supply Value Chain; and (iii) programme management and coordination. The programme will invest along the agricultural value chains by improving access infrastructure for sustainable agricultural intensification and production of staple crops (such as rice, maize, cassava, banana, sesame, horticulture, mango, cashew, coffee, woodlot, vegetables among others), roads construction, access to water and small-scale agricultural water management (AWM) infrastructure such as drip irrigation and gravity irrigation and equipped boreholes. At the post-harvest and processing stages, the program will invest in improving the resilience of the SCPZs through the use of low-carbon energy technologies such as biodigester and solar PV for processing, lighting and storage while at the marketing and transportation stages, the existence of Agro-Processing & Allied Industries Hubs(APHs) and Agricultural Transformation Centres (ATCs), offers adequate infrastructure, logistics and specialized facilities and services for improved marketing and competitiveness.

[5] Togo's updated NDC (2021). https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Togo%20First/CDN%20Revis%C3%A9es_Togo_Document%20int%C3%A9rimaire_rv_11%2010%2021.pdf

[6] Senegal's updated NDC (2020). https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Senegal%20First/CDNSenegal%20approuv%C3%A9e-pdf-.pdf

## B. PROJECT/PROGRAMME INFORMATION

### B.1. Climate context (max. 1000 words, approximately 2 pages)

***Climate Rationale***

Togo, Senegal and Guinea, which are situated in the part of Sahelian and Sudano agro-ecological zones (AEZs) of Africa, are among the major climate 'hotspots' in the continent. These countries are already suffering the effects of climate change as a result of historical warming and increased precipitation intensity trends. Future climate scenarios also indicate general warming and reduced precipitation trends with increased extremes[7,8]. Their vulnerabilities to climate extremes are high due to the strong dependence of the economies on rain-fed agriculture, largely dominated by smallholder farmers (about 80%) for daily subsistence. Reliance on rain-fed farming and pastoralism for income and subsistence mean that livelihoods and food security in these regions and countries are strongly affected by climate variability and change. According to WFP reports that, during the lean season (June to August), more than 21 million people across West Africa including Togo, Senegal and Guinea, will struggle to feed themselves as a result of recurrent food crises attributed to extreme weather events (prolonged droughts and floods). For example, the 2012 food crisis in Senegal caused by prolonged drought events left over 800,000 households' food insecure[9]. Similarly, in Togo, reduced grain quality and yields of maize, sorghum, millet and rice among peasant farmers have been reported, especially in the northern part of the country as a result of heat stress, water stress, flooding, erosion, and water logging.[10]

With a combined population of over 135 million people (about 11.1% of the continent's total population), which is projected to double by 2050[11], these three countries are committed to achieving the goals of the Paris Agreement, as expressed in their updated Nationally Determined Contributions (NDCs) (conditional and unconditional), by strengthening the mitigation and adaptation capacity of the agricultural sector (Table B.1.1).

**Table B.1.1: Countries NDCs in the agricultural sector, land use and forestry and energy by 2030.**

| Country | Commitment | Sectors for mitigation and enhanced removals | Budget |
|---|---|---|---|
| <u>**Togo**</u><br>***LUCF - highest emitting sector in 2014 based on WRI CAIT data (6.6* MtCO2e)** | Unconditional – Reduce GHG emissions by 20.51% by 2030, i.e. 6,236.02 Gg CO2-eq, compared to 2030 BAU emissions<br><br>Conditional – Reduce GHG emissions by 30.06% by 2030, i.e. 9,305.59 Gg CO2-eq, compared to 2030 BAU emissions | Energy, Agriculture and LULUCF<br><br>**Agriculture**<br>Identification and promotion of varieties of rain-fed rice, and support and guidance in the better use of organic matter in the paddy fields<br>Reduction of GHG emissions (optimal waste management from livestock and harvest remnants, promotion of low emission land use planning practices, etc.).<br>**Energy**<br>Promotion of energy-efficient stoves (which can yield 50-60% savings in wood and charcoal).<br>Introduction of solar equipment at the household level | **~USD$5.4 billion** (mitigation and adaptation measures are USD$2.7 and USD$2.6 billion, respectively) |
| <u>**Senegal:**</u><br>***Agriculture - highest emitting sector in 2014 based on WRI CAIT data (11.0* MtCO2e)** | Unconditional – Reduce GHG emissions by 5% (i.e. 30,987 GgCO2eq) and 7% (i.e. 35,106 GgCO2eq) by 2025 and 2030, respectively, compared to the BAU emissions in those years.<br><br>Conditional – Reduce GHG emissions by 23% (i.e. ~24,883 GgCO2eq), and 29% (i.e. ~26,611 GgCO2eq) in 2025, and 2030, compared to the BAU emissions in those years. | Agriculture, Forestry and Energy<br><br>**Agriculture**<br>Implementation of projects and programs to reduce GHG emissions<br><br>**Forestry**<br>Emission reduction related to the consumption of firewood and charcoal (unconditional and conditional actions) | **USD$13 billion** (USD$8.7 billion dedicated to mitigation with USD$3.4 and $5.3 billion unconditiona l and |

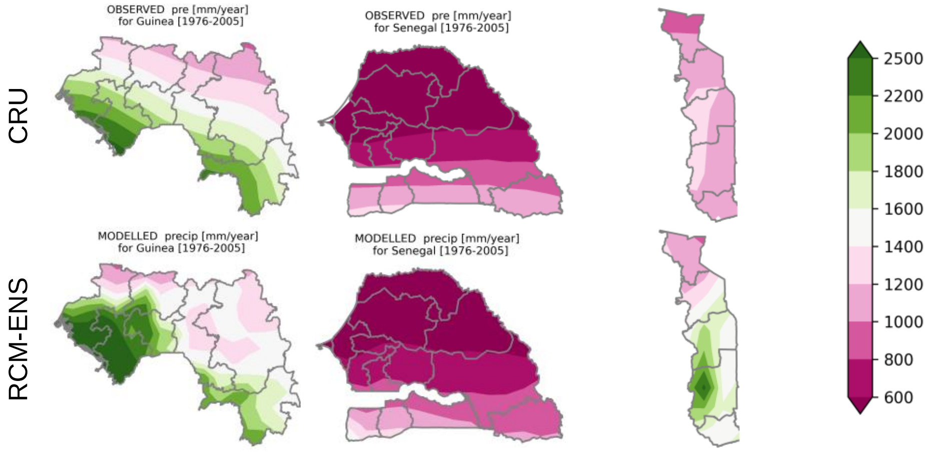
8USAID 2018. Fact Sheet: Climate Change Risk Profile, West Africa. USAID, December 2017.
9 https://reliefweb.int/report/senegal/wfp-launches-support-more-800000-people-hit-food-crisis-senegal
10USAID 2018. Fact Sheet: Climate Change Risk Profile, West Africa. USAID, December 2017.
11The Montpellier Panel (2013).

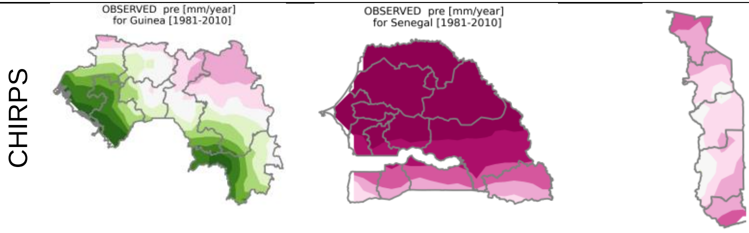
| | | | |
|---|---|---|---|
| | | Emission reduction from deforestation and Forest degradation (unconditional and conditional actions)<br>**Energy**<br>Provision of 160 MW of Solar PV<br>392 villages provided with Solar or hybrid mini-grid<br>Installation of 27,000 biodigester at the household level. | conditional, respectively, and USD$4.3 billion for adaptation, including USD$1.4 and $2.9 billion unconditional and conditional, respectively) |
| <u>**Guinea:**</u><br>***LUCF - highest emitting sector in 2014 based on WRI CAIT data (14.0 MtCO2e)*** | Excluding LULUCF<br>Unconditional - 2,056 ktCO2eq/year, i.e. a 9.7% reduction in emissions by 2030<br>Conditional – Excluding storage capacity from LUCF, reduce 3,929 ktCO2 eq/year, or 17.0% GHG emissions by 2030 compared to the BAU scenario<br><br>On LULUCF<br>Unconditional - 23% reduction in emissions by 2030<br>Conditional – Excluding reforestation actions, reduce 49% GHG emissions by 2030 compared to the BAU scenario | Energy, Land-Use Change and Forestry (LUCF) and Agriculture<br><br>Produce 30% of its energy (excluding wood-energy) from renewable energy sources (cumulative GHG reduction of 34 MtCO2e by 2030)<br>Support the dissemination of technologies and practices that are energy- efficient or use alternatives to wood-energy and charcoal (cumulative GHG reduction of 23 MtCO2e by 2030) | **USD$13.8 billion** |

**Source:** Adopted from USAID 2019 and Updated NDCs.

These countries have a long history of confronting serious crises. Key human-influenced drivers of climate change include rising temperatures caused by global GHG emissions and changes in land use land cover, as result of deforestation (clearing land for agriculture, extended periods of shifting cultivation, mono-cropping, over exploitation of the already vulnerable land and forests capes and cutting trees for firewood and as an alternative source of income), and overgrazing of pastureland. These combine to contribute to alterations in rainfall patterns and increase the frequency of extreme weather events (floods including riverine floods, wind and dust storms, prolonged dry spells or droughts), resulting in loss of assets (infrastructure, productive land livestock, etc), crop losses, lower productivity levels and even famine and desertification. All these climatic together with non-climate factors contribute to growing food insecurity and poverty experienced in the countries.

The spatial distribution of precipitation in the host countries shows distinct patterns with average annual rainfall ranging from about 600 mm/year to above 2500 mm/year (Figure B.1.1). Both observations and model agree in terms of reproducing the south to north decreasing rainfall signal in Guinea, the driest northern region in Senegal, and wettest western Plateaux sector and drying conditions further north of the Kara region in Togo. On the average, Senegal receives the lowest amount of annual total rainfall. Although the RCM response to topographic features such as the Plateaux in Togo and Ocean effects in Guinea is strong, the mean pattern of the observed rainfall in conserved highlight the usefulness of the RCMs for investigating the possible climate situation in the future.






The major feature of the observations is a wide band of high rainfall, which spans along the Sudano AEZ in the southern parts of Senegal, Guinea, and Togo (see Section 5 of Annex 2 for details). This feature is present for most of the year, being more intense during the heart of the rainy season from July to September. The model is able to simulate most of the precipitation characteristics, particularly during June, July and August (JJA). Also, the intense precipitation region over Senegal, Togo, and Guinea, which mainly occurs during (JJA) and (SON), is well represented by the model. However, in general, simulated precipitation is overestimated with respect to observations in all seasons especially along the Atlantic coastline of Senegal, Guinea, and Togo. The position and latitudinal displacement of the precipitation band northwards from July to September, associated with both Pacific El Niño/La Niña related sea surface temperature (SST) anomalies and the Atlantic dipole,[12]is well reproduced by the model.

At project scale the RCM ensemble mean also compares well with most of the stations although with some systematic biases (Figure B.1.2). For instance, increasing rainfall was observed at Niamtougou in Togo. In Kankan, the observed and modelled trends in precipitation diverge slightly, where the observed trends show increase.

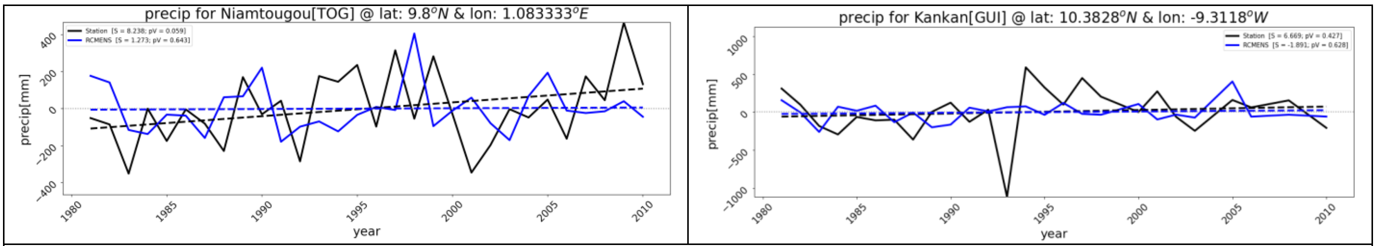
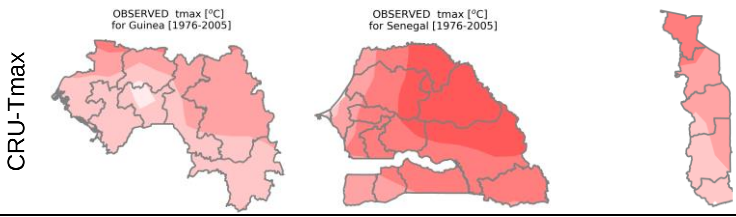


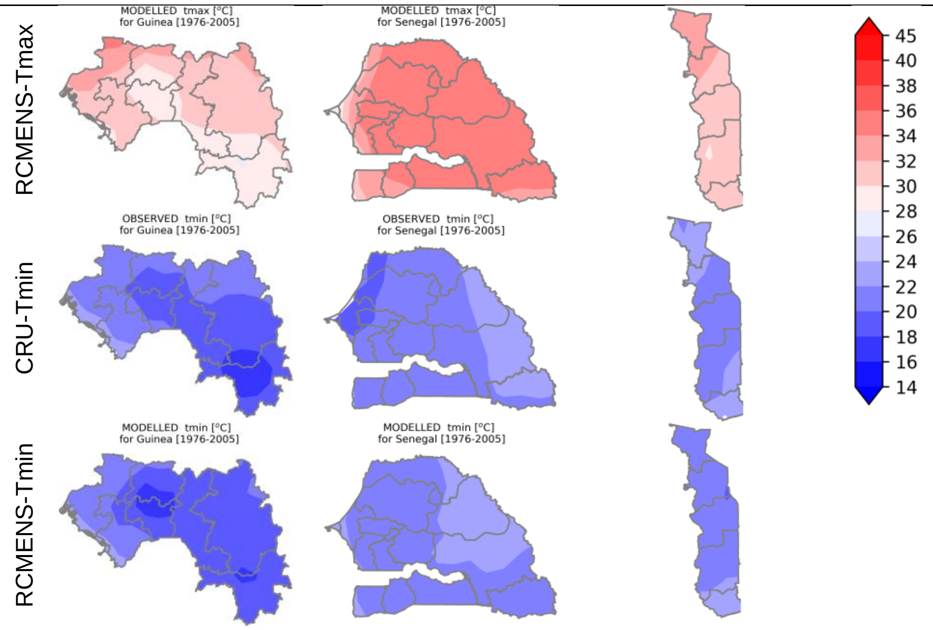

**Figure B.1.2.** Trends in observed[13] (black line) and modelled (blue line) annual total precipitation for selected SCPZ stations the period 1981-2010.

Both observed and modelled 2-metre temperature agree well in terms of spatial patterns for all seasons in the two AEZs (see details in Section 5 of Annex 2). The temperature gradient, the regional characteristics, and the seasonal variations are well reproduced by the ENS model. The ENS simulations captures well the areas of cold temperatures, in the coastal area of Guinea, and the areas of warmer temperatures such as in northern Senegal and northern Guinea. However, the ENS model tends to overestimate temperatures almost everywhere in the coastal domain. The warm bias of temperature may be contributing to the overestimation of precipitation in the coastal areas as warming results in enhanced evaporation that will favor convection trigger criterion in the ENS model depends on the near-surface air temperatures. The spatial pattern of annual mean surface temperature is conserved in the RCM both for minimum and maximum temperature (Figure B.1.3). Temperatures are largely influenced over complex terrains like the Plateaux in Togo and nearness to water bodies like the coastlines of Guinea and Senegal. The average minimum and maximum temperature ranges from 14ºC in the south of Guinea to about 38ºC in the northern region of Senegal.




[12] Bette L. Otto-Bliesner, 1999, "El Niño/La Niña and Sahel precipitation during the Middle Holocene", Geophysical Research letters, , vol. 26, issue 1, January. Available from https://agupubs.onlinelibrary.wiley.com/doi/epdf/10.1029/1998GL900236

[13] The observed data are computed from daily rainfall data retrieved from the archives of the respective meteorological services for selected stations within the SCPZs




Local trends in temperature show a general increase for minimum and maximum temperature in both station observed and model simulated data (Figure B.1.4). The warming is statistically significant, especially with maximum temperature where the mean annual deviation from the climatological mean is about 1 to 2ºC. The observed warming partly contributes to soil degradation which could result in low crop yields in the SCPZs. Consequently, decline in agricultural productivity discourages the farmers and may lead to change in livelihood especially in the rural settings of the host SCPZs.

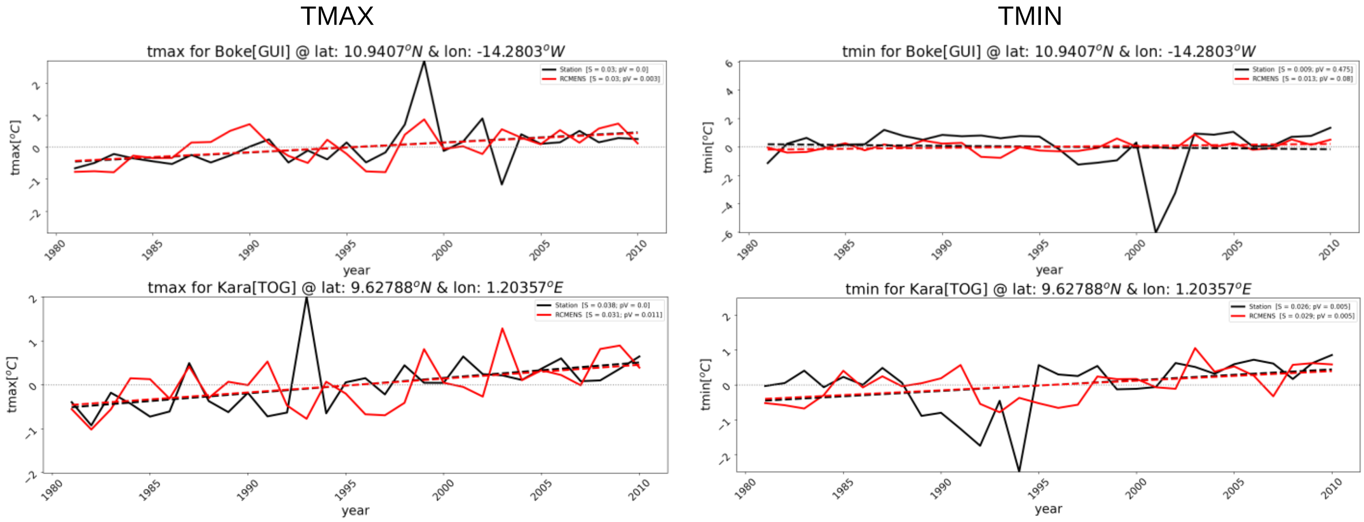



**Figure B.1.4.** Trends in observed[14] (black line) and modelled (red line) minimum and maximum surface temperature (ºC) for selected SCPZ stations the period 1981-2010.

Table B.1.2 shows the spatial correlation between the simulated precipitation pattern by the ENS model, the RCMs and CRU. DJF is the season when the 3 RCMs and the ENS-model precipitation exhibit the highest correlations. Results indicate that the ENS provides added value to the simulations over the three RCMs. In all seasons, the spatial pattern of temperature is better reproduced by the ENS model when compared to the individual model runs. For the highly variable precipitation, the pattern correlation is low and comparable between the CCLM model and ENS. Although the pattern correlation between ENS model simulations and CRU observations are lower for precipitation than for temperature, these correlations are still much higher than with the individual model simulations, except for DJF, MAM, and JJA when the CCLM correlates better with CRU observations.

**Table B.1.2:** Pattern correlations between the CCLM, RCA4, REMO simulations and observations, and between the three RCM ensemble (ENS) simulations and observations, for DJF, MAM, JJA, and SON seasons.

[14] The observed data are computed from daily minimum and maximum temperature data retrieved from the archives of the respective meteorological services for selected stations within the SCPZs

| 2-m Temperature | | | | |
|---|---|---|---|---|
| RCMs/season | DJF | MAM | JJA | SON |
| CCLM | 0.88 | 0.82 | 0.82 | 0.85 |
| RCA4 | 0.87 | 0.82 | 0.83 | 0.85 |
| REMO | 0.81 | 0.81 | 0.82 | 0.90 |
| ENS | 0.90 | 0.88 | 0.88 | 0.93 |
| **Precipitation** | | | | |
| CCLM | 0.39 | 0.32 | 0.38 | 0.22 |
| RCA4 | 0.38 | 0.30 | 0.34 | 0.23 |
| REMO | 0.38 | 0.31 | 0.33 | 0.22 |
| ENS | 0.38 | 0.31 | 0.35 | 0.23 |

Annual cycle: Figure B.1.5 shows the annual cycle of the observed and simulated precipitation for the SCPZ programme regions. This includes; the Kara region of Togo; the Casamance Natural region of Senegal; and the Boké and Kankan regions of Guinea. The annual cycle is shown for the CRU observational datasets, and the three RCMs simulations and ENS simulations. As observed in Figure B.1.5, in all seasons, the RCMs and ENS-model simulations approach the CRU observations for all the targeted regions with the exception of the Oromia and Tigray regions, where the ENS-model simulations slightly underestimate and overestimate the precipitation with respect to CRU observations. This is mostly during the months of AMSO for the Oromia region, and JA for the Tigray region. Therefore, in general, the regional model simulations added improvement over the CRU. The annual cycle of precipitation of the project areas, shows a single cycle of precipitation for most of the regions. The onset of the rainy season, which usually occurs in May, is well captured by the ENS model simulations. In addition, the amount of rainfall is also well simulated by 3 RCMs simulations. This variability is better reproduced by the ENS model. The ENS, in general, captures better the intensity of the peaks of rainfall in most of the points in the regions.

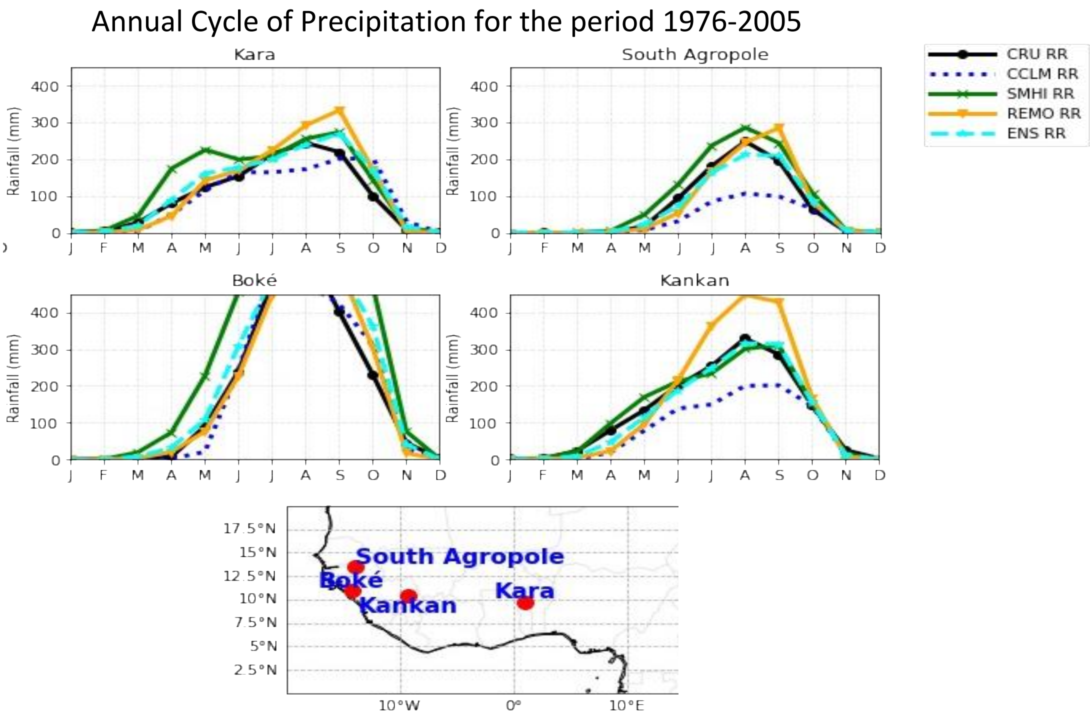



**Figure B.1.5.** Mean annual cycle of precipitation (mm/day), averaged over the baseline 30-year period (1976-2005) for SCPZ Project Areas. Observations from CRU are plotted (black line), as well as model simulations (CCLM, RCA4, REMO and ENS. The curves refer to the model grid-point that contains the SCPZ project regions in Togo, Guinea and Senegal.

The mean annual cycle of 2-metre temperature for Kara Region in Togo; the Casamance Natural Region of Senegal; and the Boké and Kankan Regions of Guinea is shown in Figure B.1.6. The observed and modelled simulations show similar trends in the mean annual cycle of 2-metre temperatures in all the regions. The shape of the annual cycle of the RCMs and the ENS model simulations closely follows the CRU temperature cycle. The ENS simulations reproduces the temperature better when compared with the regional model simulations, although some underestimate and overestimation also occurs in some points.

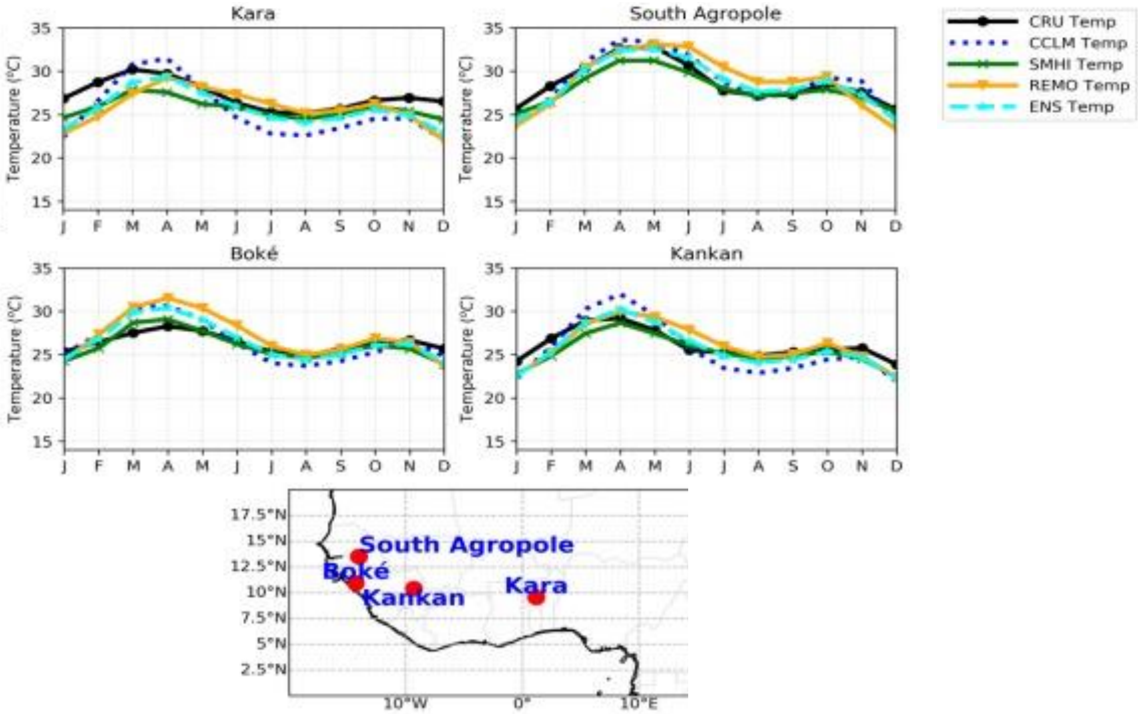



**Figure B.1.6.** Mean annual cycle of 2-metre temperature (ºC) averaged over the baseline 30-year period for SCPZ Project Areas. Observations from CRU are plotted (black line), as well as model simulations (CCLM, RCA4, REMO and ENS. The curves refer to the model grid-point that contains the SCPZ project regions in ETogo, Guinea and Senegal.

Seasonal Drought (SD): With rising temperatures as evident from the analysis of mean annual cycle of 2-metre temperature at the project specific levels, droughts occurrence are bound to be more frequent with far reaching implications for the agricultural and water related sectors. Drought monitoring is very important for irrigational planning in agriculture as well as a guide for early warning and assessment. The WMO recommends the use of the Standardized Precipitation Index (SPI) and the Standardized Precipitation Evapotranspiration Index (SPEI) as a starting point for meteorological drought monitoring. The SPEI takes temperature into account, but the SPI does not.The ability to calculated the SPI and SPEI at various timescales allows for multiple applications in drought monitoring and assessment. Depending on the drought impact in question, SPI values for 3 months or less might be useful for basic drought monitoring, values for 6 months or less for monitoring agricultural impacts and values for 12 months or longer for hydrological impacts.[15]

In Figure B.1.7, the SPI and SPEI calculated at 3, 6 and 12 months time intervals for Kara region in Togo (1976 - 2005) is presented. As observed, the 6 months and 12 months intervals, indicate that the drought frequency and intensity in Kara region were more pronounced in the early parts of the 80's and 90's than the 21st century. However, more very dry (with SPI) and moderate (with SPEI) drought events with values ≥ -1 where mostly recorded in the later parts of the 21st century, notably years 2000 and 2003.

[15]World Meteorological Organization (WMO) and Global Water Partnership (GWP), 2016: *Handbook of Drought Indicators and Indices* (M. Svoboda and B.A. Fuchs). Integrated Drought Management Programme (IDMP), Integrated Drought Management Tools and Guidelines Series 2. Geneva.

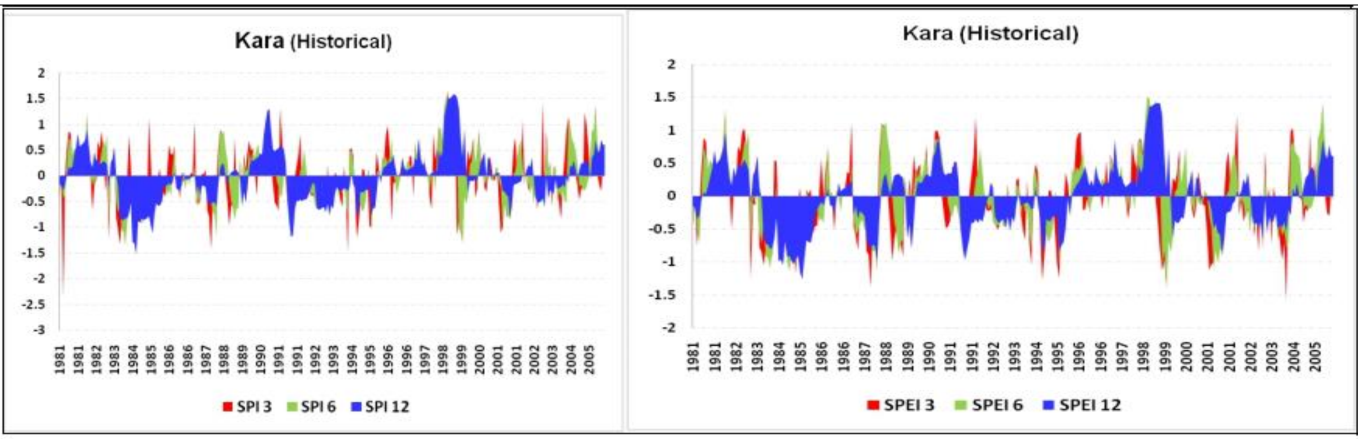



**Figure B.1.7:** The SPI and SPEI time series (for 3, 6 and 12months), Kara Region, Togo.

Also, the SPI and SPEI calculated at 3, 6 and 12 months time intervals for the South Agropole of the Casamance Natural region of Senegal and the Boké and Kankan regions of Guinea (1976 - 2005), are shown in Figure B.1.8. The SPEI captures more drought frequency with longer duration for Senegal than the SPI at 6- and 12-months intervals. Conversely, in the Boké and Kankan regions of Guinea, both graphs exhibit very similar shape, but with different duration and intensity. While SPEI measure of drought for 12 months' time scale, shows larger duration and intensity from 1987 to 2000, it was smaller and milder with the SPI during this period.

The general takeaway from the SPI and SPEI analysis so far presented above is that, drought is a recurrent drought events in the targeted areas. And also, that, the agricultural and water sectors are the two sectors with the greatest drought impacts.

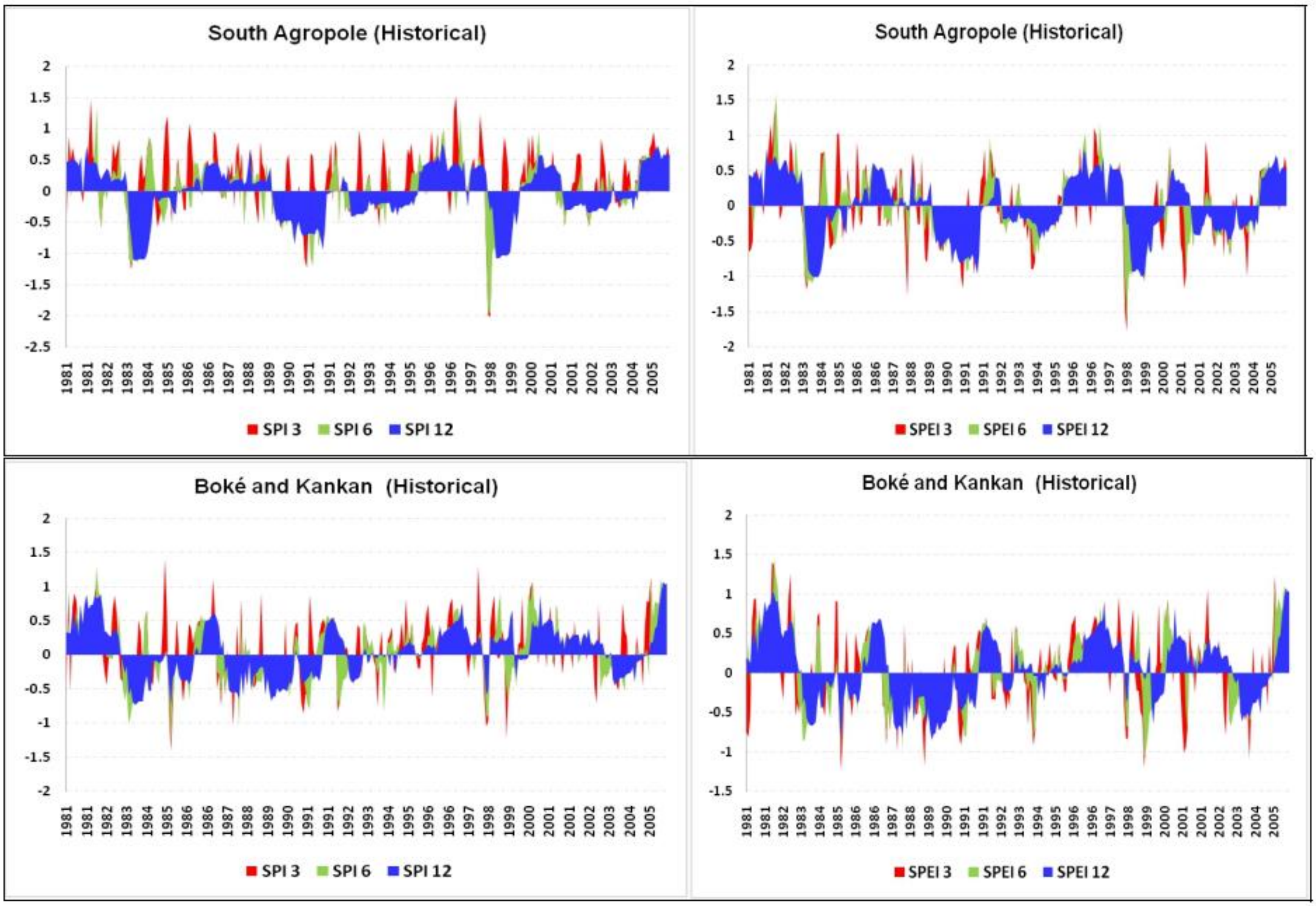



**Figure B.1.8.** The SPI and SPEI time series (for 3, 6 and 12months), SCPZ areas in Senegal and Guinea.

**Projections:** To further understand the potential physical risk associated with future climate, projections from downscaled global climate model projections are used to provide a more refined regional and country specific information especially for extremes that are particularly relevant for agriculture. The assessment of future climate change are based on the projections produced from ensemble mean (ENS) of three RCMs under the representative concentration pathways (RCP4.5) emission scenario. It is quite worthy of note that this ENS presents only one of several possible future outcomes that will manifest as a result of GHG emissions trajectory and other socioeconomic situations.

**Change in mean climate:** Figure B.1.9 shows projected change in annual total precipitation rate averaged for the climate periods 2011-2040 (current-near future), 2041-2070 (mid-future) and 2071-2100(far future) under RCP4.5 emission scenario. Statistically significant drying and wet conditions reaching 300 mm/year are expected in the three countries. For instance, drying conditions in the central region of Boké in Guinea is likely to further intensify significantly in the mid future but less in the far future. In Kankan, a pattern of high and low precipitation is expected in the southern and northern parts, respectively, of the region. A general dryness is expected to dominate most parts of Senegal and Togo, especially in the Casamance and Kara regions of the corresponding countries.

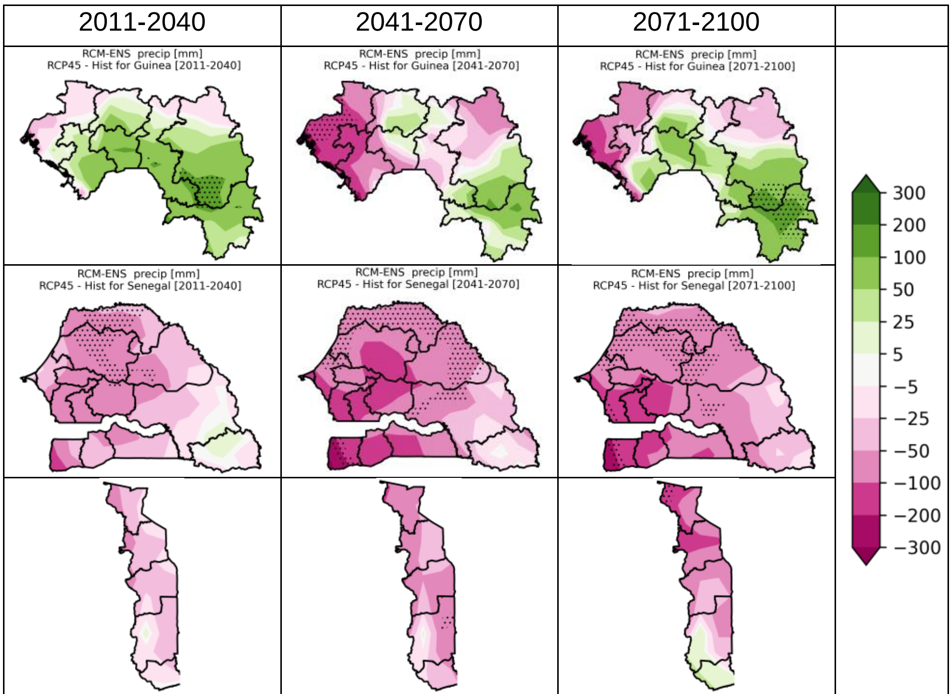
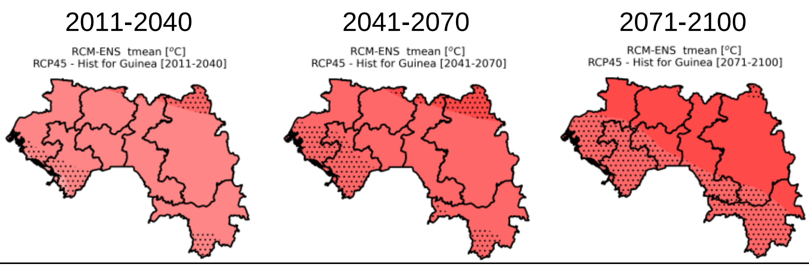



**Figure B.1.9.** Projected change in mean annual total precipitation (mm) relative to the baseline (1976-2005) for the periods 2011-2040 (current-near future), 2041-2070 (mid-future) and 2071-2100 (far future) under the RCP4.5 emission scenario. Dotted areas indicate regions where the change is statistically[16] significant at 0.05 significance level.

Climate model projects a general increase in surface temperature for all three climate periods: current-near; mid-; and far future under RCP4.5 emission scenario (Figure B.1.10). The projected warming is estimated at about 0.8°C, 1.2°C and 2.0°C for current-near, mid- and far future, respectively. To be precise, significant warming of about 1.2°C is expected to increase further in the mid- and far future. Warming in Boké and Kankan in Guinea is about 0.8°C in the current-near future climate. A change in temperature of about 2°C is expected in the mid- and far future with a high level of significance in the southern and northern areas of Boké and Kankan, respectively. Due to the effect of the Atlantic Ocean, there is a west to east increase gradient of temperature change in Senegal. Temperature change from the central to the eastern part of the country is statistically significant and could reach a value of 2.4°C. Although not statistically significant, temperatures in the Casamance region could attain values of 2°C in the future. For Kara region in Togo, larger temperature difference (greater than 2.4°C) is expected in the extreme north. The projected intense warming is a potential threat to food security as it could present far reaching implications for crops growth and productivity including livestock farming. This is particularly worrisome for these countries where the ability to prepare and plan for eminent disasters are still very low as measured by their climate vulnerability readiness index (Table B.1.3).




[16]The current and future climate conditions are compared using a two sided student t-test considering unequal variance at 0.05 significance level to determine if the mean values are significantly different between the two climates.

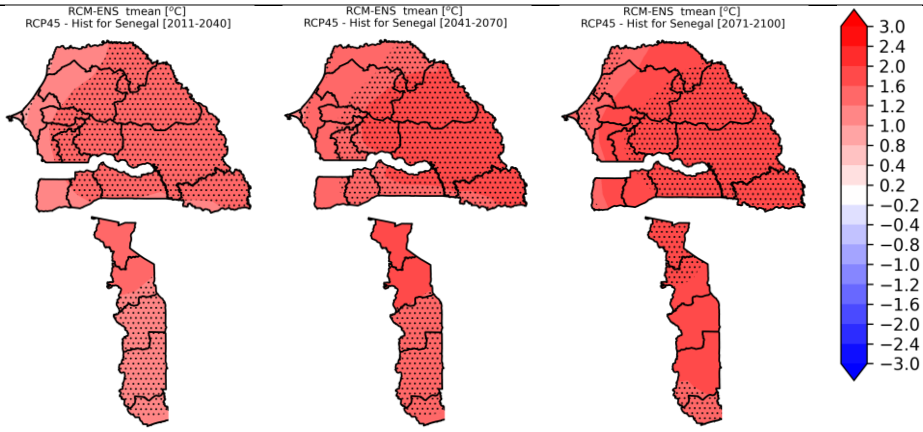



**Figure B.1.10.** Projected change in mean surface temperature (°C) relative to the baseline (1976-2005) for the periods 2011-2040 (current-near future), 2041-2070 (mid-future) and 2071-2100 (far future) under the RCP4.5 emission scenario. Dotted areas indicates regions where the change is statistically[17] significant at 0.05 significance level.

**Table B.1.3:** Vulnerability and readiness scores for each country

| **Countries** | **Adaptive Capacity Scoring** | **Readiness Ranking Scoring** |
|---|---|---|
| Togo | 0.521 (Low) | 0.301 (Low) |
| Senegal | 0.533 (Lower Middle) | 0,535 (Lower Middle) |
| Guinea | 0.528 (Low) | 0.301 (Low) |

**Source: ND-GAIN index ranking, 2020.**

**Drought projections:** In Figure B.1.11, the SPI and SPEI calculated at 3-, 6- and 12-months' time intervals for Kara, Casamance, and the Boké and Kankan regions are presented. As expected, the results are not quite different from the baseline results obtained. The frequency and duration of droughts vary between the 6- and 12-months intervals, and between the SPI and the SPEI. Notably, the SPEI predicts more frequent and longer drought periods for all the regions than the SPI. However, contrary to the baseline, the calculated 6- and 12-months drought duration, increases for all the regions. This is consistent with the rising trends in temperature and declining precipitation as projected in Figure B.1.9 and B.1.10. The growing frequency and duration suggest that investing in agriculture must be accompanied by plans to support water efficient use in the regions. This offers greater opportunity for more investment in drip irrigation technologies. It equally calls for need to promote the use of drought-tolerant varieties for seasonal crops in addition to adopting climate-smart agricultural practices for precision farming to ensure food security.

[17] The current and future climate conditions are compared using a two sided student t-test considering unequal variance at 0.05 significance level to determine if the mean values are significantly different between the two climates.

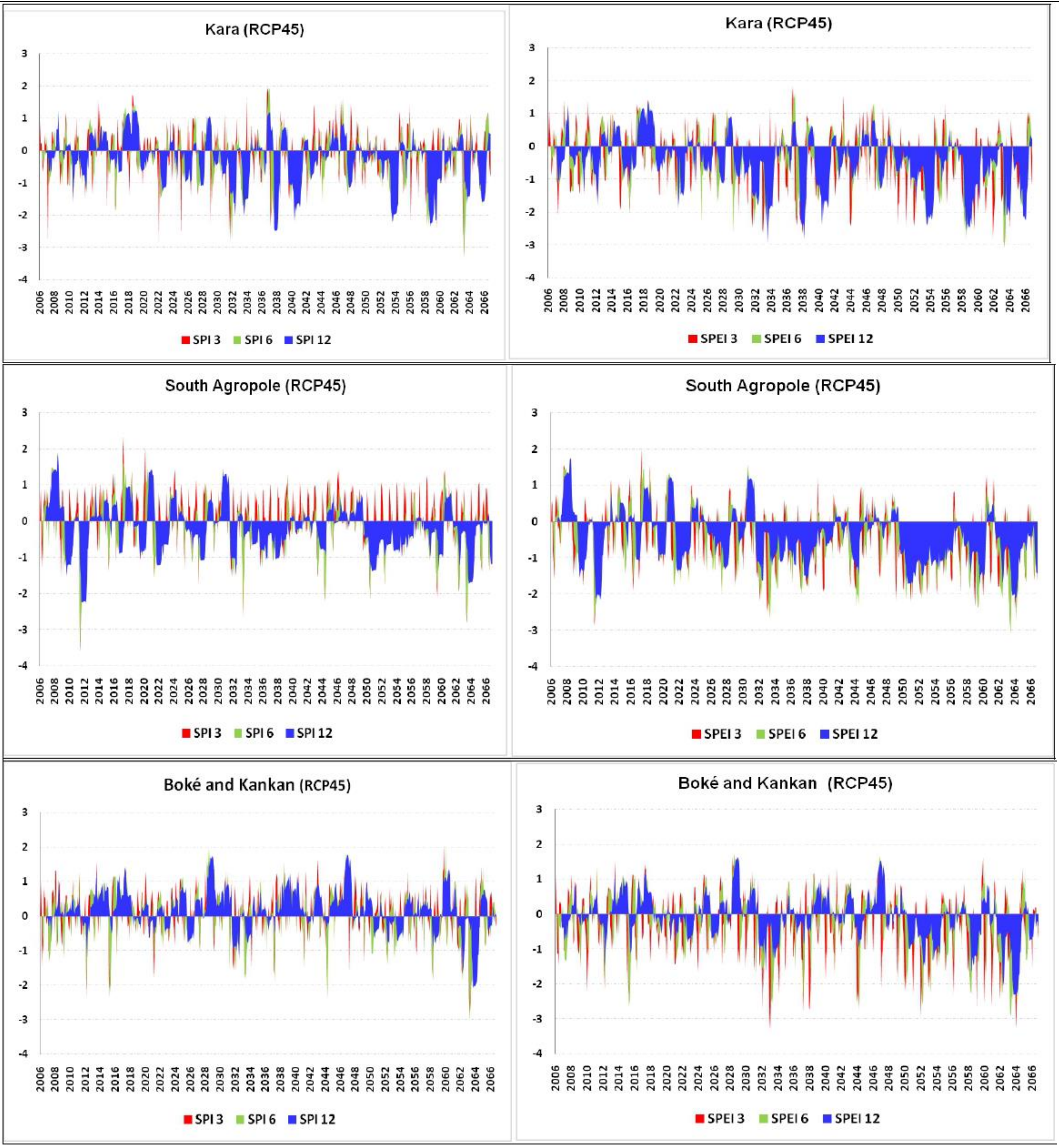



**Figure B.1.11:** The SPI and SPEI time series (for 3, 6 and 12months), Togo, Senegal and Guinea

**Consecutive Dry Days (CDD):** Another useful indicator for assessing climate change impacts on agriculture beside changes in temperature and precipitation, is the maximum number of consecutive dry days (CDD). This is defined as a day without any agriculturally meaningful rainfall, which is generally defined by a threshold of 1 mm/day[18]. This indicator is an important metric for rain-fed agriculture as it directly impacts soil moisture, and thus crop growth. CDD in the three host countries is expected to increase through the end of the 21st Century (Figure B1.12). There are chances of CDD increase by 12 days in Boké and Kankan. The increase is significant in the north eastern parts of Boké. An eastward pattern of significant increase in CDD is possible in the north-west region of Senegal. A significant increase in CDD (i.e great than 17 days) is likely around the eastern parts of

[18]https://climateknowledgeportal.worldbank.org/.

Zinguinchor. While the southern parts of Togo are expected to experiences decrease in CDD, the northern parts including the Kara region will possibly have to cope with increasing CDD. Higher number of CDD with little or no precipitation can lead to drought. When accompanied by higher temperatures, it increases evaporation thereby, adding more stress to limited water resources, with far reaching implications for irrigation and other water uses.

***Figure B.1.1*3 *shows the historic and projected CDD*** for the Kara Region of Togo, Casamance Region of Senegal, and the Boke and Kankan Regions of Guinea for the period 1981 to 2066. The black line indicates historic while the blue and red lines represent trends derived from the emissions scenarios RCP 4.5 and RCP8.5. In addition, the average annual maximum number of consecutive dry days are projected to increase for both emission scenarios. However, the effects is more pronounced in Senegal than Togo and Guinea, where it is projected that the CDD could reach up to 200 days per year for most of the years. This means more drought conditions with implications for food security.

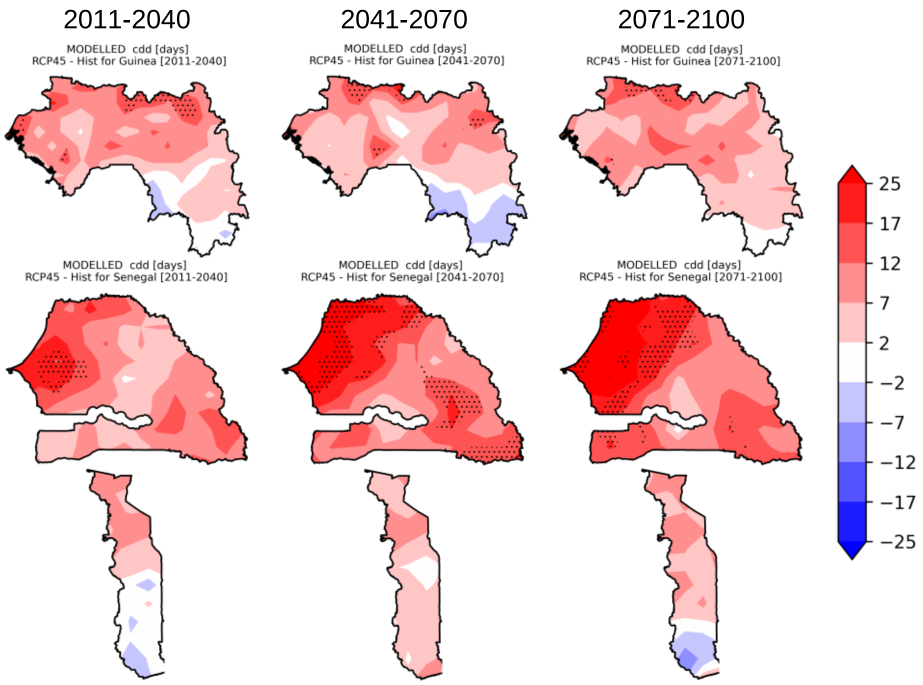



**Figure B.1.12.** Projected change in consecutive dry days (CDD) (days) relative to the baseline (1976-2005) for the periods 2011-2040 (current-near future), 2041-2070 (mid-future) and 2071-2100 (far future) under the RCP4.5 emission scenario. Dotted areas indicates regions where the change is statistically significant at 0.05 significance level.

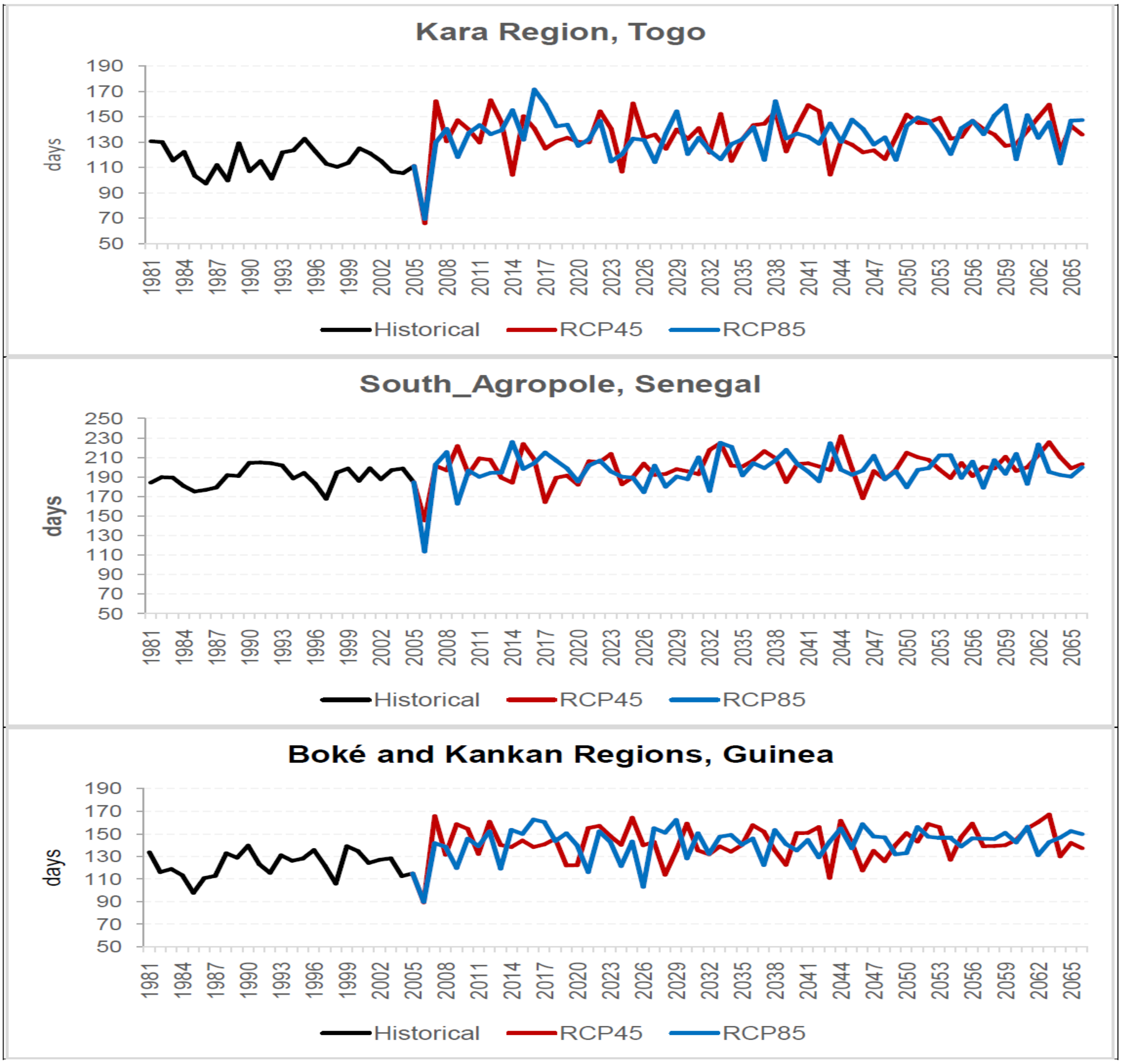



**Figure B.1.13:** *Historic and projected maximum number of consecutive dry days (CDD) in the Kara Region of Togo, Casamance Region of Senegal, and the Boke and Kankan Regions of Guinea, for the period 1981 to 2066.The black line indicates historic while the blue and red lines represent trends derived from the emissions scenarios RCP 4.5 and RCP8.5.*

**Length of Growing Period (LGP):** The Length of growing period (LGP) is defined as the period during the year when average temperatures are greater than or equal to 5$^{O}$C (Tmean>=5$^{O}$C) and precipitation plus moisture stored in the soil exceed half the potential evapotranspiration (P>0.5PET). It is an important indicator for crop forecasting, especially when planning agricultural programmes in the mist of changing climatic conditions.[19] A normal growing period is defined as one when there is an excess of precipitation over pet (i.e. a humid period). Such a period meets the full evapotranspiration demands of crops and replenishes the moisture definite of the soil profile. An intermediate growing period is defined as one in which precipitation does not normally exceed PET but does for part of the year. No growing period is when temperatures are not conducive to crop growth or P never exceeds PET.

[19] CLIMPAG: Climate Impact on Agriculture | ADVICE AND WARNINGS | Agrometeorological Crop Forecasting (fao.org)

**Table B.1.4:** Range of uncertainty between climate projections and historical (Observed vs. Model) for the LGP

| Station | Rainfall Onset (Days) | | Rainfal Cessation (Days) | | LGP (Days) | | Difference (Days) from Historical | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | LGP | | Onset | Onset | Cessa tion | Cessa tion |
| | (1) | (2) | (3) | (4) | RCP 4.5 | RCP 8.5 | RCP 4.5 | RCP 8.5 | (5) | (6) | (7) | (8) |
| **KARA** | | | | | | | | | | | | |
| *Historical (1981 - 2005)* | 118 | 109 | 304 | 309 | **186** | **200** | **-** | **-** | - | - | - | - |
| *Projections (2006 -2035)* | 106 | 105 | 310 | 310 | **204** | **205** | **18** | **5** | 12 | 4 | -6 | -1 |
| *Projections (2036 -2065)* | 110 | 109 | 312 | 314 | **202** | **205** | **16** | **5** | 8 | 0 | -8 | -5 |
| **SOUTH AGROPOLE** | | | | | | | | | | | | |
| *Historical (1981 - 2005)* | 171 | 175 | 285 | 302 | **114** | **127** | **-** | **-** | - | - | - | - |
| *Projs. RCPs (2006 -2035)* | 179 | 178 | 305 | 297 | **126** | **119** | **12** | **-8** | -8 | -3 | -20 | 5 |
| *Projs. RCPs (2036 -2065)* | 191 | 192 | 296 | 300 | **105** | **108** | **-9** | **-19** | -20 | -17 | -11 | 2 |
| **KANKAN** | | | | | | | | | | | | |
| *Historical (1981 - 2005)* | 127 | 142 | 308 | 307 | **181** | **165** | **-** | **-** | - | - | - | - |
| *Projs. RCPs (2006 -2035)* | 137 | 135 | 309 | 308 | **172** | **173** | **-9** | **8** | -10 | 7 | -1 | -1 |
| *Projs. RCPs (2036 -2065)* | 141 | 140 | 310 | 310 | **169** | **170** | **-12** | **5** | -14 | 2 | -2 | -3 |
| **BOKE** | | | | | | | | | | | | |
| *Historical (1981 - 2005)* | 137 | 138 | 326 | 325 | **189** | **187** | **-** | **-** | - | - | - | - |
| *Projs. RCPs (2006 -2035)* | 134 | 136 | 321 | 321 | **187** | **185** | **-2** | **-2** | 3 | 2 | 5 | 4 |
| *Projs. RCPs (2036 -2065)* | 139 | 140 | 324 | 325 | **185** | **185** | **-4** | **-2** | -2 | -2 | 2 | 0 |

*[a] The differences presented in rows (5) to (8) are relative to the historical period (1981 - 2005) for the two emission scenarios RCP 4.5 and RCP 8.5. And except for the historical period where (1) and (2) are observed onsets and (2) & (3) are model cessations, the projections are based on the two emission scenarios RCP 4.5 and RCP 8.5.Note that: ( - ) values show late onset or cessation while (+) values depict early onset or cessation for the two emission scenarios RCP 4.5 and RCP 8.5.*

In Figure B.1.14a, the average length of growing period over the Kara region for the periods (a) 1981-2005 (top), (b) 2006-2035 (middle) and (C) 2036-2065 (bottom), during historical climate and under different emission scenarios (i.e. RCP4.5 and RCP8.5) is presented. Again, the major feature of the observations is a slight shift in the rainfall cessation outwards in both the (b) 2006-2035 (middle) and (C) 2036-2065 (bottom) periods, leading to a longer growing period as depicted in Table B.1.4. This is consistent with the projections reported in Figure B.1.10, which predicts early onset in April and with a cessation period in early October. This justifies the use of drip irrigation techniques in agriculture coupled with the adoption of climate resilient agricultural (CRA) practices for precision farming to ensure food security in the country.

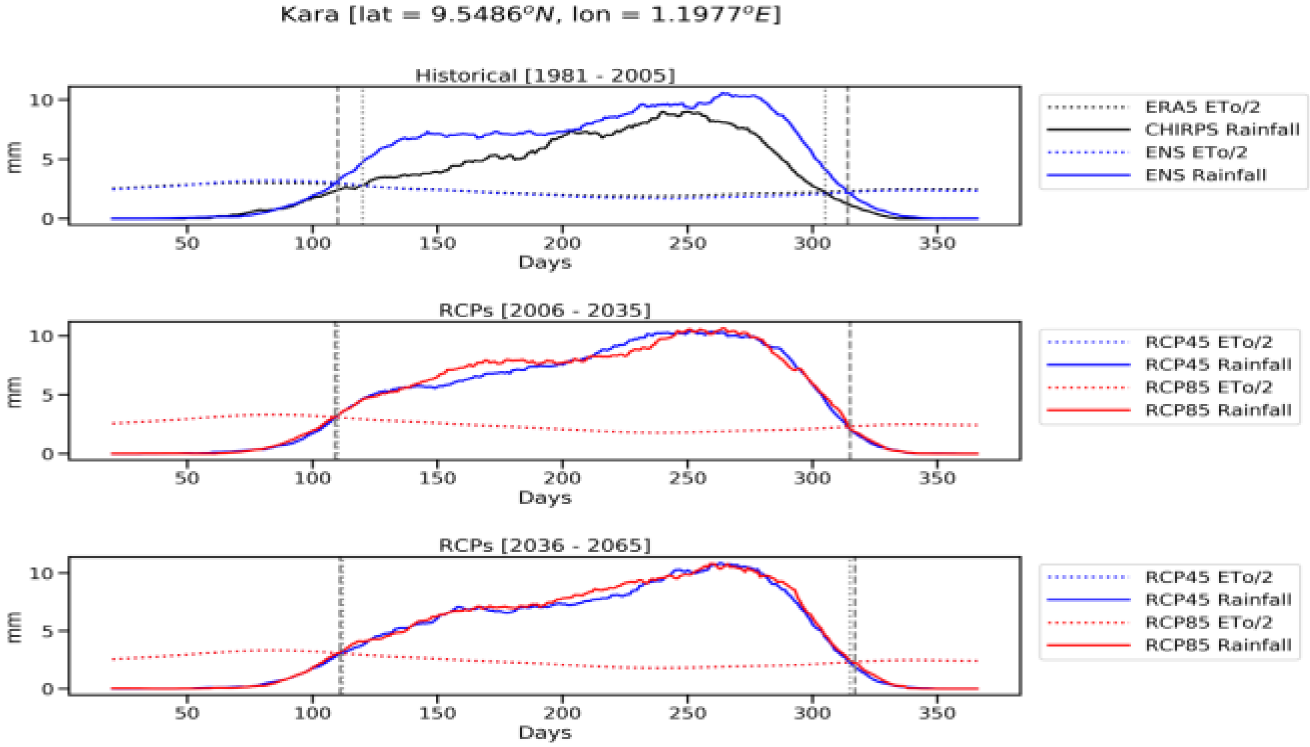



**Figure B.1.14a.** Average length of growing period over the Kara Region for the periods (a) 1981-2005 (top), (b) 2006-2035 (middle) and (C) 2036-2065 (bottom), under different emission scenarios (i.e. RCP4.5 and RCP8.5). **The continues lines indicate rainfall and the dotted line is 50% (half) of the evapotranspiration (ETo). Both lines are 20days running mean values. The vertical dotted and dashed lines mark the point of intersection between rainfall and half of ETo for observation (and RCP4.5) and model (and RCP8.5), respectively. (NOTE: The ETo is computed from minimum and maximum surface temperature following the Hargreaves method as described in the FAO56 documentation (http://www.fao.org/3/X0490E/x0490e08.htm#eto%20calculated%20with%20different%20time%20steps.)**

In the Casamance region of Senegal (Figure B.1.14b), during the historical period, the observed onset of rainfall is earlier compared to the modelled onset, but the cessation is late in the ENS model runs than the observed. This is consistent with the projected scenario presented in Figure B.1.9. For the period 2006-2035, rainfall cessation in RCP8.5 is early compared to RCP4.5 therefore making the former scenario to shorter growing period of about 7 days. There is, however, no significant difference in the growing period between RCP4.5 and RCP8.5 in the period 2036-2065, although the growing length is shorter compared to the historical and 2006-2035 periods by 9 days and 19days respectively (Table B.1.4).

Figure B.1.14c shows the average length of growing period over Kankan for the periods (a) 1981-2005 (top), (b) 2006-2035 (middle) and (C) 2036-2065 (bottom), under different emission scenarios (i.e. RCP4.5 and RCP8.5). As noted earlier, the time between the first and second intersection defines the growing period. During the historical period, the observed onset of rainfall is earlier compared to the modelled onset but the cessation is early in the model. For the period 2006-2035, onset in RCP8.5 is early compared to RCP4.5 therefore making the former scenario to longer growing period (about 16days). While the cessation is almost the same in both scenarios, the growing period is shorter relative to the historical period as illustrated in Table B.1.4. There is, however, no significant difference in the growing period between RCP4.5 and RCP8.5 in the period 2036-2065, although the growing length is shorter compared to the historical and 2006-2035 periods as shown in Table B.1.4.

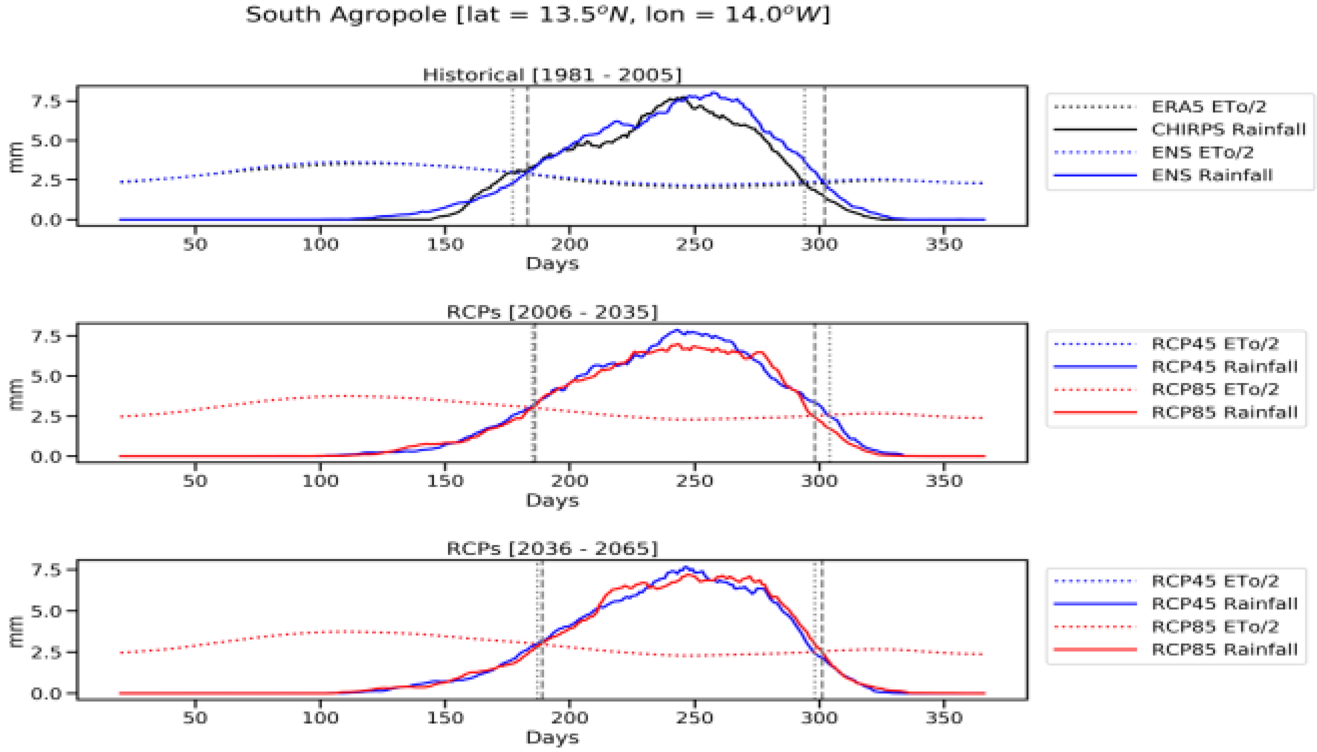



**Figure B.1.14b.** Average length of growing period over the Casamance Region of Senegal for the periods (a) 1981-2005 (top), (b) 2006-2035 (middle) and (C) 2036-2065 (bottom), under different emission scenarios (i.e. RCP4.5 and RCP8.5).

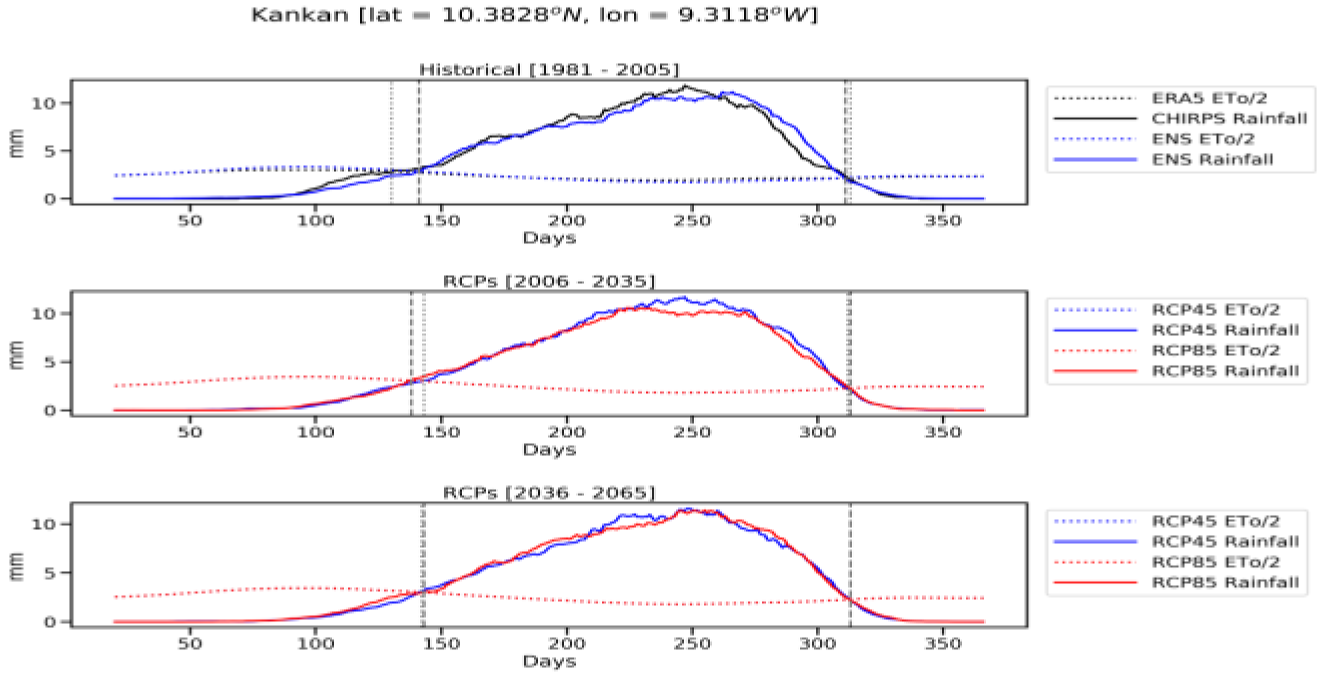



**Figure B.1.14c:** Average length of growing period over Kankan for the periods (a) 1981-2005 (top), (b) 2006-2035 (middle) and (C) 2036-2065 (bottom) during historical climate and under different emission scenarios (i.e. RCP4.5 and RCP8.5).

Lastly, the average length of growing period over Boke for the periods (a) 1981-2005 (top), (b) 2006-2035 (middle) and (C) 2036-2065 (bottom) during historical climate and under different emission scenarios (i.e. RCP4.5 and RCP8.5) is presented in Figure B.1.14d. Contrary to Figure B.1.16, during the historical period, the observed onset of rainfall is early compared to the modelled. For the period 2006-2035, onset in RCP8.5 is early compared to RCP4.5 therefore making the former scenario to longer growing period (about 2days). While the cessation is almost the same in both scenarios, the growing period is shorter relative to the historical period. There is, however, no significant difference in the growing period between RCP4.5 and RCP8.5 in the period 2036-2065, although the growing length is shorter compared to the historical and 2006-2035 periods by 4 days for the RCP 4.5 scenario and about 2 days for the RCP 8.5 scenario as illustrated in Table B.1.4.

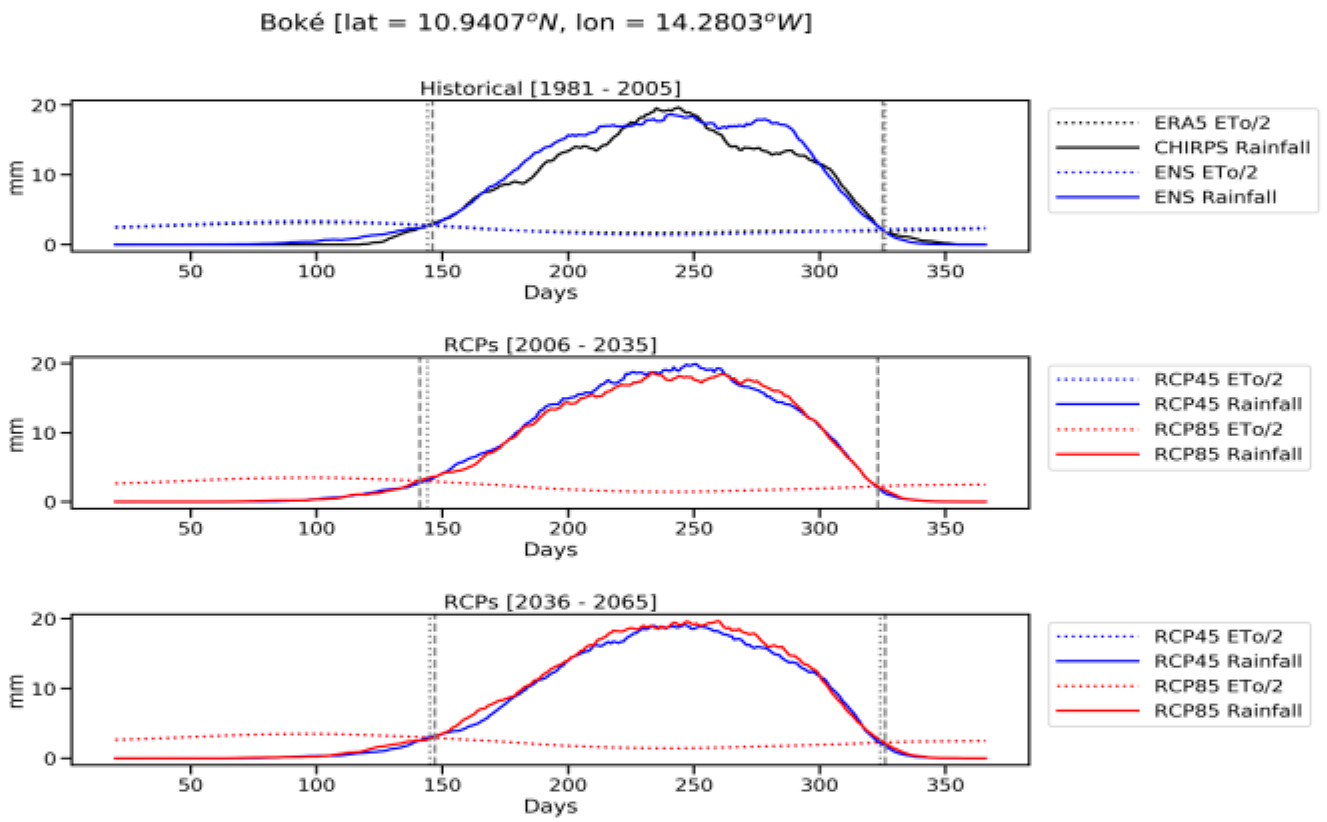



**Figure B.1.14d.** Average length of growing period over Kankan for the periods (a) 1981-2005 (top), (b) 2006-2035 (middle) and (C) 2036-2065 (bottom) during historical climate and under different emission scenarios (i.e. RCP4.5 and RCP8.5).

**Crop Water Requirements (CWR):** Information on crop water requirement of crops is vital for irrigation water planning in agriculture, especially in periods of water shortage and scarcity exacerbated by climate change and its variability. The water need of a crop is usually expressed in mm/day, mm/month or mm/season. Based on the different climatic variabilities of the regions, it is evident that one crop grown in different climatic zones will have different water needs, which is directly corrected to yields. For instance, a certain maize variety grown in a cool climate will need less water per day than the same maize variety grown in a hotter climate. Even in the absence of measured data on pan evaporation, the Blaney-Criddle method as recommended by FAO[20] can be used to calculate the reference crop evapotranspiration ETo (daily water needs of the reference grass crop), sometimes referred to as potential evapotranspiration (PET). This can then be used to compute the ETo in different climatic conditions. With the standardized reference crop evapotranspiration ETo computed, calculations of the crop water requirement ETcrop is relatively straight forward. It is simply calculated by multiplying the crop factor (kc)[21], by the reference crop evapotranspiration, ETo. For more details on this approach, see pages 103 -114 of the programme feasibility study report (PFS).

In Figures B.1.15a - B.1.15c, we present the computed differences between the approximate value (in mm/total season) of crop water need for major staple crops grown in the three pilot countries. The project areas are grouped under the following climatic conditions: Semi-Arid for Togo; Humid and Semi-Arid for Senegal; and Humid/Sub-humid for Guinea. Two categories of crops were used in the computations, that is, long cycle growing crops with longer growing days denoted by (LGC), and more drought tolerant crops with shorter growing days, denoted by (SGC). For each crop, the approximate crop water need (CWR) in mm per total season was estimated for the historic period (1976 - 2005), and the future (2006 - 2045) with the ENS model from the 3 RCMs (CCLM, SHMI and REMO), using the RCP 4.5 scenario. The figures presented below show the approximate crop water need for each crop in in mm per total growing season for the future period minus the baseline period.
The major feature of the observations is a wide disparity in the approximate crop water requirement between the LGC crops and the SGC crops under the present and future scenarios. This disparity is present for all the countries and regions, but being more intense for Senegal and Togo than the other countries. Another interesting feature of the observations is that, as these countries continue to experience global warming in the mist of declining precipitation, LGC crops will even need more water than the SGC crops: suggesting that the deployment of crops with LGC in these countries would not be profitable in terms of water needs in

[20]http://www.fao.org/3/u3160e/u3160e04.htm.

[21] Since the crop factor (or "crop coefficient") varies according to the growth stage of a given crop, seasonal average has been widely used.

the future. This again, calls for careful and meticulous planning in the introduction of seed varieties in the programme, including the adoption of more efficient water management and conservation techniques such as drip irrigation. It also calls for the adoption of good climate resilient agricultural (CRA) management practices such as mixed agro-forestry management, improved nutrient recycling, and improve genetic resources among many other techniques. These CRA management practices have be shown to not only improving the yields of crops and soil health, but also responsible for reducing green house gases (GHGs). In all three countries, agriculture and forestry are the main culprits of GHG emissions.

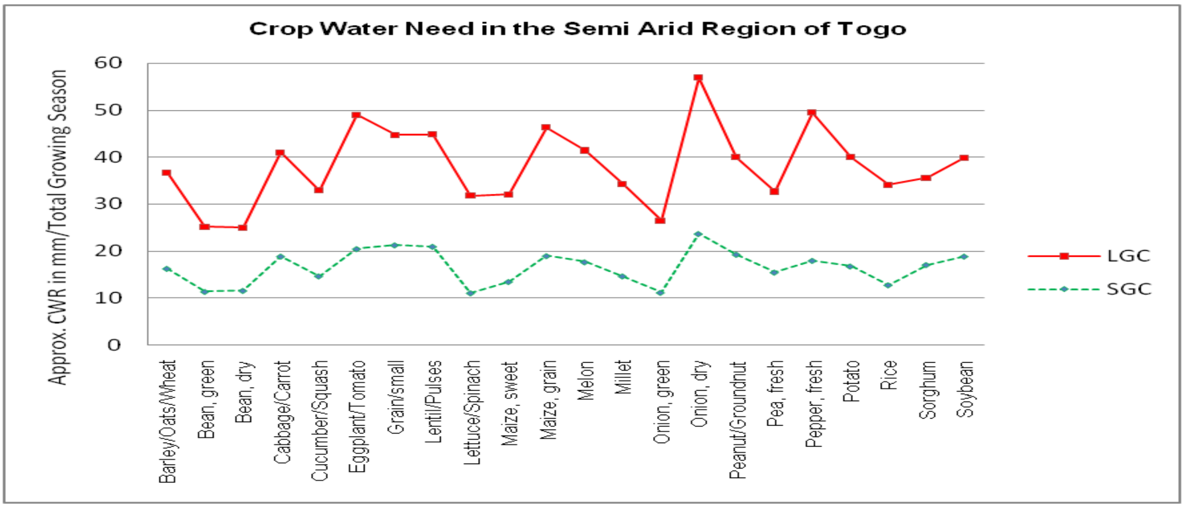



**Figure B.1.15a: Diffirence in crop water need for LGC and SGC in the Semi-Arid Region of Togo.**

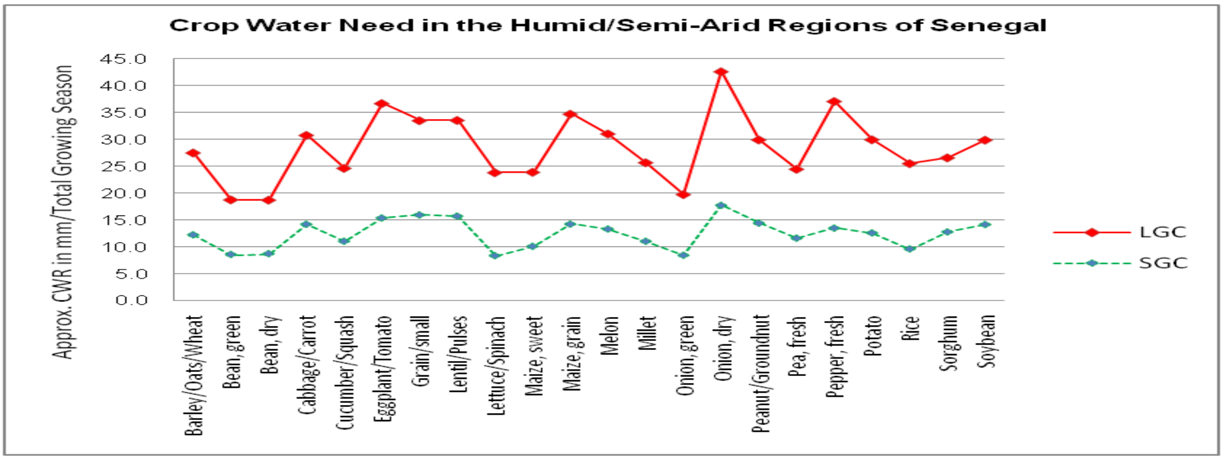



**Figure B.1.15b:** Difference in crop water need for LGC and SGC in the Humid/Semi-Arid Regions in Senegal

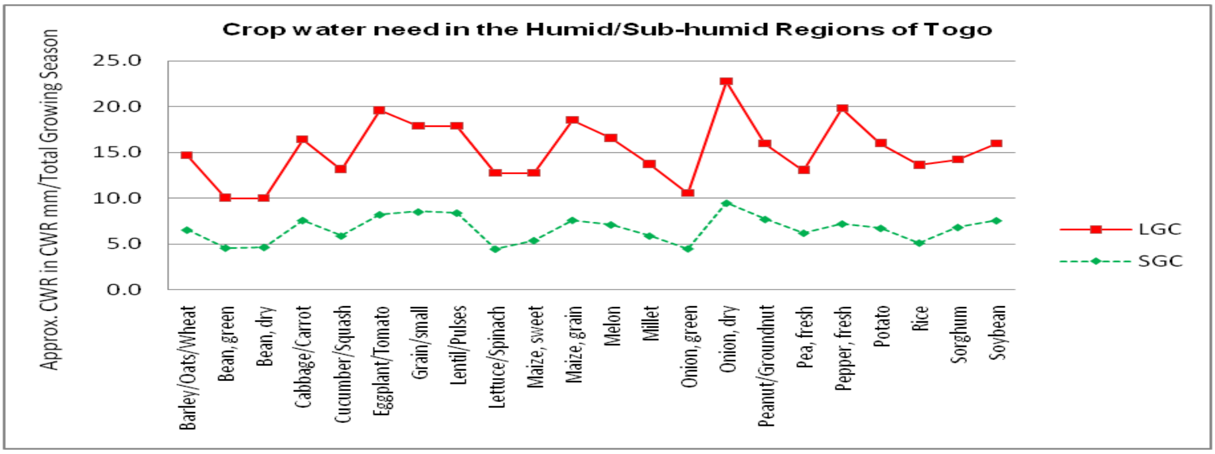



**Figure B.1.15c:** Difference in crop water need for LGC and SGC in the Humid/Sub-humid Regions in Guinea

While Figures B.1.15a to B.1.15c show the difference between the approximate crop water need for each crop (in mm per total growing season) for the future period minus the baseline period, Tables B.1.5a to B.1.5c show the seasonal crop water need and rainfall availability in the past, present and future in these regions for the two types of crops considered (LGC and SGC). This is basically done in order to demonstrate the impacts of climate change and its variability on seasonal crop water need. As observed (Tables B.1.5a-c), the water requirement for each crop increases steadily for each period with significant differences between the traditional LGC crops and the more drought tolerant SGC crops. With substantial evidence of declining precipitations coupled with late rainfall onset and early cessation (Figures B.1.14a - d), causing a shift in the crop calendar, subsistence agriculture is at great risks in these countries and region. This invariably calls for the use of precision farming in agriculture that requires less water consumption at every stage in the crop seasons such as drip irrigation. It equally warrants the introduction of proven climate resilient agricultural techniques that can be deploy alongside drip irrigation. The groundwater and surface water resources potentials for irrigation at the project-specific regions are presented in Tables B.1.7a and B.1.7b.

**Table B.1.5a:** Approximate range of values of seasonal crop water requirements in the Humid zone of Togo

| | | | | | | Short Cycle (mm/total growing period) | | Long Cycle (mm/total growing period) | |
|---|---|---|---|---|---|---|---|---|---|
| | kc Factor | LGS (Short) | LGS (Long) | ETo (Historic) | ETo (Future) | Historic (1976- 2005) | Future (2006-2045) | Historic (1976 -2005) | Future (2006-2045) |
| Barley/Oats/Wheat | 0.68 | 120 | 150 | 5.1 | 5.3 | 416.2 | 432.5 | 520.2 | 540.6 |
| Bean, green | 0.76 | 75 | 90 | 5.1 | 5.3 | 290.7 | 302.1 | 348.8 | 362.5 |
| Bean, dry | 0.61 | 95 | 110 | 5.1 | 5.3 | 295.5 | 307.1 | 342.2 | 355.6 |
| Cabbage/Carrot | 0.79 | 120 | 140 | 5.1 | 5.3 | 483.5 | 502.4 | 564.1 | 586.2 |
| Cucumber/Squash | 0.7 | 105 | 130 | 5.1 | 5.3 | 374.9 | 389.6 | 464.1 | 482.3 |
| Eggplant/Tomato | 0.79 | 130 | 180 | 5.1 | 5.3 | 523.8 | 544.3 | 725.2 | 753.7 |
| Grain/small | 0.71 | 150 | 165 | 5.1 | 5.3 | 543.2 | 564.5 | 597.5 | 620.9 |
| Lentil/Pulses | 0.7 | 150 | 170 | 5.1 | 5.3 | 535.5 | 556.5 | 606.9 | 630.7 |
| Lettuce/Spinach | 0.74 | 75 | 140 | 5.1 | 5.3 | 283.1 | 294.2 | 528.4 | 549.1 |
| Maize, sweet | 0.84 | 80 | 110 | 5.1 | 5.3 | 342.7 | 356.2 | 471.2 | 489.7 |
| Maize, grain | 0.76 | 125 | 180 | 5.1 | 5.3 | 484.5 | 503.5 | 697.7 | 725.0 |
| Melon | 0.74 | 120 | 160 | 5.1 | 5.3 | 452.9 | 470.6 | 603.8 | 627.5 |
| Millet | 0.7 | 105 | 140 | 5.1 | 5.3 | 374.9 | 389.6 | 499.8 | 519.4 |
| Onion, green | 0.8 | 70 | 95 | 5.1 | 5.3 | 285.6 | 296.8 | 387.6 | 402.8 |
| Onion, dry | 0.79 | 150 | 210 | 5.1 | 5.3 | 604.4 | 628.1 | 846.1 | 879.3 |
| Peanut/Groundnut | 0.74 | 130 | 140 | 5.1 | 5.3 | 490.6 | 509.9 | 528.4 | 549.1 |
| Pea, fresh | 0.86 | 90 | 100 | 5.1 | 5.3 | 394.7 | 410.2 | 438.6 | 455.8 |
| Pepper, fresh | 0.75 | 120 | 210 | 5.1 | 5.3 | 459.0 | 477.0 | 803.3 | 834.8 |
| Potato | 0.8 | 105 | 145 | 5.1 | 5.3 | 428.4 | 445.2 | 591.6 | 614.8 |
| Rice | 0.71 | 90 | 150 | 5.1 | 5.3 | 325.9 | 338.7 | 543.2 | 564.5 |
| Sorghum | 0.71 | 120 | 130 | 5.1 | 5.3 | 434.5 | 451.6 | 470.7 | 489.2 |
| Soybean | 0.7 | 135 | 150 | 5.1 | 5.3 | 482.0 | 500.9 | 535.5 | 556.5 |

**Table B.1.5b:** Approximate Range of Values of Seasonal Crop Water Need in the Humid/Semi-AridZones of Casamance, Senegal

| | | | | | | Short Groiwing Cycle (mm/total growing period) | | Long Growing Cycle (mm/total growing period) | |
|---|---|---|---|---|---|---|---|---|---|
| | kc Factor | LGS (Short) | LGS (Long) | ETo (Historic) | ETo (Future) | Historic (1976- 2005) | Future (2006-2045) | Historic (1976 -2005) | Future (2006-2045) |
| Barley/Oats/Wheat | 0.68 | 120 | 150 | 8.2 | 8.3 | 669.1 | 677.3 | 836.4 | 846.6 |
| Bean, green | 0.76 | 75 | 90 | 8.2 | 8.3 | 467.4 | 473.1 | 560.9 | 567.7 |
| Bean, dry | 0.61 | 95 | 110 | 8.2 | 8.3 | 475.2 | 481.0 | 550.2 | 556.9 |
| Cabbage/Carrot | 0.79 | 120 | 140 | 8.2 | 8.3 | 777.4 | 786.8 | 906.9 | 918.0 |
| Cucumber/Squash | 0.70 | 105 | 130 | 8.2 | 8.3 | 602.7 | 610.1 | 746.2 | 755.3 |
| Eggplant/Tomato | 0.79 | 130 | 180 | 8.2 | 8.3 | 842.1 | 852.4 | 1166.0 | 1180.3 |
| Grain/small | 0.71 | 150 | 165 | 8.2 | 8.3 | 873.3 | 884.0 | 960.6 | 972.3 |
| Lentil/Pulses | 0.70 | 150 | 170 | 8.2 | 8.3 | 861.0 | 871.5 | 975.8 | 987.7 |
| Lettuce/Spinach | 0.74 | 75 | 140 | 8.2 | 8.3 | 455.1 | 460.7 | 849.5 | 859.9 |
| Maize, sweet | 0.84 | 80 | 110 | 8.2 | 8.3 | 551.0 | 557.8 | 757.7 | 766.9 |
| Maize, grain | 0.76 | 125 | 180 | 8.2 | 8.3 | 779.0 | 788.5 | 1121.8 | 1135.4 |
| Melon | 0.74 | 120 | 160 | 8.2 | 8.3 | 728.2 | 737.0 | 970.9 | 982.7 |
| Millet | 0.70 | 105 | 140 | 8.2 | 8.3 | 602.7 | 610.1 | 803.6 | 813.4 |
| Onion, green | 0.80 | 70 | 95 | 8.2 | 8.3 | 459.2 | 464.8 | 623.2 | 630.8 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Onion, dry | 0.79 | 150 | 210 | 8.2 | 8.3 | 971.7 | 983.6 | 1360.4 | 1377.0 |
| Peanut/Groundnut | 0.74 | 130 | 140 | 8.2 | 8.3 | 788.8 | 798.5 | 849.5 | 859.9 |
| Pea, fresh | 0.86 | 90 | 100 | 8.2 | 8.3 | 634.7 | 642.4 | 705.2 | 713.8 |
| Pepper, fresh | 0.75 | 120 | 210 | 8.2 | 8.3 | 738.0 | 747.0 | 1291.5 | 1307.3 |
| Potato | 0.80 | 105 | 145 | 8.2 | 8.3 | 688.8 | 697.2 | 951.2 | 962.8 |
| Rice | 0.71 | 90 | 150 | 8.2 | 8.3 | 524.0 | 530.4 | 873.3 | 884.0 |
| Sorghum | 0.71 | 120 | 130 | 8.2 | 8.3 | 698.6 | 707.2 | 756.9 | 766.1 |
| Soybean | 0.70 | 135 | 150 | 8.2 | 8.3 | 774.9 | 784.4 | 861.0 | 871.5 |

**Table B.1.5c:** Approximate Range of Values of Seasonal Crop Water Requirements in the Humid/Sub-Humid Regions of Guinea

| | | | | | | Short Growing Cycle (mm/total growing period) | | Long Growing Cycle (mm/total growing period) | |
|---|---|---|---|---|---|---|---|---|---|
| | kc Factor | LGS (Short) | LGS (Long) | ETo (Historic) | ETo (Future) | Historic (1976- 2005) | Future (2006-2045) | Historic (1976 -2005) | Future (2006-2045) |
| Barley/Oats/Wheat | 0.68 | 120 | 150 | 5.1 | 5.18 | 416.2 | 422.7 | 520.2 | 528.4 |
| Bean, green | 0.76 | 75 | 90 | 5.1 | 5.18 | 290.7 | 295.3 | 348.8 | 354.3 |
| Bean, dry | 0.61 | 95 | 110 | 5.1 | 5.18 | 295.5 | 300.2 | 342.2 | 347.6 |
| Cabbage/Carrot | 0.79 | 120 | 140 | 5.1 | 5.18 | 483.5 | 491.1 | 564.1 | 572.9 |
| Cucumber/Squash | 0.70 | 105 | 130 | 5.1 | 5.18 | 374.9 | 380.7 | 464.1 | 471.4 |
| Eggplant/Tomato | 0.79 | 130 | 180 | 5.1 | 5.18 | 523.8 | 532.0 | 725.2 | 736.6 |
| Grain/small | 0.71 | 150 | 165 | 5.1 | 5.18 | 543.2 | 551.7 | 597.5 | 606.8 |
| Lentil/Pulses | 0.70 | 150 | 170 | 5.1 | 5.18 | 535.5 | 543.9 | 606.9 | 616.4 |
| Lettuce/Spinach | 0.74 | 75 | 140 | 5.1 | 5.18 | 283.1 | 287.5 | 528.4 | 536.6 |
| Maize, sweet | 0.84 | 80 | 110 | 5.1 | 5.18 | 342.7 | 348.1 | 471.2 | 478.6 |
| Maize, grain | 0.76 | 125 | 180 | 5.1 | 5.18 | 484.5 | 492.1 | 697.7 | 708.6 |
| Melon | 0.74 | 120 | 160 | 5.1 | 5.18 | 452.9 | 460.0 | 603.8 | 613.3 |
| Millet | 0.70 | 105 | 140 | 5.1 | 5.18 | 374.9 | 380.7 | 499.8 | 507.6 |
| Onion, green | 0.80 | 70 | 95 | 5.1 | 5.18 | 285.6 | 290.1 | 387.6 | 393.7 |
| Onion, dry | 0.79 | 150 | 210 | 5.1 | 5.18 | 604.4 | 613.8 | 846.1 | 859.4 |
| Peanut/Groundnut | 0.74 | 130 | 140 | 5.1 | 5.18 | 490.6 | 498.3 | 528.4 | 536.6 |
| Pea, fresh | 0.86 | 90 | 100 | 5.1 | 5.18 | 394.7 | 400.9 | 438.6 | 445.5 |
| Pepper, fresh | 0.75 | 120 | 210 | 5.1 | 5.18 | 459.0 | 466.2 | 803.3 | 815.9 |
| Potato | 0.80 | 105 | 145 | 5.1 | 5.18 | 428.4 | 435.1 | 591.6 | 600.9 |
| Rice | 0.71 | 90 | 150 | 5.1 | 5.18 | 325.9 | 331.0 | 543.2 | 551.7 |
| Sorghum | 0.71 | 120 | 130 | 5.1 | 5.18 | 434.5 | 441.3 | 470.7 | 478.1 |
| Soybean | 0.70 | 135 | 150 | 5.1 | 5.18 | 482.0 | 489.5 | 535.5 | 543.9 |

Table B.1.5d shows the projected change in crop suitability for selected crop types for present-day (PD, 2011-2040) and mid-century (MC, 2041-2070) climates relative to the baseline (1981-2010). The crop suitability was derived from the Eco-crop model implemented in R. This model uses the FAO Eco-crop (FAO, 2000) database alongside climatic variables such as temperature and precipitation as inputs to determine the crop suitability index (See Annex 2 for details). The projected change shows that the suitability of wheat, sorghum, maize, and cassava will decline in the SCPZs based on the climate variables under the two scenarios (SSP245 and SSP585). Wheat is expected to be the most affected by climate change, especially in Kara and Kankan, where the change is statistically significant at 0.05 level under the two scenarios, while Sorghum and Maize will likely become less suitable particularly in Kara during the mid-century under SSP585 (Table B.15d). On the other hand, future climate conditions are expected to favor the production of rice, yam, and millet in the SCPZs. Rice mostly shows a general increase across the zones for the different scenarios and climate epochs. The production of yam and millet show almost similar pattern in Kara and Zinguinchor. The suitability of these crops is likely to reduce in other zones especially in Boke and Kankan.

***Table B.1.5c. Projected change in crop suitability for selected crop types in the SCPZs for present-day and mid-century climate relative to the baseline (1981-2010).***

| Crop | Station | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Kara | | | | Zinguinchor | | | | Boke | | | | Kankan | | | |
| | SSP245 | | SSP585 | | SSP245 | | SSP585 | | SSP245 | | SSP5855 | | SSP245 | | SSP585 | |
| | PD | MC | PD | MC | PD | MC | PD | MC | PD | MC | PD | MC | PD | MC | PD | MC |
| Wheat | *-0.054 | *-0.098 | *-0.043 | *-0.106 | -0.001 | -0.001 | -0.001 | -0.001 | -0.009 | -0.009 | -0.009 | -0.009 | *-0.039 | *-0.064 | *-0.036 | *-0.072 |
| Sorghum | -0.008 | -0.01 | -0.014 | *-0.032 | -0.027 | -0.013 | -0.005 | -0.000 | -0.020 | -0.021 | -0.01 | -0.018 | -0.026 | -0.025 | -0.012 | -0.026 |
| Maize | -0.037 | -0.033 | -0.047 | *-0.084 | -0.021 | -0.009 | -0.01 | -0.006 | -0.005 | -0.006 | -0.009 | -0.016 | -0.030 | -0.025 | -0.017 | -0.025 |
| Rice | 0.037 | *0.032 | 0.04 | 0.069 | 0.013 | 0.007 | 0.025 | *0.005 | *0.001 | -0.008 | 0.009 | 0.009 | 0.023 | 0.0186 | 0.021 | 0.016 |
| Millet | 0.01 | -0.001 | 0.003 | -0.005 | -0.002 | -0.005 | 0.005 | *0.013 | -0.016 | -0.015 | -0.008 | *-0.007 | -0.01 | -0.017 | -0.002 | -0.019 |
| Yam | 0.015 | -0.00 | 0.006 | 0.002 | 0.002 | -0.001 | 0.000 | *0.027 | *-0.002 | 0.001 | -0.003 | 0.002 | -0.01 | -0.03 | -0.008 | -0.027 |
| Cassava | -0.001 | -0.008 | -0.008 | *-0.024 | -0.018 | -0.027 | -0.009 | -0.021 | -0.016 | -0.018 | -0.011 | -0.018 | -0.017 | -0.027 | -0.011 | -0.033 |

PC = Present-day climate (2011-2040); MC = mid-century climate (2041-2070)

| **Increase** | **Decrease** | |
|---|---|---|
| | | **Not significant** |
| * | * | **Significant at 0.05 level** |

**Key Sectors at Risk to Climate Change**

The agriculture, biodiversity, health, infrastructure, energy and water sectors of Senegal, Guinea and Togo are highly vulnerable to climate change. The need for adaptation in these countries sectors should be prioritized in the GCF funded activity portfolio. Based on literatures and national publications, this section provides a summary of key natural hazards and their associated socioeconomic impacts in the host country. This gives a useful insight into the most vulnerable areas through the spatial comparison of natural hazard data with development data, thereby identifying exposed livelihoods and natural systems in the SCPZ countries. The summary of climate risk, impacts and likely interventions are described below and provided in Annex 2B.

Senegal[22]

Senegal remains vulnerable to environmental shocks given the recurring incidences of natural disasters, such as, floods and droughts, that threaten the national stability maintained over time. Such type of destabilizing events will increase in magnitude and extent due to increased climate variability thereby disrupting the steady economic growth recorded over the last decade. For instance, the prolonged droughts recorded between 1970 and 2000 worsen the rural-urban migration experienced in the country. Historical climate records show that average surface air temperatures have warmed by 0.9°C, especially in the north and between October and December. This led to a corresponding increase in 'hot' nights (27 days per year) since 1960. Although rains have partially recovered since the mid-1990s, the observed rainfall decrease with slight recovery below the pre-1970 levels and 15 % the long-term average. Rainfall decline is most significant in the southern region during the wet season (June – September). Climate models project a higher warming ranging from 1.1 to 3.1°C, more hot days, highly variable rainfall and sea level increase of about 1 meter. Climate change will also impact climate sensitive sectors such as agriculture (70 percent of production is rainfed), livestock and fisheries, which
account for 20 percent of GDP and employ a majority of the workforce. This negatively impact food security, which is already stressed due to low yields and high population growth. Over 15 percent of rural households and
8 percent of urban households are estimated to be food insecure, and the country national production with imports of approximately 60 percent of
its cereal requirements, mostly rice, the main staple crop in the country.
Climate risks in the water sector include reduced freshwater supply, increased evaporation

[22]USAID, 2017: Senegal's Climate Change Risk Profile, Fact Sheet, https://www.climatelinks.org/sites/default/files/asset/document/2017_USAID%20ATLAS_Climate%20Change%20Risk%20Profile%20-%20Senegal.pdf

of surface water and reduced recharge of groundwater impacting water quality and reduced hydropower production.

Guinea[23]

The climate of Guinea is characterized by highly variable rainfall. Decrease in rainfall over Guinea has offset income flow, influenced agricultural calendar, and altered river regimes. The country also records droughts, which is expected to be the highest climate risk with negative effects on streamflow in rivers between 1961 and 1990. Water scarcity is another implication of reduced streamflow. Also, droughts are projected to contribute to loss of biodiversity, degrade headwaters, increase the proliferation of diseases and plant pests, increase water scarcity, and contribute to more bushfires in the country. Rainfall variability have serious implications on the main economic activity of 80% of Guineans as well as agricultural production and food security. Flood events are among other natural disaster in Guinea that can impact many different aspects of the country's socioeconomic activities. These flood events occur frequently and negatively impact the already poor sewage and water systems as well as sanitation facilities, leading to inadequate disposal of human waste and contributing to the transmission of diseases such as cholera, typhoid fever, malaria, and/or polio. Sea level rise has been occurring along Guinea's coast and will cause increased salinization and flooding in coastal regions, impacting agriculture, shortage in drinking water, destroying infrastructure, destruction of mangrove ecosystems, and proliferation of diseases.

Given these range of climate risk, it is paramount to invest in Improved water management, early warning systems, and saltwater tolerant crop varieties to reduce risks in Guinea which are associated with sea level rise. This will further help to alleviate water shortage risks and help the population prepare for these extreme events. In addition, there is the need for improved technologies in agricultural, introduction of new crop species, and other likely interventions to decrease the country's populations and communities' vulnerability to rainfall variability.

Togo[24]

Togo being a low-income country with high poverty rate will further experience untold hardship from climate risk and associated impacts. Seasonal rainfall in this region significantly vary on inter-annual and inter-decadal timescales, partly due to variations in the movements and intensity of the Inter-Tropical Convergence Zone (ITCZ), and variations in timing and intensity of the West African Monsoon[25]. Togo is characterized by high temperatures and mean annual temperature has increased by 1.1°C since 1960, at

an average rate of 0.24°C per decade[26]. Hot days was observed to have increased by 15.5% while cold nights decrease since 1960. This has significantly impacted on livelihoods, human and animal health and natural resources. Variable precipitation pattern is observed across the country. Climate change trends in Togo are likely to increase the risk and vulnerability of local communities to the intensity of extreme events, coastal storms, and natural hazards such as heat waves, droughts, floods, and wildfires[27]. The country's agriculture including crop production, livestock and fisheries, is highly sensitive to climate conditions. Climate models project a rising temperatures, changing seasonal rainfall patterns, increased duration of dry spells, and increased aridity and drought, which is serious consequences on the country's agricultural sector.

As highlighted in Togo's adaptation action for the Agriculture Forestry and Other Land Use (AFOLU) sector, there is the need for: (i) capacity building in AFOLU sector focusing on building technical capacity of key research institutions on modeling; (ii) strengthening the resilience of crops, livestock and forests through breeding of crops and livestock for resistance to pests and diseases and promoting small scale irrigation; (iii) promoting sustainable forest management through promoting fast growing tree species for wood energy; and (iv) sustainable land management including the development of a national monitoring program for land use. Furthermore, the agricultural sector will benefit from the strengthening of Integrated Soil Fertility Management (ISFM), the construction and/or improvement of reservoirs for micro-irrigation and livestock watering in rural areas throughout all regions, support for the mapping of areas vulnerable to climate change, and the promotion of rice production systems with very low water consumption and low GHG emissions[28]. The country has committed to the Climate-Smart Agriculture process outlaid in the

[23] https://climateknowledgeportal.worldbank.org/country/guinea/vulnerability
[24] https://climateknowledgeportal.worldbank.org/sites/default/files/2021-06/15859-WB_Togo%20Country%20Profile-WEB.pdf
[25] UNDP (2019). Togo Climate Change Adaptation Overview. URL: https://www.adaptation-undp.org/explore/western-africa/togo
[26] https://climateknowledgeportal.worldbank.org/sites/default/files/2021-06/15859-WB_Togo%20Country%20Profile-WEB.pdf
[27] UNECA (2015). Assessment Report on Mainstreaming and Implementing Disaster Risk Reduction in Togo. URL:https://www.uneca.org/sites/default/files/uploadeddocuments/Natural_Resource_Management/drr/drr_west-africa_english_fin.pdf
[28] Togo (2015). Third National Communication to the UNFCCC. URL: https://unfccc.int/sites/default/files/resource/tgonc3.pdf

framework of the implementation of the agricultural policy laid out by ECOWAS and in the National Policy for the Agricultural Development of Togo 2013–2022[29].

Based on these literature review and national publications the proposed SCPZ programme interventions align with the adaption measures highlighted in the individual country adaptation action plan.

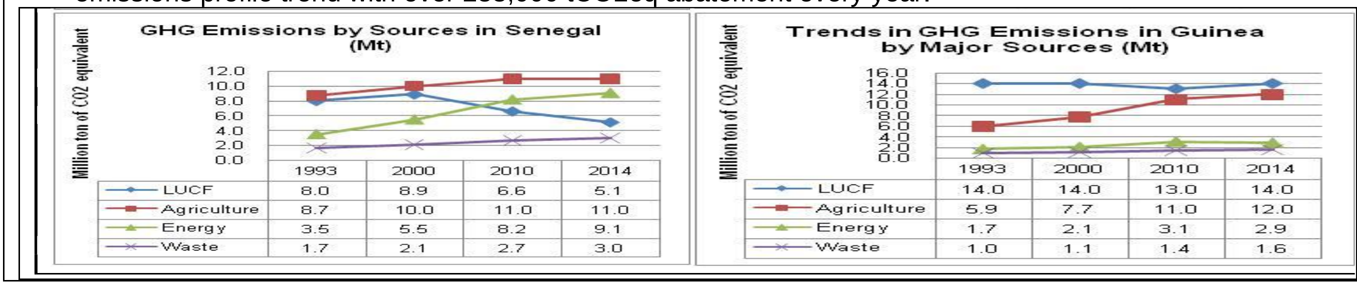
**Baseline situation:** Today, more frequent and prolonged droughts, shrinking of the crop growing season due to late onset and early cessation, shift in crops calendar, increasing demand for more water in agriculture, and poor yields from traditional crops (LGC), pose considerable challenges to subsistence agriculture that is widely carried out by peasant farmers in these countries. In addition to this, pandemic and disease outbreaks, such as the COVID-19 or the previous Ebola crisis not only pose threats to the population's health, but also cause major disruptions in entire agricultural value chains. The Covid-19 pandemic has had severe impacts on the already vulnerable economies in the region and undermined efforts to strengthen the resilience of smallholder farmers to climate change[30]
This makes smallholder agriculture more vulnerable than ever. Without adequate measures to strengthen and improve the entire agriculture value chain from the production stage up to the processing stage, the effects of climate change on agriculture will persist, driving more vulnerable rural communities and people further into poverty severity. With oven 90 per cent of the population in Togo, Senegal and Guinea, depending on rainfed agriculture for daily subsistence, the situation could become even more precarious. The unanticipated shock of COVID-19 underscores the need for a shift from "business as usual" practices to a more forward-looking approach that invests in the productivity, sustainability and resilience of food systems.[31] Strengthening the agricultural value chain in these countries is extremely important, as climate change predictions indicate that there will be more frequent and prolonged drought events, late onset of rainfall and early cessation, shrinking of the length of the growing season for most regions (Table B.1.4), altering of the crop calendar, increasing crop water requirement. Hence, the need for an integrated approach to agriculture that combines adaptation gains with mitigation.

The integrated approach is paramount as agriculture has been identified as the number one culprit of GHG emission in three of the three countries as shown in Figure B.1.16.

- As at 2014, Togo's total GHG emissions stood at 13.7 $MtCO_2e$ and largely driven by emissions from LUCF (6.6 $MtCO_2e$), followed by energy (2.68 $MtCO_2e$), agriculture (2.6 $MtCO_2e$), waste (0.8 $MtCO_2e$) and Bunker Fuels (0.30 $MtCO_2e$). The programme will curb Togo´s emissions profile with a reduction of over 380,000 tCO2eq annually.

- For Senegal, between 1990 and 2014, total GHG emissions grew by over 43%, and largely dominated by emissions from the agricultural sector (36%) and energy (27%), followed by emissions from land-use change and forestry (LUCF) (22%), waste (9%), and industrial processes (7%). The programme will act on each of these key contributors and reduce annually over 145,000 tCOe from the country's emissions profile.

- In Guinea during same period, total GHG emissions grew by 40%, and mainly associated with activities from the land-use change and forestry (LUCF) sector, which accounted for 46.9% of total GHG emissions. This is closely followed by the agricultural sector that emitted 37.4% of total GHG emissions, energy (9.5%), and finally emissions from waste and industrial processes that together accounted for less than 6.1% of total emissions. Here again, the programme will reduce the country's emissions profile trend with over 288,000 tCO2eq abatement every year.




[29] Republic of Togo (2016). Nationally Determined Contributions to the UNFCCC. URL: https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Togo%20First/INDC%20Togo_english%20version.pdf
30 4 CIRAD (2021), COVID and Food Security: How African Agriculture could hold its own in the face of the crisis, https://www.cirad.fr/en/ciradnews/news/2020/science/covid-food-security-african-agriculture-resilient-in-face-of-crisis
31 FAO (2021), The impact of disasters and crises on agriculture and food security.

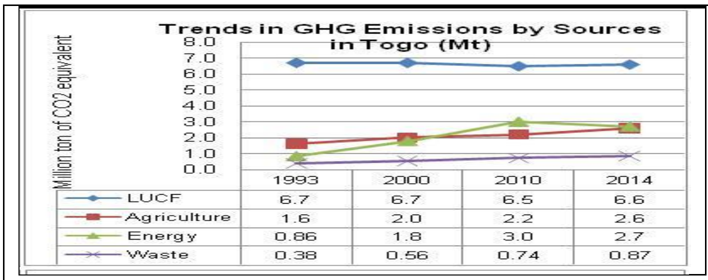



**Figure B.1.16:** GHGs Emissions Profiles for Togo, Senegal and Guinea (1993 – 2014)

Four main aspects (barriers) define the baseline of the proposed programme, serving as foundational elements that characterize the initial conditions and challenges that need to be addressed and transformed:

1) *Increasing number of extreme climate events especially droughts, late onset of rain and early cessation, increasing demand for water by crops, shrinking of the length of the growing season, in countries with very limited resilience and adaptive capacities:* According to the ND-GAIN vulnerability index, the three countries in the proposal rank among the world's least resilient countries to climate change and in the first quartile in terms of resilience, with limited adaptive capacities (see Table B.1.3). Furthermore, these countries experience high demographic growth with an estimated population of 135 million people (11% of the entire African continent), and projected to double by 2050. This implies significant increases in demand for food, as well as fragile socioeconomic conditions as shown in Table D.4.1. The increasing impact of the extreme climate-related events mentioned above, are affecting a larger range of sectors as described above (see Annex 2B for details), and in particular, agriculture, water and energy. This can make populations more vulnerable to other hazards and socio-political changes. In the affected countries, where agriculture is a key sector (more than 40 percent of GDP in 2018), these impacts can reduce national GDP by up to four percent per year.

2) *Lack of promotion and use of effective water management techniques in agriculture in the presence of increasing climate change challenges*: Even though drought and other climate-related extremes requiring better water management for agricultural production are well-known phenomenon in these countries, the use of water efficient techniques such as irrigation among smallholder farmers is still very low. According to FAO, expanded irrigation development and improved water management are keys to increasing agricultural production, under water scarcity conditions. In Togo, the figure is slightly lower with only 0.8 percent of the arable land of a smallholder is under irrigation. The figure is slightly higher for Senegal, with about 5 *percent* of land under *irrigation*[32]. Though the water sector is at risk from climate change which limits the sector potentials for irrigation, many studies carried out in the three countries show mainly uncertainties and complexities in the climate change research, and also uncertainties associated with the impacts of future climate changes on water resources[33,34]. Even with future climate change uncertainty on water resources, these countries have abundant surface and ground water potentials quite suitable for irrigation. Table B.1.6 shows the estimated ground water storage capacities for these countries, while Figure B1.17 shows the aquifer productivity of ground water resources. Aquifer productivity simply describes the likely interquartile range for boreholes drilled and sited using appropriate techniques and expertise. The assessment of ground and surface water resources potentials for irrigation in the regions are presented in Tables B.1.7a and B.1.7d.

   As observed (Figure B.1.17), the aquifer productivity for Togo, Senegal and Guinea ranges from high to moderate and therefore, very suitable for drilling of boreholes to support agriculture during periods of water scarcity. The ground water storage capacities are very moderate and quite suitable for extraction.

**Table B.1.6:** Countries estimated groundwater storage capacity

| **Groundwater Storage ($km^2$)** | | |
|---|---|---|
| Country | Best Estimate | Range |
| Togo | 541 | 133 – 1,935 |
| Senegal | 12,500 | 8,280 – 29,100 |
| Guinea | 297 | 102 – 879 |

**Source:** MacDonald et al. (2012)

[32] Word Bank (2017): World Development Indicator

[33] van der Wijngaartet al.(2019):Irrigation and irrigated agriculture potential in the Sahel: The case of the Niger River basin.

[34] Altchenko and Villholth (2015):Groundwater irrigation potential

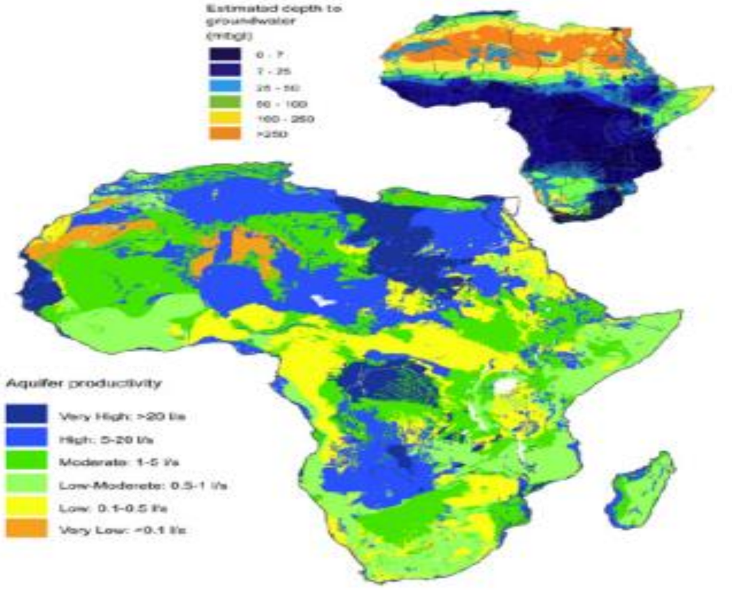



Figure B.1.17. Aquifer productivity for Africa. The inset shows an approximate depth to groundwater
Source: MacDonald et al. (2012)

As observed in Table B.1.7a, the groundwater resource is in general abundant in each of the region and most of the physicochemical properties are within acceptable ranges (only few parameters, for some regions, have to monitor when using these resources for irrigation purposes). The same can be said for surface water resources (Table B.1.7b), there are more or less major rivers in each region, implying important surface water potential. For drip irrigation purposes, this surface water could complement the groundwater resource in each region.

A study on the impact of projected climate change on water resources in the SCPZs was conducted as part of the Global Center on Adaptation (GCA's) Technical Assistance Program (TAP) to "Access and Leverage Climate Finance" (TAP) through the Africa Adaptation Acceleration Program (AAAP). The study investigates the impact of projected climate on the SCPZs water availability using RCA4 RCM under three emission scenarios (i.e., RCP2.6, 4.5 and 8.5) and compared the changes that are likely to occur in these zones with the baseline conditions of 1981-2010 (**See Section 5.4.2 of Annex 2 for details**). The study corroborates results found in previous studies that most SCPZs currently have abundant water resources. However, there will be a reduction in future water availability for both surface and potential groundwater recharge particularly for the Casamance, and Kara SCPZs. The net potential PET is also projected to increase in all the SCPZs compared to the baseline conditions which means more water will be required to meet crop water demands. It will thus be necessary for efficient water conveyance and distribution systems such as drip irrigation to meet crop water requirements, encourage on-farm water storage and monitor both surface and groundwater for sustainable utilization.

**Table B.1.7a:** Groundwater availability, properties and degree of restriction for irrigation at project-specific levels

| | | **Togo** | **Senegal** | **Guinea** | |
|---|---|---|---|---|---|
| | | **Kara** | **Casamance** | **Boké** | **Kankan** |
| **Groundwater (GW) potential** | | Moderate[o] | High to very high[n] | Probably low to high[p] | Low to moderate[p] |
| **Depth to GW (mbgl)** | | 0-25[l] | 0-25[l] | 0-25[l] | 0-25[l] |
| **Groundwater property** | **Temp. (ºC)** | 26-32[k] | | | |
| | **pH** | 5.4-7.9[k] | 3.8-8.3[m] | 6.31-6.39[r] | 6.31-6.39[r] |
| | **Electrical conductivity (µS/cm or µmho/cm)** | 42-982[k] | 38.7-7160[m] | | |
| | **Total alkalinity (mg/l)** | | | | |
| | **Total dissolved solids (ppm or mg/l))** | | 20.6-3900[m] | | |
| | **Turbidity (NTU)** | | | 0.40-050[r] | 0.40-050[r] |
| | **Nitrogen (mg/l)** | 0[k] | 0.0[m] | 3.34-4.01[r] | 3.34-4.01[r] |
| | | 10[k] | 342.0[m] | | |
| | **Na+ (mg/l)** | 1.8-96[k] | 4.5[m] | | |
| | | | 1726[m] | | |
| | **Sodium adsorption ratio** | | | | |
| | | | | | |

***Note: Degree of restriction expressed following Ayers and Westcot 1985; and Müller and Cornel 2017.Blue box represents no restriction, Green box is for slight to severe (i.e., use on sensitive crops must be cautioned.)***

**Table B.1.7b:** Potential Groundwater Recharge in the SCPZs for the baseline and 2011-2040 Period

| | Baseline MCM/yr | RCP 2.6 MCM/yr | RCP 4.5 MCM/yr | RCP8.5 MCM/yr |
|---|---|---|---|---|
| **Kankan** | 46.177 | 50.777 | 48.127 | 46.103 |
| **Boke** | 47.023 | 47.233 | 46.292 | 53.598 |
| **Kara** | 20.366 | 18.176 | 17.786 | 17.786 |
| **Cassamance** | 3.562 | 2.216 | 2.051 | 2.051 |

**Table B.1.7c** Surface Water availability in the SCPZs for the baseline and 2011-2040 Period

| | | **Baseline** | | **rcp2.6** | | **rcp4.5** | | **rcp8.5** | |
|---|---|---|---|---|---|---|---|---|---|
| **SCPZ** | **Basin** | **m3/s** | **BCM/yr** | **m3/s** | **BCM/yr** | **m3/s** | **BCM/yr** | **m3/s** | **BCM/yr** |
| Kankan | Niger | 1271 | 40.1 | 1396 | 44.1 | 1326 | 41.8 | 1273 | 40.2 |
| Boke | Cogon | 312 | 9.8 | 317 | 10.0 | 311 | 9.8 | 444 | 14.0 |
| | Tingulinta | 226 | 7.1 | 231 | 7.3 | 229 | 7.2 | 285 | 9.0 |
| Kara | Kara | 54 | 1.7 | 48 | 1.5 | 47 | 1.5 | 152 | 4.8 |
| Cassamance | Cassamance | 21 | 0.7 | 13 | 0.4 | 12 | 0.4 | 13 | 0.4 |

*[k]Zoulgami et al. 2015 (Physico-chemical study of groundwater in the Northeast of Kara region (Togo) DOI: 10.4314/ijbcs.v9i3.49)*
*[l]Ouedraogo 2017 (Mapping groundwater vulnerability at the pan-African scale, Thesis UCL Louvin).*
*[m]Ndoye et al. 2018 (Groundwater Quality and Suitability for Different Uses in the Saloum Area of Senegal, https://www.mdpi.com/2073-4441/10/12/1837)*
*[n]Africa Groundwater Atlas- Hydrogeology by country-Hydrogeology of Senegal http://earthwise.bgs.ac.uk/index.php/Hydrogeology_of_Senegal#Groundwater_Quantity)*
*[o]Africa Groundwater Atlas- Hydrogeology by country-Hydrogeology of Togohttp://earthwise.bgs.ac.uk/index.php/Hydrogeology_of_Togo*
*[p]Africa Groundwater Atlas- Hydrogeology by country-Hydrogeology ofGuineahttp://earthwise.bgs.ac.uk/index.php/Hydrogeology_of_Guinea*
*[r]Sidibe and Oulaye (L'eauenGuiée, Presentation https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&ved=2ahUKEwjYxNDZrtDuAhWBKewKHeYNB94QFjAAegQIARAC&url=https%3A%2F%2Funstats.un.org%2Funsd%2Fenvironment%2Fenvpdf%2FUNSD_TogoWorkshop%2FSession%25207b_Guin%25C3%25A9e_L%2527Eau%2520en%2520Guin%25C3%25A9e.pdf&usg=AOvVaw0u4y4HcXhogQteNFetzSyl)*

With the introduction of drip irrigation, the use of scares water resources become even more advantageous in these countries. Besides, it has been identified as one water management technology to draw useful policy lessons for the design of phase II of the programme in other RMCs. It is less expensive to installed and used compared to other AWM technologies as shown in the programme feasibility study. Besides, It can use ground or surface water resources at minimal additional cost. A research by Bajwa et al.(2018) on Design and Implementation of an IoT System for Smart Energy Consumption and Smart Irrigation in Tunnel Farming[35], indicate that drip irrigation has between 80 to 90% water efficiency compared with sprinkler irrigation, overhead irrigation, sub irrigation, level basin and surface irrigation that have relatively lower water efficiency. Drip irrigation also has higher energy efficiency than these systems. Drip irrigation is also noted to "save water by reducing the size of the wet soil surface, thus decreasing the amount of direct evaporation and excess percolation through the root zone. Unlike sprinklers, drip irrigation is practically unaffected by wind conditions, nor is it affected by soil surface conditions. Soil is maintained in a continuously moist condition. Nutrients can be applied through the drip systems, thus reducing use of fertilizers and improving quality of returned water. Increases in water use efficiency in drip irrigation, compared to conventional basin/furrow irrigation, are attributed to both water savings and the increase in yields resulting from favorable soil moisture and nutrient regimes. Due to the relative suitability to drought conditions that are typical of these countries, applicability on small-scale and complementarity with solar systems, drip irrigation is gaining widespread use in Africa. For example, IRENA cited a solar-powered drip irrigation project by Solar Electric Light Fund installed for Solar Market Gardens in the Kalale district of northern Benin, on an average of half hectare farm operated by a co-operative composed of 35-45 women. The solar-powered drip irrigation system provided safety net for farmers especially during drought period. It also reduced their daily task to a weekly or bi-weekly activity since it takes only a few minutes to water each plant bed, saving each woman up to four hours a day.[36]

3) *Knowledge of the adaptation and mitigation co-benefits associated with the use of agro-wastes to generate clean and reliable energy at the processing stage is still too weak: A*ll the three countries produced large quantities of agricultural wastes beside waste generated by the large livestock population in these countries (cattle, sheep, poultry and goats etc.,). These waste products (liquid and solid waste), food and agro wastes that are likely to pose sanitation and waste management challenges in these countries are actually sustainable "resources" for biogas technology **(Refer to Chapter 6 of the Feasibility study for the livestock size and proportion of organic matter that can be used for bio digestion).**

[35]Bajwa (2018). https://www.researchgate.net/project/http-wwwmdpicom-1996-1073-11-12-3427-pdf.
[36] IRENA (2018). ttps://www.irena.org/-/media/Files/IRENA/Agency/Publication/2016/IRENA_Solar_Pumping_for_Irrigation_2016.pdf

Biogas technology presents great opportunities for decentralized (off-grid) and diversified (multiple uses such as for stoves, refrigerators and generators) energy use. It is also the only renewable energy technology that is able to address multiple challenges such as waste management, access to energy, and improving agricultural productivity. Other products of anaerobic digestion include effluent, sludge or bio slurry that are suitable by-products for implementing climate resilient activities. Compared to composting, anaerobic process reduces nitrogen loss from 50-80% to just 5-12%. It also increases ammonia content by 120% and quick-acting phosphorus by 150% and increases agricultural yield by 11% per year.[37] Other benefits of biogas technology include improved sanitation, sustainable land use, improved environmental health, climate change mitigation by reducing atmospheric emission of methane, costs savings and optimizing water use through recycling and reuse and providing green jobs (for example for masons, carpenters, electricians).

In Togo for example, the energy sector is characterized by the dominance of biomass (71% of final energy consumption) and heavy dependence on energy imports (100% of petroleum product needs and 79% of electricity needs covered by imports). Moreover, the country has not yet been able to scale the development of renewable energy sources (solar, wind, mini and micro hydropower). Currently, electricity is generated primarily from five thermal power plants which jointly represent an installed capacity of 128 MW, including a 100 MW HFO plant, and the Nangbeto hydroelectricity power plant with a capacity of 30 MW. In the rural areas where most farming is happening, the reliance on fossil fuel is exacerbated. Togo can benefit from additional cheap and clean energy to bridge its current renewable energy gap, by helping the country adopt clean energy practice in farms irrigation and processing of agriculture products. This will reduce Togo's carbon footprint in energy generation. In Senegal, charcoal and firewood consumptions as well as illegal logging are the main drivers of increasing GHG emissions. According to Senegal's Third National Communication (TNC) to the UNFCCC, 2016, over 55.5% of households in Senegal use fuel wood for cooking and 11% use charcoal. The report further states that in 2013 alone, households consumed over 1,735,219 tons of fuel wood and 482,248 tons of charcoal, equivalent to a total harvest of 3,778,326 m3 of forest wood resources. From 2001 to 2019, Senegal lost close to 3,710 ha of tree cover, equivalent to an 9.4% decrease since 2000, and 719 kt of $CO_2$ emissions[38]. Also, about 15% of all tree cover loss occurred in areas where the dominant drivers of loss resulted in deforestation and degradation. With the countries' huge ruminant and agro wastes potentials including those to be generated at the AIPs, this trend can be reversed. However, without GCF support, it may be far-fetched.

In Guinea, sequestration through agroforestry will be achieved with other crops, while investments in anaerobic digestion of waste will provide an opportunity to achieve further climate mitigation, alongside solar energy.

4) *The capacity of smallholder farmers in these countries to adapt to climate change and variability is extremely low:* Today, most smallholder farmers in Togo, Senegal and Guinea, engage in unsustainable agricultural practices that are not only vulnerable to climate change and variability, but also contribute to it. Existing local coping strategies coupled with agricultural and land use practices in these countries (e.g., rain-fed agriculture, deforestation, overgrazing of livestock, logging and hunting) in the context of climatic stress, are clearly inadequate and only add to the numerous challenges such as food insecurity, malnutrition and conflicts over resources. As has been illustrated under the climate context, climate variability and change put heavy burdens on farmers and local communities that exceed their capacities to adapt. The paradigm shift that this programme seeks to promote is to move from implementing agriculture in the normal "business as usual" way to one that integrated the entire value chain from the production to processing stages. In otherwords, the resilience of each aspect of the value chain including its actors will be made more climate resilience. First, at the production stage, the programme will invest in the development of agricultural access infrastructure and support for the development of agro-industrial parks (AIPS) such as Agro-Processing Centers (APCs) and Agricultural Transformation Centres (ATCs). These are purposely built shared facilities, to enable agricultural producers, processors, aggregators and distributors to operate in the same vicinity to reduce transaction costs and share business development services for increased productivity and competitiveness.

It will also involve the construction of access facilities such as roads that are resilient to extreme weather events, administrative building and storage warehouses and facilities to support production. Also at this stage, the programme will support the rehabilitation/upgrading of small-scale agricultural water management (AWM) technologies such as equipped boreholes, wells and small reservoirs/dams including rain water harvesting technologies. Furthermore, the use of drip irrigation technology will be widely promoted especially among the most vulnerable group (women), to increase production of vegetables and fruits, as well as engaging in climate resilient agricultural management practices such as mixed agro-forestry management, improved nutrient recycling and improve genetic resources among many other techniques. These CRA management practices have be shown to not only improving the yields of crops and soil health, but also responsible

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[37]McGarry, Michael G. and Jill Stainforth. 1978. Compost, Fertilizer, and Biogas Production from Human and Farm Wastes in the People's Republic of China. Ottawa: International Development Research Centre.
Gregory R. 2010. China Biogas. http://www.ecotippingpoints.org/our-stories/indepth/china-biogas.html. Accessed December 2016

[38] Global Forest Watch. 2019. https://www.globalforestwatch.org/dashboards/country/SEN?category=forest-change&dashboardPrompts=eyJvcGVuIjp0cnVlLCJzdGVwSW5kZXgiOjAsInN0ZXBzS2V5Ijoidmlld05hdGlvbmFsRGFzaGJvYXJkcyJ9&map=

for reducing greenhouse gases (GHGs). In all three countries, agriculture and forestry are the main culprits of GHG emissions.

Second, at the processing stage, linkages between farmers and cooperative societies will be established including the rehabilitation and /or establishment of new ones to facilitate cooperatives' cross-border and trans-national activities. Equally at this stage, renewable energy efficient drying, processing and packaging technologies will be introduced to improve post-handling of staple crops to reduce losses and waste after harvest, and add market value especially for cashew, mango, coffee and other cash crops. At each investment stage, women will constitute at least 54% of the target beneficiaries.

**Programme Rationale:** The Governments of Togo, Senegal and Guinea have requested the support of the African Development Bank (AfDB) to implement the Staple Crops Processing Zones (SCPZs) Program, as part of the 'Flagship' of the AfDB under its 'Feed Africa' Agenda. The goal of the entire SCPZs program in Africa is to: (i) increase food production capacity and efficiency by promoting amongst others, the use of climate resilient practices and technologies adoption; (ii) enhance value addition to agriculture; (iii) promote local, regional and international trade; (iv) promote investments in agribusiness; and, (v) increase the contribution of the agriculture sector to Gross Domestic Product (GDP), wealth and employment creation. The initial focus of the SCPZs program is on 18 priority commodities across the five agro-ecological zones of Africa considered important in these zones, with set targets of raising Africa's 2015 food production by an estimated 174 million metrics. The first phase of the program is currently being implemented in three Regional Member Countries (RMCs) at the request of these countries (Togo, Senegal and Guinea). The next phases of the SCPZs program are replication and out-scaling in other RMCs requesting the interventions of AfDB.

Consistency with Countries' Climate Change Strategies:

- In Togo, the countries revised NDC target to unconditionally reduce GHG emissions by 20.51% by 2030, i.e. 6,236.02 Gg CO2-eq, compared to 2030 BAU emissions and conditionally reduce GHG emissions by 30.06% by 2030, i.e. 9,305.59 Gg CO2-eq, compared to 2030 BAU emissions. At the adaption side, over 42% of the country's population (males and females) will build their resilience to climate change.

- For Senegal, the target is to use available internal resources to reduce GHG emissions by 5% (i.e. 30,987 GgCO2eq) and 7% (i.e. 35,106 GgCO2eq) by 2025 and 2030, respectively, compared to the BAU emissions in those years. In the situation when international support is received, the intention is to reduce GHG emissions by 23% (i.e. ~24,883 GgCO2eq), and 29% (i.e. ~26,611 GgCO2eq) in 2025, and 2030, compared to the BAU emissions in those years. At the adaption side, over 20% of the country's population (males and females) will build their resilience to climate change.

- Guinea's NDC seeks to unconditionally cut down 2,056 ktCO2eq/year, i.e. a 9.7% reduction in emissions by 2030 while excluding LULUCF and conditionally reduce 3,929 ktCO2 eq/year, or 17.0% GHG emissions by 2030 compared to the BAU scenario when storage capacity from LUCF is excluded. With LULUCF, the NDC targets to unconditionally reduce emissions by 23% by 2030 and conditionally reduce 49% GHG emissions by 2030 compared to the BAU scenario when reforestation actions are excluded. It will also increase the resilience to climate change of over 29.2% of the entire population (males and females).

Programme Targeted Areas and Beneficiaries:

- In Togo, the SCPZ project covers the Kara region, which consists of seven (7) Prefectures namely; Assoli, Bassar, Bimah, Dankpen, Doufelgou, Kéran, and Kozah, with an estimated population of approximately 769,940 inhabitants. The programme area of intervention is estimated at about 165,000ha of which 15,420 ha will be farmed during the programme lifetime. The project in Togo will directly benefiting more than 484,350.0 males and females and over 1,857,346.5 indirectly. The key commodities of focus are rice, maize, soybean, cashew nuts, sesame, and horticulture that are widely cultivated by both peasant and commercial farmers.
- In Senegal, the program activities will focus in the Casamance Natural Region comprising of the administrative regions of Ziguinchor, Sédhiou and Koldaqui, inhabited by over 1,872,668 people. The South Agropole project in Senegal covers a total land area of 105,000ha, of which 11,940 ha will be farmed during the project's lifetime. Th project in Senegal will directly benefiting more than 624,775.0 males and females and over 6,567,628.0 indirectly. The priority commodities of focused are mango, cashew maize, rice, horticulture, banana and non-timber forest products.
- In Guinea, the project covers two Agro-industrial Parks (AIPs) located at Boké and Kankan regions, with a total land area of about 110,000 ha, of which 11,810 ha will be farmed during the project's lifetime. This will directly benefit more than 327,425.0 males and females and over 1,733,259.2 indirectly. The goal of the project is on the sustainable intensifications of: (i) cash crops such as cashews, oil palms, sesame and soya; (ii) food products such as rice, fonio, groundnuts, maize, potatoes, yams and cassava; (iii) fruit such as oranges, mangoes and pineapples; (iv) market garden products, in particular, tomatoes, onions, okra and chillies; (v) poultry and sheep breeding; (vi) fishery products from fish farming; (vi) non-wood forest products such as honey, leaves, wild seeds and fruit, oils and resins among others.

A summary of the programme target beneficiaries is presented in Table B.1.8 while, the criteria for the selection of sub-projects to be executed under the programme and beneficiaries are extensively discussed under section B.3 (See Annexes 22c and d for more details). For projects site selection in each country, see Section 4.1 of the programme feasibility study (Annex 2).

*Table B.1.8: Direct and Indirect Programme Beneficiaries*

| **Programme Beneficiaries (Direct Plus Indirect)** | | | | | | | |
|---|---|---|---|---|---|---|---|
| Country | **Direct (Baseline)** | **Direct (GCF)** | **Indirect (Baseline)** | **Indirect (GCF)** | **Total(D+I)** | **Male** | **Females** |
| Togo | 353.000,0 | 75.853,0 | 309.557,8 | 437.520,0 | 1.175.930,8 | 538.576,3 | 637.354,5 |
| Senegal | 150.000,0 | 83.215,0 | 2.462.860,5 | 735.160,0 | 3.431.235,5 | 1.571.505,9 | 1.859.729,6 |
| Guinea | 320.000,0 | 122.660,0 | 1.155.506,1 | 511.812,0 | 2.109.978,1 | 966.370,0 | 1.143.608,1 |
| **Total** | **823.000,0** | **281.728,0** | **3.927.924,4** | **1.684.492,0** | **6.717.144,4** | **3.076.452,1** | **3.640.692,2** |
| ***% of Total Population*** | ***2,4%*** | ***0,8%*** | ***11,5%*** | ***4,9%*** | ***19,7%*** | ***9,0%*** | ***10,7%*** |

Synergies and Complementarities between AfDB Baseline Investments and GCF Intervention in the Programme: Figure B.1.18 best describes the existing synergies and complementarities between AfDB baseline Investments and GCF Interventions.

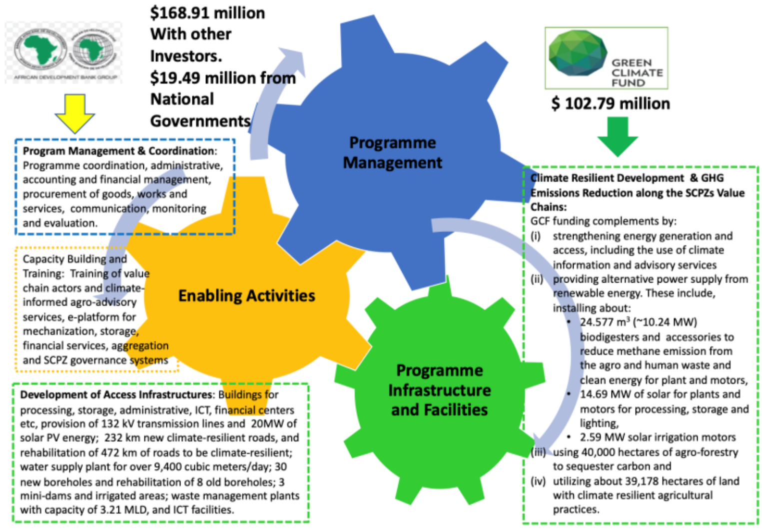


**Figure B.1.18:** Synergies and Complementarities between AfDB baseline Investment and GCF Interventions

Financing from the GCF will help convert human waste and agro-waste especially from post-harvest losses and processing that could have emitted methane and generate over 424,879 $tCO_{2eq}$ every year. The availability of biogas will also help generate over 44,867 kWh/yr clean power to replace diesel fuel for generators, motors and other plant systems. GCF financing will also help reduce about 14,738 $tCO_{2eq}$ of emissions every year by replacing diesel-based irrigation and energy systems with solar (taking advantage of the great solar PV conditions in these countries). Moreover, GCF financing will help integrate agro-forestry activities in the SCPZ programme to help sequester over 5.7million tCO2eq from agro forestry practices that will also enhance resilience of the production systems. These emission reductions will significantly contribute to the efforts being made by the programme countries to meet their NDC commitments under the Paris Agreement. Furthermore, the programme will help to strengthen the generation, access and use of climate information services within the SCPZs to ensure that agro-advisory services and farm management practices are climate-informed. It will also contribute to rationalizing agricultural value chains to be more low-carbon and climate resilient. Lessons from this programme will be applied when replicating similar programme in several African countries that rely heavily on the agriculture sector and as a result also generates significant percentage of their carbon emissions. In a post COVID-19 context, contributing to the elimination of these technical and financial barriers would allow these countries to finance transformational change towards climate compatible investments at a time when local financial institutions are facing significant challenges.

Other Complimentary Initiatives that the SCPZ Programme Builds on:
**Senegal:** The programme builds on the following initiatives carried out in Senegal:

- Ecosystem and Communities-based Resilience Building in Senegal, financed by the GCF
- Integrated Urban Flood Management Programme in Senegal, financed by the GCF
- Building Resilient of Food Insecure Smallholder Farmers in Senegal, financed by the GCF

- Promotion of climate-friendly cooking in Kenya and Senegal, financed by the GCF.
- ASER Solar Rural Electrification Project, financed by the GCF.
- Training Programme on Climate Change Adaptation and Disaster Risk Reduction in Agriculture financed by the Italian Ministry of Foreign.
- Increasing Productivity and Livelihoods in the Nioro du Rip Basin in Senegal financed by USAID-CIAT.
- Senegal Disaster Risk Management and Climate Change Project financed by the World Bank.
- Senegal National Adaptation Plan financed by GEF and UNDP.
- Prog Innovative Finance and Community-based Adaptation in Communities Surrounding Community Natural Reserves financed by UNDP and GEF.
- Mainstreaming ecosystem-based approaches to climate resilient rural livelihoods in vulnerable rural areas through the Farmer Field School methodology, financed by GEF and FAO.
- Collaborative Management for a Sustainable Fisheries Future (COMFISH) and COMFISH, financed by USAID.
- Climate Change Adaptation Project in the Areas of Watershed Management and Water Retention, financed by GEF and IFAD.
- Adaptation to Coastal Erosion in Vulnerable Areas, financed by the Adaptation Fund.
- Climate Information Services for Increased Resilience and Productivity in Senegal, financed by USAID.
- The Future Climate for Africa Program involving Burkina Faso and Senegal, financed by CDKN, UKAid, and NERC.
- Senegal River Basin Climate Change Resilience Development Project, financed by the GEF and the World Bank.
- Building Resilience and Adaptation to Climate Extremes and Disasters (BRACED),involving Senegal financed by DFID.
- Other AfDB initiatives provided in Annex 3C of the FP.

**Guinea:** In Guinea, the programme builds on the following initiatives:
- Ecosystem-Based Adaptation Targeting Vulnerable Communities of the Upper Guinea Region, financed by GEF
- Strengthening Resilience of Farming Communities' Livelihoods against Climate Changes in the Guinean, financed by GEF.
- Improving IWRM, Knowledge-based Management and Governance of the Niger Basin and the Iullemeden-Taoudeni/Tanezrouft Aquifer System, financed by GEF.
- West Africa Biodiversity and Climate Change Project (WA-BiCC) (regional), financed by USAID.
- Other AfDB initiatives listed in Annex 3D of the FP.

**B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)**

Programme objectives against baseline: In the selected SCPZ countries, various fragmented initiatives are being implemented to help build the resilience of smallholder farmers and rural communities to climate change as discussed above. However, achieving this is not easy and may not be entirely possible unless key technical, financial, policy and regulatory obstacles that lead to fragmented and ineffective interventions in key sectors like agriculture, land use and forestry, water and energy are carefully addressed. Unless these obstacles are overcomed, implementing programmes in the normal "business as usual" way in the context of climatic stress, are clearly inadequate and only add to the numerous challenges such as increasing emissions of GHG from agriculture, land use and forestry, food insecurity, malnutrition and heightened conflicts over resources use. To address these obstacles, this programme's ultimate outcome is: *strengthen climate resilience and reduce GHG emissions within the agricultural value chain to help stimulate productivity and value addition, competitiveness, employment and incomes generation in three selected African countries.* The outcome contributes to the overall programme impact, which is: improved climate resilience and increased adaptive capacity of 2,400 Agricultural Cooperative Societies (ACSs), Women-led Agribusiness Enterprises (WABEs), Farm-based Associations (FBAs),79,178 smallholder farm households (SHFHs) and over 79,000 ha of land made more resilient to climate change.

**Programme Goal Statement:** IF the required infrastructure, technical and financial support as well as climate-based agro-advisory services for staple crops processing zones (SCPZs) are provided, THEN the sustainability of agriculture value chains in 3 countries will be ensured BECAUSE critical SCPZ value chain infrastructure are strengthened and made more climate resilient, and adoption of sustainable climate resilient practices, technologies and innovations among smallholder farmers are promoted, thus leading to improved agricultural productivity and value addition, better forest and land use management practices, access to renewable energy and generation contributing to avoid over 10.87 million tCO2eq in GHG emission of the countries' NDCs targets by 2030. The proposed programme supports a more sustainable development pathway towards climate resilience for vulnerable smallholder farmers and rural communities, as described in each country's long-term vision for climate resilient and low emission agriculture. This can be achieved through strategic investments and interventions in the entire agricultural value chain from the production stage to processing. Since the COVID-19 pandemic began, governments have had to face even greater budget constraints, as they were forced to direct record amounts of spending on national social, economic and humanitarian support plans and social safety net programmes to help their most vulnerable populations cope and recover from the impacts of COVID-19. As a result, countries now need even more support to ensure an inclusive and climate resilient recovery, which includes the participation of the private sector, and progress towards the achievement of NDCs.
The programme's integrated approach is organized around two mutually connected components as follows:

**Component 1: Strengthening of critical SCPZs value chain infrastructure and management governance -** Under this component, the programme will invest in improving the climate resilience of critical infrastructure along the agricultural value chain in the three selected countries. First, at the production stage, the programme will invest in the development of SCPZs access infrastructure and support for agro-industrial parks (AIPS) management governance such as Agro-Processing Hubs (APHs), Agricultural Transformation Centres (ATCs). It will also involve the construction of access facilities such as roads, administrative building and storage warehouses, rehabilitation/upgrading of small-scale agricultural water management (AWM) technologies such as equipped boreholes, wells and small reservoirs/dams including rain water harvesting technologies, while the use of drip irrigation technology powered by solar pumps will be widely promoted especially among the most vulnerable group (women) to increase production of vegetables and fruits. Second, at the processing stage, linkages between farmers and cooperative societies will be established including the rehabilitation and /or establishment of new ones to facilitate cooperatives' cross-border and trans-national activities. Also, at this stage, renewable energy efficient drying, processing and packaging technologies will be introduced to improve post-harvest handling of staple crops to reduce losses after harvest, and addmarket value especially for cashew, mango, coffee and other cash crops. At each investment stage, women will constitute at least 50% of the target beneficiaries.

**Component 2: Promotion of climate resilient agricultural practices and technologies adoption among smallholder farmers -** Under this component, the programme will invest in the promotion and use of CRA practices and technology uptake among agricultural value chain actors in the three selected SCPZ countries. At the production stage, the programme will invest in the adoption of CRA practices and technology uptake such as, use of drought resistance or SGC crops, integrated nutrient management, and sustainable agro-forestry practices. It will also involve capacity building of value chains agents/communities' and development of agricultural processing support infrastructure such as: 1) O&M learning and training, management, business planning, stakeholder engagement, contract negotiation, PPP arrangements, governance, project performance monitoring, reporting and evaluation (M&E), investment mobilization and support, and investment promotion among others; and, 2) training and capacity building of value chain actors in the use of 'Good Agricultural Practices' in product quality, standardization and conformity requirements, as well as developing linkages and responding to commodity demand in terms of quality and quantity.Lastly,under this component, the programme will support investment in biogas technologies to improve diversification, value addition, productivity and profitability across the agricultural value chain.

The detailed Theory of Change (ToC) is presented below in Figure B.2.1:

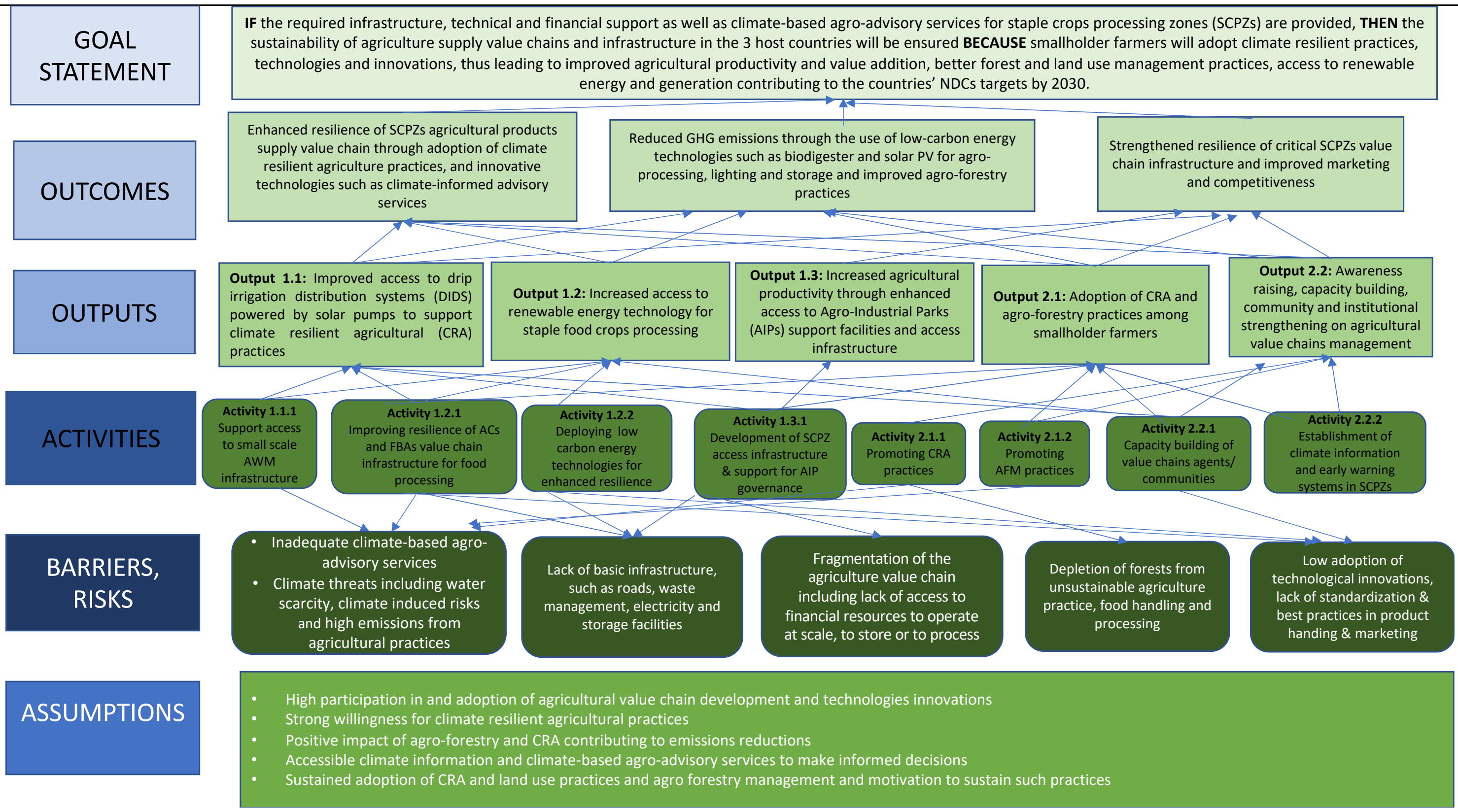


**Figure B.2.1:** Programme Theory of Change

## B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

***Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.***

| **Outcome number** | **GCF Mitigation Results Area (MRA 1-4)** | | | | **GCF Adaptation Results Area (ARA 1-4)** | | | |
|---|---|---|---|---|---|---|---|---|
| | **MRA 1** Energy generation and access | **MRA 2** Low-emission transport | **MRA 3** Building, citiies, industries, appliances | **MRA 4** Forestry and land use | **ARA 1** Most vulnerable people and communities | **ARA 2** Health, well-being, food and water security | **ARA 3** Infrastructure and built environment | **ARA 4** Ecosystems and ecosystem services |
| **Outcome 1** Enhanced resilience of SCPZs agricultural products supply value chain through adoption of climate resilient agriculture practices, and innovative technologies such as climate-informed advisory services | ☐ | ☐ | ☐ | ☐ | ☒ | ☒ | ☐ | ☐ |
| **Outcome 2** Reduced GHG emissions through the use of low-carbon energy technologies such as biodigester and solar PV for agro-processing, lighting and storage and improved agro-forestry practices | ☒ | ☐ | ☐ | ☒ | ☐ | ☐ | ☐ | ☐ |
| **Outcome 3** Strengthened resilience of critical SCPZs value chain infrastructure and improved marketing and competitiveness | ☐ | ☐ | ☐ | ☒ | ☒ | ☐ | ☐ | ☐ |

***If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.***

| | **Co-benefit** |
|---|---|

| Co-benefit number | Environmental | Social | Economic | Gender | Adaptation | Mitigation |
|---|---|---|---|---|---|---|
| Employment opportunities created through the programme | ☐ | ☒ | ☒ | ☐ | ☐ | ☐ |
| Co-benefit 2 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Co-benefit 3 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Co-benefit 4 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Co-benefit 5 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

## B.3. Project/programme description (max. 2500 words, approximately 5 pages)

The proposed GCF financed Staple Crop Processing Zones (SCPZs) Programme, mobilizing USD 102.79 million of GCF funding will trigger a paradigm shift towards low-emission, climate-resilient development pathways across the agricultural value chains in Togo, Senegal, and Guinea. It builds onto the AfDB financed baseline programme – the Staple Crops Processing Zones Programme: Phase 1, with a total budget allocation of USD 253.25 million. The entire programme consists of three complementary components, of which are:

- **Component 1:** Strengthening the resilience of critical SCPZs values chain infrastructure ;
- **Component 2:** Promotion of climate resilient agricultural practices and technologies adoption among smallholder farmers; and,
- **Component 3:** Programme Management and Coordination

***Programme Components, Outputs and Activities***

**Component 1: Strengthening the resilience of critical SCPZs values chain infrastructure**

Under this component, the programme will invest in improving the climate resilience of critical infrastructure along the agricultural value chain in the three selected countries. First, at the production stage, the programme will invest in the development of SCPZs access infrastructure and support for agro-industrial parks (AIPS) management governance such as Agro-Processing Hubs (APHs), Agricultural Transformation Centres (ATCs). It will also involve the construction of access facilities such as roads, administrative building and storage warehouses, rehabilitation/upgrading of small-scale agricultural water management (AWM) technologies such as equipped boreholes, wells and small reservoirs/dams including rain water harvesting technologies, while the use of drip irrigation technology will be widely promoted especially among the most vulnerable group (women) to increase production of vegetables and fruits. Second, at the processing stage, linkages between farmers and cooperative societies will be established including the rehabilitation and /or establishment of new ones to facilitate cooperatives' cross-border and trans-national activities. Also, at this stage, renewable energy efficient drying, processing and packaging technologies will be introduced to improve post-harvest handling of staple crops to reduce losses after harvest, and add market value especially for cashew, mango, coffee and other cash crops. At each investment stage, women will constitute at least 54.2% of the target beneficiaries. The following outputs are expected of this component**.**

**Output 1.1: Improved access to drip irrigation distribution systems (DIDS) powered by solar pumps to support climate resilient agricultural (CRA) practices (GCF Financed)**

This additional programme interventions financed by the GCF, aims at improving the climate resilience of critical small-scale AWM infrastructure along the agricultural value chain in the three selected countries. First, at the production stage, the programme will support access to drip irrigation distribution (DIDS) systems powered by solar pumps to increase water efficient use, time, and labour saving in agriculture (see page 138 - 141 of the programme feasibility study for the technical viability of using DIDS in the programme). Second, at the processing stage, linkages between farmers and cooperative societies will be established. Old cooperatives will be rehabilitated and new linkages established to facilitate cooperatives' cross-border and trans-national activities. Under a business-as-usual scenario (BAU), that is, without GCF interventions, the emissions baseline of using 100% Diesel for solar irrigation, amounts to 3,023 tCO2eq annually. With GCF support however, emissions abatement attributed to solar irrigation systems could reach as much as 1,007 tCO2eq annually, for about 39,179 ha. Additionally, this intervention will directly benefit more than 475,068 smallholder farmers including their household members, and create employment in the farming sector (casual labourers, handlers, produced buyers etc., for an additional 395,890 individuals.

***Activity 1.1.1: Support access to small-scale agricultural water management (AWM) infrastructure***
Under this activity, GCF will support smallholder farmers, women-led agribusiness enterprises (WABEs), agricultural cooperation societies (ACSs) and farm-based associations (FBAs), in the three SCPZs pilot countries, to invest in drip irrigation technology powered by solar pump. This will support horticulture and market gardening of vegetables and fruits including other cash crops, all year round in at least 31,179 ha.

**Output 1.2: Increased access to renewable energy technology for staple food crops processing (GCF Financed)**
One of the biggest challenges confronting the agricultural sector in SSA is post-harvest handling of staple food crops, which contributes up to 30-70% loss depending on the crop type. This is due to the non-availability of efficient food processing technologies, storage facilities, and the inability of farmers to readily have access to markets. In Senegal for instance, losses associated with post-harvest handling of Maize, Cassava, Yam and Rice amounts to over 35.1 percent, 34.6 percent, 24.4 percent and 6.9 percent respectively. These losses are as a result of ineffective food processing technologies for cleaning, sorting, storage and packing. It is also due to damage during transportation because of bad roads, inappropriate market practices and inadequate storage facilities. The situation is not much different for Togo and Guinea where losses range between 40% and 60% respectively.

Furthermore, existing food processing technologies in these countries rely heavily on biomass-based energy sources. In Togo, approximately 80% of households, micro and small-scale agribusiness enterprises rely largely on biomass-based sources of energy for consumption, which comes from the forest. It is estimated that at least 15,000 hectares of forest are cleared annually, which leads to forest degradation and deforestation, with resulting loss in carbon sinks. The situation is not different for Senegal, where over 55.5% of households including micro and small-scale agribusiness enterprises rely heavily on biomass-based sources of energy. In 2013 alone, households consumed over 1,735,219 tons of fuel wood and 482,248 tons of charcoal, equivalent to a total harvest of 3,778,326 m3 of forest wood resources. In Guinea, between 1990 and 2014, over 1.2 $MtCO_2e$ emissions was attributed to biomass burning for energy consumption. According to FAO, as of 2000, 33% of Guinea was tree cover. However, just 18 years later, the country lost over 16% of tree cover, resulting to a total release of 306 $MtCO_2e$ (17.0Mt per year) into the atmosphere.[39]

Apart from the use of biomass-based sources of energy for consumption, these pilot countries equally rely heavily on fossil-based energy systems. Fossil-based energy systems are relatively inefficient, not reliable and contributes significantly to GHG emissions. For instance, in 2014 alone, fossil-based energy systems emitted close to 26$MtCO_2e$ into the atmosphere, while in Togo, the figure stood at 2.7$MtCO_2e$. The same can be said for Senegal and Guinea, where fossil-based energy systems emitted about 9.1 $MtCO_2e$ and 2.6 $MtCO_2e$, respectively into the atmosphere.

Under this output, the programme will promote the use of off-grid solar to power agro-equipment in the SCPZs programme areas such as milling machines and to light warehouses and agribusiness facilities, including solar irrigation systems. This will significantly reduce smallholder farmers, micro and small-scale agribusiness enterprises reliance on biomass-based and fossil-based energy systems for processing. Under BAU, the emissions baseline of using 100% Diesel and biomass-based energy systems for lighting and processing in the programme could amount to over 14,738 tCO2eq annually. However, with GCF support, an amount of 13,731tCO2eq from this baseline could be avoided annually, an abatement which will be attributed to solar lighting and processing systems. In addition, GCF support for this output is expected to directly benefit over 2,400 agricultural cooperative societies (ACSs), and farm-based associations (FBAs), including their employees and families (578,400).

***Activity 1.2.1:*** *Improving resilience of agricultural cooperatives and FBAs value chains infrastructure for food processing*
Under this activity, the programme will invest in the installation of 14.69MW of solar energy for lighting, processing, drying and packaging of staple food crops.
To be consistent with AfDB's procurement procedures for competitive bidding, no particular technology is stated as preferable. Besides, there is technical feasibility for all the technologies under consideration in the Programme countries with adequate solar irradiation.

[39] Global Forest Watch, Guinea.

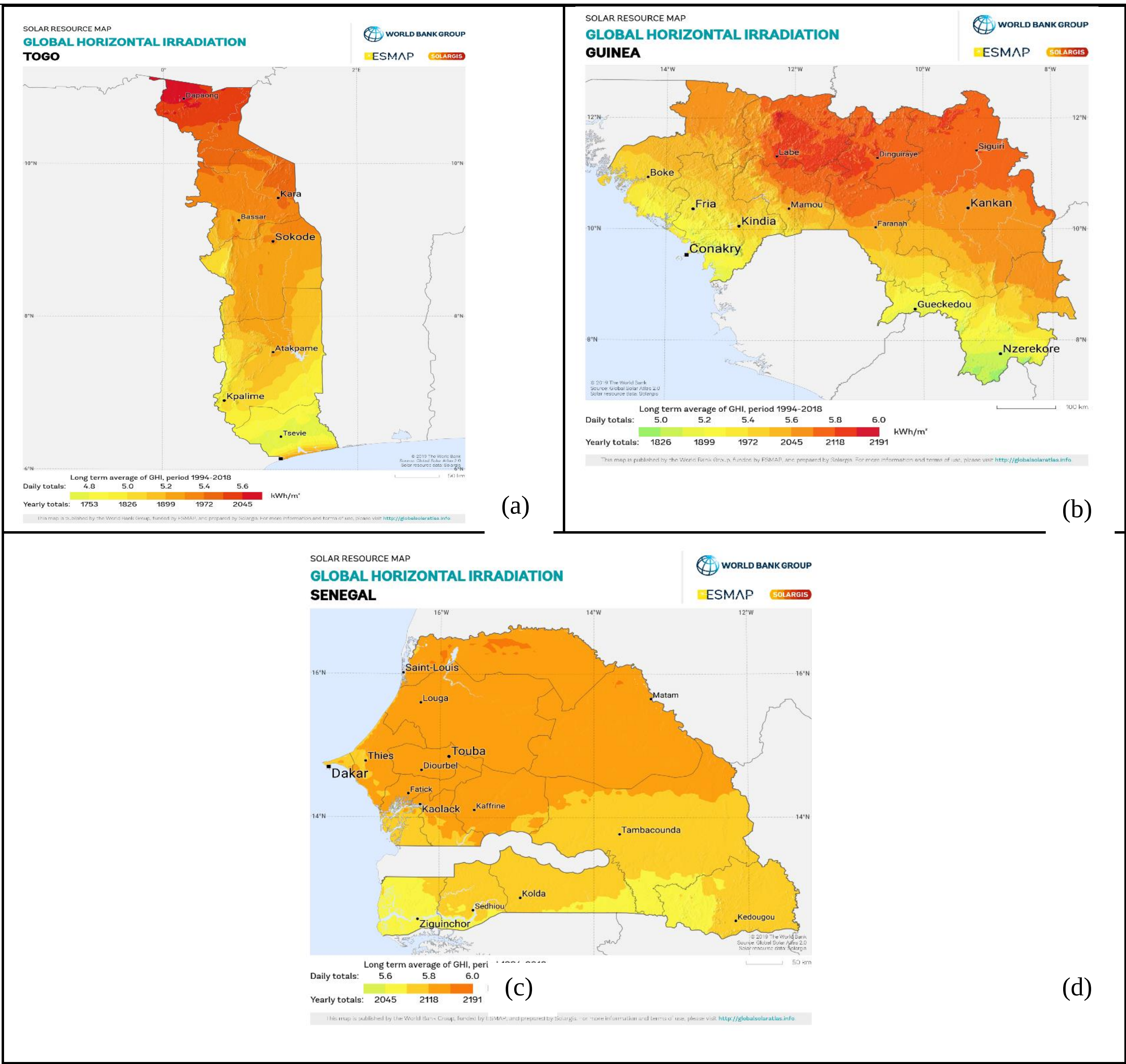


**Figure B.3.1.** Solar Irradiation Map of (a) Togo, (c) Guinea and (d) Senegal

- In Togo, solar irradiations are such that 1 m² of installed photovoltaic can yield between 4.8 and 5.6 kWh energy daily. Long term data collected between 1994 and 2018 shows the potential to collect between 1753 and 2045 kWh of energy per square meter. The Kara region in the north is particularly blessed with abundant solar potential. The solar irradiation map of Togo is provided in Figure B.3.1a.
- In Guinea, solar irradiations are such that 1 m² of installed photovoltaic can yield between 5 to 6 kWh energy daily. Long term data collected between 1994 and 2018 shows the potential to collect between 1826 and 2191 kWh of energy per square meter. The solar irradiation map of Guinea is provided in Figure B.3.1b.
- In Senegal, solar irradiations are such that 1 m² of installed photovoltaic can yield between 5.6 and 6 kWh energy daily. Long term data collected between 1994 and 2018 shows the potential to collect between 2045 and 2194 kWh of energy per square meter. The solar irradiation map of Senegal is provided in Figure B.3.1c.
- The solar potential confirms that the solar arrays will sustainably and predictably be able to provide the required energy without concerns about local weather patterns.

***Activity 1.2.2:*** *Deploying low-carbon energy technologies for enhanced resilience*

As extensively discussed in the programme feasibility study, the over dependence on charcoal and firewood as a major source of energy by households in the three countries, including the use of diesel-based equipment for agricultural processing, further compound the problems of deforestation, degradation and increased carbon emissions. Also, the use of low-intensive technologies and unsustainable agricultural practices equally increases GHG emissions for these countries. For instance, enteric fermentation (3.9Mt$CO_2$e), rice cultivation (3.5Mt$CO_2$e), manure left on pasture (3.3Mt$CO_2$e), burning Savannah (0.6Mt$CO_2$e), and other types of agricultural activities (0.6Mt$CO_2$e), was responsible for over 117% increase in GHG emissions from agriculture in Guinea between 1990 and 2014. The same can be said about Senegal, where the growth of GHG emissions in agriculture have been largely attributed to enteric fermentation (4.0 Mt$CO_2$e), burning savannah (3.9 Mt$CO_2$e), manure left on pasture (2.8 Mt$CO_2$e), and others (0.5 Mt$CO_2$e) as illustrated in Figure B.1.5. On the other hand, waste products (liquid and solid waste), food and agro-wastes that are likely to pose sanitation and waste management challenges in these countries are actually sustainable "resources" for clean energy generation such as biogas technology. As shown in Figure B.3.2, the biogas generated from these "waste" usually referred to as feedstocks could provide energy for diversified use such as for cooking, electricity, cooling, heating and even for transportation. Without GCF support, baseline emissions from; 1) methane discharge from bio-waste; and 2) Diesel equivalent for biogas generators, would amount to over 477,392 tCO2eq annually. With GCF support however, emissions abatement for energy and waste; 1) bio-waste digested; and 2) avoided diesel using biogas for electricity, amounts to more than 424,879 tCO2eq annually. In addition to this, output 1.2 will directly benefit over 2,400 agricultural cooperative societies (ACSs) and farm-based associations (FBAs), and more than 578,400 firms' employees and household members.

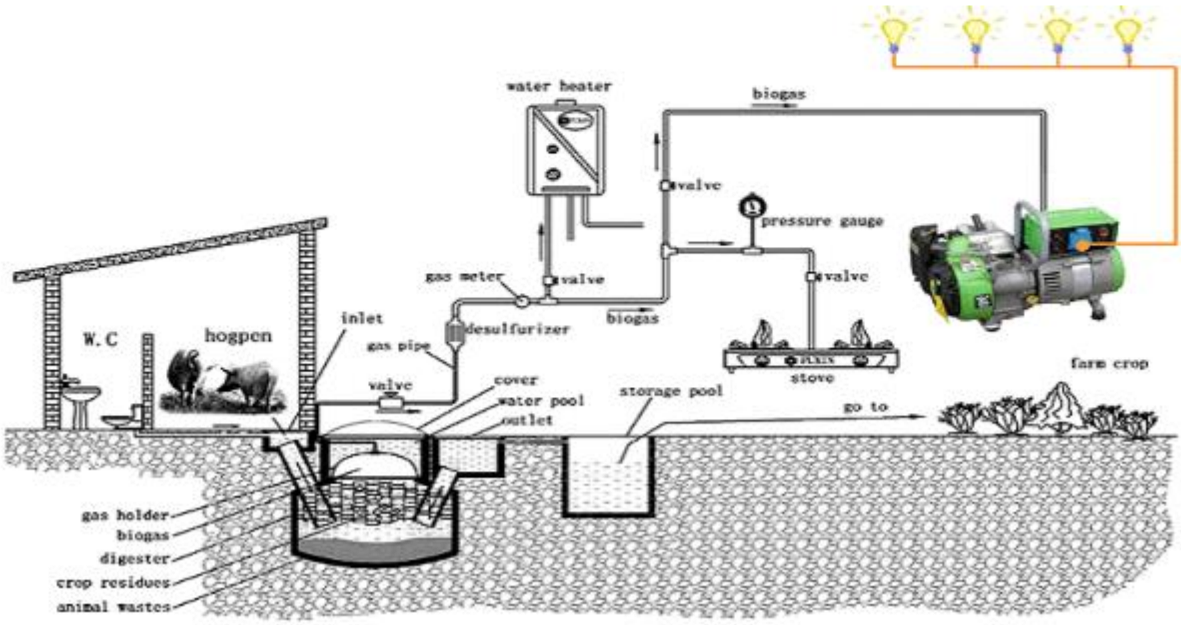


**Figure B.3.2.** Biogas generation from human and agro-waste for diversified energy use.

Source: Greenpower.com cited in Elmar Dimpl/GTZ 2010[40].

Under this activity, the programme will invest in the installation of about 24,576[41] m3 of biogas digester to treat livestock manure and produce biogas for power generation (See programme financial and economic models for the justification of Biodigester volume). The genset output for the total biodigesters to be installed is about 10.2 MW of renewable energy from biogas generation. The energy provision will support processing of farm produce, powering of cold storage facilities, and where applicable, to ensure that farmers can offer their produce with the required market standards for packaging. The biogas energy systems will also provide them with opportunities to diversify their sources of incomes through the sale of agriculture wastes,

[40] Elmar Dimpl (2010). Small-scale Electricity Generation from Biomass. Part II Biogas. Published by GTZ-HERA – Poverty-oriented Basic Energy Service. https://energypedia.info/images/4/43/Smallscale_Electricity_Generation_From_Biomass_Part-2.pdf

[41] See biogas estimation model for the basis of this figure.

which otherwise would have been left in the nature and would have been the source for more GHG emissions as highlighted above.

The temperature in all the Programme Countries provides great advantage for thermophilic conditions favourable to anaerobic digestion. Organic material degrades more rapidly at higher temperatures so thermophilic digestion in could lead to methane production rate that is approximately twice the rate of mesophilic digestion. Because of this favourable condition, reactors for thermophilic digestion can be half of the volume of mesophilic digesters, and still deal with the same volume of materials, while maintaining the same gas production with similar efficiency levels. Due to the thermophilic conditions, all the key biodigester plant models such as Fixed dome plant (e.g., hemisphere design, Deenbandhu design, and Chinese design), and Floating drum plant could be used in the project areas. So, preferring one single technology will not be consistent with the AE procurement procedures. At the bidding, the private ESCOs will demonstrate why a particular technology is chosen as the specific costs.

**Output 1.3: Increased agricultural productivity through enhanced access to Agro-Industrial Parks (AIPs) support facilities and access infrastructure (Financed by AfDB and Partners).**

There are two essential components that make up a well-functioning Staple Crops Processing Zone (SCPZ) Programme. That is: i) Agro-Processing Hubs (APHs)[42] that offer adequate infrastructure, logistics and specialized facilities and services (e.g. electricity, water, cold chain facilities, laboratory and certification services, business services, ICT, waste treatment, etc.), required for agro-industrial activities; and ii) Agricultural Transformation Centres (ATCs)[43], which are strategically located within the production areas to serve as aggregation points to accumulate products from the community to supply the Agro-Processing Hub. These two essential components must be supplemented with access to basic infrastructure such as feeder roads, electricity, ware houses, water and water storage facilities, cold chain facilities, produced storage facilities, laboratory and certification services, business services, ICT, waste treatment, among others.

*Activity 1.3.1: Development of SCPZs Access Infrastructure & Support for AIP Management Governance*

Under this activity, AfDB and partners will finance under the Agro-Processing Hubs (APHs) facility: 1) the construction of about 10 regional modules comprising of; administrative building, installation of ICT facilities such as fibre optics (50km length) and market platforms (hubs), financial centres, water supply plant for over 15,000 cubic meters/day; waste management plants with capacity of 3.21 MLD, 8 new boreholes and the rehabilitation of 30 old boreholes, 3 mini-dams and additional gravity irrigated areas spanning over 15,000 ha; provision of 243 kV transmission lines to, and around the IAIPs areas; 230 km new access roads that are resilient to extreme weather events, including the rehabilitation of 472 km of old roads. Under the Agricultural Transformation Centres (ATCs) facility, AfDB will finance: 2) construction of more than 40 ATCs) that are strategically located within the production areas of farming communities to serve as aggregation points to accumulate products from farmers and communities; and, 3) provision of cold stores, storage warehouses for fresh-farm produces from the farming communities.

**Component 2: Promotion of climate resilient agricultural (CRA) practices and technologies adoption among smallholder farmers**

Under this component, the programme will invest in the promotion and use of CRA practices and technology uptake among agricultural value chain actors in the three selected SCPZ countries. At the production stage, the programme will invest in the adoption of CRA practices and technology uptake such as use of drought resistance and improved yield seeds, integrated nutrient management, sustainable agro-forestry practices, and livestock management. This component will also focus on activities to enhance climate information services with for example, the establishment and expansion of agrometeorological and rain gauge stations network and by promoting the generation of, access to and use of seasonal and sub-seasonal weather forecast and agricultural relevant weather products and climate-smart agro-advisory services in the SCPZs.

**Output 2.1: Adoption of CRA and agro-forestry practices among smallholder farmers**

[42] An APH is a well-defined, centrally managed tract of land developed, subdivided and dedicated to supporting firms and other stakeholders engaged in agro-processing and related activities located throughout the production area surrounding the hub.

[43] An ATC is a physical complex of facilities centrally located in the middle of a farming community, where required services are offered to farmers, including crop drying facilities, cold stores and warehouses, farm equipment rental and maintenance services, crop handling, grading, storage, and processing for increased shelf life; livestock handling, slaughtering and meat packing; fish handling, grading and processing; food quality and safety control and certification; distribution and marketing platforms.

According to FAO[44] and USAID[45], climate-resilient practices can reduce the intensity of climate impacts on agriculture productivity and generate additional benefits by increasing resilience to floods and droughts. Certain CRA practices such as, use of drought resistance and improved yield seeds, sustainable agro-forestry practices, and livestock management, have been shown to improve soil quality and potentially double the yield per hectare (see programme feasibility study for possible benefits that can be derived from adopting CRA practices). On the other hand, agro-forestry practices will provide an alternative and more sustainable source and many other benefits for smallholders as it offers compelling synergies between adaptation and mitigation. According to Mbow et al. (2014)[46], agroforestry is a source of income from carbon and wood fuels, it improves soil fertility and creates micro-climates and it provides ecosystem services and reduces the intensity of human impacts on natural forests. In general, agroforestry improves the economic and resource sustainability of agriculture while sequestering greenhouse gases. It provides a particular set of innovative practices that are designed to enhance productivity in a way that often contributes to climate change mitigation through enhanced carbon sequestration, and that can also strengthen the system's ability to cope with adverse impacts of changing climate conditions (Torquebiau, 2013).[47]

***Activity 2.1.1: Promote CRA among smallholder farmers***

Under this activity, the programme will invest in the promotion and use of selected CRA practices among smallholder farmers, ACSs and FBAs, covering about 39,178 ha of land (see pages 69 - 71 of the programme feasibility study for selected CRA practices). It will equally involve the deployment of climate-resilient seed varieties, seed production and multiplication, in collaboration with the National Food Crop Research Institutes in the respective countries.These national food research centres have the mandate to develop and introduced new seed varieties and Grafted-Soin seedlings that are adaptable to the local climates and conditions. Such seeds and Grafted seedlings are an important agricultural requirementsfor improving crop productivity and ensuring food security. Adapted and improved seed varieties and Grafted seedlings made available to farmers at the right time will make a significant contribution to the yield increase necessary to meet the market demands. High quality seeds of adapted and improved varieties are essential to: (i) Increase the efficiency in use of nutrients and water; (ii) Increase resistance to insect pests and diseases, and provide greater tolerance to drought, flood, frost and higher temperatures associated with climate change; (iii) Improve nutritional value and quality; (iv) Improve provision of ecosystem services. The programme will attract the private sector, which will set up seed and Grafted seedlings distribution companies, and support the introduction of the most appropriate varieties for the project areas such as those with short growing cycles (SGC). Compared to the traditional baseline seeds used in these countries, these improved seed varieties have SGCs with less crop water requirements as discussed under crop water requirement in section B.

***Activity 2.1.2: Promote agro-forestry practices among smallholder farmers***

Under this activity, the programme will assist farm households to create sustainable managed forests (about 40,000 ha) for income generation from woodlot, cashew nuts, mango and coffee orchards for carbon sequestration. The introduction of this activity will provide an alternative and more sustainable income source, including many other benefits for small holders as it offers compelling synergies between adaptation and mitigation.

**Output 2.2: Awareness raising, capacity building, community and institutional strengthening on agricultural value chains management (GCF, AfDB and Partners)**

The adoption of new climate technologies as well as climate resilient agricultural (CRA) practices, require massive awareness raising and knowledge transfers among value chain actors. This is particularly important to increase the rate of adoption of these technologies and innovations in agriculture, including operations and maintenance (O&M) of assets. Besides, in the event of possible up-scaling and out-scaling of these technologies and agricultural innovations and management practices, a critical mass of well-trained value chain actors, community agents and institutions are needed.

***Activity 2.2.1: Capacity building of value chains agents/communities'& development of agricultural processing support infrastructure***

Under this activity, the following training and capacity building exercise will be conducted. 1) Capacity building and training of relevant staff of the Executing Entities (EEs), the Project Implementing Units (PIUs), the Agro-industrial Parks (AIPs), value chain

[44]FAO. (2012). Identifying opportunities for climate-smart agriculture investments in Africa. Rome: Food and Agriculture Organization of the United Nations - Economics & Policy Innovations for Climate-Smart Agriculture.
[45] USAID (2017). Cost and Benefit Analysis for Climate-Smart Agricultural (CSA) Practices in the Coastal Savannah Agro-Ecological Zone (Aez) of Ghana. Working Paper, September 2017. file:///C:/Users/akt105093/Downloads/Cost_and_Benefit_Analysis_for_Csa_Practices_in_the_Coastal_Savannah_AEZ_of_Ghana.pdf
[46]Mbow, C., Smith, P., Skole, D., Duguma, L., Bustamante, M. (2014). Achieving mitigation and adaptation to climate change through sustainable agroforestry practices in Africa. Current Opinion in Environmental Sustainability, 6, 8-14.
[47]Torquebiau, E. (2013). Agroforestry and climate change. FAO webinar. http://www.fao.org/climatechange/36110-0dff1bd456fb39dbcf4d3b211af5684e2.pdf

actors and ESCOs. This will include; O&M learning and training, management, business planning, stakeholder engagement, contract negotiation, PPP arrangements, governance, project performance monitoring, reporting and evaluation (M&E), investment mobilization and support, and investment promotion among others. 2) Training and capacity building of value chain actors in the use of‘Good Agricultural Practices’ in product quality, standardization and conformity requirements, as well as developing linkages and responding to commodity demand in terms of quality and quantity. 3) Training of young potential entrepreneurs including women to seize business opportunities along the value chains. 4) Massive sensitization and training of private sector investors, Ministries, Departments and Agencies (MDAs) on governance systems reforms such as land reforms and export promotion initiatives.

***Activity 2.2.2: Establishment of climate information and early warning systems in SCPZs***

This activity will help to enhance climate information services by promoting the generation of, access to and use of seasonal and sub-seasonal weather forecast and agricultural relevant weather products in the SCPZs. Collection of climate data and transmission of climate information at community-level will be promoted. The activity will improve climate-smart agro-advisory services, such as, planting time, good farm management practices, irrigation schedules, use of farm manure and choice of inorganic fertilizers, suitable crop types and varieties to be planted, weeding regimes, the available seed suppliers, crop pests and diseases prevention and control measures. This activity will provide technical assistance on preparatory mechanism including analysis of vulnerability, gathering of data, decisions on contingency plan and/or adaptation measures, analysis of gender and socioeconomic inequalities; seasonal information prior to major rainfall season including information on irrigation requirements during growing season; improved dissemination technique; Capacity building - adaptation and contingency & anticipatory plans.

Specifically, the activity will deliver on the following: ***(i) Expansion of agrometeorological and rain gauge stations network within the SCPZs -*** The GCF intervention will establish and expand observing network with at least ten (10) automated agrometeorological and fifty (50) self-recording rain gauge stations in each of the SCPZ region (see Annex 23 for technical description of sensors). The local weather stations will be established following the World Meteorological Organization guidelines[48] to ensure best practices and standards. The activity will also procure and maintain two (2) four-wheel project vehicles per region (i.e., a total of sixteen (16) project vehicles). One of the project vehicles will be fitted with a calibration unit to monitor the operation and conduct regular maintenance of all equipment and facilities installed across the region; **(ii) *Deployment of technologies to strengthen climate information services and early warning systems in SCPZs and Business Model Innovation for Togo, Guinea, & Senegal*** - This sub-activity will promote rural economic empowerment in the SCPZs through digital and financial inclusion. A cost-effective reseller model will be used to deploy and expand climate-smart agro-advisory services to targeted users within the SCPZs. These services will be deployed following the prototype model from ESOKO, an African-based digital agribusiness company, in collaboration with local partners including the National Meteorological Services (NMSs) and mobile network operators (MNO) in the 4 proposed countries (Figure B.3.3).

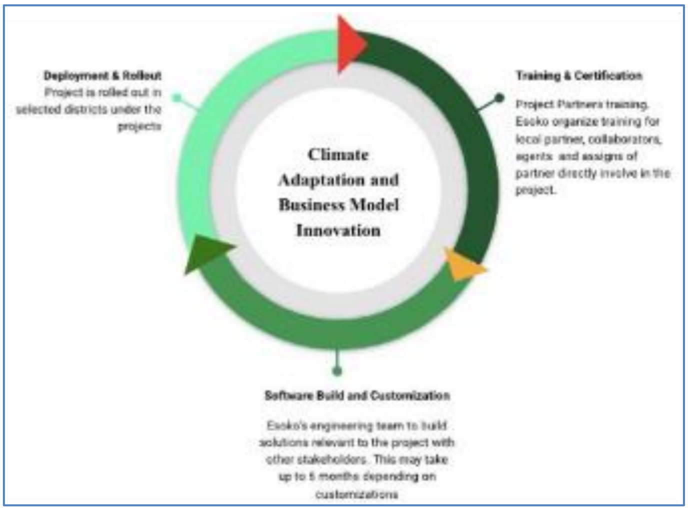


*Figure B.3.3. A typical partners onboarding process in specific project site.*
*Source: ESOKO (https://esoko.com/)*

[48] World Meteorological Organization (WMO). 2010. Guide to Agricultural Meteorological Practices. WMO-No. 134. WMO, Geneva, Switzerland.

Automated solutions will be used to profile and disseminate critical information to a base target of 20,000 beneficiary farmers in the inception year of set-up in each of the 3 host countries. The project approach will be to grow this beneficiary farmers by an average of 50% annually in subsequent years. In addition, the business model innovation will create about 30 direct jobs in the first 2 years of implementation in each of the proposed countries. There will also be the opportunity to create additional short-term jobs for about 60 young people to be engaged as field enumerators for the initial profiling of the 20,000 beneficiary farmers. The Tech Services to be provided in this phase is illustrated in Figure B.3.4, consist of rainfall forecast, climate-smart agro-advisory contents, market prices, digital tools to support end-to-end digitization, and digital literacy skills:


Figure B.3.4. Illustration of Tech Services to be provided
Source: ESOKO (https://esoko.com/)

Digital Tech Services in Critical Information Delivery includes: (a) Seasonal Rainfall forecasts; (b) Onset and end of the rainy season; (c) Climate Smart Agronomic Advisory; (d) Creating Market linkages through Market Price Advisory from all the districts/provinces/counties of all the 3 proposed countries of deployment; and (e) Farmer crop/livestock/fishery/poultry protocol advisories that is intended to enable farmers understand and apply received critical information in the best possible channels as well as health, nutrition and financial inclusive services.

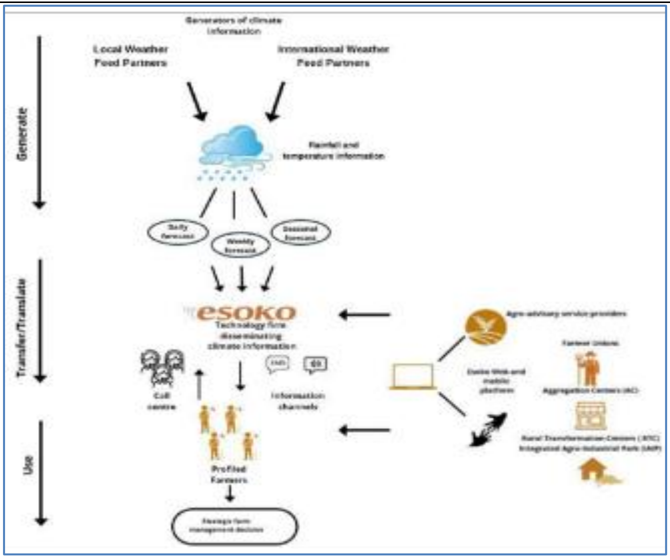


Figure B.3.5. Automated system for the dissemination of Climate and Weather-related Agriculture Advisories

Further details on the automated system for the dissemination of climate and weather-related agriculture advisories (Figure B.3.5) is presented in Section 7 of Annex 2; and **(iii)** ***Capacity building and awareness raising –*** This activity will focus on awareness and training on how to use and maintain the observing networks. The activity will conduct stakeholders' engagement meeting with prospective farmers, aggregators, Famer Unions (FUs), Famer Based Organizations (FBOs), and other interested bodies with the focus on creating awareness on the importance of the installed agrometeorological and gauge stations and the need to be responsible for the security of the networks. Farmers and other interested stakeholders will take ownership of the self-recording rain stations. Furthermore, selected volunteers will be engaged in training on how to maintain the rain gauge stations in order to ensure accurate rainfall measurements. Bi-annual exercise on calibration will be organized to re-calibrate the weather sensors to ensure measurement accuracy.

**Component 3: Programme management and coordination**

***Output 3.1: Project management and coordination (AfDB, Partners and Governments Contributions)***

This output is entirely financed by AfDB, other project financiers and counterpart funds from the governments of Togo, Senegal and Guinea.

***Activity 3.1.1: Project management, evaluation and coordination***

Under this activity, project management, evaluation and coordination will consists of: 1) baseline survey (e.g., energy use and penetration, socioeconomic conditions and quality of life, agricultural baseline etc.); 2) project implementation, management and coordination at the individual country level; 3) procurement and disbursement arrangements for goods, services, and consulting works at the projects and sub-projects levels; 4) organisation of procurement and implementations at the levels of the TRP of the Executing Entities (TRP/EEs) and Project Implementing Units (PIUs) or Project Management Units (PMUs); 5) financial management arrangements at the levels of the TRP/EEs and PIUs; 6) preparation of internal and external audits and audit follow-ups, production of financial statements and reporting; and 7) monitoring and evaluation of project and sub-project performance measurement frameworks (more details are provided in the programme operation manual –POM, supplied as an Annex 21 to the funding proposal).

**Selection of Sub-projects to be Executed under the Programme and Beneficiaries[49]**

As extensively discussed above and further highlighted in Table B.4.1, a number of sub-projects have been identified in each countries PAR and the program feasibility study, primarily related to: (1) small–scale agricultural water management (AWM) infrastructure such as, (a) drip irrigation distribution systems (DIDS) powered by solar pumps, (b) improving agricultural cooperatives, FBAs value chains infrastructure. (2) Promotion of CRA practices and technologies adoption to include; (i) prog CRA and agro-forestry practices, and (ii) deployment of low-carbon energy technologies such as Biodigester and Solar PV for renewable energy generation.

[49] Details the sub-projects selection criteria for the baseline component financed by the Bank and partners, are summarised in the programme feasibility study (PFS) report supplied as annex 2 to the funding proposal (FP).

Concerning all sub-projects, in accordance with AfDB implementing arrangements, the first step of the process involves the Issuance of International Competitive Bidding (ICB), for the recruitment of national or international consulting firm(s). After the selection process (see POM for more details on selection procedure), the firm(s) will visit each pilot country to agree on the detailed scope and timeline of activities, carry out a stock taking exercise on existing knowledge and gaps with respect to the proposed interventions, and developed an inception report. Next, the selected firm(s) will liaise with each country's Technical Responsible Party (TRP) of the Executing Entity (EE) PIU, PSC, value chains actors, community members and other key stakeholders (at the national and regional levels), to agree and to finalise the programme selection criteria to be used for enlisting programme beneficiaries. Also at this stage, a detailed mapping exercise will be undertaken to identify both programme beneficiaries and the 39,178 ha of land for the GCF funded activities. Equally, at the stage, the firm is expected to work with the respective TRP of the EE, PIU and PSC/PMSC, to finalize the process of constitution a sub-project management committee for each activity (within target beneficiaries), and also to develop a field finding report of the mapping exercise. At the last stage, the firm(s) is expected to conduct training of the programme beneficiaries in conjunction with line ministries, departments and agencies (MDAs) involved in the programme. This will be in close collaboration with the TRP of the EE, PIU, PSC/PSMC and other key stakeholders. In addition, the selected firm(s) is expected to organize a 'Training of Trainers'– (ToTs) workshop for selected programme beneficiaries, community members, and other value chain actors. The aim is to train and produced a critical mass of 'project experts' that will support the up-scaling and out-scaling of project activities, especially technology and agricultural management innovations within the SCPZ Phase 1, and to other regions of the country. The firm(s) is equally expected to produce a workshop report that summarizes the key lessons from each training, developed a ToT manual covering each activity, and prepare a final report highlighting the main findings of the assignment (see POM for sample Request for Expression of Interests (REOIs) to be issued under each sub-project activities).

***Activity 1.1.1: Support access to small-scale agricultural water management (AWM) infrastructure***

- All activities related to drip irrigation will be carried out within the SCPZs designated programme areas in all three countries (31,179 ha out of the total 1.5 million ha designated for the programme).
- 80% preference for women and girls who are the most vulnerable groups in the society;
- Beneficiaries must be a formally registered ACS, FBA, women-led agribusiness enterprise (WABE), or smallholder farm household (SHFH) willing to work with the programme to implement climate resilient agricultural practices associated with on-farm surface water catchment;
- Preference for ACS, FBA, WAE, or SHFH that are willing to belong to a farmer water user group (FWUG) or farmer water user community (FWUC) to promote water harvesting through on-farm catchment (small ponds and reservoirs), for irrigational use, including necessary O&M;
- Preference for ACSs, FBAs, WABEs, and SHFHs that commits to contribute to necessary O&M of on-farm water catchment ponds and reservoirs;
- Preferences for ACSs, FBAs, WABEs, and SHFHs practicing any form of water coping strategy in agriculture such as water harvesting, use of boreholes, small reservoirs and dams;
- Preferences for ACSs, FBAs, WABEs, and SHFHs located in communities around the IAIPs more prone to climate hazards, risks and potential disasters. Here, community involvement would be crucial in the selection of programme beneficiaries;
- Preference for ACSs and FBAs that have women farmers as potential beneficiaries;
- Preferences for ACSs, FBAs, WABEs, and smallholder farmers that engaged in dry season farming as a coping strategy;
- Being in production of fresh agro produces within the last two years;
- Willing to participate in the programme and to accept technological innovations such drip irrigation, including necessary O&M;
- Willing to accept new fresh agro varieties that may be introduced by the programme; and,
- Committed to work with the IAIPs, PIUs and other regional agricultural institutions and line ministries.

***Activities 1.2.1 and 1.2.2: Providing solar and biogas energy for staple food crops processing***

**The selection criteria for the eligible for solar PV and bio-digester includes those listed under activity 1.2.1 and *1.2.2* as well as the following additional criteria:**

- The Design, Construction, Operational and Maintenance (DCOM) will be handled by Private Sector Energy Service companies (ESCOs), which will be selected through a public procurement process in collaboration with the Project Implementing Units (PIUs) in each country;
- ESCO must show through formal commitment the responsibility of handling the operational and maintenance (O&M) costs of the equipment throughout the project lifespan. ESCOs were contacted during a field mission in 2022 and were questioned about their interest to participate in the project. Most responded positively provided sufficient incentives were provided (refer to Chapter 6 of the Feasibility Study)
- Direct programme beneficiaries must be in production of at least 100 kg – 200 kg of manure daily;
- Has no objection for ACS, WAE or FBA to be used as a training and demonstration location for the duration of the project;

- Commitment to adopt climate resilient and good agricultural practices such as composting, mulching, organic agriculture, climate resilient varieties;
- Capacity to contribute funds either by having own funds or taking loans;
- Commitment to use bio-slurry as a fertilizer substitute.

***Activity 1.3.1: Development of SCPZs Access Infrastructure & Support for AIP Management Governance[50]***

- The agricultural potential of each region with respect to the targeted staple food crops for sustainable agricultural intensification.
- The existence of other ongoing activities in the regions complement the project.
- Proximity to local and international markets, transport networks and telecommunications.
- Stakeholders' consultations and viewpoints.
- Willingness of communities to participate in the project.
- Proximity to raw materials.
- Existing agro-processing infrastructures that are in place in each region.
- Energy and water requirements.
- Furthermore, consideration was given to identifying sites that have access to existing commercial and support services such as universities, research centres, technical vocational education and training centres; farmers' cooperatives and unions; and financial institutions.

- A formally registered agricultural cooperative society (ACS), FBA or WABE with the Government for more than 1 year;
- A formally registered SHFH with a cooperative society, FBA or WABE
- Main focus of activities are along the agricultural value chains involving targeted crops under the programme;
- Currently practicing any form of climate resilient activities in its daily operations such as use of drought resistance and improved yield seeds, among others;
- Willing to accept new innovations that may be introduced by the programme such as clean and renewable energy sources; including necessary O&M
- Preference ACSs, FBAs that utilises diesel-based energy sources for agro-food crops processing;
- The size of the cooperative society or FBA must not be less than 20 member with at least 20 -30% of the members being women;
- Preference for ACSs and FBAs that have women farmers as potential beneficiaries;
- Has a strong business plan to develop a trading agribusiness with potential downstream value chain linkages;
- Should have strong commitment to implement climate change measures to reduce vulnerability to impacts of climate change, and reduce GHG emissions in its operations through prog renewable energy and improving energy efficiency uptake; and,
- The area under cooperative or FBA primarily grows any of the 18 staple food crops considered under the 'Feed Africa' Initiative such as rice, maize, potato, yam, cassava, mango, cashew, coffee, sorghum, millet, cowpea, wheat, soybean, livestock, poultry, and fish among others.

***Activities 2.1.1 and 2.1.2: Promote CRA and Agro-forestry Practices among Smallholder Farmers***

- All activities related to CRA and agro-forestry management practices will be carried out within the SCPZs designated programme areas in all three countries (79,178 ha out of the total 1.5 million ha designated for the programme).
- A formally registered SHFHs with an ACS, or FBA for more than 1 years
- A formally registered ACS, WAE, FBA with the Government for more than 1 year;
- 80% preference for women and girls who are the most vulnerable groups in the society;
- The size of the ACS, WAE, or FBA must not be more than 15 member with at least 20 -30% of the members being women;
- Preference for ACSs and FBAs that have women farmers as potential beneficiaries;
- Preference for beneficiaries that are currently practicing any form of climate resilient activities in daily farming operations such as water harvesting, mixed agro-forestry practice, livestock rearing, use of drought resistance and improved yield seeds, among others;
- Preference for beneficiaries that are willing to accept new technological innovations and new farming practices that may be introduced by the programme such DIDS powered by solar pumps; improved nutrient management, and improved genetic resources
- Activities are mainly carried out along the agricultural value chains involving targeted crops under the programme;

[50] The selection criteria for this activity differ for the different project-specific regions and extensively discussed in the various ESS reports, and summarised in pages 44 - 51 of the programme feasibility study (PFS) report.

- Willing to accept new innovations that may be introduced by the programme such as improved nutrient management, improved genetic resources, and the adoption of clean and renewable energy sources;
- Willing to accept new seed varieties and technologies that may be introduced by the project;
- Has a strong business plan to develop a trading agribusiness with potential downstream value chain linkages;
- Should have strong commitment to implement climate change measures to reduce vulnerability to impacts of climate change, and reduce GHG emissions in its operations through prog the adoption of good agricultural practices and technology and innovations uptake; and,
- The area under cooperative or FBA primarily grows any of the 18 staple food crops considered under the 'Feed Africa' Initiative such as rice, maize, potato, yam, cassava, mango, cashew, coffee, sorghum, millet, cowpea, wheat, soybean, livestock, poultry, and fish among others**.**

***Activity 2.2.1: Capacity building of value chains agents/communities'& development of agricultural processing support infrastructure***

The selection criteria for activity 2.2.1 includes:

- All program beneficiaries of activities 1.1.1 – 2.2.1;
- Preference for SHFHs, ACSs, WABEs, FBAs who are non-beneficiaries of the program, but located within the SCPZ program areas;
- Staff of the EEs, PIU, AIPs, and relevant staff of Ministries, Departments and Agencies (MDAs) working on the agro-industrial domain and other key sectors involved in agricultural value chain development (AVCD);
- Representatives of organised Civil Society Organizations (CSOs), Environmental Non-Governmental Organizations (NGOs), and the prospective private sector investors;
- Representative of Public and Private Financial Institutions (PFIs);
- Preference for young potential entrepreneurs, SMEs, MSMEs working on the agro-industrial domain; and,
- Relevant representatives of the donor community.

**Activity 2.2.2: Establishment of climate information and early warning systems in SCPZs**

For climate-resilient and sustainable farm practices within the SCPZs, this activity will help to enhance climate information services by promoting the generation of, access to and use of seasonal and sub-seasonal weather forecast and agricultural relevant weather products in the SCPZs.

- The intervention will expand automated agrometeorological and self-recording rain stations to improve collection of climate data and transmission of climate information at community-level
- The activity will use innovative tech services to improve on climate-smart agro-advisory services, such as, planting time, good farm management practices, irrigation schedules, use of farm manure and choice of inorganic fertilizers, suitable crop types and varieties to be planted, weeding regimes, the available seed suppliers, crop pests and diseases prevention and control measures.
- This activity will (i) provide technical assistance on preparatory mechanism including analysis of vulnerability, gathering of data, decisions on contingency plan and/or adaptation measures, analysis of gender and socioeconomic inequalities; (ii) produce seasonal information prior to major rainfall season including information on irrigation requirements during growing season; (iii) improve dissemination technique; and (iv) build capacity - adaptation and contingency & anticipatory plans

**The technical approach to expand business in the 4 proposed project countries is summarized as follows:**

- To deploy a Tech Team to support the local partner team to conduct an initial field survey for the set-up of the technology infrastructure, engagement with local MNOs for the platform integrations and configurations and licensing bodies within the countries of deployment.
- To deploy a Call Centre Team for the set-up of the Call Centre's
- Deploy a Business Development Team to train the local partner business team on the commercialization of the solutions in their respective countries and beyond.
- To engage Agricultural Experts in the proposed project countries to use the developed digital platforms to collect and design tailored climate smart advisory, weather alerts, market price information, health and financial inclusive information services
- To use appropriate digital channels to communicate the generated information and warnings to target beneficiaries including smallholder farmers for their innovative farm management decision making vis-a-vi market linkage
- To create a digital database of profiles of the initial 20,000 beneficiary farmers in selected value chains to roll out the services

***Commercial sustainability***

Once deployed, the project aims to be the go-to for market linkages and financial inclusive service in the agricultural for beneficiary farmers and other actors/stakeholders in the value chain. The project will enable service providers to grow by helping farmers to grow their yield and with their businesses. The range of productivity tools to be deployed is expected to increase the beneficiary farmers' revenues by at least 20%. Such an opportunity highlights the responsibility to sustain the tech innovations and gives back to the farmers within the proposed project countries the kind of support they need to leap out of the base of the pyramid.

***Choice of selecting the programme interventions over alternative solutions and potentials***

The selection of all interventions in the SCPZ case studies where guided by each countries National Adaptation Programme of Action (NAPA), their respective NDCs, the Special Agro-Industrial Processing Zones Development Programme for each country, their different Country Strategy Papers (CSPs), AfDB's Strategy for the Industrialization of Africa (2016-25), and through several consultative processes with key stakeholders. The SCPZ program is therefore in line with the African Development Bank vision of transforming African Agriculture through its 'Feed Africa' agenda. The 'Feed Africa' agenda focuses on 18 priority commodities, which are consistent with the SCPZs structure with a set target of raising Africa's food production by an estimated 174 million metric tons. Thus, the SCPZ initiative is an important growth driver of the 'Feed Africa' agenda. It is expected that the SCPZ initiative will transform the African rural landscape into economic zones of prosperity, thereby lifting million of Africans out of poverty while simultaneously reducing GHG emissions in the agricultural value chain.

## B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

The African Development Bank Group (AfDB) is the Accredited Entity (AE). AfDB is the premier pan-African development institution prog economic growth and social progress across the continent. It is composed of three institutions, namely 1). The African Development Bank (ADB) which was established in 1964, 2). The African Development Fund (ADF) which was established in 1972 and 3) the Nigeria Trust Fund (NTF) which was established in 1976. Currently, there are 81 members - made up of 54 independent African countries (regional members) and 27 non-African countries (non-regional members).

The Bank Group focuses on five development priorities within the framework of its Ten-Year Strategy (2013–2022) aimed at prog inclusive and green growth. Usually referred to as the "High 5s", the development priorities are (i) Light Up and Power Africa, (ii) Feed Africa, (iii) Industrialize Africa, (iv) Integrate Africa, and (v) Improve the Quality of Life for the People of Africa.

This program is intended to support the realization of the "Feed Africa" and the "Improve the Quality of Life for the People of Africa" development priorities. The AfDB has supported several transformative activities and investments in these priority areas. These include, provision of financing for rural infrastructure (such as roads, irrigation, electricity, storage facilities, and access to markets) to improve agricultural productivity and competitiveness. AfDB has also supported several initiatives to strengthen agriculture and food security to improve the livelihoods of particularly the rural populations in Regional Member Countries (RMCs). Because agricultural production in rural communities largely depends on rain-fed irrigation, they are critically vulnerable to extreme weather conditions and uncertainties. It will be unfortunate to pursue an integrated approach to improved agricultural production and food security without the appropriate risk sharing products, access to affordable finance especially for women and services to mitigate the impacts of climate risks.

The AfDB has financed several related projects and programmes in Togo, Senegal and Guinea. For example, as of June 30, 2019, the AfDB's active portfolio in Senegal includes twenty-nine transactions for a total net commitment of 1,030 million Units of Account (MUC). This amount is distributed as follows: 864.29 MUC for the public sector and 167.14 MUC for the private sector. The sectoral breakdown of national projects is as follows: Transport (58.3%); Water and sanitation (16.4%); Rural (9.6%); ICT (6.4%); Social (5%); Energy (3.8%) and Governance (0.5%). The performance of the portfolio of national public sector projects remains satisfactory. The project turnaround time has decreased by an average of 9.4 months to 5 months from 2011 to 2019 and the average time between approval and fulfilment of the pre-payment conditions has improved from 9.6 months in 2013 to 7 months in 2019. The disbursement rate is 26% for an average age of the portfolio of 3.5 years. Finally, the public sector portfolio comprises 4 regional operations for a total amount of UA 83.52 million (Senegal) and a disbursement rate of around 35%.

In Togo, the AfDB's Ongoing portfolio includes 14 operations for a total net commitment of UA 231.85 million. It consists of 11 national operations of UA 112.47 million, accounting for 48.5% of the total budget, and three multinational operations of UA 119.38 million, accounting for 51.5%. Transport infrastructure (69.7%) dominates this generally satisfactory portfolio (with a rating of 3.2 on 6 in 2016 compared to 3 in 2016). The portfolio is relatively young, with an average age of 3 years; more than 64% of the projects approved between 2015 and 2018. Disbursement rate is 55.3%. Difficulties encountered mainly concern: (i) mobilizing counterpart funding; and (ii) cumbersome bureaucratic procedures related to disbursement and procurement. A close monitoring is required to accelerate the start-up of approved projects.

In Guinea, as of 30 June 2018, the AfDB's portfolio comprised 16 operations amounting to a total commitment of UA 282.95 million. The portfolio is dominated by regional public projects (64.23% of commitments) comprising nine operations. National public projects represent only 11.33% of commitments with six operations. The portfolio also includes a non-sovereign operation with 24.44% of commitments. The sector breakdown of the portfolio shows a predominance of the energy sector with six operations representing 39.39% of commitments, followed by transport with three operations (25.25%); industry with one operation (24.44%); governance with three operations (6.24%), the social sector with one operation (3.47%) and agriculture with one operation (0.35%).

AfDB as the AE will receive GCF proceeds in trust and sign the funded activity agreement with the GCF. It will sign loan and grant agreements with the Ministries of Finance for Guinea, Senegal and Togo. The EEs as indicated in Figure B.4.5 are Government of Guinea acting through Authority for the Development and Administration of Special Economic and Industrial Zones (ADAZZ) for Guinea, Government of Senegal acting through the Ministry of Industrial Development and Small and Medium Industries (MDIPMI) for Senegal, and Government of Togo acting trough Agency for the Promotion and Development of Agropoles in Togo (APRODAT) for Togo.

*Provide information on governance arrangements (supervisory boards, consultative groups among others) set to oversee and guide project implementation. Provide a composition of the decision-making body and oversight function.*

***Togo***

Figure B.4.1, shows the governance and implementation arrangements of the SCPZ project in Togo (more details can also be found in the POM supplied as Annex 21 to the funding proposal).

- At the top of the figure B.4.1 (Box 1), is the Ministry of Agriculture (MOA) that has the oversight function over the Kara Integrated Agro-Industrial Park (IAIP) in Togo. The Minister of Agriculture receives frequent project executing updates from the Agency for the Promotion and Development Agropoles (APRODAT), which is the technical responsible party of the - EE (Box 2). APRODAT was established by Decree No. 2018-036/PR of 27 February 2018, to steer the implementation of the Togo Agropoles Strategic Development Plan.
- The activities of APRODAT will be supported with the advises of two organs also serving as the JSC for the project in Togo: i) a Board of Directors, chaired by the Prime Minister, and comprised of seven active members (public and private) and observer members; and (ii) a Directorate General (DG) made up of five Directorates. That is, the Directorate of Legal, Administrative and Financial Affairs; the Directorate of Forecasting, Planning, Monitoring and Evaluation; the Directorate of the Promotion of Financing and Investments; the Directorate of Facilities and Works Control; and the Directorate of Training, Research and Development. The role of the Board of Directors is to meet every quarter to review, recommend, amend and approve the work programme and activity reports of APRODAT. On the other hand, the DG role is to support APRODAT in partially covering equipment and personnel costs, and in setting up the administrative, financial and accounting management system required for the successful implementation of the project (see POM for more details).
- Box 4, shows the Project Implementing Unit (PIU), which has as its main functions to: i) manage the construction of the Kara IAIPs (Box 5); ii) administer rent, sell or lease the park facilities; and iii) perform sensitization and promotional work to attract investors, especially private sector investors among others.

APRODAT was created by virtue of Decree No. 2018-036/PR of 27 February 2018 establishing the institutional framework for agro processing zones. It is an entirely governmental agency created in accordance with the principles and rules of the internal legal order. The assessment of country, sector and project risks as well as of the capacity of the technical responsible party of the EE on procurement matters was conducted for the programme and the outcomes served to guide the decision on the choice of procurement system (Togo or AfDB) used for specific activities or for all similar programme activities. APRODAT has been fully operational since 2018 with the recruitment of nine (9) specialists, including specialists in Procurement, Financial Management, Monitoring and Evaluation, Rural Infrastructure, Environment, Capacity Building and Gender, Agro-industry, Legal aspects and Internal Audit. In June 2021, with the agreement of the AfDB and BOAD, a Project Management Unit (PMU) was set up to boost the implementation of activities. Since then, this team of specialists has been expanded to include specialists in civil engineering, social development, agronomy and communication. The team of specialists is supported by two accountants, an agricultural assistant, an administrative assistant and drivers. The Togolese Government and the Bank will agree on a procurement plan for the first 18 months of the Programme, which will be updated annually or as needed, in order to reflect the actual programme implementation needs. Any revision of the plan will be subject to prior approval by AfDB.

Some of the actionable recommendations proposed for Togo after the FMA are:

- Update of the organization chart in accordance with the principles of internal oversight, particularly the separation of incompatible functions, and the appointment of the heads of services making up the directorates mentioned in the Decree, especially those of the directorates involved in the implementation of the SCPZ (Administrative and Financial Officer, Accountant, Procurement Officer, etc.);
- Review of the administrative, financial and accounting procedures manual, on which its internal oversight system will be based, covering all management cycles: expenditure management cycle (procurement procedures for goods and services),

asset management cycle (fixed assets, inventories, etc.), cash management cycle (scheduling, disbursement, cash monitoring and control, etc.), resource mobilization cycle (fund raising, direct payments, etc.), accounting, financial and budgetary information cycle, personnel management cycle, control and internal audit cycle;

- Development, where appropriate, of an operation manual for facilitation and financial intermediation activities (access of processing units to financing);
- Establishment of an integrated multi-project and multi-user management system to ensure the keeping of budgetary, analytical and general budget accounts. General accounting has to be a private accrual accounting system, applying the SYSCOHADA standards and taking into account the specific features of development projects;
- Review of internal audit mechanisms, with the recruitment of an Internal Auditor who will ensure that the SCPZ control system is operational and effective; and
- Review of external audit mechanisms through the hiring of independent external auditors on a competitive basis and in accordance with the terms of reference agreed with the Bank, in line with the Bank's requirement that audit reports be submitted no later than 30 June of the year following the year audited.

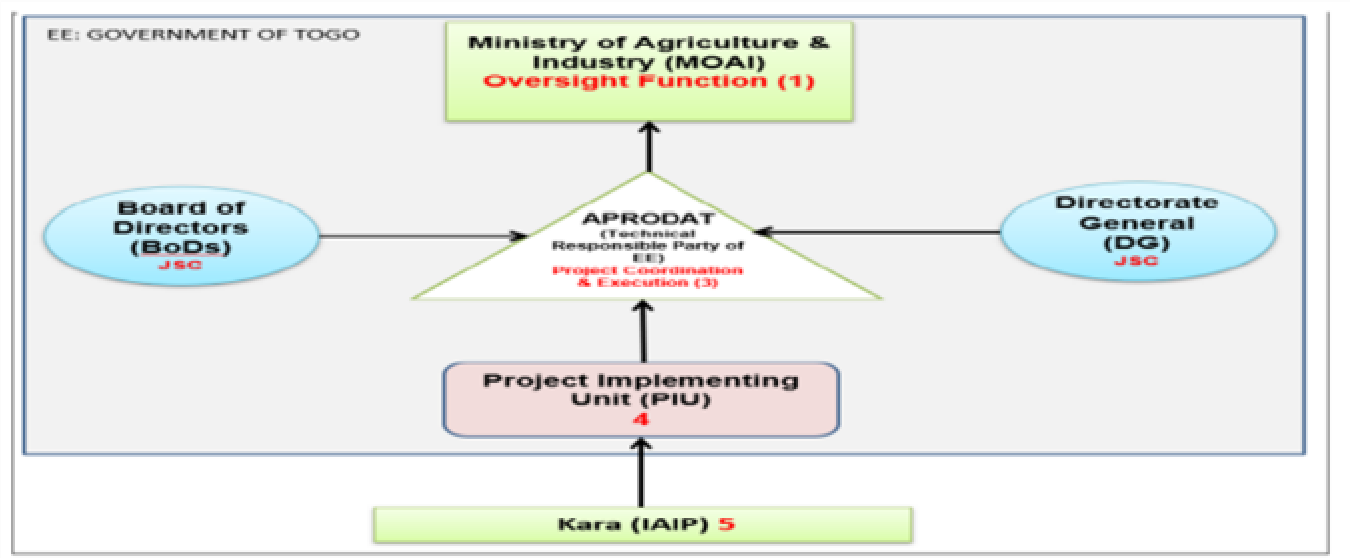


**Figure B.4.1. Governance and Implementation Arrangement in Togo**

***Senegal***

Figure B.4.2, illustrates the governance and implementation arrangements of the SCPZ project in Senegal (more details can also be found in the POM supplied as Annex 21 to the funding proposal).

- At the top of the figure B.4.2 (Box 1), is the Ministry in charge of Industry - MOI which has an oversight role in the project in Senegal. The Minister in charge of Industry receives frequent project execution updates from the technical responsible party of the Executing Entity, which is the Ministry of Industrial Development and Small and Medium Industries (MDIPMI) shown in Box 2. The activities of MDIPMI are supported by a Project Steering Committee - PSC (Box 3), which among others provides technical support and advisory services. These include among others, overall strategic advisory such as; review and approval of the annual work plan and budget (AWPB), and the analysis of activity reports (including financial reports and audit reports). The JSC will also provide policy advisory on matters related to the IAIPs development (see POM for the composition of the JSC).
- Box 4, shows the Project Implementing Unit (PIU), which is based in Dakar, Senegal. The PIU personnel, which will be recruited on a competitive basis, will include among others a coordinator, an administrative and financial manager, an accountant and an accounting assistant (see POM for the composition of the PIU). Among its numerous functions, are: i) development, management, and coordination of the three IAIPs (South IAIP, Centre IAIP, and North IAIP); ii) rents administration at the parks, selling or leasing of parks facilities; and iii) performing sensitization and promotional work to attract investors, especially private sector investors among others (see POM for more details).

An assessment of the existing country financial and accounting management system was also carried out for Senegal during the programme appraisal mission and the required mechanism specified for the programme. The current accounting system is accrual accounting on software enabling the recording of project/programme operations. The software has the potential to handle multiple projects with budgeting, analytics and general modules, up to the production of financial statements. The programme's internal control will be formalized in an administrative, financial and accounting procedures manual to be drawn up once the programme is put into effect. PZTA-Sud will produce quarterly project monitoring and annual financial statements and submit

them to AfDB after approval by the Programme Steering Committee (PSC). The programme financial and accounting audit will be conducted by an independent firm registered on the role of a national or regional order of the AfDB member countries. The audit firm will be recruited for three years based on terms of reference approved by AfDB, and paid out of AfDB funds.
Moreover, the MDIPMI has a well established procurement unit for the development and execution of the department's annual consolidated procurement plan each year including liaising with external missions, in particular contract audit or inspection missions,

Some key Recommendations of the FMA in Senegal:

- Prepare and forward to AfDB the administrative, financial and accounting procedures manual covering all management cycles, latest four (4) months following the effectiveness of the loan agreement; and
- Set up a multi-programme and multi-chapter integrated management system on which the programme budgetary, analytic and general accounting modules shall be kept, at most 6 (six) months following first disbursement of the loan or grants.

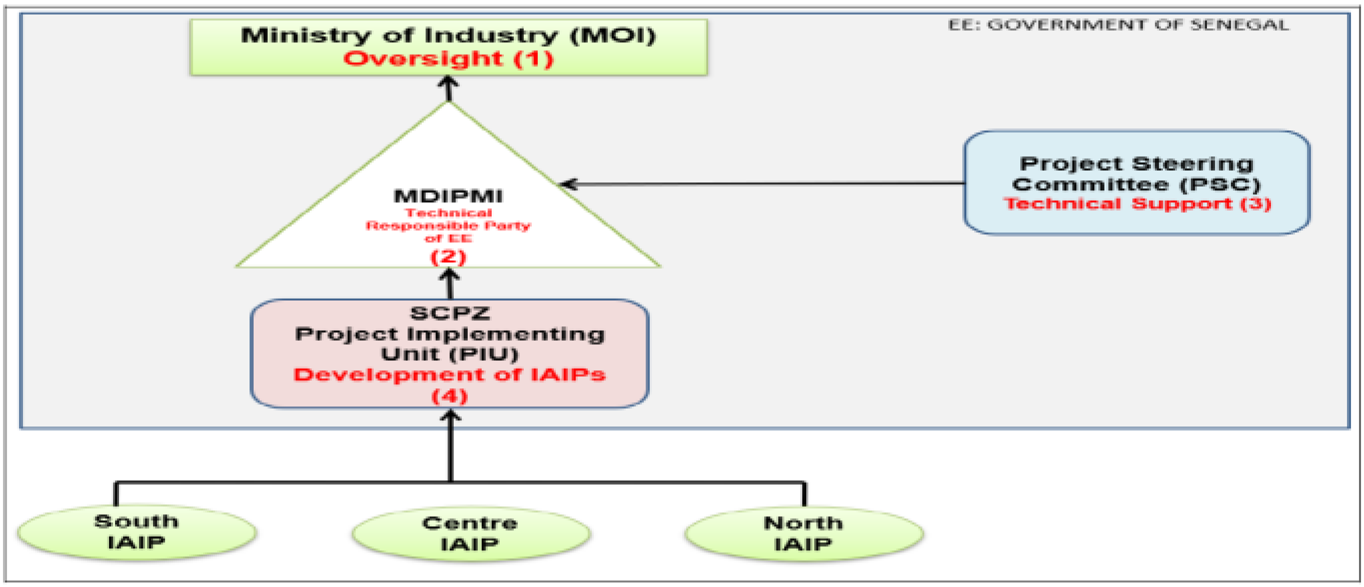


**Figure B.4.2. Governance and Implementation Arrangement in Senegal**

***Guinea***

The governance and implementation arrangements of the SCPZ project in Guinea is presented in Figure B.4.3

- At the top of the figure B.4.3 (Box 1), is the Presidency of the Republic, which has direct oversight function over the IAIP project in Guinea. This institutional arrangement is due to the multi-sector nature of the project and the need for strong leadership to provide coordination among all the project actors. The Office of the Presidency of the Republic receives day-to-day updates about project execution from the Technical Responsible Party (TRP) of the Executing Entity, which is Authority for the Development and Administration of Special Economic and Industrial Zones (ADAZZ) in Guinea (Box 2). The activities of ADAZZ, will be supported by a Project Steering and Monitoring Committee (PSMC). Attached to Box 3, are periodic technical assistance (Box 3A and 3B) in the form of international and national consultants to support and strengthen the activities of the PSMC.
- Box 4, shows the Project Implementing Unit (PIU), which composition is described in the POM. It is directly responsible for; i) development, management, and coordination of the two IAIPs in Guinea (Boke IAIP and Kankan IAIP); ii) rents administration at the two parks, selling or leasing of parks facilities; and iii) performing sensitization and promotional work to attract investors, especially private sector investors among others (see POM for more details).

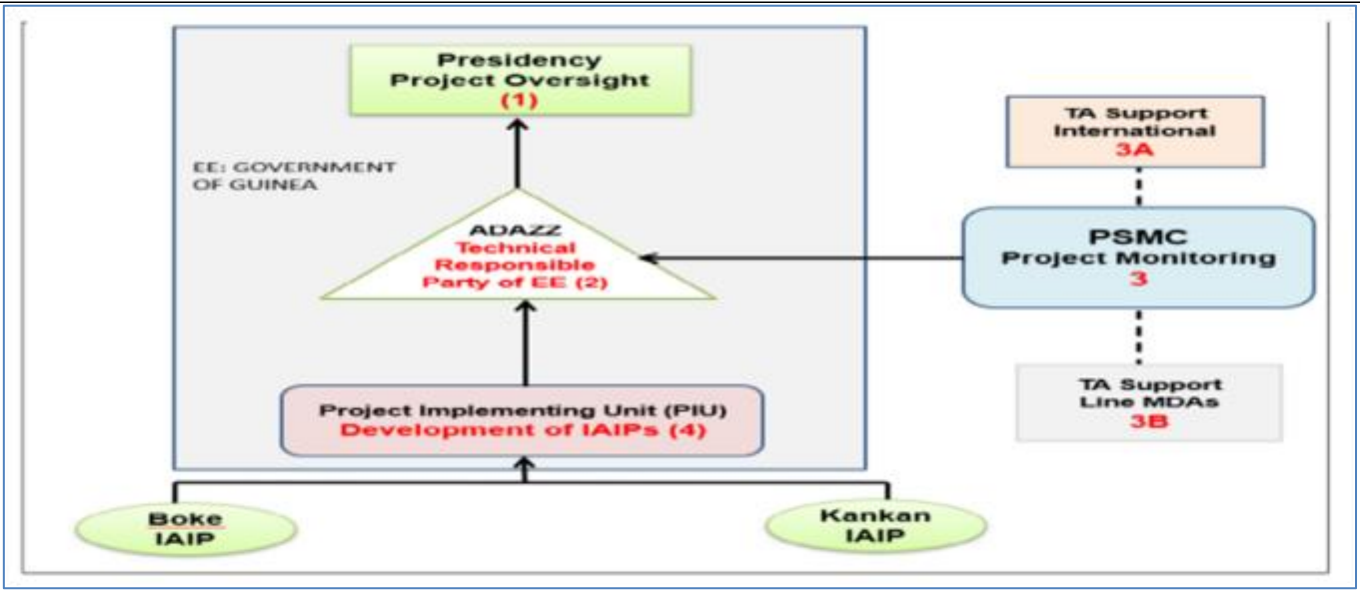


**Figure B.4.3.** Governance and Implementation Arrangement in Guinea

The FMA of the technical responsible party of the EE (ADAZZ) has recently appointed a Financial coordinator as well as Head of procurement. was found not satisfactory on the whole and the overall fiduciary risk was considered substantial because of the absence of: (i) a financial coordinator and personnel; (ii) an administrative, financial and accounting procedures manual; (iii) operational accounting and financial management software; and (iv) FM tools and reference framework. As a result, the following recommendations were made:

- To finalize the recruitment of all necessary staff to the programme implementing unit (PIU);
- As soon as the programme is launched, prepare a budget plan coupled with the indicative activities schedule;
- After the launching of the programme, a consultant should be recruited to prepare the administrative, financial and accounting procedures manual (AFAPM), and another responsible for establishing the computerized accounting and financial management system (SIGCF);
- Assign financial tasks as soon as the financial staff are recruited; and,
- To establish the financial management framework and instruments in accordance with AfDB's requirements.

**General Implementation Arrangements for Renewable Energy Investments**

Implementation arrangements on the renewable energy aspects of the programme (Activities 1.2.1 & 1.2.2) across the three countries which include solar irrigations schemes, solar systems for lighting and processing, as well as bio-gas based generators and digesters, are going to be Public Private Partnerships (PPP) arrangements. The model which has been retained under the PPP is the DCOM model, which stands for Design, Construct, Operate and Maintain. Under this model, the Executing Entity which is a government body will enter through its Technical Responsible Party for execution into an arrangement with Energy Service Companies (ESCOs) for the design, construction, operation and maintenance of the solar and biogas assets. ESCOs will be selected through a public procurement process and selected ESCOs will enter a financing arrangement with the Executing Entity under which they will receive financing in form of grant and loans. ESCOs will then design and construct the assets and will have the right to operate and maintain the same assets and generate revenue while re-paying the loan granted by the Executing Entity. The annual repayment which the ESCOs will pay to the Executing Entities will come from the revenues the ESCOs will generate with the assets.The ESCO business model sensitivity analysis which has been conducted shows that without a significant portion  of grant in the beginning combined with sufficient concessional loan, the ESCOs won't have an economic incentive to take part into the programme or operate any renewable energy asset (negative NPV), which in return translates into end beneficiaries not having the opportunity to access any energy service.

Section 6 of the Feasibility study assesses the conduciveness of ESCO solution in the beneficiary countries.

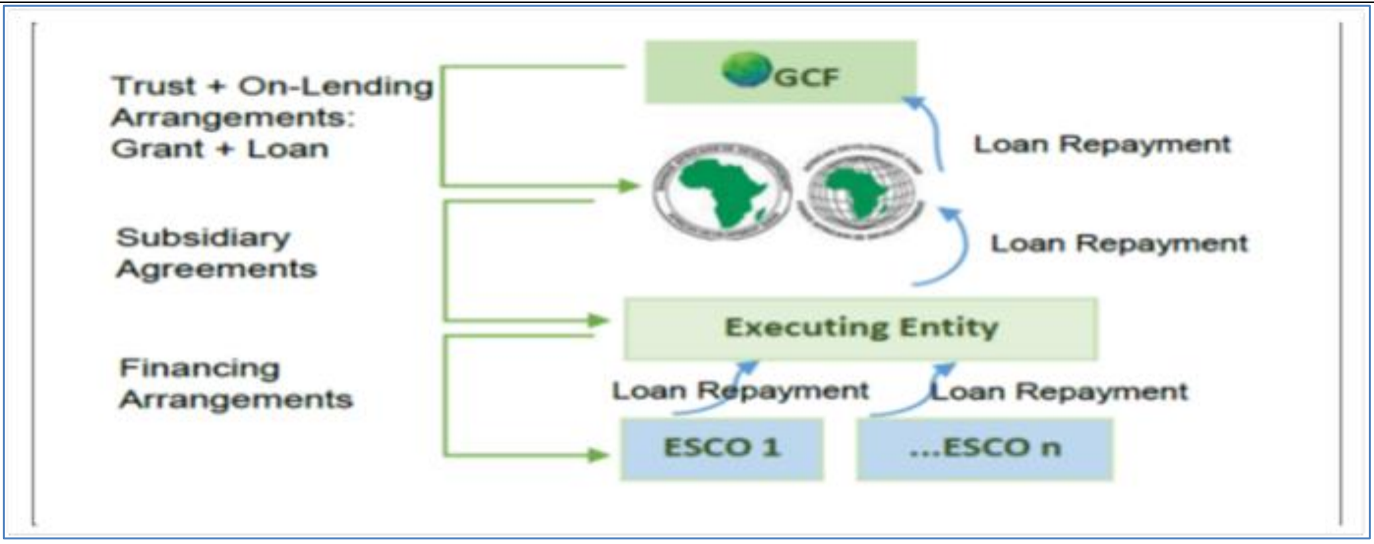


**Figure B 4.5: Implementation Arrangements for Renewable Energy Assets.**

## Use of Programme Proceeds and Assets Ownership

The programme financial instruments consist of reimbursable funds (loans) and non-reimbursable funds (grants) from the GCF. Both funds will pass through AfDB that will channel the funds to the various governments via the Ministry of Economy and Finance (MoEF) as described in Figure B.4.1 below. Upon agreed quarterly/annual work plans and budgets (AWPB) forecasts for each output/activity approved by the project steering committee (PSC), and signed off by AfDB, the quarterly budget requests will be disbursed to the executing entity (EE) in each country (see POM for disbursement methods to be used). As indicated in Figure B.4.5, the EE will sign loan and grant agreement with and then pass the funds received (reimbursable and non-reimbursable) to selected private sector provider(s), the Energy Service companies (ESCOs), which would have been selected through a public procurement process in collaboration with the Project Implementing Unit (PIU) for the delivery of services for the drip/solar irrigation, solar PV and biogas (for Activity 1.2.1, and Activity 1.2.2) as well as for CRA and agro-forestry practices (related to Activity: 2.1.1 and Activity 2.1.2). The ESCOs will also make direct repayment to the governments and the governments will make subsequent repayments to the AfDB for reflow to the GCF. The Ministry of Finance for each country will disburse GCF non-reimbursable proceeds to the Technical Responsible Party (TRP) of the Executing Entity of the various Governments, for the intervention following AfDB policies and procedures such as for procurement, disbursement, ESS and gender.

Questions on assets ownership will be solved in the financing arrangement as follow: during the initial years of the loan agreement with ESCOS where a repayment is required, fixed and core assets are owned by the Executing Entity, while movable and non-core assets are owned by the ESCO. If the ESCO does not perform its obligation, the right to operate all fixed and core assets can be re-tendered by the EE. Once the loan is repaid ownership will be transferred to the ESCOs who can continue operating the renewable energy assets without being required to pay money to the EE from that moment on.

**Table B.4.1: Summary Roles and Responsibilities**

| ***RACI Matrix for SCPZs Programme*** | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Project Leadership** | | | | | **Project Team Members** | | | | | | | **External Resources** | | | | |
| | **Programme Executing Entity** | **Project Executing Entity (PEE)** | **Project Implementing Unit** | **Project Steering Committee (PSC)** | **Project Oversight** | **National Project Coordinator** | **Project Coordinator** | **M&E Officer** | **Procurement Officer** | **Environmental Specialist** | **Admind. & Finance Officer** | **Project Accountant** | **Consultant** | **Line Ministry, Department &** | **Private Sector** | **Community Agents** | **Value Chains Agents** |
| ***Activity 1.1.1 Support access to small-scale agricultural water management (AWM) infrastructure (Irrigation of a total of 39,178 ha of land with 2-inch-diameter drip irrigation distribution systems (DIDS)*** | | | | | | | | | | | | | | | | | |
| Issuance of International Competitive Bidding (ICB), short listing, and selection of firm for | **R/A** | **I/S** | | **P/I** | **P** | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| training, supply and installation of DIDS | | | | | | | | | | | | | | | | | |
| - Mapping exercise and selection of DIDS sub-project beneficiaries and Identification of *59,726.3* ha of land | **I/S** | **R/A** | **P/I** | **P/I** | | | | | | | | | | | | | |
| - Grant work plan for DIDS sub-project beneficiaries including management committees determined | **I/S** | **R/A** | | **P/I** | **P** | | | | | | | | | | | | |
| - Capacity building and training of sub-project beneficiaries in the use of DIDS in four pilot countries | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| - Distribution and installation of DIDS to selected sub-project beneficiaries in four pilot countries | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| ***Activity 1.2.1 Improving agricultural cooperatives, FBAs value chain infrastructure for food processing (Installation of 14.69MW of solar energy for lighting, processing, drying and packaging of staple food crops)*** | | | | | | | | | | | | | | | | | |
| Issuance of ICB, short listing, and selection of Energy Service Companies (ESCOs) for design, construction, operation and maintenance/supply of of solar energy | **R/A** | **I/S** | | **P/I** | **P** | | | | | | | | | | | | |
| - Mapping exercise and selection of sub-project beneficiaries of 20MW of solar energy installations | **I/S** | **R/A** | **P/I** | **P/I** | | | | | | | | | | | | | |
| Arrangements for sub-project loans and grants including work plan as determined by management committee | **I/S** | **R/A** | | **P/I** | **P** | | | | | | | | | | | | |
| - Capacity building and training of sub-project beneficiaries of 20MW of solar energy | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| Distribution and installation of of Solar PV equipment to sub-project beneficiaries | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| ***Activity 1.2.2: Deploying low-carbon energy technologies (installation of 24, 577m3 of biogas digester with an estimated genset power output of about 14 MW)*** | | | | | | | | | | | | | | | | | |
| - Issuance of ICB and selection of firm for the installations of 42 MW of clean energy | **R/A** | **I/S** | | **P/I** | **P** | | | | | | | | | | | | |
| - Mapping exercise for the selection of eligible beneficiaries of bio-digesters in four pilot countries | **I/S** | **R/A** | **P/I** | **P/I** | | | | | | | | | | | | | |
| - Biogas grants and loans work plan, including biogas management committees determined | **I/S** | **R/A** | | **P/I** | **P** | | | | | | | | | | | | |
| - Capacity building and training in Biogas digesters, Bio-slurry, safety standards etc., | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| - Construction and bio-digester handover to sub-project beneficiaries in pilot countries | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ***Activity 2.1.1: Promoting CRA practices among smallholder farmers (A total of 39,178 ha will be used for CRA practices such as minimum tillage, use of drought resistance and improved yield seeds, crop rotation, integrated nutrient management, and mixed cropping).*** | | | | | | | | | | | | | | | | | |
| - Mapping exercise and selection of sub-project beneficiaries of GCF loans and grants | **I/S** | **R/A** | **P/I** | **P/I** | | | | | | | | | | | | | |
| - Work plan for sub-project grants and loans to promote CRA practices determined | **I/S** | **R/A** | | **P/I** | **P** | | | | | | | | | | | | |
| - Capacity building and training of sub-project beneficiaries in CRA practices | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| - Implementation of CRA activities | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| ***Activity 2.1.2: Promoting agro-forestry practices among smallholder farmers (A total of 40,000 ha will be used for mixed agro-forestry and livestock management)*** | | | | | | | | | | | | | | | | | |
| - Mapping exercise and selection of sub-project beneficiaries of GCF loans and grants for agro-forestry practices | **I/S** | **R/A** | **P/I** | **P/I** | | | | | | | | | | | | | |
| - Work plan for sub-project grants and loans to promote agro-forestry practices determined | **I/S** | **R/A** | | **P/I** | **P** | | | | | | | | | | | | |
| - Capacity building and training of sub-project beneficiaries in agro-forestry practices | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| - Implementation of agro-forestry activities | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| ***Activity 2.2.2: Establishment of climate information and early warning systems in SCPZs ((i) Expansion of agrometeorological and rain stations by establishing 10 automated agrometeorological and 50 rain gauge stations in each of the SCPZ region; (ii) Procure and maintain 2 four-wheel project vehicles per region (i.e., a total of eight (8) project vehicles); (iii) Deploy technologies to strengthen climate information services and early warning systems in all the SCPZs; (iv) Capacity building and awareness raising)*** | | | | | | | | | | | | | | | | | |
| Issuance of ICB for site identificationa and procurement of agrometeorological and rain gauge stations in the SCPZs | **R/A** | **I/S** | | **P/I** | **P** | | | | | | | | | | | | |
| Issuance of ICB, short listing and selection of firm for the establishment of climate information and early warning systems in SCPZs | **R/A** | **I/S** | | **P/I** | **P** | | | | | | | | | | | | |
| Procurement of project vehicles for site supervision and sensor maintenance | **R/A** | **I/S** | | **P/I** | **P** | | | | | | | | | | | | |
| Mapping exercise for the selection of eligible beneficiaries of CIEWS | **I/S** | **R/A** | **P/I** | **P/I** | | | | | | | | | | | | | |
| CIEWS grants and loans work plan, including CIEWS management committees determined | **I/S** | **R/A** | | **P/I** | **P** | | | | | | | | | | | | |
| Capacity building and training on key aspects of the CIEWS | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| Setting up and commissioning of CIEWS | **P/I** | **P/S** | **R/A** | **P/I** | | | | | | | | | | | | | |
| ***Activity 3.1.1: Project Management, Evaluation and Coordination*** | | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| - Management, Evaluation and Coordination | R/ A | I/S | P | P | P | | | | | | | | | | | |

| Code: Every activity should have 1 and only 1 R | |
|---|---|
| Accountable - A | Accountable for success/failure of this activity |
| Participant - P | Actively participates in the activity |
| Responsible - R | Responsible for this activity |
| Input Required - I | Project Team needs input from this person in this activity |
| Sign-off Required - S | Must sign-off the appropriate document |
| Programme Executing Entity (PEE) - AfDB | Overall programme coordination, management and reporting to GCF |
| Project Executing Agency (PEA): Togo - APRODAT; Senegal - MDIPMI; Guinea - ADAZZ | Coordination, management and reporting at the country level to PEE |
| Project Implementing Unit (PIU) | Coordination and management of project activities at the country level and reporting to PEA |

***Financial flows and implementation arrangements (legal and contractual) between the AE and the EE, between the EE and third parties and beneficiaries.***

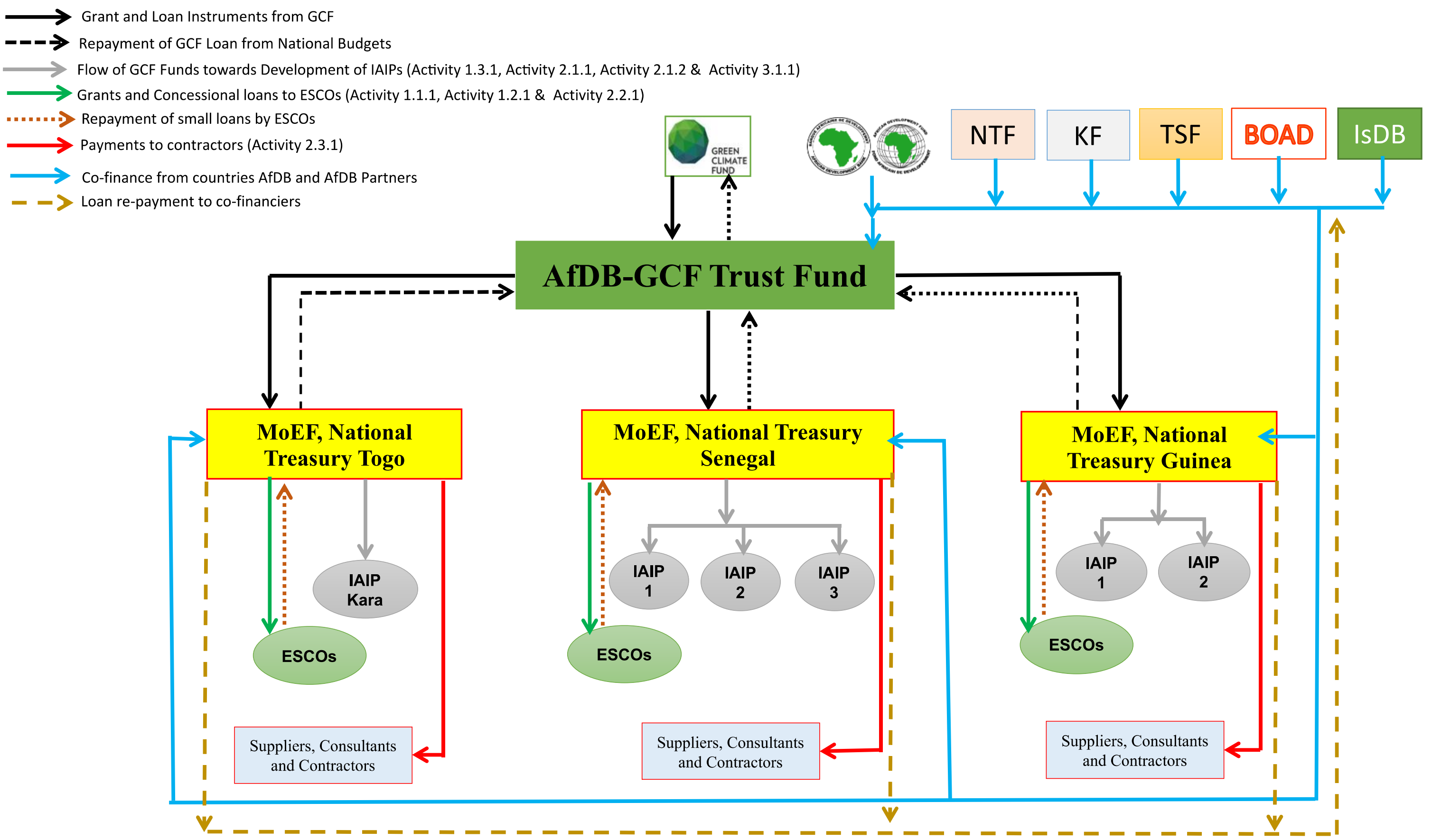


**Figure B.4.6:** Programme Fund Flow

***Note:*** *MoFEP = Ministry of Finance and Economy Planning; MoEF = Ministry of Economy and Finance; MFB = Ministry of Finance and Budget; IAIP = Integrated Agro-Industrial Park; ESCOs = Energy Services Companies; EU= European Union; KEB = Korean Exim Bank; KSU = Korea Samaeul Undong; BOAD = West African Development Bank; IDB = Islamic Development Bank;*

As shown in Figure B.4.6, GCF proceeds will be received by the AfDB after signing the Funded Activity Agreement (FAA) and meeting all requirements for execution, effectiveness of the FAA as well for disbursements in accordance with the schedules. The AfDB will sign the loan and grants agreements with the specific Ministry of Finance (MoF) of the country in accordance with procedures and policies of the AfDB in financing sovereign operations, and in consideration of the terms and conditions of the term sheet for the SCPZ programme with the GCF. The Ministry of Finance for each country will disburse GCF non-reimbursable proceeds for the CRA (related to Activity: 2.1.1 and Activity 2.1.2) as well as the co-financing for Activity 3.11 related to programme management and coordination to the Technical Responsible Party (TRP) of the Executing Entity of the various Governments, for the intervention following AfDB policies and procedures such as for procurement, disbursement, ESS and gender. The EEs will work with the project implementation or management units (PIUs/PMUs) to procure goods and services in accordance with AfDB procurement procedures and policies for the implementation of the programme. For the non-reimbursable and reimbursable funds for the drip/solar irrigation, solar PV and biogas (related to Activity 1.2.1, and Activity 1.2.2), the Ministry of Finance for each country will sign grant and loan agreement with the private sector providers/ESCOs and disburse the proceeds to the selected private sector providers/ESCOs through competitive bidding process under the direct responsibility and accountability of the TRP of the EE as governed by AfDB's Disbursement Handbook[51]. The private sector providers will add their own equity and debt resources to design, construct, operate and maintain (DCOM) the assets to provide the expected services for agreed user fees. They will use the cash flow from the user fees to service repayment to the government through the Ministry of Finance. The Ministry of Finance makes repayment on behalf of the government to AfDB and AfDB further reflows to the GCF as illustrated in Figure B.4.6. For Activity 2.2.2 related to capacity building of value chains agents, the EE through the ministries in charge of finance and the TRP will oversee the procurement and payment for goods and services with suppliers, consultants and contractors in accordance with AFDB procurement policies.

The Countries-specific conditions precedent to AfDB first disbursement of funds are provided in pages 49-51 in the Programme Operational Manual (Annex 21)

The flow of funds for each country is presented in Figures B.4.6.1, B.4.6.2, and B.4.6.3. The narratives will be the same. The only difference will be the co-financiers for individual country project.

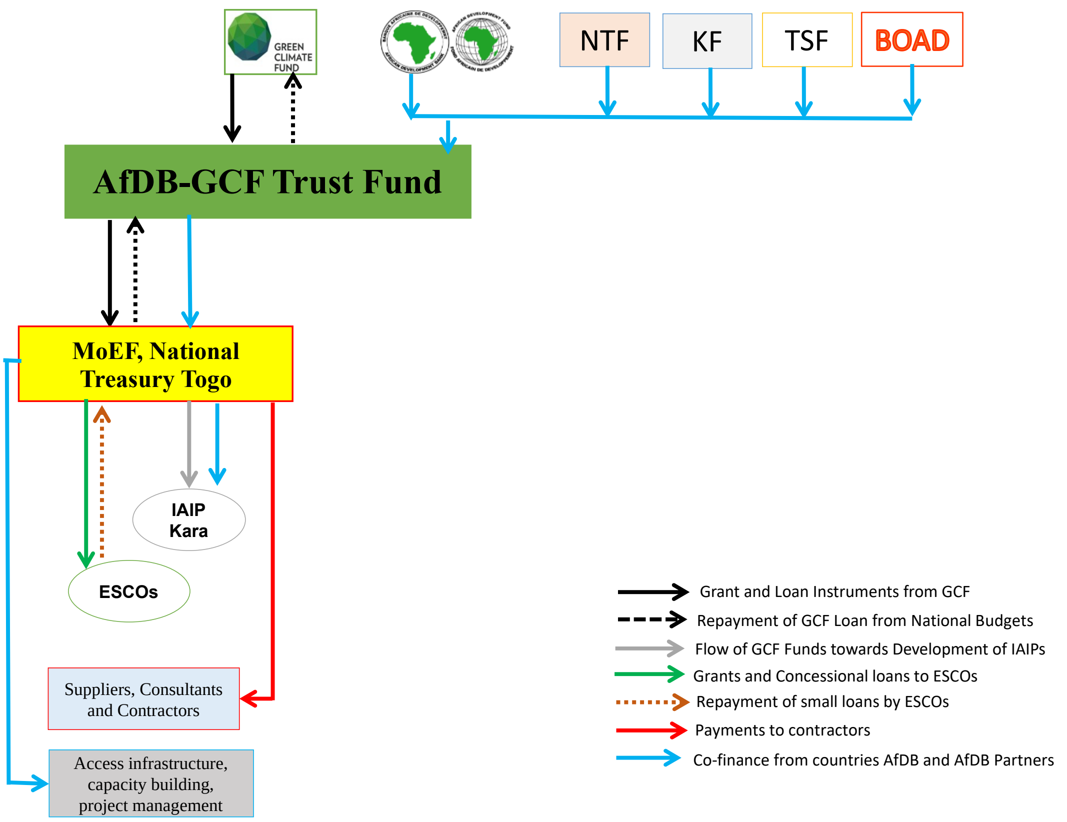


**Figure B.4.6.1:** Programme Fund Flow for Togo

[51]Available at: https://www.afdb.org/fileadmin/uploads/afdb/Documents/Financial-Information/Disbursement%20Handbook%202012%20version.pdf

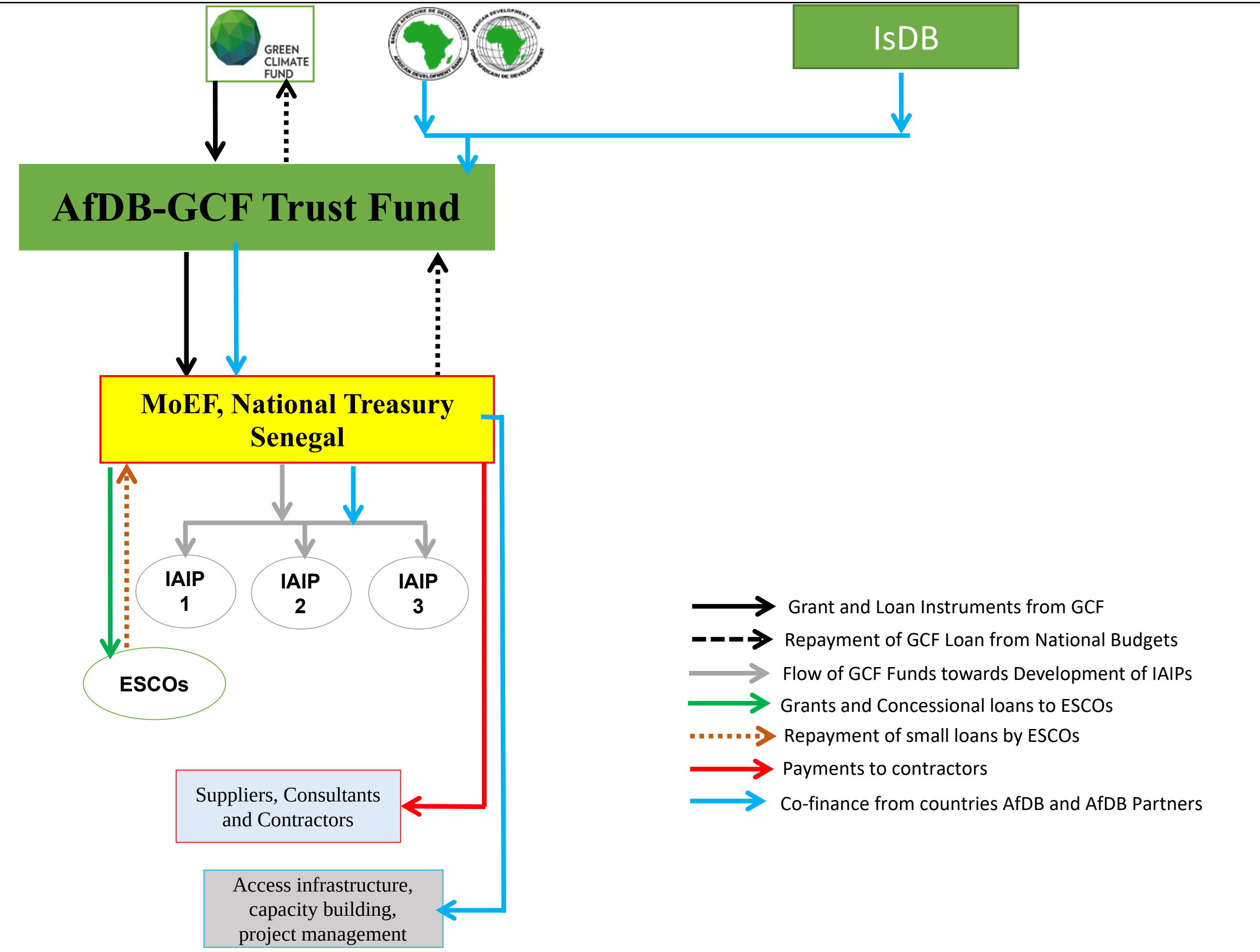


**Figure B.4.6.2:** Programme Fund Flow for Senegal

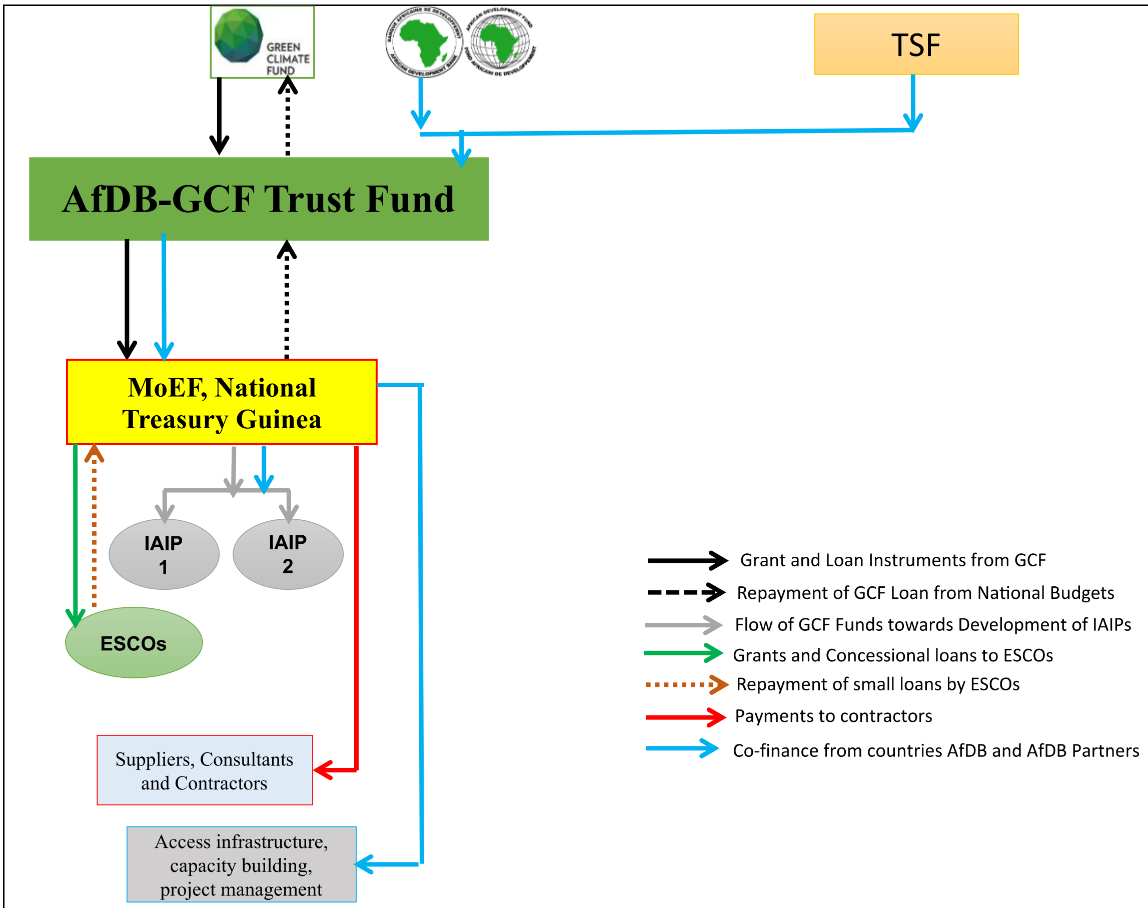


**Figure B.4.6.3:** Programme Fund Flow for Guinea

## B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Firstly, the GCF was established to finance programs and projects for climate change adaptation and/or mitigation. With its huge potential to spark a paradigm shift towards resilient and low-carbon sustainable development, the project proposed herein captures the essence of the GCF mission and mandate globally. The proposed program contributes directly towards reducing GHG emissions from the agro foresry sector. It will also strengthen the resilience of communities and people to climate change and in particular, their most vulnerable sector to climate change and its variability, namely; agriculture. By supporting this project, the GCF will contribute towards increasing resilience and enhancing livelihoods of the vulnerable people and communities; enhancing food and water security; and improving access to low-carbon energy technologies, especially the switch from firewood and charcoal consumption to cleaner energy, which is considered the greatest driver of forest degradation and deforestation in Togo, Senegal and Guinea. Therefore, the GCF finance will make a real contribution to move the traditional agriculture development efforts in the selected countries, towards a more climate resilient and low-carbon trajectory. Many of the project outputs to be produced under the GCF financed components, have very high potentials for scaling up and scaling out in other African countries including prospective SCPZ countries like Mali, Zambia, DRC, Niger, and Nigeria. Thus, GCF involvement in the SCPZ initiative is worth undertaken and reflects best value for their investment.

Secondly, for all the three pilot SCPZs countries considered in this first phase of the program, two key sectors that featured most in their unconditional targets for mitigation and enhanced removals of GHG emissions are agriculture and energy. For instance, Senegal's unconditional action in energy is to strengthen the distribution of electricity and fuel at the household level by: (i) installing 27,500 Biodigester at the household level; (ii) generate 160MW from the use of Solar PV; and, (iii) providing over 392 villages solar or hybrid mini-grid. Same for Guinea, with key energy targets of: (i) producing 30% of its energy (excluding wood-energy) from renewable energy sources (cumulative GHG reduction of 34 $MtCO_2e$ by 2030); and (ii) supporting the dissemination of technologies and practices that are energy- efficient or use alternatives to wood-energy and charcoal (cumulative GHG reduction of 23 $MtCO_2e$ by 2030). Similarly, for Togo, some of their unconditional actions in energy include: (i) prog the use of energy-efficient stoves (which can yield 50-60% savings in wood and charcoal); and (ii) introducing solar equipment at the household level and capacity-building for the various actors concerned. The greatest mitigation potentials from the SCPZ project under GCF financed activities are: (i) avoided emission from the use of renewable energy; (ii) avoided emission from climate resilient agricultural practices; and (iii) carbon sequestration from agroforestry practices. Thus, the GCF finance will make a real contribution in meeting the respective NDC targets of these countries. Additionally, many of the sub-projects under the SCPZ project (in particular, activities related to *Deploying Low-carbon Energy Technologies and Promote CRA and Agro-forestry practices*), and financed by the GCF, have very high potentials for catalysing climate change impacts beyond one-off projects. These sub-projects can therefore be scaled up, out scaled, and used for knowledge and learning in many other parts of Africa and the least developed countries.

Thirdly, and finally; one of the most important aims of the GCF is to lever additional money for the projects/programs it funds to shift global economic development to more sustainable pathways. To close the financing gap towards climate resilient economies, countries need public concessional finance that will leverage additional private sector investment to create a viable green business market in the region. Covid-19 has further impacted investment activity in the selected countries, as institutional investors - particularly those interested in smallholder agriculture - become all the more risk averse. With a total co-financing of about $168.9 million made available for the pilot SCPZ program, GCF involvement in the program is paramount to take advantage of this existing co-financing to shift global economic development in these countries to more sustainable pathways. Also, the involvement of the GCF in financing the adoption of CRA and agroforestry practices and low-carbon energy technologies along the agricultural value chain will help to 'crowd in' long-term investments from the private sector beyond GCF investments. This is besides the 'transformational change' potentials of the program.
*Justification of the GCF financial instruments concessionality to make the investment viable.*

*Togo*
Togo's average score from the Country Policy and Institutional Assessment (CPIA) of the AfDB and the World Bank has never exceeded 3.2 out of 6 from 2008 to 2016. This score means that Togo is still a country facing conditions of fragility. From 2011 to 2016, the Government of Togo made an extensive use of debt to fund major capital expenditure especially in the area of transport infrastructures. As a result, the country's debt level increased from 46.9% of GDP in 2011 to 79.2% in 2016. According to the IMF/World Bank joint analysis of debt sustainability, the country's debt distress risk is high since 2016. In order to mitigate this vulnerability, the program pursued by the Government of Togo with the IMF support as part of the Extended Credit Facility (ECF 2017-2019) aims at reducing the level of indebtedness gradually from 79.2% in 2016 to 69.9% in 2019, i.e. below the WAEMU convergence threshold, (less than 70%) after 3 years. According to this ECF program, the government has committed to no longer contract any new nominal external debt on non-concessional terms and to impose substantial supervision to concessional borrowing. In these circumstances, grant financing would be the best GCF instrument to accommodate the current program under ECF and to further mitigate the effects of country's fragile situation while accelerating its transition to resilience. Based on this analysis, the GCF financing is requested in the form of a grant ($30.7 million) and loan (($10.0 million) to make the investment viable and to avoid a further debt overhang situation for Togo. The abatement cost of this programme in Togo is estimated at US$ 18.47/tCO2eq while the GCF contributes at US$ 6.68 for each abatement unit achieved.

*Senegal*
Despite relative political stability, Senegal still ranks 162 out of 187 countries in the 2016 Human Development Index. Some 39 percent of the population lives below the poverty line and 75 percent of households suffer from chronic poverty and recurrent food insecurity.[52] Though agriculture is the main growth driver of the economy, it is largely subsistence-based in nature and highly vulnerable to recurrent climate shocks such as droughts and extreme temperatures. For example, the 2012 food crisis caused by drought left over 800,000 people food insecure in the country. The government had to recourse to food imports especially rice (approximately 60 percent) to ameliorate the food crisis, which continues to deteriorate the country's current account deficit. Average annual trade deficit represented 18% of GDP over the period 2011-2015 as a result of very poor export performances which represented an annual average of 19% of GDP[53]. At the same time, public debt in Senegal rose from less than 23% of GDP in 2007 to over 54% in 2015 but remains below the standard of 70% retained under the WAEMU convergence criteria. The Government's objective is, however, to maintain that ratio at 53% under the programme with the IMF entitled 'Economic Policy Support Instrument' (2015-2017 EPSI). Based on this, the government is committed to no longer contracting additional external debt on non-concessional terms and to imposing substantial supervision to concessional borrowing. Given these circumstances, a 74% grant financing and a 26% loan financing would be the best GCF instrument required to make the investment viable. The abatement cost of this programme in Senegal is estimated at US$ 18.49/tCO2eq while the GCF contributes at US$ 5.44 for each abatement unit achieved.

*Guinea*

52 World Food Programme 2018. Senegal Country Strategic Plan (2019–2023).https://docs.wfp.org/api/documents/5b0e7061163e4ba98d6348b150e588e2/download/?_ga=2.248193219.594354054.1569774348-665238984.1569774348

53 AfDB 2016. Bank Group's Country Strategy Paper For Senegal, 2016-2020. African Development Bank, Abidjan, Cote d'Ivoire.

The economy of Guinea is still recovering from the 2014–2015 Ebola outbreaks that slashed the country's growth rate from 5.9% in 2012 to 3.8% in 2015. Though slowly recovering, the economy is still characterized by a high poverty rate of about 55% coupled with a persistent unemployment rate and poor crop yields (approximately 1.2 tons per hectare for rice). Additionally, Guinea is grappling with fragility factors that discourage inclusive growth. In recognition of this, the country's new strategy seeks to broaden the basis for inclusive and sustainable growth by improving the supply of energy to communities and productive activities, and developing agriculture and agricultural product processing. However, significant infrastructure spending has aggravated the country's budget deficit. The overall budget deficit (net of grants) deteriorated from 1.3% of GDP in 2016 to over 3.6% in 2017. Equally, significant public investment, financed by loans, raised the debt level from less than 27.2% of GDP in 2012 to over 37.2% in 2017. Domestic debt surged from 9.3% of GDP in 2012 to over 17.6% in 2017. During the same period, the external debt increased from 18% of GDP to 19.8%. According to the IMF, the public debt should remain below the threshold (70% of GDP) set in the convergence framework of the Economic Community of West African States (ECOWAS) until 2022[54].

Against this background, a 74% grant financing (26% loan) would be the best GCF instrument required to accommodate the current debt situation in Guinea if the threshold should remain below the 70%. Besides, in its Intended Nationally Determined Contribution (INDC), Guinea pledged to undertake GHG mitigation actions, conditional upon receipt of international support. Further, the abatement cost of this programme in Guinea is estimated at US$ 7.04/tCO2eq while the GCF contributes at US$ 4.55 for each abatement unit achieved.

## B.6. Exit strategy (max. 500 words, approximately 1 page)

The SCPZ initiative is such transformational that the AE has already received several requests from governments including Cote d'Ivoire, Kenya, Madagascar, Malawi, Nigeria, and Tanzania to prioritize their allocations for similar projects in their countries. Senegal is already scaling up the SCPZ with a new agro-industrial park to be financed together with the AE in another part of the country. So, the scale up and contextualization potential of the SCPZ is really great and it will be essential to partner GCF to ensure climate measures are integrated to rationalize agricultural value chains towards low carbon and climate resilient activities. Moreover, the exit and sustainability measures were already embedded in the design of the agricultural transformation centres and agro-industrial parks where shared facilities are provided to producers, processors, aggregators and distributors to operate in the same vicinity to reduce transaction costs and share business development services for increased productivity, diversification and competitiveness. The PPP arrangements for the operation and management of the facilities for user fees will ensure that there is adequate revenue generated to maintain the facilities including the solar (lighting and irrigation) and biogas assets financed with GCF financing. Moreover, maintenance of the facilities is part of the PPP agreements with the ESCOs. The concentration of infrastructure (energy, water, roads, ICT) in the agro-industrial parks will spur economies of scale, forward and backward linkages for efficient and integrated operation that will maximize profitability to sustain the enterprises along the agricultural value chain. The linkages and integration will also create induced investment with significant multiplier effects where investors, developers, producers and other actors will pursue complementary investments that will facilitate the exit of government and partner interventions.  to rural areas of high agricultural potential, they attract investments from private agro-industrialists/entrepreneurs to contribute to development consistent with Nationally Determined Contributions (NDCs).

Besides, other measures such as the training of value chain actors, financial and advisory services, e-platform for mechanization and as well as the SCPZ governance systems are all intended to ensure continuity of operation after the intervention from the government and partners including the AE and GCF.

Additional exit and sustainability strategies discussed below are aimed at eliminating the following barriers to sustainability:

**a)** Economic and financial barriers: The possibility of unavailability of funding to support the activities after the completion of the project.
**b)** Social barriers: The identified social barriers include: (i) marginalization of the poor, women and entrenched inequities;(ii) limited awareness about climate change threats; (iii) Inadequate interaction between the private sector and government institutions; and, (iv) Insufficient incentives for the private sector to pursue sustainable development.
**c)** Political barriers: These include inadequate economic, social and environmental methods for policies, plans and projects.
**d)** Poor monitoring and evaluation systems: A basic problem is lack of specific targets (nationally and at local level), measurement and data to track progress, resulting in a lack of information to decision-makers.
**e)** Institutional barriers: This is based on lack of institutional experience to operate in a transparent and democratic system.

Economic and Financial strategy: Market development is one of the main strategies envisaged to remove economic and financial barriers to sustainability. The existence of the Special Agro-industrial Processing Zones and the fact that farmers will be organized in communities coupled with the improved infrastructure and quality of produce will ensure that the farmers get sufficient profits. In that regard the project intends to use the following approaches:

[54] AfDB 2018. Guinea Country Strategy Paper 2018-2022. African Development Bank, Abidjan, Cote d'Ivoire.

**a)** Improving incentives for investing in sustainable natural resources use through higher and more stable producer prices and reduced input costs;
**b)** Avoiding market-driven resource-degradation by relating market development with gender-sensitive management of property rights and governance especially for common pool resources;
**c)** Limiting exclusionary practices and structures by improving access to marketing infrastructure (information, storage, transportation, equipment, financing etc.), for poor and marginalized groups or reducing the costs of standards compliance;
**d)** Reducing market-related risks and vulnerability, through measures that help stabilize the price of food staples and those that reduce pressure for livestock owners to destock during drought years by increasing non-livestock forms of savings, improving access to dry-season pastures and water sources among others; and,
**e)** Providing sustainable market opportunities for the poor, particularly through targeted efforts to broaden and improve the participation of small producers(male and female) in value chains serving markets for food staples and high-value products by addressing constraining gender norms that hinder women seeking new market opportunities.

The sub-projects in the SCPZ program are designed in such a way that all parties involved can make reasonable profits from their investments in comparison with the next available alternatives. This will ensure continued interest in the activities after the completion of the project. For biogas digesters and waste management projects, the designs will be such that the private partners can make sufficient profits from their activities.

Social strategy: Ideologies that rigidly limit opportunities, participation and autonomy for some members of the population cause whole groups of people to be'left behind. Patriarchy – the belief that power and decision making should reside with some men, permeates lives, relationships and policies at the family, community, national and international levels. In doing so, fundamentalist beliefs systematically limit women's participation in every aspect of private and public life. Planned capacity building activities for the SCPZ include elements that will specifically support the marginalized and disadvantaged groups to ensure that by the end of the project, they have gained the financial and technical knowledge to manage their investments. They will also incorporate training courses that are aimed at informing the people about the threats that the change in their environment could have on them and their investments. That way, climate and environmental issues will be considered more seriously even after the project. Management structures will be put in place that will include both the local communities, government organizations at different levels and the private sector. They will include relevant representatives from the different sections of the value chain and will help in breaking the barriers that exist in the relationships between the local communities, government officials and the business community. To further enable the private sector's involvement, information on climate change impacts relevant to the private sector will be packaged and communicated in a relevant format to the private sector. A number of responsibilities associated with capacity building will be anchored in existing permanent institutions in the programme countries to ensure the continuity of access after the project life.

Political strategy: In order to overcome political barriers, sensitization of political leaders, private sector involvement, and participation of local communities will be concrete in the program approach through the use of Programme Management Unit (PMU) and PSMC. Capacity-building is also necessary across the board. In addition, this funding proposal has governments as the executing entities because of the strong political support and alignment with their national development and NDC goals. This strong political support will ensure the continuity of the project and replication in other areas of the nation as demonstrated by Senegal.

Monitoring and evaluation strategy: At all levels of stakeholders, the program will establish well-presented protocols and well-presented, intuitive and communicable data to strengthen decision making, progress measurement, transparency and accountability. Relevant online resources on adaptation in relevant and usable formats to local communities, private and government institutions will be provided through least cost means like mobile phone technology.

Institutional strategy: Social capital is critical for poverty alleviation and sustainable human and economic development. Social capital can improve environmental outcomes through decreased costs of collective action, increase in knowledge and information flows, increased cooperation, reduced resource degradation and depletion, more investment in common lands and water systems, improved monitoring and enforcement. Through enabling collective actions and putting in place community institutional structures, the program will build social capital in the project areas that will contribute strongly to the sustainability of the project. Furthermore, actions will be put in place to create permanent and mutually beneficial links between the community social structures, the private sector and government agencies. The project will carefully analyse the existing institutions to ensure that capacities and responsibilities are developed/given to the most appropriate institutions that are in a position to sustain the responsibilities after the completion of the project and program.

*Role of public, private sector or civil society as a consequence of the programme implementation for scaling up and continuing best practices*

Firstly, the institution of a Programme Management Unit (PMU) as well as a Programme Steering and Monitoring Committee (PSMC), involving all major stakeholders along the agricultural value chain such as the public and private sectors including civil society organizations, implies that knowledge gathering and learning is widely shared among different stakeholders and institutions. The program will provide avenues for knowledge sharing and learning and the continuing application of best practices among target beneficiaries.

Secondly, the program has inbuilt training activities that are aimed at providing: i) capacity building and training in CRA practices, climate information and technology uptake; (ii) training in mixed agroforestry practices and livestock management; (iii) training on design, construction, operation and maintenance of biogas digesters; (iv) capacity building on waste recycling processes; and, (v) training on regulatory framework strengthening for private sector investment in Agriculture. These training activities will strengthen the potential for knowledge transfer and learning throughout the programme lifespan, especially among these target groups. Currently, it is envisaged that the awareness and skills of over 2.2 million people will be raised throughout the programme in these three countries. It is expected that this number will have a strong multiplier effect in terms of knowledge transfer about low carbon and climate resilient measures in agricultural value chain development both upstream and downstream.

## C. FINANCING INFORMATION

### C.1. Total financing

| (a) Requested GCF funding (i + ii + iii + iv + v + vi + vii) | Total amount | | Currency | |
|---|---|---|---|---|
| | 102.791 million | | million USD ($) | |
| **GCF financial instrument** | **Amount** | **Tenor** | **Grace period** | **Pricing** |
| (i) Senior loans | $26,999,831.98 | 40 years | 10 years | 0.00 % |
| (vi) Grants | $75,791,156.08 | | | |

| (b) Co-financing information | Total amount | Currency |
|---|---|---|
| | $168,912,536.95 | million USD ($) |

| Name of institution | Financial instrument | Amount | Currency | Tenor & grace | Pricing | Seniority |
|---|---|---|---|---|---|---|
| **African Development Bank** | **Senior Loans** | **$85,622,259.82** | million USD ($) | 40 years | 0.0% | senior |
| -AfDB | Senior Loans | *$47,448,790.0* | | | | |
| *- African Development Fund (ADF)* | Senior Loans | *$15,802,152.82* | | | | |
| *- Nigeria Trust Fund (NTF)* | Senior Loans | *$5,619,480.0* | | | | |
| *- Transition Support Facility (TSF)* | Senior Loans | *$16,751,837.0* | | | | |
| | | | | | | |
| *- African Development Fund (ADF)* | Grant | **$10,140,378.13** | million USD ($) | | | |
| Korea Fund (KF) | Grant | **$5,000,000.0** | million USD ($) | | | |
| West African Development Bank (BOAD) | Senior Loans | **$17,600,000.0** | million USD ($) | 15 years | 0.0% | senior |
| Islamic Development Bank (IDB) | Senior Loans | **$31,063,890.0** | million USD ($) | 25 years | 0.0% | senior |
| **Governments Contributions** | **In-kind** | **$19,486,009.0** | million USD ($) | | | |
| *Government of Togo (GoT)* | *In-kiind* | *$10,856,000.0* | | | | |
| *Government of Senegal (GoS)* | *In-kiind* | *$5,000,000.0* | | | | |
| *Government of Guinea (GoG)* | *In-kind* | *$3,630,009.0* | | | | |

| (c) Total financing (c) = (a)+(b) | Amount | Currency |
|---|---|---|
| | **$271,703,525.01** | million USD ($) |

| | |
|---|---|
| **(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)** | *N/A* |

### C.2. Financing by component

| Component | Output | Indicative cost million USD ($) | GCF financing | | Co-financing | | |
|---|---|---|---|---|---|---|---|
| | | | Amount million USD ($) | Financial Instrument | Amount million USD ($) | Financial Instrument | Name of Institutions |
| **Component 1:** Strengthening of critical SCPZs value chain infrastructure and management governance | **Output 1.1:** Improved access to drip irrigation distribution systems (DIDS) powered by solar pumps to support climate resilient agricultural (CRA) practices | 21,508,722.00 | 5,729,451.56 | Senior Loans | - | - | - |
| | | | 15,779,270.44 | Grants | | | |
| | **Output 1.2:** Enhanced access to renewable energy technology for staple food crops processing | 22,063,265.64 | 5,686,342.41 | Senior Loans | - | - | - |
| | | | 16,376,923.23 | Grants | | | |
| | **Output 1.3:** Increased agricultural productivity through enhanced access to Agro-Industrial Parks (AIPs) support facilities and access infrastructure ~~Enter amount~~ | 98,318,205.78 | Choose an item. - | - | 15,802,152.8 | Grant | ADF |
| | | | | | 9,810,109.90 | Senior Loans | TSF |
| | | | | | 5,203,638.48 | Senior Loans | NTF |
| | | | | | 11,457,600.00 | Senior Loans | BOAD |
| | | | | | 5,000,000.00 | Grant | KF |
| | | | | | 27,520,298.20 | Senior Loans | AfDB |
| | | | | | 23,049,406.38 | Senior Loans | IsDB |
| | | | | | 475,000.00 | In kind | GoS |
| **Component 2:** Promotion of climate resilient agricultural practices and technologies | **Output 2.1:** Adoption of CRA and agro-forestry practices | 42,580,470.42 | 12,004,102.44 | Senior Loans | - | - | - |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| adoption among smallholder farmers | among smallholder farmers | | 30,576,367.98 | Grants | | | |
| | **Output 2.2:** Awareness raising, capacity building, community and institutional strengthening on agricultural value chains management | 50,269,543.66 | 3,579,935.57 | Senior Loans | 2,858,142.46 | Grant | ADF |
| | | | 13,017,594.43 | Grants | 2,527,873.74 | Senior Loans | TSF |
| | | | | | 4,118,400.00 | Senior Loans | BOAD |
| | | | | | 4,570,376.00 | In kind | GoT |
| | | | | | 14,187,188.21 | Senior Loans | AfDB |
| | | | | | 3,914,050.14 | Senior Loans | IsDB |
| | | | | | 240,000.00 | In kind | GoS |
| | | | | | 1,255,983.11 | In kind | GoG |
| **Component 3:** Programme management and coordination | **Output 3.1:** Project management and coordination | 36,963,317.51 | 41,000.00 | Grants | 7,282,235.67 | Grant | ADF |
| | | | | | 4,413,853.36 | Senior Loans | TSF |
| | | | | | 2,024,000.00 | Senior Loans | BOAD |
| | | | | | 415,841.52 | Senior Loans | NTF |
| | | | | | 6,285,624.00 | In kind | GoT |
| | | | | | 5,741,303.59 | Senior Loans | AfDB |
| | | | | | 4,100,433.48 | Senior Loans | IsDB |
| | | | | | 4,285,000.00 | In kind | GoS |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | 2,374,025.89 | In kind | GoG |
| **Indicative total cost (USD)** | | **$271,703,525.01** | **$102,790,988.06** | | **$168,912,536.95** | | |

| | |
|---|---|
| **C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)** | |
| C.3.1 Does GCF funding finance capacity building activities? | Yes ☒ No ☐ |
| C.3.2. Does GCF funding finance technology development/transfer? | Yes ☒ No ☐ |

Activity 2.2.2 of the proposed GCF intervention will support capacity building in the aspect of establishing climate information and early warning systems that will enhance climate-smart agro-advisory services in the SCPZs. The overall cost estimates including grants and loans amounts to $16.597.530,00. This activity involves a number of sub-activities targeted at improving access to CIEWS. These include: 1) establishing and expanding climate observing network facilities with at least ten (10) automated agrometeorological and fifty (50) self-recording rain gauge stations in each of the zones; 2) employing the services similar to that of ESOKO, a private organization focusing on rural economic empowerment in Africa through digital and financial inclusion, as an implementing partner to use the most cost-effective reseller model to deploy and expand climate-smart agro-advisory services to targeted users within the SCPZs; and 3) awareness raising and training on how to use and maintain the climate observing networks.

## CTED PERFORMANCE AGAINST INVESTMENT CRITERIA

*This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.*

### D.1. Impact potential (max. 500 words, approximately 1 page)

***Mitigation impacts.***

In terms of direct mitigation impacts, the greatest mitigation potential from the SCPZs programme under GCF financed activities is the avoided emission from the use of renewable energy. That is (a) Solar PV; and (b) Biogas. The indirect emissions calculation is based on (i) avoided emission from the use of climate resilient agricultural practices; and (ii) carbon sequestration potential from improved agroforestry practices as illustrated in Table D.1.1 below. The emission profiles of road construction and rehabilitation have also been estimated and are negligible in relation to the overall abatement that will be achieved by the use of renewable energy, CRA and sequestration from agroforestry.

GHG Accounting SCPZ

| | | | | |
|---|---|---|---|---|
| Color code: | Technology | | | |
| red cells: Assumptions | Solar PV for irrigation (MW): | 2.59 | | |
| black cells: Calculations | Solar PV for lighting and processing (MW) | 14.69 | | |
| | (m3) | 24,576.66 | | |
| **Assumptions** | | | | |
| **Yearly irrigation hours (h)** | 1825 | | | |
| **Yearly hours for lighting and processing (h)** | 4380 | | | |
| **Yearly hours running Diesel equivallent biogas plants** | 4380 | | | |
| **Efficiency of solar PV systems** | 0.8 | | | |
| **Ratio tCH4/tCO2 eq avoided** | 25 | | | |
| **Country digesters ratio compared to overall programme** | 100% | | | |
| **Energy Balance** | | **Annual** | | |
| Annual solar energy generation for irrigation (MWh) | | 3,775 | | |
| Annual Solar energy Generation for lighting and processing (MWh) | | 51,475 | | |
| Annual Energy generation from biogas plants (MWh) | | 44,867 | | |
| **Emissions Baseline** | | | | |
| Baseline 100% Diesel for solar irrigation and lighting (tCO2eq) | | 14,738 | | |
| Baseline methane discharge from bio-waste (tCO2eq) | | 477,392 | | |
| Baseline Diesel equivallent for biogas generators | | - | | |
| Baseline non sequestration by agro forestry (6 Years maturity) | | 5,743,266 | | |
| Baseline non sequestration by crops | | - | | |
| **Baseline emissions including CRA and Agroforestry (tCO2eq)** | | **6,235,396** | | |
| **Abatement (Energy & Waste)** | | **Annual** | **2030 NDC (Ten Years)** | **Lifetime (25 Years)** |
| Abatement attributed to solar irrigation systems (tCO2eq) | | 1,007 | 5,035 | 5,035 |
| Abatement attributed to solar lighting and processing systems (tCO2eq) | | 13,731 | 82,388 | 343,285 |
| Abatement attributed to the proportion of bio-waste digested (tCO2eq) | | 424,879 | 2,549,273 | 10,621,972 |
| Abatement attibuted to avoided diesel using biogas for electricity (tCO2eq) | | 11,969 | 71,812 | 299,219 |
| **Agroforestry abatement (Total Ha: 40,000)** | | | | |
| Baseline non sequestration by agro forestry (5 Years maturity) | | 5,743,266.00 | 5,743,266.00 | 5,743,266.00 |
| Ex ante estimates abatement by roads construction and rehabilitation (tC | -58894.4 | | | |
| **Total Estimated Emission Reductions** | | 6,194,852 | 8,392,881 | 16,953,882 |
| | | | | |
| **Total funding (USD)** | **$271,703,524.59** | | | |
| **GCF (USD)** | **$102,790,987.64** | | | |
| **Estimated cost per tCO2e** | **$16.03** | | | |
| **Estimated GCF cost per tCO2e** | **$6.06** | | | |

**Table D.1.1:** Avoided Carbon Emission from Major Activities of Programme (GCF Financed Components)

**Core Mitigation Impact Indicators:**

1) Avoided emissions from the use of renewable and waste. The total emission reduction attributed to energy and waste is forecast at 451,586tons of CO2 annually, and to reach 2,708,509 tons of CO2 in 2030, and 11,269,510 tons of CO2 in 25 years. It is noteworthy that a 25 year lifespan was assumed for the biodigestor and PV systems for lighting and processing but only 5 years for the solar irrigation PV systems (pico systems)

2) Avoided emission from carbon sequestration as a result of new agro-forestry management practices. It is forecast that a total of 5,743,266 during the duration of the accounting period.
3) Estimated cost per t $CO_2$eq. $16.03
4) Estimated GCF cost per t$CO_2$eq removed $6.06

***Adaptation impacts***

**Core Adaptation Impact Indicators:**

The computation of beneficiaries for the SCPZ programme involves a comprehensive analysis of both direct and indirect beneficiaries, disaggregated by country (see Annex 22c & d for further details). This analysis is essential to understand the program's reach, impact, and distribution across different populations. The beneficiaries are further categorized by gender, allowing for a deeper understanding of gender dynamics and inclusivity within the programme. The data presented in Table B.1.8 provides a summary of this computation for the three countries: Togo, Senegal, and Guinea.The core adaptation benefits are presented in Annex 22c and d. The bases for the computations are presented in Annex 22d.

**Other Core Adaptation Impact Indicators:**

- Improved water availability and efficient water supply for crop production and processing by providing; (i) Over 15,000 cubic meters/day from efficient and portable water supply systems, 30 new boreholes and rehabilitation of 8 old boreholes and 3 mini-dams with reduced sedimentation, flood control mechanisms, and reduced maintenance requirements and economic life; and (ii) 2.6 MW solar and drip irrigation systems to improve irrigation efficiency up to 90%, labor, energy and water savings of up to 30%, and reduction in post-harvest losses of about 20% especially for market gardening and cereal crops.
- Over 15% increase in agricultural profit for growing horticulture practices especially in the dry season benefiting at least 1.2 million women directly and over 19 million indirectly.

Over 20% increase in the yields of maize, soybean, rice, cassava, banana, sesame, horticulture, mango, cashew and coffee, directly benefiting at least 2.4 million households and about 48 million indirectly.

**D.2. Paradigm shift potential (max. 500 words, approximately 1 page)**

*Potential for scaling up and replication*

As indicated earlier, the SCPZs programme is an agricultural transformation agenda of the AfDB and its RMCs under its 'Feed Africa' Initiative, aimed at boosting national productivity and integrating production, processing and marketing of stable food crops along the agricultural value chains. The first phase of the Initiative is currently being piloted in Togo, Senegal and Guinea with the expectation of scaling up and replication in other regional member countries (RMCs). Presently, preliminary field visits have begone in Nigeria, Cote D'Ivoire, Senegal Central, Tanzania, and Zambia, at the request of the various Governments for the implementation of Phase II of the SCPZs programme in these countries. Thus, the success of Phase 1 is highly crucial for the scaling up and replication of the programme in other RMCs. On this note, a number of strategies have been inbuilt into Phase I of the programme with respect to scaling up and replication in other RMCs of AfDB. First, the programme has been designed to train a 'critical mass' of project expects (Training of Trainers -ToT) across all programme interventions, especially for the GCF Financed components that can easily be scale up and replicated in other regions and countries. This critical mass of experts will be drawn upon to provide hands-on-experiencing during the implementation of Phase II as international consultants. Second, for each activity financed under Phase I by the GCF, a comprehensive training manual will be prepared and developed to facilitate the replication of similar activities, especially Phase II of the programme. Third, several training workshops have been factored into the programme to facilitate knowledge sharing among value chain actors. It is expected that during these workshops and training courses, a knowledge product will be developed that will serve as a guide in subsequent Phases of the SCPZs Initiative. Fourth, working with community members, value chains actors, and other key stakeholders in the selection of programme beneficiaries, is a strategy that has proven very useful in many targeting programme. It is expected that the lessons learned from such a mapping exercise will greatly enrich AfDB's knowledge in conducting similar activities under Phase II. Fifth, the use of sub-project management committees for each activity in Phase I, is meant to draw lessons on the pros and cons of each new innovations introduced into the programme such as DIDS, Biogas and Solar technologies etc. It is highly expected that the finding of the committees will generate new insight concerning the use of these new climate resilient technologies in different cultural context. Such useful project information will greatly minimize the risks of failure in other similar programme context as Phase II. Sixth, and finally; greater participation of private sector investors into the programme will largely depend on the current business model employed in Phase I. That is, a combination of semi-autonomy of the TRP/EEs, and allowing the management of the IAIPs at the hands of the PIUs. A lot is expected to be learned from this current business model for future engagement of the private sector in Phase II.

***Potential for knowledge sharing and learning***

Firstly, the institution of a Program Management Unit (PMU) as well as a Program Steering and Monitoring Committee (PSMC), involving all major stakeholders along the agricultural value chain, implies that knowledge gathering and learning is widely shared among different stakeholders and institutions.
Secondly, as discussed under the potential for scaling up and replication of the programme, a number of initiatives have been designed into the programme to facilitate knowledge sharing and learning. Key to this are: i) the training of a critical mass of project experts for knowledge sharing and dissemination; ii) development of training manuals specific to each programme/sub-project activity to help in disseminating knowledge and share best practices in the adoption of new climate resilient technologies and climate resilient agricultural practices; iii) preparations of program operational manual and other tool kids such as drip irrigation operational manual for use by prospective agric-business investors.
Thirdly, the programme has inbuilt training activities and capacity development initiatives that are aimed at providing: i) capacity building and training in CRA practices and technology uptake; (ii) training in mixed agroforestry practices and livestock management; (iii) training on design and construction of biogas digesters; (iv) capacity building on waste recycling processes; and, (v) training on regulatory framework strengthening for private sector investment in Agriculture. These training activities will create avenues for knowledge sharing and learning on global best practices in agricultural value chains development. Direct beneficiaries of these training initiatives constitute potential sources of knowledge sharing and learning for other value chain actors not directly involved in the programme.

***Contribution to the creation of an enabling environment***
The program will bring knowledge resources and capacity building to the private sector, particularly potential agribusiness investors. This will help in creating an enabling environment for ʻcrowding inʼ additional private sector financing into agriculture. Equally, the project will strengthen the technical capacity of key government agencies, universities, local NGOs and other key stakeholders on climate resilient practices and low carbon emission technologies, which deliver both adaptation and mitigation benefits. This will provide the necessary critical mass of technical personnel that will enable the sustainability of the technology even after the completion of the project. Also, women access to finance to invest in CRA practices has been identified as a major impediment to the growth of women-led agribusiness ventures in these countries. With affordable access to finance to invest in CRA practices, women will be empowered to act as key change agents for climate change adaptation in these vulnerable communities.

***Contribution to the regulatory framework and policies***
Some of the key challenges that are faced in mainstreaming mitigation and adaptation policies into development planning, policy and investments that are common to the three countries can be outlined as follows:
- Absence of a unified vision and approach
- Lack of understanding of the challenges of climate change at different levels of government;
- Lack of comprehensive and localized risk and vulnerability assessments;
- Absence of coherent research programs to identify and describe impacts associated with near term, long term and abrupt global climate change;
- Absence of organized and coordinated efforts across local, state and federal agencies as well as absence of a strong link between indigenous communities and other local partners;
- Absence of strategies for evaluating and applying lessons learnt;
- Limited availability of reliable, useful and usable climate change information;
- Limited integration among sectoral agencies;
- Limited financial resources to mainstream and manage climate risks and adaptation;
- Lack of due diligence procedures.

The program will contribute to reducing the challenges and thereby contributing to the regulatory frameworks and policies through the following:
- Accelerating the process of education, training and awareness of climate change and its impacts to speed up implementation of response actions;
- Ensuring the cooperation and buy in of all stakeholders to climate change response to facilitate coordinated national programs;
- Harnessing the efforts of all stakeholders to achieve the objectives of energy efficiency strategies to promote a sustainable development path;
- Adapting agricultural, rangeland and forestry practices appropriately;
- Sharpening the connection between project activities and climate change vulnerability and change;
- Strengthening monitoring of and reporting on mainstreaming climate change;
- Integrating science into decision-making, improving information about risks and opportunities, enhancing communication and capacity building among relevant stakeholders, defining process of coordination and collaboration among stakeholders, among others;
- Improving the capacity of government agencies, infrastructure to support multi-sectoral platforms and inter-organizational networks, and adequate resources.

The programme intervention will contribute to climate-resilient development pathways consistent with relevant climate change adaptation strategies and plans of each nation. A summary of the individual countries' commitments described in the NDCs and their alignment with the SCPZ interventions are shown in Table B.1.1.

## D.3. Sustainable development (max. 500 words, approximately 1 page)

As illustrated in Table D.2.1 above, the program is well aligned with, and contributes to, national priorities and strategies in addition to those that are specific to climate change. It is therefore expected to generate important economic, environmental and social co-benefits at all levels.

***Environmental co-benefits***

The environmental benefits that will accrue from the project include;

- The reduction of 16.9Mt $Co_2$ eq. of greenhouse gas emissions during programme lifespan;
- The reduction in the degradation of 40000ha of farmland from tree planting under agroforestry practices;
- Improvements to soil fertility of 31,179 million ha of farmland.
- Additional irrigation of 31,179 ha of farmland.
- 5.7Mt $Co_2$eq of carbon sequestered from agroforestry practices.

Other environmental benefits will include: a) soil conservation and reduction of erosion, b) improved tree cover in farmlands, catchment areas and other components of the cascade ecosystems such as silt barriers, salt/mineral balancing areas etc. that will have several interlinked environmental benefits such as improved micro-climate, improved soil structure, bioremediation for irrigation and drinking water quality, increased biodiversity; c) restoration of ecosystem integrity, goods, and services; and d) preservation of biodiversity in farmlands and agro-forests. Soil conservation measures introduced in upstream farm fields will not only prevent siltation of reservoirs but also support soil biota and improve ground-water yields by allowing greater percolation. Improved water yields will have a positive impact on water availability during the dry season to both humans and wildlife. Ecological agricultural practices introduced by the project will reduce the harmful effects of agro-chemicals use on soil, animals and plants; and increase the diversity of soil biota. The reduction of chemical inputs, especially nitrogen fertilizers through agroecological agriculture, will contribute to reduced emissions from agriculture as well. Agro-forestry in the watersheds and catchment areas will also contribute to enhanced emissions reduction.

***Social co-benefits including health impacts***

The social benefits that will accrue from the project will include:

- improved food security and nutrition for about 40 million people;
- improved access to 27MW of renewable energy that will translate to significant knock-on socio-economic and health benefits such as access to communication, decrease indoor pollution from biomass and charcoal burning among others,
- Reversal of migration as wage labour opportunities will increase by as much as 20%.
- Improved nutrition security through the increased availability of nutrient-rich diverse foods throughout the year;
- Sustained sovereignty and autonomy of farmers through their own preservation and reproduction of local seed varieties and other genetic resources, while increasing the control of consumers on how they feed themselves;
- Empowering small-scale producers, women and youth by building on their knowledge and experience, creating spaces for challenging barriers, like patriarchal values, and recognizing the role of women and disadvantage people.

***Economic co-benefits***

The project will provide economic activities both for the small holder farmers and processing sector in the SCPZs that will include:

- On the average, at least 50% increase in agricultural yield, which will result to a 40% income for smallholder farmers (target of 46% men, 54% women);
- improved resilience of at least 5.8 million smallholder farmers including over 48 million indirect beneficiaries from climate change vulnerable (target of 46% men, 54% women);
- improved access to renewable energy from low-cost sources;

- Over 5 million jobs created for from CRA and agroforestry practices, biogas generation process targeting 46% men and 54% women;
- Trainings provided to over 2.3 million smallholder farmers (targeting 50% men and 50% women); and,
- Job creation for women that empowers them and increase their participation in household decision

***Gender-sensitive development impact***
Men and women often have different opportunities to access agricultural support and information, and agricultural land is generally owned by men. Recognizing these differences, this project will seek to understand and respond to the different needs, concerns, and challenges that men and women face in investing in ecological agriculture and clean energy adoption.
Specific areas where gender equity and women's empowerment can be improved include:

- Ensuring that both male and female smallholder farmers (and their families) have equitable access to the information, services, technology, and support that this project will provide on climate-resilient agricultural livelihood options,
- Targeting to reach equally male and female smallholders (50-50) for deployment of finance;
- Targeting to reach equally male and female inhabitants (50-50) for activities related to access to energy and water;
- Providing additional outreach, education, specialized services, etc., as needed, to ensure that female farmers are able to participate and benefit from this project's activities;
- Incorporating discussion and reflection about the gender implications of climate change and small-scale agriculture into all training and educational materials produced through this project.

Training in agroecological methods will contribute to enabling women to take on leadership functions as they will be empowered with knowledge that will provide them the confidence to take on responsibilities. The agroecological approach will provide women with women-friendly spaces that will be important for the exchange of ideas, and marketing.

Food security as a priority will contribute to enabling women's empowerment since they hold, in most cultures, a central role in providing nutrition for the household. Gender inequalities are clearly critical for affecting child nutrition. If women have unequal labour and low decision-making power, there is likely to be high barriers to improved child nutrition. The agroecological approach taken will help to build farmer knowledge and draw on local resources, strategies which are important for long-term sustainability as well as building men's support and interest in child nutritional outcomes.

More details are provided in the gender assessments and gender action plans of the 4 countries of this programme. Indeed, as there are substantial differences in terms of gender mainstreaming priorities according to the different areas targeted by the programme, it would be confusing to list here all the proposed activities mentioned in the gender action plans, knowing that they don't cover all regions.

## D.4. Needs of recipient (max. 500 words, approximately 1 page)

***Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)***
The type of economic activity on which households depend crucially determines the vulnerability of their livelihoods. This is not only the type of main source of income but also how sensitive it is to climate change related impacts. The gender of the head of a household also plays an important role in deciding household vulnerabilities. Female-headed households are more vulnerable to the impacts of climate change in comparison to their male counterparts because of the differences in access to basic resources such as land and finance services. The distribution of household members in rural areas in all three countries shows that the majority of household members are engaged in farming and farm labour. The fact that farming is the major source of income in these countries exposes households to frequent climate change shocks like droughts and flooding typically experienced in the countries. In these countries, women, are more engaged in small businesses compared to men, and the reduction of household income forces women to come out of their households to earn additional income. This leaves children without care, which in turn affects their wellbeing in terms of health, nutrition and education. In addition, the children are forced to work outside the household which comes at the expense of their education. The impact of climate change is not only manifested in the loss of farm income as a result of reduced crop yields, but also in family labour. The loss of income forces households to allocate their labour to non-farm income that triggers short to long term forced migration with far reaching implications on labour force supply. Some key livelihood profile indicators for the pilot countries are presented in Table D.4.1.

**Table D.4.1** Selected Livelihood Profile Indicators

| Average 1990-2017 | |
|---|---|
| Countries | Domestic Food Price Level Index |

| | |
|---|---|
| Togo | 2,5 |
| Senegal | 2,0 |
| Guinea | 2,3 |
| Value of Food Imports in Total Merchandise Exports (Percentage) | |
| Togo | 20% |
| Senegal | 53% |
| Guinea | 22% |
| Domestic Food Price Volatility Index | |
| Togo | 340,1 |
| Senegal | 24,4 |
| Guinea | 62,9 |
| Cereal Import Dependency Ratio (Percentage) | |
| Togo | 20,3% |
| Senegal | 51,4% |
| Guinea | 17,7% |
| Prevalence of Undernourishment (Percentage) | |
| Togo | 25% |
| Senegal | 22% |
| Guinea | 19% |
| Depth of the Food Deficit (kcal/person/day) | |
| Togo | 131,0 |
| Senegal | 124,0 |
| Guinea | 104,0 |
| Prevalence of Food Inadequacy (Percentage) | |
| Togo | 28,9% |
| Senegal | 28,0% |
| Guinea | 23,6% |

Source: FAO - Food Security Indicators

***Economic and social development level of the country and the affected population***

Institutional services and access to infrastructure affect the adaptation responses of households through incentive structures. Moreover, they are an intermediary through which external interventions such as finance, information and skills are channelled to the local economy. Extension services are one of the strategies of transferring knowledge and skills to users to build their adaptation capacity. While access to physical resources such as land and water is essential for adaptation, the use of technology is one of the most important ways of building the adaptation capacity of communities. Technology ranges from the use of efficient irrigation methods to plant breeding for drought tolerance. The areas under irrigation and access to information about improved irrigation practices are important inputs to alleviate constraints of adaptation. Lack of access to finance can constrain or affect the adaptive capacity of producers or actors along the value chain. The majority of household sources of credit in the countries are relatives and friends who can afford to lend out very limited amounts.

A variety of infrastructure can constrain the adaptive efforts of a particular community. Access to road transport, market, education and other public services including health administrative and agricultural offices are among the key infrastructure with important implications for adaptation capacity. Access to basic social services such as portable water, sanitation and energy are integral parts of the adaptive capacity of households or communities to climate change impacts. Table D.4.2 shows the level infrastructure deficiencies in the pilot countries.

**Table D.4.2:** State of infrastructure in the pilot countries

| Average 1990-2017 | |
|---|---|
| Countries | Percent of Paved Roads over Total Roads |
| Togo | 29,2% |
| Senegal | 29,4% |
| Guinea | 15,5% |
| Road Density (per 100 square km of land area) | |
| Togo | 15,6% |
| Senegal | 7,3% |
| Guinea | 14,8% |
| Percentage of Population with Access to Improved Drinking Water Sources | |
| Togo | 64% |
| Senegal | 66% |
| Guinea | 54% |
| Cereal Import Dependency Ratio (Percentage) | |
| Togo | 20,3% |
| Senegal | 51,4% |
| Guinea | 17,7% |
| Percent of Arable Land Equipped for Irrigation | |
| Togo | 3,6% |
| Senegal | 3,2% |
| Guinea | 0,3% |

Source: FAO - Food Security Indicators 2017

*Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)*

***Togo***

Togo's average score from the Country Policy and Institutional Assessment (CPIA) of the AfDB and the World Bank has never exceeded 3.2 out of 6 from 2008 to 2016. This score means that Togo is still a country facing conditions of fragility. Between 2011 and 2016, the Togolese government made extensive use of debt to fund major capital expenditure especially in the area of transport infrastructures. As a result, the country's debt level increased from less than 46.9% of GDP in 2011 to over 79.2% in 2016. According to the IMF/World Bank joint analysis of debt sustainability, the country's debt distress risk is high since 2016. In order to mitigate this vulnerability, the program pursued by the Government of Togo with the IMF support as part of the Extended Credit Facility (ECF 2017-2019) aims at reducing the level of indebtedness gradually from 79.2% in 2016 to 69.9% in 2019, i.e., below the WAEMU convergence threshold, (less than 70%) after 3 years. According to this ECF program, the government has committed to no longer contracting any new nominal external debt on non-concessional terms and to imposing substantial supervision to concessional borrowing. Under such circumstances, mobilizing private international capital will play a fundamental role, but public finance, such as climate finance, will contribute significantly to closing the funding gap.

***Senegal***

The average annual trade deficit represented 18% of GDP over the period 2011-2015 as a result of very poor export performances which represented an annual average of 19% of GDP[55]. At the same time, public debt in Senegal rose from less than 23% of GDP in 2007 to over 54% in 2015 but remains below the standard of 70% retained under the WAEMU convergence criteria. The Government's objective is, however, to maintain that ratio at 53% under the programme with the IMF entitled 'Economic Policy Support Instrument' (2015-2017 EPSI). Based on this, the

[55] AfDB 2016. Bank Group's Country Strategy Paper For Senegal, 2016-2020. African Development Bank, Abidjan, Cote d'Ivoire.

government is committed to no longer contracting additional external debt on non-concessional terms and to impose substantial supervision to concessional borrowing. Under these prevailing circumstances, mobilizing private international capital will play a fundamental role, but public finance, such as climate finance, will contribute significantly to closing Senegal's debt burden.

***Guinea***
In Guinea, significant infrastructure spending has aggravated the country's budget deficit. The overall budget deficit (net of grants) deteriorated from 1.3% of GDP in 2016 to over 3.6% in 2017. Equally, significant public investment, financed by loans, raised the debt level from less than 27.2% of GDP in 2012 to over 37.2% in 2017. Domestic debt surged from 9.3% of GDP in 2012 to over 17.6% in 2017. During the same period, the external debt increased from 18% of GDP to 19.8%. According to IMF recommendations, Guinea public debt should remain below the threshold (70% of GDP) set in the convergence framework of the Economic Community of West African States (ECOWAS) until 2022[56]. Against this background, the government is committed to no longer contracting additional external debt on non-concessional terms but to mobilizing private international capital as well as international public finance, such as climate finance to close the funding gap.

***Need for strengthening institutions and implementation capacity***
As reported in the program feasibility study, the current public financial management (PFM) systems in place for the three pilot countries are not very satisfactory and as such, the overall fiduciary risks of the project implementation are quite substantial. There is therefore a need for institutional reforms targeted at institutional strengthening of the PFM systems in these countries.
Secondly, donor funds are usually managed outside the annual budget cycles in these countries. This equally calls for institutional reforms aimed at integrating donor funds within the annual budget cycles.
Third, climate finance is a relatively new area especially for many of these countries participating in the SCPZs program. Under these circumstances, there is a need for institutional strengthening to deal with climate finance flows and reporting

**D.5. Country ownership (max. 500 words, approximately 1 page)**

***Existing national climate strategy***
All three pilot SCPZs countries are parties to the United Nations Framework Convention on Climate Change (UNFCCC) and have signed and ratified the Kyoto Protocol. By ratifying the UNFCCC, the pilot countries have committed to the implementation of measures to adapt to climate change. By the same token, upon ratification of their respective Paris agreement in September 2016, all the three pilot SCPZ countries have committed to the implementation of measures to conditionally and unconditionally reduced GHGs by 2030 compared to the BAU emissions scenarios. SCPZs interventions are therefore well aligned with operationalization of national commitments on climate change mitigation and adaptation through NAPAs, NDCs, National Climate Change Policies and Strategies, National Communications (NCs), SDGs, Disaster Risk Reduction, and National Agricultural Investment Plans and Strategies.

***Existing GCF country programme***
Apart from Senegal that have ongoing country-led GCF financed projects/programme in the adaptation thematic area[57], Togo and Guinea are yet to have GCF financed projects that are country-led. Besides, all 4 the ongoing GCF Funded projects in Senegal are adaptation-related in nature.

***Alignment with existing policies such as NDCs, NAMAs, and NAPs***
A summary of each country's commitments to their respective INDC consistent with the SCPZ Interventions are presented in Table D.2 under section D.2.

[56] AfDB 2018. Guinea Country Strategy Paper 2018-2022. African Development Bank, Abidjan, Cote d'Ivoire.

[57]These include: Ecosystem and Communities-based Resilience Building in Senegal (FP003); Integrated Urban Flood Management in Senegal (FP021); Building Resilient of Food Insecure Smallholder Farmers in Senegal (FP048)

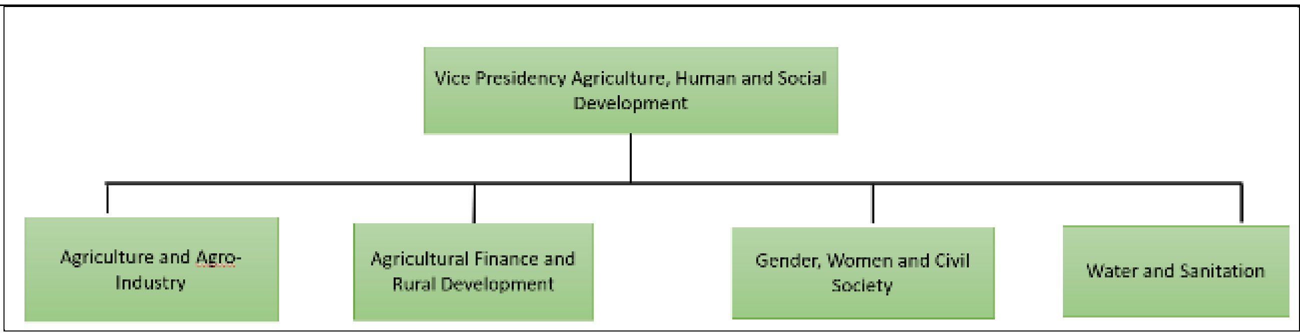


***Figure D.5.1.*** *Capacity of Accredited Entity.*

The AfDB has considerable experience in supporting African countries in developing their agricultural sector. The Bank has long recognized that agriculture is central for the socioeconomic development of the continent which is now intertwined with the Bank's efforts to contribute to the climate change adaptation and mitigation agenda. This is through measures aimed at addressing climate change issues as well as climate change mainstreaming in every project design and operations of the Bank. As indicated in Figure D.5.1, the AfDB has departments with complementary funding operations and initiatives to mutually reinforce impacts for sustainable agricultural development in Africa.

Between 2006 and 2014, the Bank Group carried out 198 operations in agriculture and agribusiness, amounting to 4.9 billion USD, including 181 of sovereign operations and 17 non-sovereign operations. The bulk of the sovereign projects in Agriculture (124, or approximately 69% of all projects) has been allocated to ADF-classified countries (UA3.1 billion), representing 80% of the total agricultural portfolio. A total of 25 projects (14% of the total) have been allocated to ADB-classified countries, amounting to 496 million USD, 10% of the total portfolio. Blend and transition countries have received approximately 132 million USD and 298 USD million respectively (financing 15 projects in total), amounting to a cumulative 9% of the overall portfolio. Projects considered to be multinational (15) have received approximately 300USD million in financing. Between 2006 and 2014, the majority of sovereign projects were concentrated in the West Africa region (67); North Africa (16); East Africa (27); South Africa (30); and Central Africa (26).

Recent Bank interventions (under the 2010-2014) have primarily focused on building infrastructure for sustainable agricultural development, including rural roads, irrigation, storage facilities, and markets – its areas of comparative advantage. The Bank has collaborated with other specialized agencies such as the Food and Agriculture Organization (FAO), the International Fund for Agriculture Development (IFAD) and others better positioned to intervene in different parts of the value chain, such as seeds, fertilizer, and research and extension. The Bank has also made efforts to leverage funds from partnerships and trust funds to complement its traditional funding sources in order to meet the demands of its RMCs in the agriculture sector.

The Bank has also mobilized resources from global funds such as the Global Agriculture and Food Security Program (GAFSP) and Climate Investment Funds (CIF), including the Pilot Program on Climate Resilience (PPCR) and the Forest Investment Program (FIP). While being proactive in supporting the demands of the Bank's RMCs, the Bank has aimed to remain selective, focused, and innovative. More than 80% of the agriculture sector's approvals between 2010 and 2012 were allocated to rural and value-chain development. Given the Bank's comparative advantage, it is particularly successful in leveraging resources through partnerships, co-financing, and initiatives. Mainstreaming gender through the Bank's interventions in the agriculture sector was identified as a major shortfall in this sector. This made the Bank come up with the AFAWA bold initiative which identified the need for better handling of gender issues in agriculture projects, beyond disaggregated indicators by gender. A pilot programme under the AFAWA Initiative was recently approved by the GCF during the B.24 Board meeting *(Program on Affirmative Finance Action for Women in Africa (AFAWA): Financing Climate Resilient Agricultural Practices in Ghana (FP0114).*

Bank Group support to agriculture has been predominantly from the public sector window and use of sector budget-support instruments, for example, has been limited.

*Role of National Designated Authority*

The role of the NDA is to provide technical input into the draft funding proposal and ensure that the approved funding proposal is in line with climate change mitigation and adaptation targets of the country. Also, the NDA is responsible for issuing a 'No Objection' letter to the Accredited Entities once it is satisfied with the content of the funding proposal.

*Engagement with civil society organizations and other relevant stakeholders, including indigenous peoples, women and other vulnerable groups*

The project has undergone extensive steps to endure country ownership in the three countries. An engagement was carried out during the identification stage in the three countries where all stakeholders from government institutions, the private sector, farmers, farmers associations, civil society organization, among others, were all involved (see PARs). Also, different workshops and field visits were held in the three different countries during the identification stage and during the project appraisal to bring on board all stakeholders and capture their input in making the project a success. The draft proposal was shared with the various NDAs and their inputs have been taken into account during the revision process.

**D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)**

As indicated earlier under section D.4 describing the needs of the recipient countries Togo, Senegal and Guinea are heavily incepted poor countries with public debt burden constituting a significant portion of the GDPs of these countries. Under these circumstances, it is highly advisable for these countries not to contract additional external debts on non-concessional terms and to impose substantial supervision of concessional borrowing. Mobilizing climate finance will contribute significantly to closing these gaps. Besides, in the Intended Nationally Determined Contributions (INDCs) of the involved SCPZs countries, each pledged to undertake GHG mitigation actions, conditional upon receipt of international support such as GCF resources. With GCF resources Togo, Senegal and Guinea can triple their emissions reduction targets as shown in Table B.1.1.

**Mitigation Impact**

In Table D.6.1 we present the cost per $tCO_2e$ in the SCPZs program. As observed, the estimated cost per $tCO_2e$ is about $16.03. To this abatement cost, the GCF only contributes $6.06 per $tCO_2eq$ achieved, which makes a respective contribution of 59% and 41% in terms of financing the mitigation potential for respectively the mobilized co-finance by AfDB and the GCF funding.

**Table D.6.1: Cost per $tCO_2e$ with and without GCF Support**

| | |
|---|---|
| **Total funding (USD)** | **$271,703,525.01** |
| **GCF (USD)** | **$102,790,988.06** |
| **Estimated cost per tCO2e** | **$16.03** |
| **Estimated GCF cost per tCO2e** | **$6.06** |

**Adaptation Impact**

In Table D.6.2 we present the cost per adaptation in the SCPZs program. As observed, the estimated cost per beneficiary to adapt is about $245.95 To this adaptation cost, the GCF only contributes $93.05 per programme beneficiary, which makes a respective contribution of 72.6% and 27.4% in terms of financing the adaptation potential for respectively the mobilized co-finance by AfDB and the GCF funding.

**Table D.6.2**: Cost of Adaptation per Beneficiary with and without GCF Support

| | |
|---|---|
| **Total Funding (USD)** | **$271.703.525,01** |
| **GCF (USD)** | **$102.790.988,06** |
| **Programme Direct Beneficiaries** | **1.104.728,00** |
| **Estimated Cost per Beneficiary** | **$245,95** |
| **Estimated GCF Cost per Programme Beneficiary** | **$93,05** |

***Expected economic rate of return based on a comparison of the scenarios with and without the project.***

The return parameters for renewable energy (Solar PV and Biogas) and CRA practices are presented in Tables D.6.3, D.6.4, D.6.5. For detailed calculations and methodologies used, please refer to the Financial and Economic Models (FEMs) supplied **as Annex 3 and the sensitivity analysis presented in the feasibility study**.

Emissions reduction is estimated based on the following total effort (for a budget of $102.79 million):

- Solar irrigation (15.4% of the budget): 2.69 MW of capacity
- Biogas (for electricity: 6% of the budget): 10.24MW of capacity.
- Agro-forestry practices (for carbon sequestration: 20.37% of the budget): 40,000Ha

For the comparative analysis of electricity consumption, solar, biomass to electricity, diesel power generators and electricity purchased from the grid are considered. The financial assessment was carried out using a cost of capital of 0.75% (0% interest rate and 0.75% of fees) as an implicit discount rate, 12% social discount rate, and 20% private sector ESCO discount rate. The analysis reveals the following key findings:

Data on price of cubic volume of biogas varies widely depending on the country and specific system of collection, packaging, delivery and utilization. It's therefore useful to rely on benchmark price for expected revenue, the analysis used price of biogas production in the form of methane for the transport sector based on IRENA (2017)[58] report as the surrogate to calculate the expected revenue. IRENA (2017) indicate that the price of producing biogas typically ranges between USD 0.22 and USD 0.39 per cubic meter of methane for manure-based biogas production, and USD 0.11 to USD 0.50 per cubic meter of methane for industrial waste-based biogas production. IRENA also notes that the byproduct of biogas is the bio-slurry which is usually used to replace commercial agriculture. According to Smith (2011[59]), bio-slurry is an effective organic fertilizer. The emission impacts from GHGs such as nitrous oxide, were significantly lower from crops treated with bio-slurry than from those treated with urea. Furthermore, using bio-slurry as a fertilizer can increase crops yield by 25% and save the farmers more than 50 US$ on the cost of chemical fertilizers. The expected annual revenue was discounted to reflect the time value of money.

The analysis also considered the potential cost savings from replacing kerosene and wood charcoal that are fossil- and biomass-based systems with significant climate and deforestation impacts with biogas for cooking and lighting. Biogas based cookstoves and lamps are very efficient. Data on 2019 export quantity (tonnes/yr) and value for wood charcoal was obtained from FAOSTAT[60]. The estimated value was $278/tonne. For kerosene, Tracy and Jacobson (2012)[61] provided in their study the average price of kerosene for rural areas of Senegal that was used for the analysis. As indicated in Table D.6.3, switching to biogas from kerosene and wood charcoal could result in saving about $2.3million every year and over $13 million even at the high ESCO discount rate.

**Table D.6.3.** Summary Results of financial Analysis of biodigester system and direct use of biogas

| **Summary.** | **Values** |
|---|---|
| Annual Cost Savings from Replacement of Kerosene and Wood Charcoal with biogas ($) | **1,655,154.75** |
| Total Discounted (Soc Disc) Cost Savings from Replacement of Kerosene and Wood Charcoal with biogas ($) over project lifespan | **14,539,402.02** |
| Total Discounted (ESCO Disc) Cost Savings from Replacement of Kerosene and Wood Charcoal with biogas ($) over project lifespan | **9,826,826.58** |

| | |
|---|---|
| IRR (Y1 - Y25) | 3.15% |
| NPV(Y1 - Y25) at Discount (0.75% implicit for cost of capital) | $2,162,520.57 |

[58]IRENA 2017. Biogas Cost Reductions to Boost Sustainable Transport. https://irena.org/newsroom/articles/2017/Mar/Biogas-Cost-Reductions-to-Boost-Sustainable-Transport#:~:text=Typically%20the%20price%20of%20producing,industrial%20waste%2Dbased%20biogas%20production.

[59]Smith, J. U. 2011. The Potential of Small-Scale Biogas Digesters to Alleviate Poverty and Improve Long Term Sustainability of Ecosystem Services in Sub-Saharan Africa. DFID NET-RCA06502. https://assets.publishing.service.gov.uk/media/57a08ad9e5274a31e00007ec/FinalReport_Biogas-Digesters-in-Sub-Saharan-Africa.pdf

[60]FAOSTAT. http://www.fao.org/faostat/en/#data/FO

[61]Jennifer Tracy and Arne Jacobson. The True Cost of Kerosene in Rural Africa https://www.lightingglobal.org/wp-content/uploads/2012/04/40_kerosene_pricing_Lighting_Africa_Report.pdf

| | |
|---|---|
| NPV (Y1 - Y25) at Social Discount | ($3,310,059.12) |
| NPV (Y1 - Y25) at ESCO Discount | ($4,079,516.36) |

The financial analysis also indicated a negative NPV at the social discount rate of 12% and ESCO discount rate of 20%. The NPV is only positive with a discount rate representing the implicit costs of capital.

The assumption for capital costs was based on IRENA (2012)[62]. The analysis of electricity from biogas used the same assumptions of diesel emission factor, diesel and grid price as solar because they are all replacing solar and are off-grid solutions. The Levelized Costs of Energy (LCOE) was calculated with the net system cost represented by present value of costs and net generation represented by net energy generation. Avoided Electricity grid costs is costs of grid electricity price minus LCOE of biogas or solar multiplied by annual generation. Avoided social costs of carbon were estimated as annual emission from grid electricity multiplied by the social costs of carbon (SCC) per ton. The SCC is an estimate usually in dollars, to represent the economic damage that would result from emitting one additional ton of greenhouse gases into the atmosphere. Although the Biden Administration has raised the social costs of carbon to $51 the analysis uses a conservative figure of $31.

**Table D.6.4.** Summary Results from Financial Analysis of Electricity Generation with Biogas

| | | |
|---|---|---|
| IRR (20X0-25) | 7.6% | |
| IRR with externalities (20X0-25) | 5.3% | |
| NPV (20X0-25) | Cost of Capital Imp Rate | $23,363,229.07 |
| | Social Discount | -$5,839,611.33 |
| | ESCO Discount Rate | -$10,510,080.77 |
| NPV with externalities (20X0-25) | Cost of Capital Imp Rate | $14,933,862.41 |
| | Social Discount | -$8,749,660.49 |
| | ESCO Discount Rate | -$12,345,789.82 |

The annual grid cost savings by undertaking this off-grid option is about $9 million with annual emission avoided at 424,879 $tCO_{2eq}$. The annual avoided social costs of carbon by switching from diesel and grid to biogas-based plants for electricity is over $6 million. The incremental costs of replacing diesel with biogas is a cost to the developer that is usually externalized. However, the benefit is enjoyed by the global society and in the absence of reliable carbon market these incremental costs should justify the level of concessionality being requested from the GCF. As shown in Table D.6.4, the use of biogas for gas generation is financially viable only at the implicit costs of capital with GCF financing both with externalities and without externalities.

The analysis for solar generation for irrigation, lighting and processing relied on benchmark costs of $1.5/watts (including accessories) for solar irrigation systems and $1/watts for solar PV for lighting and processing. Assumption based on capacity factor was based on information from SunMetrix[63]. As with the analysis for the biogas for electricity, the grid emission factor was an average calculated IFI Dataset[64] on grid emission factors for the Programme Countries. The figure of 640 g/kWh was equivalent to the average of the grid EF for the 3 Programme Countries. The assumptions

[62]"IRENA 2012. RENEWABLE ENERGY TECHNOLOGIES: COST ANALYSIS SERIES. Volume 1: Power Sector. Issue 5/. Biomass for Power Generation.
. https://www.irena.org/publications/2012/Jun/Renewable-Energy-Cost-Analysis---Biomass-for-Power-Generation"

[63] SunMetrix. What is capacity factor and how do solar and wind energy compare?. https://sunmetrix.com/what-is-capacity-factor-and-how-does-solar-energy-compare/#:~:text=What%20is%20capacity%20factor%20and%20how%20do%20solar,%20%2070%25%20%202%20more%20rows%20

[64] The IFI Dataset of Default Grid Factors v.2.0.
https://unfccc.int/sites/default/files/resource/Harmonized_Grid_Emission_factor_data_set.pdf

for electricity tariff and price of diesel were based on Trimble et al/World Bank (2016)[65] and IEA (2020)[66] respectively just as was used for the biogas electricity generation analysis.

**Table D.6.5.** Summary Results from Financial Analysis of replacement of diesel-based systems with solar

| | | |
|---|---|---|
| IRR (20XX-20XX+5) | 9.9% | |
| IRR with externalities (20XX-20XX+5) | 6.0% | |
| NPV (20XX-20XX+5) | Cost of Capital Imp Rate | $21,215,096.60 |
| | Social Discount | -$1,904,991.04 |
| | ESCO Discount Rate | -$5,800,164.03 |
| NPV with externalities (20XX-20XX+5) | Cost of Capital Imp Rate | $11,391,689.73 |
| | Social Discount | -$5,356,142.36 |
| | ESCO Discount Rate | -$7,999,583.09 |

The annual grid cost savings by undertaking this off-grid solar option is over $2 million with annual emission avoided at 13,731 $tCO_{2eq}$. The annual avoided social costs of carbon by switching from diesel and grid to solar plants for electricity is over $2 million. The incremental costs of replacing diesel with solar PV is a cost to the developer that is usually externalized. However, the benefit is enjoyed by the global society. As shown in Table D.6.5, the use of biogas for gas generation is financially viable only at the implicit costs of capital with GCF financing both with externalities and without externalities.

Besides the global benefits from using this low carbon energy, investing in solar energy and biogas also reduces the risk of capacity outages and makes the SCPZ countries power generation more secured hence increasing the economic resilience of farmers through reducing the economic risks they could face in the long run as a result of climate change. In addition, by reducing the reliance on fossil fuels and global oil prices, using solar power leads to more stable and predictable electricity costs in the longer run. In addition to the economic and environmental benefits described above, renewable energy is more labor intensive and would hence generate more jobs in production, construction and operations and maintenance. Particularly for biogas energy systems, the reliance on local materials for construction of biodigesters provides significant opportunities for green jobs in the Programme Countries.

***Climate Resilient Agriculture***

In order to assess the potential impact of GCF support for the implementation of climate resilience agriculture interventions in the SCPZ countries, we have built on previous experience and the literature, and have carried out the following analysis using Cost-Benefit Analysis (CBA). It has been used to assess the net benefits generated by CRA practices compared to the current baseline. Hence:

- We have considered cereal, fruits and vegetables, legumes and nuts, roots and tubers, and cash crops extensively grown in these countries, and an area of 1 hectare for implementation (as well as for cost and revenue calculation).
- We have assumed, in a baseline scenario, that yield will progressively decline year after year, reaching a reduction of 20% by 2035. The timeline of the analysis is 20XX – 20XX+25.
- We have analysed five climate resilience agriculture (CRA) interventions: (1) improved nutrient management, and (2) improved genetic resources.
- We have created a project financing model to estimate the internal rate of return (IRR), net present value (NPV) and debt coverage ratio (DCR) for each of the 5 interventions analysed. Currently the increase in productivity of

[65] "Chris Trimble, Masami Kojima, Ines Perez Arroyo, Farah Mohammadzadeh. Financial Viability of Electricity Sectors
in Sub-Saharan Africa. Quasi-Fiscal Deficits and Hidden Costs. World Bank publication. https://documents.worldbank.org/en/publication/documents-reports/documentdetail/182071470748085038/financial-viability-of-electricity-sectors-in-sub-saharan-africa-quasi-fiscal-deficits-and-hidden-costs"

[66] IEA. Energy Prices. May 2020. https://www.iea.org/reports/energy-prices-2020

crops is calibrated based on the NPVs obtained from the USAID (2017) study and the AFAWA funded project (FP0114).

- For the programme financing model, we use a discount rate of0.75% (0% interest and 0.75% in fees) which is the threshold cost of capital in order to incentivize ESCOs to venture into providing their services. All assumptions (including yield, market prices, and costs of each intervention) are presented in the Excel files attached and can be easily adjusted for replication elsewhere.

Results from the CBA show that the IRR is positive for all interventions with improved nutrient management being the most profitable intervention. These results indicate that there is strong potential for avoiding climate impacts when using CRA interventions, and that these are also economically viable investments for farmers especially smallholders' farmers. If support were to be provided to increase access to financing, the risk of a loan being repaid would be low, given the strong upside of productivity and production, and hence revenues. The payback time generally ranges between 1 and 3 years with the DCR looking positive almost immediately for most crops.

***Drip Irrigation Technology (DIT)***

Concerning DIT, in order to assess the financial and economic viability of drip supported irrigation technology within the programme context, a drip irrigation investment manual has been prepared (see annex 1 of program feasibility study). This can be contextualised for use in many parts of Africa to help inform irrigation water management policies. Particularly, the investment guide seeks to provide potential farm investors, smallholder farmers, cooperative societies; farmer-based associations and practitioners among others, with an overall guide on drip irrigation systems, particularly in terms of water requirements and required irrigated plot sizes for profitability.

The initial investment for drip irrigation technology is based on a budget of $1,500 per hectare (covering climate resilient activities for women engaged in off season farming including purchase of drip irrigation kits).

The underlying assumptions used are;

- A sample plot with a total surface area of 2,278 $m^2$ including 1,400 $m^2$ of vegetable plots and 200 $m^2$ reserved for nursery.
- The suggested irrigation system used is composed of drip irrigation kits (iDE[67] technology) adapted to the water requirements of vegetable crops in most parts of the Sahel and SSA (Togo, Senegal and Guinea). These include: (1) water tanks; (2) two kits of 500 m2 for a total of 1000 m2; (3) four kits of 100 m2 for a total of 400 m2; and, (4) a water tower of up to 5,000 litres of capacity, which must be supplied by a regular water supply such as a borehole initially expected to have a discharge potential of at least 5 $m^3$ $h^{-1}$
- The following vegetable crops are irrigated during the dry season (onion, tomato, local eggplant, cabbage, pepper, carrot, lettuce, cucumber and okra), all widely cultivated by small farm households in the three pilot countries.
- Variable costs include; drip irrigation kits, solar powered pumps with capacity of 66 kW/ha, water storage tank and small reservoir of with capacity of 5 m3, small agricultural tool kids, labour (plot installation), cost of site clearing and preparations, fencing of plot, seeds costs, fertilizer, pesticides and disease control, permanent labor (daily labor and follow-up during harvesting), cost of transportation and procurement of supplies for marketing, and a seasonal changes of 6% on total variable cost.
- Fixed cost includes interest payment on part of variable costs incurred (6%), and non-fixed cost includes a flexible maximal annual depreciation charge of 5% and total intermediate consumption by households.

Based on this, a cost-benefit analysis (CBA) was carried out to assess the profitability of investing in drip irrigation technology under the programme (2021 - 2030). The cash flow outlay generated by the investment (see Annex 3 B) was used to calculate the Extended Internal Rate of Return (XIRR) and the Net Present Value (NPV) under different Discount Rates. The results are presented in Table D.6.5 below. As observed, results from the CBA indicate a moderate XIRR of 68% for the investment with an XNPV value of **$81,221** assuming a 0.75% interest rate from the GCF (cost of borrowing from GCF). However, when the discount rates are alternated (i.e., using 12% social discount rate and 20% ESCO Discount rate (private sector), the XNPV values drops to$39,832 and $24,898 respectively, with a payback period of approximately 3.72 years or 44.7 months for the initial investment or $1,500. Even at different discount rates applied for the investment, the results indicate that there is strong potential for investing in drip irrigation technology in the programme in combination with other CRA interventions, and that these are also economically viable investments for farmers especially smallholders' farmers. If support were to be provided to increase access to financing, drip irrigation offers the greatest opportunity for smallholder farmers in these countries. With evidence of recurrent drought and increasing demand for agricultural water, supporting smallholder farmers to invest in drip irrigation technology offers the most promising climate resilient pathway to development. With a minimal investment capital of only $1,500, in five years, women in agriculture in the three pilot countries can earned up to $20,000.0 as shown in the Figure D.6.1.

[67] International Development Enterprises

**Table D.6.5: Economic Rate of Return (ERR), DIDS**

| | |
|---|---|
| **XIRR** (20XX-20XX+5) | 69% |
| **XNPV** (20XX-20XX+5) **at Discount (0.75% implicit for cost of capital)** | $81,221.61 |
| **XNPV** (20XX-20XX+5) **at Social Discount Rate (12%)** | $39,832.81 |
| **XNPV** (20XX-20XX+5) **at ESCO Discount Rate (20%)** | $24,898.19 |
| **Payback Period (Years)** | 3.72 |

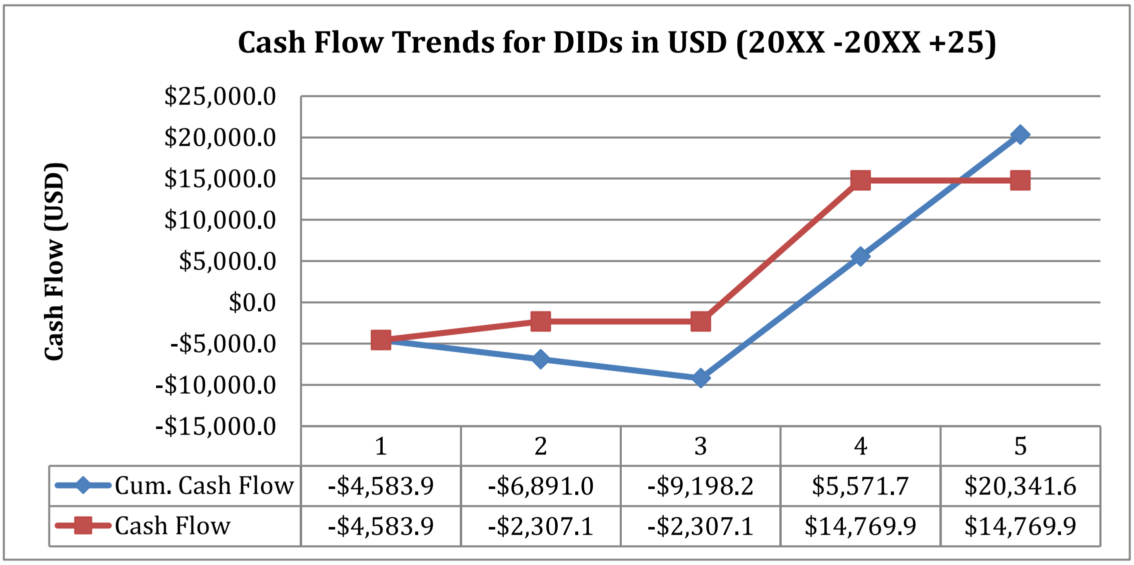


**Figure D.6.1:** Growth in Cash Flows from Dry Season Vegetable Cultivations using DIDS.

**Agroforestry Management Practices.**
The economic and financial analysis for agroforestry is based on a per hectare budget of $1,500 each for Cashew, Mango (Kent or Keitt) Arabica Coffee. Based on this, a simple economic budgeting for the three agroforestry practices was carried out. Cashew and Mango (Togo, Senegal, and Guinea). All prices used in the calculations were based on field information gathered from all three countries and calibration using studies so far carried out in these countries. Three components typically characterised an economic budgeting for agroforestry practices; 1) expected revenues; 2) variable costs; and 3) fixed costs.

***Mango (Kent or Keitt mango) - Senegal, Togo and Guinea***
Key Assumptions

- The mango plantation lifespan is for 25 years since after 25 years, most mango orchards are no longer very profitable.
- Yield is optimal for 10 to 15 years but starts at year 4
- Average per ha yield of 18.1kg although yields of 20 - 30 tons per ha have been recorded in Senegal
- The farm gate prices per/kg ranges from 151XAF in Senegal, 130 XAF in Togo and 120 XAF in Guinea
- Can be intercropped with cabbage, okra or onions from year 3
- An initial budget of $1,500 per ha is assumed covering drip irrigation and CRA practices for vegetable farming (if decided).
- Variable costs for per ha production include: site preparations cost (land clearing for liming, digging of holes and fencing the farm); costs of fertilizing holes with MOP and FYM, cost of planting (seedlings -Grafted scion, cost of transplant (July - August), labour cost for planting, and replanting- 1/50th of a hectare each year for first three years); maintenance cost: Fertilizer (October, June - July with Nitrogen, Phosphate & Potash, application of Pesticides/Fungicides, Herbicides, Weed Control (May to September), and pruning (Once every 2 years) and starts at year 3), and labour cost for maintenance, and initial investment in drip irrigation); lastly, labour for harvesting of fruits, which takes place from January to May. In economic prices, the variable cost per hectare under irrigated farming conditions is **$3,288.80** before yield starts at year 3. (Annex 3B).
- Fixed and non-fixed costs per hectare include: interest payment on part of variable costs incurred (6%) and depreciation charges (5%), which is **$348.42** prior to year 3 when yield starts.

**Developing a Mango cash flow plan:** All figures for the economic budgeting where calibrated based on the following source references [68,69,70,71,72,73,74,75]. On the basis of this, the per hectare net profit and cash flow streams of Mango were computed throughout the lifespan of the investment using CBA under irrigated farming conditions. This is because drip irrigation is highly recommended for improved mango yields, schedule cropping, and management of climate change issues. Though mangoes are considered drought resistant, extended drought periods and more erratic rainfall patterns can create stress for the trees and diminish fruit quantity and quality. The use of drip irrigation will ensure that the mango trees have adequate moisture throughout the year and are not subject to excessive drought stress. Irrigation for mangoes increases yields, improves the overall quality of the fruit produced, and enhances the results of floral manipulation (used to produce fruit out of season).

**Economic Analysis**: Positive income flows for Mango Orchards start only in year 4 (Annex 3B). The expected cash flows were then used to compute the Extended Internal Rate of Return (XIRR) and the Net Present Value (NPV) for the initial investment of $1,500 per hectare, using three different discount rates (cost of borrowing from GCF - 0.75%, 12% Social Discount rate and private sector ESCOs Discount rate of 20%). The CBA shows that investing in Mango production in the programme is the most profitable of the agroforestry management practice for smallholder farmers, ACSs and FBAs. The Extended Internal Rate of Return (XIRR) is 81% with a Net Present Value (XNPV) of $261,714 per hectare (0.75% implicit for cost of capital). However, when a Social Discount Rate of 12% is used as in all AfDB investments, the XNPV drops to $77,270 and further down to $42,159 (at social discount of 12%) and with a payback period of 3.21 years. Of all the agroforestry management practices proposed under the programme, the analysis shows that investing in Mango is most profitable. The Kent or Keitt Mango form Senegal has the highest export value in the whole of the West African region. Recorded yields of 20 - 30 tons per ha have been reported for small farms in Senegal under irrigated farming conditions[76]. This to say that the per hectare profit may even be higher within this programme with the addition of irrigation and adoption of CRA at the farm level.

**Table D.6.7:** Economic Rate of Return (ERR). Mango

| | |
|---|---|
| XIRR (2XX0-20XX +5) | 85% |
| XNPV (2XX0-20XX +5) at Discount (0.75% implicit for cost of capital) | $261,714.60 |
| XNPV (2XX0-20XX +5) at Social Discount | $77,270.55 |
| XNPV (2XX0-20XX +5) at ESCO Discount | $42,159.05 |
| Payback Period (Years) | 3.20 |

***Cashew - Togo, Senegal and Guinea***

Key Assumptions.

- The plantation lifespan is 25 years
- Yield starts at year 3 and increases at a rate of 3% starting from year 4 to 10. but productivity declines after 25 years of production
- An average of 4.5kg of raw nuts is produced per Cashew tree and 1 ha has approximately 625 Cashew trees.
- Minimum government farm gate price per/kg ranges from 450 - 550 XAF in West Africa.
- Can be intercropped with cabbage, or okra from year 3
- Minimum expected yield per hectare is about 2,812.5 kg/ha
- An initial budget of $1,500 per ha is assumed also covering drip irrigation and CRA for vegetable farming.
- Variable costs for per ha production include: site preparations cost (land clearing for liming), digging of holes and fencing the farm); costs of fertilizer (Lime, gypsum and N-P-K); planting cost (seedlings -Grafted scion, cost of transplant, labour cost for planting, and replanting- 1/50th of a hectare each year for first three years); maintenance cost (fertilization (November & December, application of Pesticides/Fungucides, Herbicides, Weed Control (May

[68] FAO: FAOSTAT (1991 – 2019): Producer Prices. www.fao.org/faostat/en/#data/PP.
[69] https://www.rvo.nl/sites/default/files/2020/12/201204%20SVC%20Export%20Mango%20Farm.pdf.
[70] https://wire.farmradio.fm/farmer-stories/senegal-farmer-develops-method-for-growing-mango-trees-with-little-water/.
[71] (PDF) Mango-based orchards in Senegal: Diversity of design and management patterns (researchgate.net).
[72] https://www.researchgate.net/publication/274626027_Mango-based_orchards_in_Senegal_Diversity_of_design_and_management_patterns.
[73] https://www.togofirst.com/en/agriculture/1601-7101-mango-production-in-togo-latest-figures-show-improved-production.
[74] West African Mango Producers Smiling Again – CORAF.
[75] www.coraf.org/2017/10/16/coraf-takes-on-fruit-flies-to-save-millions-in-mango-losses/.
[76] https://www.rvo.nl/sites/default/files/2020/12/201204%20SVC%20Export%20Mango%20Farm.pdf.

to September), training (first 4 years) August – September, labour cost for maintenance, and initial investment in irrigation). In economic prices, the variable cost per hectare under irrigated farming conditions is **about** $2,550.05 before yield starts at year 4. (Annex 3B).
- Fixed and non-fixed costs include: interest payment on part of variable costs incurred (6%) and depreciation charges (5% on depreciable assets), which is about **$276** per/ha.

**Developing a Cashew cash flow plan: All** figures for the economic budgeting were calibrated based on the following source references[77], [78], [79]. Based on this, the per hectare net profit and cash flow streams for Cashew orchards were also computed for the lifespan of the investment using CBA (Annex 3B). Though the first two years recorded net losses, as soon as the Cashew orchard starts producing in year 3, net profit or income starts flowing in steadily. As observed in the table below, overall, investing in Cashew is highly profitable and a viable business in West Africa. The Extended Internal Rate of Return (XIRR) is 68% with a Net Present Value (XNPV) of $52,335 per hectare at 0.75% implicit for cost of capital. When a Social Discount value of 12% is considered, the XNPV falls to $14,885 and further down to $7,735 (at ESCO Discount Rate of 20%) and with a payback period of 3.84 years. Also note that these values can be increased if intercropped with more profitable seasonal CRA practices such as tomato farming and carrots (the choice is up to the farmers, however, advisory services will be provided to guide these SHFHs)

Table D.6.8: Economic Rate of Return (ERR), Cashew

| | |
|---|---|
| **XIRR (20XX-20XX+5)** | 68% |
| **XNPV (20XX-20XX+5) at Discount (0.75% implicit for cost of capital)** | $52,335.13 |
| **XNPV (20XX-20XX+5) at Social Discount Rate (12%)** | $14,885.57 |
| **XNPV (20XX-20XX+5) at ESCO Discount Rate (20%)** | $7,735.49 |
| **Payback Period (Years)** | 3.84 |

*Please explain how the best available technologies and practices have been considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.*

African Development Bank will bring on-board innovative solutions in Agric-business in a climate smart way. The facility will make use of the best practices of the Bank's operations in the agricultural sector, which has accumulated over the past 50 years on the African continent. These innovative approaches will not only be limited to:

- Country ownership that is aligned with the various Governments national plans and strategies on agriculture, climate change and energy – agriculture-food nexus.
- Leveraging the private sector investment by 'crowding in' more private sector operators.
- Developing management for results and ensuring enough value addition.
- Inclusivity sustainability and partnerships collaboration among the different stakeholders (both global and national) operating in the agriculture space in the countries where the interventions are taking place.

**Agroecological systems:** Agroecological systems approaches conserve biodiversity and natural resources such as soil organic matter, water, crop genetic diversity and natural enemies of pests. It equally improves economic stability with more diverse sources of income, spread of labour needs and production over time, and reduced vulnerability to commodity price swings among others.

**Mainstreaming gender**: Recognizing these differences, this project will seek to understand and respond to the different needs, concerns, and challenges that men and women face in investing in ecological agriculture and clean energy adoption. The program will apply techniques that will continuously empower women and improve their socio-economic status.

**Reduction in Climate change induced migration**: By reducing the production risks associated with climate change, the project will considerably reduce climate change induced migration that is very common in countries like Zambia and Togo.

[77]FAO: FAOSTAT (1991 – 2019): Producer Prices. www.fao.org/faostat/en/#data/PP.
[78]https://www.cbi.eu/sites/default/files/vca-cashew-west-africa_0.pdf.
[79]https://www.researchgate.net/publication/271743844_Cashew_from_seed_to_market_A_review.

**Private Sector in Waste Management:** The project will promote the involvement of the private sector in the management of waste. This approach will involve assisting the private sector to invest in businesses that are based on recycling waste and production of energy from waste materials in different forms.

## LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the* ***GCF's Integrated Results Management Framework*** *to which the project/programme contributes as a whole, including in respect of any co-financing.*

### E.1. Project/Programme Focus

*Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.*

☒ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

### E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

*This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.*

*Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).*

| Assessment Dimension | Current state (baseline) | | Potential target scenario (Description) | How the project/programme will contribute (Description) |
|---|---|---|---|---|
| | Description | Rating | | |
| **Scale** | The Staple Crops Processing Zone (SCPZ) programme is designed to promote sustainable agricultural value chains in three African countries including Senegal, Guinea and Togo. These countries which rely heavily on agriculture, lie within the major climate 'hotspots,' and experience the deleterious impacts of climate change. Although | <u>Medium</u> | Paradigm shift will involve the inclusive adoption of climate resilient agro processing practices and shifting from fossil fuels to renewable technologies to improve resilience of agrosystems, agricultural assets and beneficiaries while contributing to reducing GHG emissions from the processing and agroforestry practices. With further support from Climate Information and Early Warning Systems, beneficiaries will be better prepared to react proactively to climate hazards rather than facing the consequences in the aftermath. | *Describe key applicable outputs and or resulting outcomes relevant to increasing (scaling up) quantifiable results within and beyond the scope of the intervention.*<br><br>This programme will strengthen climate resilience and reduce GHG emissions within the agricultural value chain to help stimulate productivity and value addition, competitiveness, employment and incomes generation by facilitating access to technical assistance and concessional financial resources to communities within the three countries which rely heavily on agriculture |

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| | there are various initiatives to help build the resilience of smallholder farmers and rural communities to climate change in the selected countries, these activities are fragmented. However, achieving this is not easy and may not be entirely possible unless key technical, financial, policy and regulatory obstacles that lead to fragmented and ineffective interventions in key sectors like agriculture, land use and forestry, water and energy are carefully addressed. Unless these obstacles are overcomed, implementing programmes in the normal "business as usual" way in the context of climatic stress, are clearly inadequate and only add to the numerous challenges such as increasing emissions of GHG from agriculture, land use and forestry, food insecurity, malnutrition and heightened conflicts over resources use | | | The Intervention is projected to deliver:<br>• About 2.9 million direct beneficiaries and 10 million indirect beneficiaries of which 54% will be females.<br>• Over 79,000 ha of land made more resilient to climate change.<br>• A direct emission reduction of at least 16.19 million $tCO_2eq$ by the end of the project lifetime from an integrated approach that combines (i) solar PV for irrigation, lighting and processing, (ii) replacement of fossil-based and biomass-based energy systems with solar and biogas technologies and (iii) agroforestry practices.<br>• Contribute to achieving over 9% of conditional mitigation targets of Nationally Determined Contribution (NDCs) in the 3 countries.<br>• Over 15% increase in agricultural profit for growing horticulture practices especially in the dry season benefiting at least 79,000 smallholder farmers, of which 50% will be females.<br>• Over 20% increase in the yields of maize, soybean, rice, cassava, banana, sesame, |

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| | | | | horticulture, mango, cashew and coffee.<br>Given the prominent importance of agriculture in these three countries and in other countries in Africa, successful implementation of this programme and lessons learnt from same will act as a blueprint to scale up the programme within the beneficiary countries and neighbouring countries as well. |
| **Replicability** | Given growing population and increasing economic activity, many countries have approached the AfDB to set up SPCZ in other RMCs to help them improve and optimise their agro processing capabilities therefore increasing food security and improving economic prospects. Moreover with current geopolitical instability and increasing price of basic commodities on the global market, the demand for modernized and climate resilient infrastructure is bound to increase globally | Medium | If successfully implemented, farming communities in the beneficiary countries will have enhanced climate resilience while being able to sustainably power their operations. Moreover, the project will bring along social co-benefits including positive health impacts and gender sensitive development impacts. Therefore, the solution could be replicated across the city, to other regions in the country, and even internationally. | *Describe key applicable outputs and resulting outcomes that will be replicated to other sectors, markets, geographical regions, or countries.*<br><br>The programme will contribute through lessons learned which can then be adapted to other regions and countries where the farming and pastoral profile is different. The community driven approach as well as private sector engagement will also contribute in bettering the design and execution of similar projects in the future. |
| **Sustainability** | The governments of the three countries have committed to very ambitious NDCs whereby adaptation to climate particularly in the | Medium | The sustainability measures were already embedded in the design of the agricultural transformation centres and agro-industrial parks where shared facilities are provided to producers, processors, aggregators and distributors to operate in the same vicinity to | *Describe key applicable outputs and resulting outcomes that will be sustained beyond the project/programme period.*<br><br>The project will support the private sector through concessional loans and grants for the deployment of low carbon |

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| | agricultural sector is strongly emphasized as is the increased electrification of rural communities using renewable energy. This provides strong foundations for the ongoing management and development of greener agricultural infrastructure. | | reduce transaction costs and share business development services for increased productivity, diversification and competitiveness. The PPP arrangements for the operation and management of the facilities for user fees will ensure that there is adequate revenue generated to maintain the facilities including the solar (lighting and irrigation) and biogas assets financed with GCF financing. | technologies, and Climate information and Early Warning systems. Moreover, GCF intervention will promote an inclusive approach and will strengthen institutional structures to promote accountability and good governance |

**E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)**

*Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).*

| GCF Result Area | IRMF Indicator | Means of Verification (MoV) | Baseline | Target | | Assumptions / Note |
|---|---|---|---|---|---|---|
| | | | | Mid-term | Final[80] | |
| Total GHG emission reduction | Core 1: GHG emissions reduced, avoided or removed/sequestered | | 0 Mt $CO_2$ eq. | Reduction of 3,567,210 tons of CO2eq (3rd year of project after | Reduction of 8,493,358 tons of CO2eq (5th Year of project implementation) | Implementation targets for solar PV and biodigesters are met as per project timelines. |
| MRA1 Energy generation and access | Core 1: GHG emissions reduced, avoided or removed/sequestered | Reports by an independent specialist as part of the yearly M&E report to be produced during project implementation. This can be cross checked against reports by project developers or against NDC/ biennial communications | 0 Mt $CO_2$ eq. | Reduction of 3,567,210tons of CO2eq (3rd year of project after | Reduction of 8,493,358 tons of CO2eq (5th Year of project implementation) after entire 10 MW of biogas systems and 14.69 MW of solar systems are in place across the 4 countries) | 65% of Renewable Energy Capacity is installed and displacing Diesel and methane emissions from waste at year 3.<br><br>100% of Renewable Energy Capacity is installed and displacing Diesel and methane emissions from waste at Year 5.<br><br>16,140,949 tons of CO2eq avoided during project lifespan for all project activities |

[80] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

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| | Supplementary 1.4: Renewable energy generated | Report of Baseline survey on power generated annually for lighting and processing in the Agro-parks | 0MWh | 102,950 MWh of solar energy generation by mid-term. (40% of final target) | 257,375 MWh of solar energy generation by end of year 5. | Based on 12 hours of solar energy generated daily for food crops processing with an efficiency of solar PV systems of 0.8 |
| | | Report of survey on renewable Energy penetration in the areas | 0MW | 9.1MW installed by mi term<br><br>Togo: 4.44MW<br>Senegal: 2 MW<br>Guinea: 2.66MW | 14.69 MW installed capacity by end of year 5<br><br>Togo: 7.17MW<br>Senegal: 3.27 MW<br>Guinea: 4.25MW | ESCOs are willing to Design Commission Operate and Maintain (DCOM): 2.59 MW installed capacity for solar irrigation & 14.69MW for solar processing |
| | Supplementary 1.3: Installed renewable energy capacity | Report of Baseline survey on renewable Energy penetration in the areas | 0MW | 4.1 MW installed capacity by mid-term (40% of final target)<br><br>Togo: 1.73MW<br>Senegal: 0.92 MW<br>Guinea: 9.38MW | 10.24 MW installed capacity at end of year 5<br><br>Togo: 4.13MW<br>Senegal: 2.18 MW<br>Guinea: 3.94MW | ESCOs are willing to DCOM biogas Biodigester, and Biodigester are operated at 30% efficiency for 12 hours daily. |

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| MRA4 Forestry and land use | Core 1: GHG emissions reduced, avoided or removed/sequestered | Reports by an independent local carbon verifier | 0 Mt $CO_2$ eq. | 0 Mt sequestrated at year 3 $CO_2$ eq. | 5,743,266 tons of CO2eq sequestrated at year 5 | 5,743,266 during the duration of the accounting period |
| Total number of programme beneficiaries | Core 2: Direct and indirect beneficiaries reached | | | 3.5 million by year 3, of which:<br>Male= 1.6 million<br>Female =1.8 million | 6.72 million by year 5, of which:<br>Male = 4.8 million<br>Female = 5.68 million | Value chain actors, communities, and institutions etc., are willing to adopt climate resilient livelihood options & technological innovations, introduced by the programme. |
| ARA1 Most vulnerable people and communities | Core 2: Direct and indirect beneficiaries reached | At the Programme levels:<br><br>Socioeconomic baseline surveys[81]& Programme progress/ completion reports submitted by MOTI, APRODAT, MDIPMI& ADAZZ | 0 | 3.5 million by year 3, of which:<br>Male= 1.6 million<br>Female =1.8 million | 6.72 million by year 5, of which:<br>Male = 3.08 million<br>Female = 3.64 million | Value chain actors, communities, and institutions etc., are willing to adopt climate resilient livelihood options & technological innovations, introduced by the programme. |

[81]This has been budgeted as part of the baseline investment in all three countries (1 at programme inception, 1 during mid year and the last one at year 5)

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| | | At National levels:<br><br>Labour force participation surveys (LFPSs), & Demographic and health surveys (DHSs)[82].<br><br>Annual M&E reports (refer to Annex 11) will include recording of data associated with beneficiaries derived from records established by project developers and cross checked with national statistical records undertaken during project implementation | | | With the following breakdown:<br>Togo: 912,98 (direct beneficiaries) and 1,238,231 (indirect beneficiaries)<br><br>Senegal: 918,375 (direct beneficiaries) and 4,925,721 (indirect beneficiaries)<br><br>Guinea: 758,047 (direct beneficiaries) and 1,733,259 (indirect beneficiaries) | Training of beneficiaries will materialized in an increased uptake of climate resilient livelihood options.<br>Uptake of climate resilient livelihood options will benefit various segment of the society, including but not limited to, family members and the growth of MSMEs.<br><br>Programme funds are disbursed as planned. |
| | Supplementary 2.4: Beneficiaries (female/male) covered by new or improved early warning systems | Annual Performance Report<br><br>Daily, weekly and seasonal publication and distribution of agriculture-specific weather forecasts | 7,800 in Senegal<br>0 in Togo<br>0 in Guinea | 80,000[83] | 200,000[84] | Farmers are trained on how to use climate information. Farmers are trained on how to maintain simple rain gauge station<br><br>There is adequate medium for communicating and disseminating climate-base agro-advisory services |
| ARA2 Health, well-being, food and water security | Core 2: Direct and indirect beneficiaries reached | | 0 | A total of 239,853 beneficiaries (32,631 Direct and 207,221 | A total of 599,631 beneficiaries (Additional 48,947 Direct and 310,832 | Enabling environments are created for strengthening of safe water supply (e.g., farmers water user groups/committees, O&M training etc.)<br><br>Access to reliable and safe water supply benefits different segments of the society |

[82] Usual reported on a yearly basis. However, the field statistics used for preparing the yearly reports are carried out quite frequently in most countries (quarterly or mid-year).
[83] At least 20,000 persons in each of the four SCPZs
[84] At least 50,000 persons in each of the four SCPZs

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| | | | | Indirect beneficiaries)<br><br>Togo (Direct - 8,884 & Indirect - 34,537); Senegal (Direct - 13,256 & Indirect - 103,611); Guinea (Direct - 10,491 & Indirect - 69,074) | Indirect beneficiaries)<br><br>Togo (Direct - 13,326 & Indirect - 51,805); Senegal (Direct - 19,884 & Indirect - 155,416); Guinea (Direct - 15,737 & Indirect - 103,610) | especially beneficiaries household members. |
| | Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security | Methodology to be based on the definition for water sources developed by the WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation and Hygiene (JMP) and carried out in the following ways:<br><br>Socioeconomic baseline surveys administered to project beneficiaries & Programme progress/ completion reports submitted by , APRODAT, MDIPMI& ADAZZ<br><br>At National levels:<br><br>Regional level data collected by the Institute of *Statistics* and Economic and Demographic Studies (INSEED) -*Togo*; Agency of Statistics and Demography | 0<br><br>Information to be updated from socioeconomic baseline survey. | A total of 239,853 beneficiaries of which:<br>Male= 110,332<br>Female = 129,521 | A total of 599,631 beneficiaries by year 5, of which:<br>Male = 275,830<br>Female = 323,801 | Enabling environments are created for strengthening of safe water supply (e.g., farmers water user groups/committees, O&M training etc.)<br><br>Access to reliable and safe water supply benefits different segments of the society especially beneficiaries household members. |

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| | | of Senegal (ANSD), &Statistical Office of Guinea | | | | |

| **E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)** | | | | | |
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| *Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.* | | | | | |
| **Core Indicator** | **Baseline context (description)** | **Rating for current state (baseline)** | **Target scenario (description)** | **How the project will contribute** | **Coverage** |
| Choose an item. Core Indicator ( Degree to which ( investments contri to technology deployment, dissemination development c transfer and innova | Only a handful of farmers have access to small scale agricultural water management (AWM) infrastructure, solar energy for lighting/ processing and bio digestion kits. Also there is no tailored CIEWS in place to help make farming communities more climate resilient | medium | By end of project the following will be achieved:<br>- 31,179 ha of land will be covered by DIDS<br>- 2.59MW of solar PV capacity for pumping water will be installed<br>- 14.69 MW o Solar Energy installed for lighting and processing<br>- 10MW of RE will be generated from bigas<br>- Farming communities will have access to Climate Information and Early Warning Systems | Through blended finance (loans and grants) and with the support of third parties hired competitively under the project, the necessary technologies to enable climate resilience will be deployed. | Multi-countries |
| Choose an item. Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning | Communities in rural areas and depending nearly exclusively from agriculture and livestock for livelihoods suffer the brunt of climate change and do not have access to adapted information on climate resilient agriculture and agro | medium | .<br>Over 79,000 ha of land made more resilient to climate change. Over 15% increase in agricultural profit for growing horticulture practices especially in the dry season benefiting at | The project will support the collection of climate data and transmission of climate information at community-level. This will Improve climate-based agro-advisory services, including disease prevention. | Multi-countries |

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| processes, and use of good practices, methodologies and standards | forestry practices as well as effective climate information and early warning systems. | | least 79,000 smallholder farmers, of which 50% will be females. Over 20% increase in the yields of maize, soybean, rice, cassava, banana, sesame, horticulture, mango, cashew and coffee.<br>Increased climate resilience of 1,600,000 farmers through access to timely climate-informed agro-advisory services | Capacity building - adaptation and contingency & anticipatory plans will also be supported under the project | |

## E.5. Project/programme specific indicators (project outcomes and outputs)

*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "**Project/programme co-benefit indicators**" section in table below. Add rows as needed.*

*Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).*

| Project/programme results (outcomes/ outputs) | Project/programme specific Indicator | Means of Verification (MoV) | Baseline | Target | | Assumptions / Note |
|---|---|---|---|---|---|---|
| | | | | Mid-term | Final | |
| | | *Sources of information and methods used to collect and report data/information to measure progress against targets* | *The starting point or current value of the indicators before the implementation of the project* | *The estimated value of the indicator at the mid-point of the implementation* | *The estimated value of the indicator at the completion of the implementation* | *Externalities and factors outside project management's control that may impact on the Component*<br><br>*Data sources and methodologies applied for estimating baseline and targets* |

| Outcome 1 | | | | | | |
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| Output 1.1: Improved access to drip irrigation distribution systems (DIDS) powered by solar pumps to support climate resilient agricultural (CRA) practices | Hectares of land under DIDS | Agricultural baseline surveys, Annual agricultural statistics, Crops surveys, Project M&E reports, FAO Statistics. National reports on area of land contributing to climate resilient livelihood under SCPZ | 0 | 27,400 additional ha by mid-term (3 years implementation )<br><br>Togo: 10,800ha<br><br>Senegal: 8,360 ha<br><br>Guinea: 8,240ha | 39,179.3 additional ha at end of year 5<br><br>Togo: 15,428ha<br><br>Senegal: 11,941/ ha<br><br>Guinea: 11,810ha | Programme beneficiaries are willing to form farmers water users groups/committees that will operate and maintain on-farm surface water catchments. |
| | Number of males and females benefiting from the adoption of DIDS for diversified, climate resilient livelihood activities | | 0 | 25,653 females & 21,853 males | 42,756 females and 36,422 males | Small farmholders are willing to participate and adopt DIDS. |
| Output 1.2: Enhanced access to renewable energy technology for staple food crops processing | Generated Solar energy for lighting and processing of food crops (MWh) | Report of<br>Baseline survey on power generated annually for lighting and processing in the Agro-parks | 0MWh | 102,950 MWh of solar energy generation by mid-term. (40% of final target) | 257,375 MWh of solar energy generation by end of year 5. | Based on 12 hours of solar energy generated daily for food crops processing with an efficiency of solar PV systems of 0.8 |
| | Capacity (in MW) of installed Solar PV | Report of survey on renewable<br>Energy penetration in the areas | 0MW | 9.1MW installed by mi term<br><br>Togo: 4.44MW<br>Senegal: 2 MW<br>Guinea: 2.66MW | 14.69 MW installed capacity by end of year 5<br><br>Togo: 7.17MW<br>Senegal: 3.27 MW | ESCOs are willing to Design Commission Operate and Maintain (DCOM): 2.59 MW installed capacity for solar irrigation & 14.69MW for solar processing |

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| | | | | | Guinea: 4.25MW | |
| | Volume of biodigester installed | Report of Baseline survey on renewable Energy penetration in the areas | 0MW | 4.1 MW installed capacity by mid-term (40% of final target)<br><br>Togo: 1.73MW<br>Senegal: 0.92 MW<br>Guinea: 9.38MW | 10.24 MW installed capacity at end of year 5<br><br>Togo: 4.13MW<br>Senegal: 2.18 MW<br>Guinea: 3.94MW | ESCOs are willing to DCOM biogas Biodigester, and Biodigester are operated at 30% efficiency for 12 hours daily. |
| | Number of ACSs, WABEs, & FBAs integrating renewable energy in their agribusiness operations | Report of survey on renewable Energy penetration in the areas | 0 | 960 by mid-term<br><br>Togo: 320<br>Senegal: 480<br>Guinea: 160 | 2,400 by year 5<br><br>Togo: 800<br>Senegal: 1,200<br>Guinea: 400 | ACSs, WABEs, & FBAs are willing to adopt solar energy in agribusiness operations, including necessary O&M |
| **Outcome 2** | | | | | | |
| Output 1.3: Increased agricultural productivity through enhanced access to Agro-Industrial Parks (AIPs) support facilities and access infrastructure | Number of Km of new access roads constructed and made more resilient to extreme weather events | At the Programme levels: AfDB annual report & statistics, AfDB Economic Outlooks; progress/ completion reports submitted by MOTI, APRODAT, MDIPMI& ADAZZ; Socioeconomic baseline surveys and reports. At National levels M&E reports/annual performance reports of the relevant Ministries of | 0 | 92 km by year 3 | 230 km by year 5 | |
| | Number of km of existing access roads made more resilient to extreme climate events | | 0 | 188 | 472 | |

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| | Number of functional boreholes with water yielding capacity of 0.6 l/s to over 1.05 /s constructed & made more climate resilient | Agriculture, Transports, Works, Communications and Water Resources.<br><br>Yearly socioeconomic baseline surveys and reports; Dedicated labor force participation surveys and annual reports, Yearly Demographic and Health surveys (DHS); Central Banks annual statistics and reports, Annual national employment promotion surveys and reports | 0 | 5 | 8 | |
| | Number of old boreholes rehabilitated & made more climate resilient | | 0 | 18 | 30 | |
| | Number of mini dams or small reservoirs with capacity of >15 M.m3, constructed for irrigational purposes | | 0 | 2 | 4 | |
| | Number of ATCs constructed to improve climate resilient livelihood options | | 0 | 30 | 40 | |
| | Number of relevant institutions trained on value chain development (VCD) | | 0 | 200 | 250 | |

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| | Number of program beneficiaries trained and empowered on agricultural value chain development (AVCD), disaggregated by gender | | 0 | 747,209 females and 636,511 males by year 3 | 1,245,348 females and 1,060,852 males by year 5 | |
| | Number of direct beneficiaries of water supply plant, new and rehabilitation boreholes, and waste management plants, disaggregated by gender. | | 0 | 747,209 females and 636,511 males by year 3 | 1,245,348 females and 1,060,852 males by year 5 | |
| | Volume of waste (mega liters per day - MLD) to be treated daily at the AIPs facilities for resource recovery reuse (RRR) | | 0 | 3.21 MLD | 3.21 MLD | |
| | Number of new waste treatment plants constructed with waste treatment capacity of 3.21 MLD | | 0 | 3 | 4 | |
| | Volume of water in Mm3 to be harnessed for irrigational purposes from new mini dams and reservoirs | | 0 | >10Mm3 in year 3 | >15Mm3 by year 5 | |

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| | Daily water availability (m3/day) for domestic and agricultural uses from new functional boreholes and rehabilitated old ones. | | 0 | 3,000-3,500 m3/day by year3 | 4,000-5,000 m3/day by year 5 | |
| | Number of new SHFHs in project areas with additional incomes derived from dry season irrigation farming | | 0 | 78,436 by year 3 | 130,726 by year 5 | |
| | Employment generated in the farming sector (casual labourers, handlers, produce buyers etc.,) | | 0 | 392,179 by year 3 of which: 211,777 = females, and 180,402 = males | 653,631 by year 5 of which: 352,961 = females, and 300,670 = males | |
| | Number of new construction jobs generated by the AIPs facilities | | 0 | 134,400 by year 3 | 224,000 by year 5 | |
| | Value of new investments (US$) to be catalysed by private sector companies operating in the AIPs | | 0 | Togo > 60 million; Senegal >160 million, and Guinea > 30 million by year 3 | Togo > 100 million; Senegal >200 million, and Guinea > 50 million by year 5 | |

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| | Increased yields of major staple food crops in t/ha. (i) irrigated rice; (ii) low land rice; (iii) rain-fed rice; (iv) corn; (v) soybeans; (vi) sesame, (vii) cashew. | | 2t/ha; (ii) 1.5 t/ha; (iii) 1.0 t/ha; (iv) 1.2 t/ha; (v) 0.5 t/ha; (vi) 1.0 t/ha, (vii) 0.5 t/ha. | 8 t/ha; (ii) 3 t/ha; (iii) 1.5 t/ha; (iv) 2.0 t/ha; (v) 0.7 t/ha; (vi) 0.7 t/ha, (vii) 0.8 t/ha. | 10 t/ha; (ii) 4 t/ha; (iii) 2 t/ha; (iv) 3 t/ha; (v) 1 t/ha; (vi) 1.0 t/ha, (vii) 1 t/ha. | |
| | Increased in average agricultural production (per 1000 T ) of key crops such as (I) mango; (ii) cashew; (iii) maize. | | (i) 43; (ii) 15.3; (iii) 45 | (i) 60; (ii) 18; (iii) 55 | (i) 72; (ii) 24.8; (iii) 63 | |
| **Outcome 3** | | | | | | |
| Output 2.1: Adoption of CRA and agro-forestry practices among smallholder farmers | *Number of climate resilient seed varieties released* | Socioeconomic baseline surveys, annual agricultural statistics, programme M&E reports, regional level agricultural data . produced by various regional offices of Statistics | 0 | 9 | 18 | SHFHs are willing to adopt new seed varieties introduced by the programme and are committed to learning about climate resilient practices.<br><br>SHFHs are committed and willing to be trained in climate resilient livelihood options |
| | *Number of SHFHs trained on climate resilient agricultural practices* | | 0 | 52,291 SHFHs trained by mid-term of which, 54% are females | 79,178 SHFHs trained by mid-term of which, 54% are females | |
| | *Number of new CRA techniques and technologies introduced to improve staple food crops yields in the program areas* | | 0 | 5 | 5 | |
| Output 2.2: Awareness raising, capacity building, community and institutional strengthening on agricultural value chains management | *2.2.1a Expected number of people trained with increased awareness on agricultural value chain development approach* | *Programme M&E Reports, & Socioeconomic baseline surveys* | 0 | 810,575 of which, 54% are females | 1,350,958 of which, 54% are females | Target beneficiaries are committed and willing to participate in value chain development training. |
| | *2.2.1b . Number of anchor training* | | 0 | 200 | 250 | |

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| | *institutions equipped to implement the agricultural value chains development approach* | | | | | |
| | *2.2.2a Number of agrometeorological and rain gauge stations installed in each SCPZs* | Purchase order and Delivery receipt for equipment<br><br>Reports from National Met services in each country | 4 in Senegal.<br>2 in Togo<br>2 in Guinea | 20 Agrometeorological and 100 rainfall stations[85] | 40 Agrometeorological and 200 rainfall stations [86] | Suitable sites are identified following the WMO guidance[87] |
| | *2.2.2b Number of farmers benefiting from Climate-smart agro-advisory services and digital technology* | Annual Performance Report<br><br>Daily, weekly and seasonal publication and distribution of agriculture-specific weather forecasts | 7,800 in Senegal<br>0 in Togo<br>0 in Guinea | 80,000 farmers (20,000 farmers per SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | 200,000 farmers (50,000 farmers per SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | Farmers are trained on how to use climate information<br><br>There is adequate medium for communicating and disseminating climate-base agro-advisory services |
| | *2.2.2.c Number of people trained on climate smart agro-advisory services and digital technology* | Training reports and user manuals | 7,800 in Senegal<br>0 in Togo<br>0 in Guinea | 640,000 people (160,000 farmers per SCPZ; South Agropole in Senegal, Boke and Kankan in | 1,600,000 people (400,000 farmers per SCPZ; South Agropole in Senegal, Boke | Farmers subscribe and are willing and ready to uptake climate-based advisory services in their agricultural activities |

[85] At mid-term, a total of 20 agromet and 100 rainfall stations in the programme countries (i.e., 5 agrometeorological and 25 rainfall stations added for each region per country Boke, Kankan in Guinea, Kara in Togo and Southern Agropole in Senegal)

[86] At final-term, a total of 40 agromet and 200 rainfall stations in the programme countries (i.e., additional 5 agrometeorological and 25 rainfall stations added for each region per country Boke, Kankan in Guinea, Kara in Togo and Southern Agropole in Senegal)

[87] World Meteorological Organization (WMO). 2010. Guide to Agricultural Meteorological Practices. WMO-No. 134. WMO, Geneva, Switzerland.

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|---|---|---|---|---|---|---|
| | | | | Guinea, and Kara in Togo) | and Kankan in Guinea, and Kara in Togo) | |
| | *2.2.2d Number of people made aware of the importance of observing networks* | Reports on awareness-raising campaign | 0 | 4000 people (1000 people in each of the four SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | 8000 people (2000 people in each of the four SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | Quality and coverage of the awareness programme |
| | *2.2.2e Number of people trained on how to use and maintain observing networks* | Training reports and user manuals | 0 | 480 trainees (120 trainees in each of the four SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | 960 trainees (240 trainees in each of the four SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | Number of people available and enrolled for the training |
| | *2.2.2f Number of people trained on recalibration of meteorological sensors* | Training reports and user manuals | 0 | 640 trainees (160 trainees in each of the four SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | 1280 trainees (320 trainees in each of the four SCPZs; South Agropole in Senegal, Boke and Kankan in Guinea, and Kara in Togo) | Number of people available and enrolled for the training |
| **Project/programme co-benefit indicators** | | | | | | |
| Co-benefit 1 | *Employment opportunities created through the programme* | Annual Performance Report and national statistic records | 0 | 158.360 of which, 54% are females<br><br>Togo: 50,860 | 395,890 of which, 54% are females<br><br>Togo: 127,140 | For each beneficiary adopting CRA or AF techniques or accessing low carbon technologies, |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | Senegal: 63,880<br>Guinea: 43,620 | Senegal: 159,700<br>Guinea: 109,050 | 5 jobs are created (79,178 x5)<br><br>Farm holders are willing to adopt CRA and AF practices and adopt low carbon solutions |
| Co-benefit 2 | | | | | | |
| Co-benefit 3 | | | | | | |

**E.6. Project/programme activities and deliverables**

*All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.*

*Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.*

| Activities | Description | Sub-activities | Deliverables |
|---|---|---|---|
| Activity 1.1.1: Support access to small-scale agricultural water management (AWM) infrastructure | The activity is specifically designed to facilitate smallholder farmers' access to drip irrigation technology powered by solar pump. This will support horticulture, market gardening of fruits and vegetables including other SCPZ staple crops in ha of irrigable land. This will directly benefit more than 1.4million small holder farmers of which, 50% would be women. | • Issuance of International Competitive Bidding (ICB), short listing, and selection of firm for the training, supply and installation of DIDS in three pilot countries.<br>• Mapping exercise and selection of DIDS sub-project beneficiaries including the identification of 31,179.27 ha of land in three pilot SCPZs countries.<br>• Grant work plan for DIDS sub-project beneficiaries including management committees determined.<br>• Capacity building and training of sub-project beneficiaries in the use of DIDS in three pilot countries.<br><br>Distribution and installation of DIDS to selected sub-project beneficiaries in three pilot countries. | Each region (Agropole du Sud, Kankan, Boke, Kara) covered under the programme will have a dedicated report for the deliverables listed below:<br><br>• Sub-project reports highlighting major gaps and opportunities for investing in drip irrigation technology in SCPZs pilots<br>• Inception reports on detailed stakeholders communication strategies<br>• Statistics on labour, energy and water use efficiency<br>• Statistics on improvement in SCPZ crop yields<br>• Well-documented country's statistics of DIDS<br>• Additional Ha of land under irrigation<br>• Reports on sub-project beneficiaries<br>• Report on total number of men and women trained on drip irrigation technology.<br>• Sub-project training manual on drip irrigation technology. |

| | | | |
|---|---|---|---|
| Activity 1.2.1: Improving resilience of agricultural cooperatives, FBAs value chains infrastructure for food processing | The activity involves the installation of 35.5MW of solar energy to support agricultural cooperative societies, women-led agric-business enterprises (WABEs), and FBAs to dry, process, and package staple food crops using clean energy. | • Issuance of ICB, short listing, and selection of Energy Service Companies (ESCOs) for design, construction, operation and maintenance supply of 14.69 MW of solar energy<br>• Mapping exercise and selection of DIDS sub-project beneficiaries and including the identification of 31,179.27 ha of land<br>• Grant work plan for DIDS sub-project beneficiaries including management committees determined<br>• Capacity building and training of sub-project beneficiaries in the use of DIDS<br><br>Distribution and Installation of Solar PV equipment to sub-project beneficiaries | Each region (Agropole du Sud, Kankan, Boke, Kara) covered under the programme will have a dedicated report for the deliverables listed below:<br><br>• Sub-project report highlighting major gaps and opportunities for investing in low-carbon energy technologies such as Solar PV in SCPZs pilots<br>• Inception reports on detailed stakeholders communication strategies<br>• Report on project beneficiaries<br>• Catalogue of existing agricultural cooperative societies, women-led MSMEs, FBAs in the pilot countries.<br>• Detailed report of sub-project feasibility studies highlighting major gaps and opportunities for improving the value chains infrastructure of cooperative societies, agric business ventures, women-led MSMEs and FBAs<br>• Training report on total number of men and women trained in post-harvest losses and handling<br>• Sub-project training manual on strengthening value chain infrastructure for agriculture cooperative societies, agric business ventures, women-led MSMEs and FBAs on post-harvest losses and handling.<br><br>Statistics on new installed capacity of renewable energy. |
| Activity 1.2.2: Deploying low-carbon energy technologies for enhanced resilience | The focus of this activity is to specifically reduce GHG emissions in the agricultural value chains in the three SCPZs pilot countries. This is through the use of low carbon energy sources. This will involve the installation of ~10.24 MW of renewable energy from biogas generation or about 24,579 $m^3$ of biogas digester to treat livestock manure and produce biogas for heating or power generation. The energy provision will support the processing of produces, powering of cold storage facilities where applicable to ensure that farmers can offer their produce with the required market standards. Energy will also provide them with | • Issuance of ICB, short listing, and selection of firm for the construction and installation of ~10.24MW of clean energy (Biogas digesters - 24,579 $m^3$).<br>• Selection of eligible beneficiaries of bio-digesters (ESCOs).<br>• Financial arrangements with ESCOS for Biogas grants and loans including work plan, as determined by management committees.<br>• Capacity building and training in Biogas digesters, Bio-slurry, safety standards etc. | Each region (Agropole du Sud, Kankan, Boke, Kara) covered under the programme will have a dedicated report for the deliverables listed below:<br><br>• Sub-project report highlighting major gaps and opportunities for investing in low-carbon energy technologies such as biogas technology in SCPZs pilot countries. |

| | | | |
|---|---|---|---|
| | opportunities that will help them diversify their sources of income. | Construction of bio-digesters by ESCOS and first year's operation and maintenance (O&M) by ESCOs | • Mapping exercise report on direct sub-project beneficiaries including gaps and opportunities for biogas investment in pilot countries.<br>• Inception reports on detailed stakeholders communication strategies<br>• Training report on total number of men and women trained in low-carbon energy technologies.<br><br>Sub-project training manual on the adoption of low-carbon energy technologies for replication in other potential SCPZ countries. |
| Activity 1.3.1: Development of SCPZs Access Infrastructure & Support for AIP Management Governance | This activity is designed to offer adequate infrastructure, logistics and specialized facilities and services such as access roads, electricity, water, cold chain facilities, warehouse for storage, laboratory and certification services, business services, ICT, waste treatment among others, required for agro-industrial activities in the programme areas. It equally involves the establishment of ATCs, which are strategically located within the production areas to serve as aggregation points to accumulate products for value chain actors for processing and marketing. | • Construction of administrative building, installation fibre optics and market platforms (hubs), financial centres.<br>• Construction of water supply plant for over 15,000 cubic meters/day; waste management plants with capacity of 3.21 MLD.<br>• Construction of 30 new boreholes and rehabilitation of 8 old boreholes, 3 mini-dams and irrigated areas spanning over 15,000 ha.<br>• Provision of 243 kV transmission lines; 232 km new access roads that are resilient to extreme climate events, and rehabilitation of 472 km of roads to be climate-resilient.<br>• Construction of several numbered Agricultural Transformation Centers.<br>• Provision of cold stores and warehouses for storage and handling of agricultural produces from farming communities. | • Well-equipped administrative buildings, ICT facilities, market platforms and financial services<br>• Completed water supply plants with capacity of over 15,000 cubic meters/day to support processing and non-processing areas in AIPs.<br>• Waste management treatment plants with capacity of 3.21 MLD.<br>• 30 new climate resilient boreholes<br>• 8 rehabilitated boreholes<br>• 234kv transmission lines around AIPs.<br>• 232 km of new access roads constructed and 472km rehabilitated.<br>• At least 19 ACTs constructed with cool store facilities and warehouses. |
| Activity 2.1.1: Promote CRA among smallholder farmers | Specifically designed to promote the use of selected CRA practices in pilot countries, to strengthen climate resilience and reduce GHG emissions along the agricultural value chains. This is with the view to stimulate productivity and value addition, create competitiveness and employment while simultaneously generating incomes in three selected African countries | • Mapping exercise and selection of sub-project beneficiaries of AfDB-administered loans and grants to engage in CRA practices among smallholders' farm households, ACSs, women-led MSMEs and FBAs.<br>• Work plan for sub-project grants and loans to promote CRA practices including CRA management committee determined.<br>• Capacity building and training of sub-project beneficiaries in CRA practices.<br><br>Implementation of CRA activities | Each region (Agropole du Sud, Kankan, Boke, Kara) covered under the programme will have a dedicated report for the deliverables listed below:<br><br>• Sub-project report highlighting major gaps and opportunities for investing in CRA and agro-forestry practices in SCPZs pilots<br>• Inception reports on detailed stakeholders communication strategies<br>• Report on project beneficiaries<br>• Additional Ha of land under CRA practices. |

| | | | |
|---|---|---|---|
| | | | • Training report on total number of men and women trained in CRA practices in the target countries.<br>• Sub-project training manual on CRA practices for replication in other potential SCPZs countries.<br><br>Report on additional number of farmers in pilot countries engaged in CRA practices. |
| Activity 2.1.2: Promote agro-forestry practices among smallholder farmers | Specifically designed to promote the use of agroforestry management practices to create sustainable income sources for farmers in the pilot countries. Additionally, the introduction of this climate resilient management practice will strengthen climate resilience and reduce GHG emissions through carbon sequestration from 40,000 ha of woodlot, cashew nuts, mango and coffee orchards. | • Mapping exercise and selection of sub-project beneficiaries of AfDB-administered loans and grants to engage in agro-forestry practices among smallholders' farm households, ACSs, women-led MSMEs and FBAs.<br>• Work plan for sub-project grants and loans to promote agro-forestry practices and agro-forestry management committees determined.<br>• Capacity building and training of sub-project beneficiaries in CRA and agro-forestry practices.<br><br>Implementation of CRA and agro-forestry activities | Each region (Agropole du Sud, Kankan, Boke, Kara) covered under the programme will have a dedicated report for the deliverables listed below:<br><br>• Sub-project report highlighting major gaps and opportunities for investing in agro-forestry practices in SCPZs pilots.<br>• Inception reports on detailed stakeholders communication strategies<br>• Report on project beneficiaries<br>• Additional Ha of land under agro-forestry practices.<br>• Training report on total number of men and women trained in agroforestry practices in the target countries.<br>• Sub-project training manual on best practices in agroforestry management practices for replication in other potential SCPZs countries.<br><br>Report on additional number of farmers in pilot countries engaged in agroforestry practices. |
| Activity 2.2.1: Capacity building of value chains agents/communities'& development of agricultural processing support infrastructure | This activity is targeted at awareness raising and capacity strengthening of value chain actors, communities and institutions on agricultural value chain development and climate resilient livelihood options. | • Capacity building and training of relevant staff of the Project Executing Agencies (PEAs), the Project Implementing Units (PIUs), and the Agro-industrial Parks (AIPs).<br>• Training and capacity building of value chain actors in the use of 'Good Agricultural Practices' in product quality, standardization and conformity requirements, as well as developing linkages and responding to commodity demand in terms of quality and quantity.<br>• Training of young potential entrepreneurs including women to seize business opportunities along the value chains. | Each region (Agropole du Sud, Kankan, Boke, Kara) covered under the programme will have a dedicated report for the deliverables listed below:<br><br>• Report on total number of men and women trained on agricultural value chain development and climate resilient livelihood options.<br>• Socioeconomic baseline survey on number of value chain actors, community members and institutions trained and exposed to |

| | | | |
|---|---|---|---|
| | | • Massive sensitization and training of private sector investors& MDAs on governance systems reforms such as land reforms and export promotion initiatives. | agricultural value chain development and climate resilient livelihood options.<br>• Production of policy briefs on SCPZ management governance for circulation among policy makers.<br>• Report on stakeholders' engagement plans for circulation.<br>• Production of training manual on project performance monitoring, reporting and evaluation (M&E). |
| *Activity 2.2.2: Strengthening climate information services and early warning systems in SCPZs* | This activity will help to enhance climate information services by promoting the generation and access to seasonal and sub-seasonal weather forecast and agricultural relevant weather products in the SCPZs.<br>This activity will focus on awareness and training on how to use and maintain the observing networks. The activity will conduct stakeholders' engagement meeting with prospective farmers, aggregators, Famer Unions (FUs), Famer Based Organizations (FBOs), and other interested bodies with the focus on creating awareness on the importance of the installed agrometeorological and gauge stations and the need to be responsible for the security of the networks. Farmers and other interested stakeholders will take ownership of the self-recording rain stations. | • Collection of climate data and transmission of climate information at community-level.<br>• Improve climate-based agro-advisory services, such as, planting time, good farm management practices, irrigation schedules, use of farm manure and choice of inorganic fertilizers, suitable crop types and varieties to be planted, weeding regimes, the available seed suppliers, crop pests and diseases prevention and control measures.<br>• Capacity building on preparatory mechanism including analysis of vulnerability, gathering of data, decisions on contingency plan and/or adaptation measures, analysis of gender and socioeconomic inequalities; seasonal information prior to major rainfall season including information on irrigation requirements during growing season<br>• Improve dissemination technique<br>• Capacity building - adaptation and contingency & anticipatory plans<br>• Bi-annual exercise on calibration will be organized to re-calibrate the weather sensors to ensure measurement accuracy. | Each region (Agropole du Sud, Kankan, Boke, Kara) covered under the programme will have a dedicated report for the deliverables listed below:<br>• Report on the generation of, access to and use of climate-based agro-advisory services<br>• Number of people trained on preparatory mechanism<br>• Coverage of climate information services |

## E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

**Project Monitoring, Reporting and Evaluation**

**Technical and Financial Monitoring**

Each SCPZ project will have a Project Monitoring and Evaluation Officer (PMEO) appointed from within the Project Implementing Unit (PIU) and charged with the responsibility of collecting, analysing and compiling information on physical achievements and financial execution. For this purpose, he/she will have a roadmap with quantitative indicators on Project progress, relating to implementation of the sub-project components. Such monitoring will make it possible to obtain, every six months, the following information for each activity: physical objective, achievement level, expected costs, actual costs, differences and explanations for possible deviations. This information will be used to prepare the half-yearly project progress reports. Moreover, in addition to the **Project Steering and Monitoring Committee (PSMC)** review work, AfDB will carry out two supervision missions per year, to which should be added a mid-term review mission.

**Project Reporting**

There will be a two-line reporting procedure. First, project feedback to AfDB from the Project Management Unit and PSMC, and the AfDB will conduct a bi-annual supervision. Secondly, reporting of AfDB to GCF will comply with the relevant GCF policies (as specified under the AMA) in reporting and evaluation arrangements for this framework. The AfDB will provide the annual performance report (APR) to the GCF during the five-year implementation period. In addition, following the arrangement under the AMA and the FAA, inception report, mid-term and final evaluation reports, and financial information reports (semi-annually throughout the life of the loan) will be submitted.

**Project Evaluation**

As per all AfDB-GCF projects, the evaluation arrangements for this SCPZs programme will comply with the related AfDB and GCF policies. Both the independent mid-term and final evaluation will be carried out by the AfDB's independent evaluation unit (IDEV). The work of the AfDB's independent evaluation work is guided by internationally accepted principles for the evaluation of development assistance, in particular, the Organization for Economic Cooperation, and Development Assistance Committee (OECD DAC) evaluation guiding principles, and the good practice standards issued by the Multilateral Development Banks' Evaluation Cooperation Group (ECG). The AE will also comply with the GCF evaluation policy during the execution of the M&E activities

**Audit**

Once a year, competent and independent firms of external auditors will verify the reliability of the consolidated financial statements prepared by the PIU in each country, and will assess the operation of the internal control system of the entire project. Such firms will be recruited in accordance with the terms of reference and bidding procedures recommended by AfDB. Audit costs will be financed by the programme in each country. The audit reports must be forwarded to the AfDB each year, no later than six months after the closure of the audited period. Audits will be performed in accordance with the international best practice in audit standards. AfDB's audit policies and comprehensive terms of reference which will be communicated to the projects will apply. Consistent with clause 17.02 of the AMA, an audited annual Financial Statement will be submitted to the GCF six months after the end of AfDB Fiscal Year.

## F. RISK ASSESSMENT AND MANAGEMENT

### F.1. Risk factors and mitigations measures (max. 3 pages)

*Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.*

*For probability: High has significant probability, Medium has moderate probability, Low has negligible probability*

*For impact: High has significant impact, Medium has moderate impact, Low has negligible impact*

*Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices*

**Selected Risk Factor 1**

| Category | Probability | Impact |
|---|---|---|
| Technical and operational | Medium | Medium |

**Description**

*Please describe the risk to the best of your knowledge at this point in time.*

This risk occurs when agriculture is affected by many uncontrollable events that are often related to the weather, biology and environment including excessive or insufficient rainfall, extreme temperatures, insects and diseases, contamination by poor sanitation/illness or through production and processing. This could be also due to obsolescence issues whereby machinery, for example, becomes unserviceable. Key factors for this risk include natural conditions, biological and environmental hazards, technological level, natural disaster, demand, and policy decisions.

**Mitigation Measure(s)**

*Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?*

The mitigation measures put in place through the program include:

- Introduction and capacity building of small-hold farmers in soil and water conservation techniques and provision of small irrigation and water storage facilities on farms;
- Provision of access to improved seed varieties, pest control and surveillance services;
- Supporting private sector in establishing farm input supply facilities coupled with capacity building in farming methods that use minimal external inputs;
- Capacity building in agroecological systems that promote crop diversification and therefore reducing risks to crop diseases and weather shocks.
- Capacity building in biogas technology and establishing construction guidelines and enforcement processes.
- Provision of initial capital for biogas digesters' construction for pilot users.
- Social security mechanisms such as weather-based crop insurance among others

On consideration of the mitigation measures by the program, the residual risk is considered to be low.

**Selected Risk Factor 2**

| Category | Probability | Impact |
|---|---|---|
| Credit | Medium | Medium |

**Description**

*Please describe the risk to the best of your knowledge at this point in time.*

This risk occurs when the borrower fails to make payments as agreed. Agricultural production is characterized by seasonality, which may influence the specific circumstances of the settlements and the cash flow distribution in a certain period. This risk is influenced by the type of legal transaction, partner's willingness to settle the debt; and partner's financial status factors among others.

**Mitigation Measure(s)**

*Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?*

The risk is mitigated by

- Organizing farmers in groups  for purchasing farm supplies and selling of their produce like cooperative societies etc;
- Establishing clear credit terms and prog the establishment of community-based credit facilities;
- Training farmers and business holders on the principles of business and financial planning and management.
- Ensuring diversified source of incomes for the participants;
- Social protection programs such as social insurance, weather-based insurance, cash and in-kind transfers, asset diversification will contribute to the reduction of this risk.

On considering the mitigation measures the residual risk is considered low.

| **Selected Risk Factor 3** | | |
|---|---|---|
| **Category** | **Probability** | **Impact** |
| Governance | Medium | Low |

**Description**

*Please describe the risk to the best of your knowledge at this point in time.*

This risk may result from changes in policies and regulations that affect agriculture like restrictions in disposal of animal manure, restrictions in conservation practices or land use, changes in income tax policy, credit policy among others.  It could also be due to uncertainty associated with socio-political instability within a country or in neighbouring countries, disruption of trade due to disputes with other countries, among others. Limited coordination among agencies and stakeholders can lead to inefficiencies in the implementation and impact of the project interventions This risk is influenced by environmental regulations, political events, business regulations, environmental protection, and food safety factors among others.

**Mitigation Measure(s)**

*Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?*

The two main mitigation options to be implemented are: (a) “early engagement” (i.e. provision of information and data to disseminate expected outcomes); (b) Strong political buy-in to ensure policy continuity even in the event of policy changes; (c)“awareness creation” through well-orchestrated events to disseminate the project and its achievements.

| **Selected Risk Factor 4** | | |
|---|---|---|
| **Category** | **Probability** | **Impact** |
| Other | Medium | Medium |

**Description**

*Please describe the risk to the best of your knowledge at this point in time.*

It is possible that the programme will be affected by the COVID-19 context, which could cause delays in the start-up and generate operational risks during implementation if there is a new wave COVID-19 in the target countries in 2023 and beyond. AfDB has prepared a comprehensive COVID-19 risks management approach to ensure that project implementation is conducted in accordance with the AE’s and the country’s protocol.

**Mitigation Measure(s)**

*Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?*

AfDB has prepared a comprehensive COVID-19 risks management approach to ensure that project implementation is conducted in accordance with the AE’s and the country’s protocol. In order to ensure that the covid impact is minimised, necessary adaptive measures will be taken (e.g. remote supervision, use of digital tools, social distancing when required etc.).

| **Selected Risk Factor 5** | | |
|---|---|---|
| **Category** | **Probability** | **Impact** |
| Other | Medium | Medium |

| Description |
| --- |
| *Please describe the risk to the best of your knowledge at this point in time.*<br>Political conflict and insecurity strongly affect some countries in this proposal like in Guinea. Guinea is currently experiencing political unreast. The implementation of the project activities in these countries can therefore be affected. |
| **Mitigation Measure(s)** |
| *Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?*<br>Guinea is the only country facing conflicts. The regions selected for the programme in Guinea are secured. However, the government of the country have pledged their support to ensure that the objectives of the SCPZ are achieved in the country (refer to the field visit report in Chapter 6 of the Feasibility Studies). Both governments acknowledge the need for climate resilient development of the farming and pastoral communities in the respective countries and as major efforts are being deployed to minimise the impact of the political instability in both countries, the governments remain keen on leveraging the support for the development of the targeted zones. |

## POLICIES AND STANDARDS

### G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

The project's approach will not be only to avoid harming the local population, but also to ensure that the indigenous population can benefit from the project. In all the three countries, the project will apply inclusivity criteria and a do no harm approach in line with the guiding principles of the Bank's strategy to address fragility and build resilience (2022-2026; please see Annex 25 on the Country Fragility notes). A social inclusion specialist will be recruited for this purpose. All projects will benefit from the implementation and deployment of social intermediation plans to mitigate the occurrence of conflicts.

**Togo**

Strategic Environmental and Social Assessment (SESA) provides for strategic environmental and social measures to create a framework for identifying, managing and monitoring project impacts. These measures concern: (i) strengthening the political, institutional, environmental and social framework (implementing texts dealing with the SESA); (ii) sound knowledge of the area (baseline situation); (iii) management of natural resources; (iv) surveillance, monitoring and evaluation; and (v) training of actors involved in environmental and social management and education of the people. The implementation of these measures will be supported by the APRODAT and the relevant State entities. Implementation of the ESMMP will be under the responsibility of the APRODAT, which will recruit an Environmental Expert and a Social Development Expert. The implementation of the ESMMP will be monitored by various technical services under the coordination of the ANGE. The total cost of the environmental and social strategic measures is **CFAF 486,000,000,** which has already been taken into account.

Resettlement action plans have been prepared for the Agro-park and rural infrastructure. Based on the estimate of land area that will be affected and the value of the land in the area, a provision of **CFAF 800 million** has been set aside to compensate affected persons. A resettlement master plan will also be prepared as part of the project. Although it is not a tool proposed by the Bank's ISS, the resettlement policy framework (RPF) is required as a precaution, to specify the procedure by which land that is not currently affected by the project will be developed subsequently.

**Senegal**

In accordance with the requirements of the Bank's Integrated Safeguards System (ISS), the project was classified as Category 1 due to its nature and potential impacts related to the proximity of sites with ecosystems, environmentally and socially sensitive natural resources such as classified forests, rivers, and settlements and considering also the fact that the project may trigger the 5 Operational Safeguards. Indeed, the project proposes to set up a central module and 3 regional modules with infrastructure for processing agricultural products and departmental platforms with collection and packaging infrastructure. The sites of the regions of Ziguinchor, Sedhiou and Kolda are identified for the implementation of these modules. In this regard, a Social and Strategic Environmental Assessment (EESS) was prepared with the support of the Bank, and the final report approved by the Department of Environment and Classified Establishments (DEEC). The summary and the EESS were published on the ***ADB website on 01 August 2019***, 120 days before the project was submitted to the Council. However, because of the categorization of the project, an Environmental and Social Impact Assessment (ESIA) with Environmental and Social Management Plans (ESMP) are required. These documents have been updated and posted on November 9th,2020. The project is likely to cause physical and economic displacement, the development of a Resettlement Action Plan (RAP) has been completed and available on AfDB's website via https://www.afdb.org/en/documents/senegal-projet-dimplantation-dune-zone-de-transformation-agroindustrielle-sud-pzta-sud-ou-agropole-sud-p-sn-aag-004-rapport-ees.

**Guinea**

The PDZTA has been classified in Category 1 and, at this stage of its formulation process, is the subject of a Strategic Environmental and Social Assessment, which can be assessed via https://www.afdb.org/en/documents/guinee-programme-de-developpement-de-zones-de-transformation-agro-alimentaire-pdzta-p-gn-aag-004-rapport-cges. Category 1 projects are likely to have significant or irreversible environmental and/or social impacts or severely impact the environmental or social impacts considered by the Bank or borrowing country to be sensitive. Boké Department is located in the administrative region of Boké which, itself, is located in **Maritime or Lower Guinea**. With the presence of the Atlantic coast, Lower Guinea is the alluvial basin for major coastal rivers such as the Kogon, Fatala, Konkouré and Kolente Rivers. The area is also covered by mangrove swamps (Rizophora racemose and Avicennianitida). This natural region of Lower Guinea is also home to the country's main mining centres: the Guinean Bauxite Company (CBG) in Boké, ACG in Fria and the Kindia Bauxite Company (Débélé), etc.

With regard to potential sites mainly for the PDZA-BK's anchor programmes, in particular the irrigation schemes in Boké Prefecture, the Denken plains covering an area of almost 600 ha located in Kolaboui rural municipality and the Kapatchez plains covering about 9,200 ha located in Kamsar rural community provide significant development opportunities. In the **Boké Administrative Region,** there are also the 9500hectare Mankountan Plain, the 4,000 hectareMonchon Plain and the 6,000 hectare in Koba Plain.

Kankan Department, the programme's target area, is located in Upper Guinea. This natural region covers the entire north-east and centre of Guinea's territory. It is a region of plains and savannah, located between 200 and 400 metres of altitude. The Niger River and its tributaries have cut moist terraced flatlands into it.

In relation to the programme, large sedimentary plains in Upper Guinea border the main water courses and cover vast areas**:** the Niger plains in Faranah, Kouroussa, Siguiri, the Milo plains, the Fié plains and the Banié-Tinkisso plains. In Kankan prefecture, the following sites have significant irrigation potential: the Bafèle plain with about 11,000 ha of irrigable land located in Moribaya rural municipality and the Samanka plain with 800 ha located in TintiOulen rural municipality.

**G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)**

***Togo***
***Gender Aspects***
The project is classified in Category 2, according to the Gender Marker System. The detailed gender analysis of the entire value chain has been carried out including recommended measures to address the main constraints faced by women in agriculture in Togo (See annex 8.B). Some of these constraints include access to affordable financing to invest in climate resilient agricultural practices, lack of capacity in climate resilient agricultural practices and technology adoption, land tenure insecurity, poor access to small water infrastructure for market gardening and horticulture, lack of access to post-harvest handling equipment to reduce losses, and poor outlets for the marketing of agricultural products.

At the agricultural value chain, the project will also finance:

- Situation analysis of Gender Equality in the IAIPs. That is mapping of stakeholders and needs assessment (identification of target women-owned companies in the agro-processing value chain and building capacity-targeted training, mentoring and coaching including involving leaders and men's gender champions).
- Access to finance for smallholder women farmers (SHWFs), WABEs, ACSs, and FBAs to invest in drip irrigation technology powered by solar pumps. This will support horticulture and market gardening of vegetables and fruits including other cash crops, all year round in at least 11,810.0 ha.
- Access to solar energy for staple food crops processing.
- Access to finance to invest in agro-forestry management practices (10,000.0 ha)
- Training and capacity strengthening of women to access low-carbon energy technologies in agriculture such as Biogas digesters to improve staple food crops processing in project areas
- Women empowerment on agricultural value chain development through awareness raising, training courses and capacity building exercises.
- Women empowerment to take up managerial responsibilities at the Integrated Agro-Industrial Parks (IAIPs).

***The budget for the implementation of the gender promotion and women's empowerment component stands at $62.51 million. GCF contribution is $30.27 million (72% of GCF Budget) and AfDB co-financing for gender promotion in Togo is $32.24 million.***

***Senegal***
***Gender Aspects***
The project runs in a social environment where gender inequalities are still alive and accentuated by the migration of men. This situation reinforces the triple role of women with its corollaries of work overload and poverty. Thus, the project will take into account the specific needs of women and men as well as young people of both sexes. The aim is to reduce the inequalities of access for women and men of all social categories to the opportunities offered by the Agropole. Thus, the actions to be carried out are: (i) equity in the implementation of activities in the project (selection of projects, training, etc.), (ii) taking into account the specific needs of women and girls in the choice of infrastructures to be financed, (iii) support to women and youth groups, (iv) recruitment of a gender expert in the PIU. Thus, the project would be classified in category 2 according to the Bank's gender marker system (Annex 8.C).

Specifically, at the agricultural value chain, the project will also finance:

- Situation analysis of Gender Equality at the IAIPs. That is mapping of stakeholders and needs assessment (identification of target women-owned companies in the agro-processing value chain and building capacity-targeted training, mentoring and coaching including involving leaders and men's gender champions).
- Access to finance for smallholder women farmers (SHWFs), WABEs, ACSs, and FBAs to invest in drip irrigation technology powered by solar pumps. This will support horticulture and market gardening of vegetables and fruits including other cash crops, all year round in at least11,940.0 ha.
- Access to solar energy for staple food crops processing.
- Access to finance to invest in agro-forestry management practices (20,000.0 ha)
- Training and capacity strengthening of women to access low-carbon energy technologies in agriculture such as Biogas digesters to improve staple food crops processing in project areas
- Women empowerment on agricultural value chain development through awareness raising, training courses and capacity building exercises.
- Women empowerment to take up managerial responsibilities at the Integrated Agro-Industrial Parks (IAIPs).

***The budget for the implementation of the gender promotion and women's empowerment component in Senegal stands at $65.22 million. GCF financing for gender is $32.98 (75.6% of GCF Budget support) and $32.24 million as AfDB co-financing for gender related aspects.***

***Guinea***
***Gender Aspects***

According to the Gender Classification System, the programme is classified in Category 2. The detailed gender analysis and gender action plan are presented as Annex 8.D.On average, Guinean agriculture contributes an average of 17.5% to 23% to GDP and occupies 50% of the work force 75% of which is in rural areas (RGPH3, 2014). It is largely dominated by family-type farms. Over half of those in the work force (52.6%) are employed in the 'agriculture, livestock, forestry and fisheries' sector, 50.9% of whom are male and 53.5% female. Women play a major role in the agricultural economy throughout the agricultural value chain from production and processing to marketing. However, despite their impact in the agricultural production chain, they still face many difficulties, including:

- Access to resources and factors of production, in particular land, inputs and fertilizers. Because of customs and traditions, women are not usually the owners of farmed land;
- In decision-making and conflict management in the agricultural sector, women have less influence;
- In terms of financial resources: the lack of resources isa real problem both in the agricultural sector and in other areas (education, health etc.); and
- Concerning training: this remains a real concern, especially regarding capacity building, entrepreneurship, management… The literacy level of some women is also a handicap.

In order to alleviate these difficulties, the Special Agro-Industrial Processing Zones Development Programme proposes the following gender promotion and women's empowerment actions throughout its implementation: i) entrepreneurship, job creation for women and young people and access to markets; ii) agricultural production capacity building; iii) skill building in agricultural production and processing, starting and managing a business, marketing and market access; iv) sensitization campaigns on behavioural changes regarding birth control and reproductive health, balanced nutrition, gender and the prevention of gender-based violence, the prevention of early marriages, HIV/AIDS, lifestyle management and environmental sanitation.

Specifically, the project will also finance:

- Situation analysis of Gender Equality at the IAIPs. That is mapping of stakeholders and needs assessment (identification of target women-owned companies in the agro-processing value chain and building capacity-targeted training, mentoring and coaching including involving leaders and men's gender champions).
- Access to finance for smallholder women farmers (SHWFs), WABEs, ACSs, and FBAs to invest in drip irrigation technology powered by solar pumps. This will support horticulture and market gardening of vegetables and fruits including other cash crops, all year round in at least 15,428.0 ha.
- Access to solar energy for staple food crops processing.
- Access to finance to invest in agro-forestry management practices (10,000.0 ha)
- Training and capacity strengthening of women to access low-carbon energy technologies in agriculture such as Biogas digesters to improve staple food crops processing in project areas
- Women empowerment on agricultural value chain development through awareness raising, training courses and capacity building exercises.

- Women empowerment to take up managerial responsibilities at the Integrated Agro-Industrial Parks (IAIPs).

***The budget allocated to gender promotion and women's empowerment actions in Guinea is $30.60 million, of which, GCF contribution is $22.06 million (72.9% GCF Budget) and AfDB co-financing is $8.54 million.***

**G.3. Financial management and procurement (max. 500 words, approximately 1 page)**

AfDB fiduciary and financial management measures for the programme started at the upstream engagement with the programme countries during the preparation of the Country Strategy Papers (CSPs) where fiduciary risks and capacity of the countries were assessed in the Country Fiduciary Risk Assessment (CFRA). This is a key document for dialogue with the country for the risk rating and agreed mitigating measures to address noted fiduciary risks including corruption and AML/CFT and prohibited practices issues. The AE has also carried out review of sanctions involving United Nations Security Council, security and exchange commissions, financial oversight authorities as well as review of Politically Exposed Persons (PEPs) and Other High Risk Relationships of the Technical Responsible Parties (TRPs).

The SCPZ is a sovereign operation so financial management planning has also been carried out to ensure that the Borrower/Recipient's financial management systems are sufficiently sound to ensure that funds provided will be used for the purposes for which they were intended in an efficient and effective manner. The institutional assessment also focused on evaluation of the financial management capabilities of the TRPs related to budgeting, accounting, internal control of funds, treasury, financial reporting and ensured that the auditing, financial management capability for transparent, accountable and efficient control of funds is in place. These measures were incorporated into the SCPZ documents for AfDB Board approval and the legal agreements for negotiation, execution and effectiveness with the Borrower/Recipient. As the AfDB is accredited to use its internal processes, policies, systems and procedures, the SCPZ went through these procedures to ensure sound fiduciary capacity of the countries and the TRPs, and measures to reduce noted fiduciary risks. A detailed information on the Financial Management Assessment (FMA) of the EEs is presented in Section 4.1 of the Operating Manual. Recommendations based on results of the FMA has also been stated in Section B4 of the funding proposal for each programme country.

CFRA and the FMA informed the procurement planning from the project identification, preparation, appraisal, and other processes for the AE Board approval. Compliance to the FMA recommendations and the procurement plans will be monitored during implementation of the programme. Procurement Planning provides an assessment of the procurement framework of the country and the capacity of the country and TRPs systems to ensure efficient and effective procurement of goods, works and services in the project countries. For example, during the project preparation stage, the procurement assessment included assessment of the TRPs (similar to EE institutional set up of the GCF) capacity to carry out project procurement and the risks at the country, sector and operations levels associated with the implementation of procurement under the operation. This was reviewed and updated during the project appraisal, approval based on additional information identified during fiduciary risks reviews to identify fiduciary issues. Based on the results of the Procurement Risks and Capacity Assessment (PRCA) specific procurement measures are outlined for the SCPZ and are detailed in Section 4.2 of the Operating Manual. The Bank Readiness Review Process also ensures that the procurement, financial management and disbursement arrangements for the SCPZ is related to the issues and risks identified.

During implementation, as part of the procedure, there will be periodic review of adequacy of the procurement and financial management arrangements specified in the FMA during on-site implementation missions to include for example, the integrity of the accounting system, the efficiency of internal control, the application of controls, and the smooth delivery of planed financial resources to the SCPZ programme. Following on-site supervision missions, the financial management performance (FMP) will be reviewed the fiduciary risk of the SCPZ in each country will be updated in the Financial Management Supervision Report. Once in a year, on receipt of the annual Audit Report, the performance of the external auditors to will be reviewed to ensure the work completed complies with the Terms of Reference, and that the audit has been performed to the agreed Auditing Standards. Depending on the performance review for financial management, other implementation support activities including capacity building and Fiduciary Clinics and Country Portfolio Performance Reviews will be undertaken.

**Financial Management**

Each Project Management Unit (PMU) will be responsible for the project overall coordination and financial management. Its financial management staff will comprise the project coordinator, the administrative and financial officer, the accountant and cashier, all of whom will be recruited in accordance with the project procurement arrangements. The financial management systems for the project at the countries level are not satisfactory on the whole and the overall fiduciary risks are substantial because of the absence of: (i) a financial coordinator and personnel as mentioned above; (ii) an administrative, financial and accounting procedures manual; (iii) operational accounting and financial management software; and (iv) financial management tools and reference framework. As a result, each PMUs would take the following measures: (i) as soon as the project is launched, prepare budget plans coupled with the indicative activities schedule; (ii) when the project is launched, recruit a consultants to prepare the administrative, financial and accounting procedures manuals (AFAPMs), and others responsible for establishing the computerized accounting and financial management systems (SIGCFs) ; (iii) assign financial tasks as soon as the financial staff are recruited; (iv) establish the financial management frameworks and instruments in accordance with the Project Appraisal Reports (PAR).

**Procurement and Disbursement Arrangements**
***Procurement Arrangements***

The procurement of goods (including services other than consulting services), works and the acquisition of consulting services financed under the project will be carried out in accordance with the Procurement Framework for Bank Group-financed operations, 2015 edition, and in compliance with the provisions of the GCF FAA. More specifically, procurements will be made in accordance with the Bank's Procurement Methods and Procedures (BPP), on the basis of the relevant standard bidding documents (SBDs) for goods and works contracts as well as consulting services for which the BPP are considered to be the most appropriate. Indeed, following an analysis of the procurement systems of the three SCPZ countries, the procurement risk was considered to be substantial. As a result, for the implementation of this programme, the Bank's Procurement Methods and Procedures will be used, since a capacity building action plan will be the subject of dialogue with the SCPZ Government Authorities in order to rapidly ensure use of the national procurement systems following the reforms identified as necessary.

***Procurement Risks and Capacity Assessment (PRCA)***: The assessment of procurement risks at country, sector and project levels as well as the capacity of the TRP/EEs & PIU were carried out and the results were used to influence the decision to choose the BPM for all activities planned under the project.

***Organization of Procurement Implementation*****:** Project procurement will be carried out by the Project Implementing Units (PIUs). In light of the conclusions of the assessment of the TRP/EEs & PIUs capacities, a management unit including a procurement expert will be recruited to implement the procurement process but also to build the capacities of the different actors involved in procurement.

**Disbursement:** The disbursement methods to be used to mobilize the project financing are: (i) the direct payment method; (ii) the special fund/revolving fund method and (iii) the reimbursement method. Direct payments will be made in respect of contracts for works, goods and services signed between the Borrowers and suppliers, in accordance with the Bank's procurement rules and procedures and national procurement legislations. The special account method will be used to settle operating expenditure, training costs, field mission costs etc. It will require the opening of a Special Account at the Central Banks of the three SCPZ pilot countries, which will, in turn, pay all the funds received from the AfDB into the PMU bank account opened in a bank acceptable to AfDB. The opening of the Bank account will be a condition precedent to the first disbursement. The reimbursement method will be used when eligible expenditure on all project resources is pre-financed with the Bank's prior approval. These disbursements will be made in accordance with the list of goods and services and the Bank's rules and procedures as described in the Project Disbursement Handbook.

**Accounting Management:** Legal Agreements require the pilot SCPZs countries to - (i) maintain accounts and books, as well as prepare financial statements for the programmes and projects financed from AfDB's resources and AfDB's administered resources; and (ii) implement accounting, administrative and financial procedures documents in manuals. These accounts and records should provide justification for the use of funds disbursed to the programme or project, and prepared in accordance with applicable accounting standards.

**Accounting Standards and Practices:** Although accounting standards and practices may differ from country to country, the accounts of the programme or sub-project under the programme shall, irrespective of the system adopted, comply with the principles of transparent management and provide exhaustive information on the following: 1) Accountability of funds provided to the programme or sub-project; such funds may comprise proceeds from the AfDB's loan or grant, or other resources from co-financiers and counterpart funds; 2) Disclosure of expenditure and specific information on components financed under the programme.

**Programme Financial Statements:** The programme or sub-projects financial statements prepared under the SCPZs programme management shall include:

- For all programmes: i) Statement of Receipts (funds received from the Bank, counterpart funding and where applicable, co-financiers' funding) and Expenditures (expenditures incurred for both the current year and accumulated to-date) showing separately AfDB's funding, those of counterparty and co-financiers and cash balances; 2) Statement of Special Account/s; 3) notes to the Financial Statements describing the applicable accounting principles used and a detailed analysis of the main accounts.

- For Revenue-Earning Projects under the programme (in addition to the requirements stated above); i) Balance Sheet showing accumulated funds of the project, bank balances and other assets and liabilities of the project as at the close of each fiscal year; ii) Income Statement (or Operating, or Income and Expenditure, or Profit and Loss); and iii) Cash Flow Statement disclosing cash flows during each fiscal year.

- For Annex to the financial statements, the following should be included as an annex to the financial statements of all projects under the programme: i) a reconciliation between the amount shown as "received from the Bank" and that shown as having been disbursed by AfDB. The reconciliation should indicate the methods used for disbursement, i.e. Special Account, direct payment or reimbursement guarantee, reimbursement methods with those recommended in the PARs appraisal report and the disbursement letter; and ii) a comprehensive list of all fixed assets purchased with given dates, value and condition of the assets.

**Financial Reporting:** The SCPZs pilot countries are required to regularly and exhaustively communicate accounting and financial information on the programme and sub-projects under the programme. The information should detail out the various uses of the loan and grant proceeds. Financial systems and financial communication reports may vary according to the nature of the sub-project, the lending instrument and the country. AfDB may not require a standard format, but will ensure that the reports submitted for verification and consideration are adequately informed.

**Keeping and Filing of Documents:** Each participating country in the SCPZs programme is required to keep up-to-date and preserve programme/sub-project documents and supporting documents, and to make them available to AfDB's representatives as needed.

**Retention of Documentation:** AfDB's General Conditions require that each participating country in the programme retain all documentation related to the loan proceeds to enable external auditors or AfDB's representative to examine the same. The documents should be retained in a manner where they are readily accessible.

**Audit:** The General Conditions applicable to loan and grant agreements require that all programme and sub-project accounts are audited each year in accordance with the relevant generally accepted standards. The auditor, in accordance with the standard Terms of Reference for external audits of AfDB financed programmes or projects, is required, among other things, to examine pertinent documents, review internal financial control mechanisms to identify deficiencies and weaknesses that could affect the efficiency of the project, form an opinion on the quality of the financial statements and confirm that the funds granted to the project have been used for their intended purpose. Each country must have the accounts of the project or programme audited by qualified and independent accountants that meet the AfDB's requirements. In countries, where auditing is entrusted to the Auditor General's Office (AGO) under the supervisory authority of the State, AfDB may approve the designation of this Office provided that the matter is discussed at programme/project appraisal and during the loan negotiations and the audit conducted in accordance with AfDB's standard terms of reference for the audit for AfDB financed projects/programmes. The audited financial statements, the auditor's report and management letter must be received by AfDB no later than six months after the end of the financial year to which they relate. Non-compliance with this requirement will result in the immediate suspension of further replenishments to the Special Account.

In Togo, an internal audit mechanism will be established with the recruitment of an Internal Auditor who will ensure that the SCPP control system is operational and effective. Also, an independent external auditor will be hired on a competitive basis and in accordance with the terms of reference agreed with AfDB, in line with its requirement that audit reports be submitted no later than 30 June of the year following the year audited. If the audit is conducted by an independent Audit firm, to be selected competitively, the audit fee will be covered by project resources. External audit is to be conducted at the end of each Fiscal Year. The Audit report with the associated Management Letter should be submitted to AfDB not later than six (6) months after the close of the Fiscal year. With the case of Senegal, the programme financial and accounting audit will be conducted by an independent firm registered on the role of a national or regional order of AfDB member countries. The audit firm will be recruited for three years based on terms of reference approved by AfDB, and paid out of AfDB funds. Finally, in Guinea, once every year, a competent and independent firm of external auditors will verify the reliability of the consolidated financial statements prepared by the PIU and will assess the operation of the internal control system of the entire programme. It will be recruited in accordance with the terms of reference and bidding procedures recommended by AfDB. Audit costs will be financed by AfDB. The audit reports must be forwarded to AfDB each year, no later than six months after the closure of the audited period. Audits will be performed in accordance with SYSCOA international audit standards and the AfDB's comprehensive terms of reference which will be communicated to the programme.

*Programme Monitoring and Evaluation (M&E)*

Simply refers to the process of procurement related M&E throughout the programme (and procurement) cycle. Information to be monitored for the programme and sub-project under the programme will be identified in the Programme Executing Agency Procurement Assessment (EAPA), drawing on sources such as the Country Procurement Assessment Report (CPAR). Capacity development as set out in the Procurement Development Action Plan (PDAP) will also be subject to M&E throughout programme implementation. AfDB's primary monitoring tools are post-procurement reviews (PPRs), procurement audits (PAs), independent procurement reviews (IPRs) and other miscellaneous reviews associated with AfDB programme or project-related oversight and monitoring.

In Togo, the programme is also scheduled for implementation over a 60-month period (Table 1.4). The M&E Officer to be appointed from within the PIU will be responsible for collecting, analysing and compiling information on physical achievements and financial execution. For this purpose, he/she will have a roadmap with quantitative indicators on programme progress, relating to implementation of the sub-components. Such monitoring will make it possible to obtain, every six months, the following information for each activity: physical objective, achievement level, expected costs, actual costs, differences and explanations for possible deviations. This information will be used to prepare the half-yearly and annual performance programme reports by the TRP/EE together with the PSMC. Moreover, in addition to the Steering Committee's review work, AfDB will carry out two supervision missions per year, to which should be added a mid-term review mission.

Similarly for Senegal, programme monitoring and evaluation will be carried out by the PIU. The PIU will collect, analyse and compile information on physical achievements and financial execution. To this end, it will have a scorecard with technical and financial indicators on the progress of the programme and sub-project components. Such monitoring will serve to obtain the following information for each activity and with a half-yearly periodicity: physical target, level of achievement, expected costs, actual costs, gaps and explanations of any gaps. This information will be used for preparing the half-yearly progress reports and APRs. Furthermore, in addition to the review work of the Steering Committee, AfDB will field two supervision missions per year, to which should be added a mid-term review mission to produced IER.

Finally, for Guinea, internal monitoring and evaluation will be carried out by the Programme Monitoring and Evaluation Unit of the PMU or PIU, while external monitoring-evaluation will be the responsibility of the Ministry of Planning and the Ministry in charge of Finance. The objective of external monitoring and evaluation is to assess the efficacy and efficiency of the programme's outputs and their contribution to the achievement of the development objectives which are the programme outcomes and impacts. It will be carried out with the involvement of the other stakeholders in addition to two annual supervision missions organized by the AfDB. The midterm review will be carried out in year 3. Following the programme's closure, AfDB and Government will produce a completion report within the required time frame.

See pages 34-54 of the Programme Operational Manual (POM) for more details on programme financial management arrangements.

**G.4. Disclosure of funding proposal**

*Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.*

*Indicate below whether or not the funding proposal includes confidential information.*

☐ <u>No confidential information:</u> The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ <u>With confidential information:</u> The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

## ANNEXES

### H.1. Mandatory annexes

| | | |
|---|---|---|
| ☒ | Annex 1 | NDA no-objection letter(s) **(template provided)** |
| ☒ | Annex 2 | Feasibility study - and a market study, if applicable |
| ☒ | Annex 3<br>Annex3A<br>Annex3B<br>Annex3C | Economic and/or financial analyses in spreadsheet format<br>SCPZ Country Level Estimations and assumptions<br>Economic and financial Analysis<br>Financial analysis report for SCPZ |
| ☒ | Annex 4 | Detailed budget plan **(template provided)** |
| ☒ | Annex 5 | Implementation timetable including key project/programme milestones **(template provided)** |
| ☒ | Annex 6 | E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)**<br>☐ Environmental and Social Impact Assessment (ESIA) or<br>☐ Environmental and Social Management Plan (ESMP) or<br>☐ Environmental and Social Management System (ESMS)<br>☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.) |
| ☒ | Annex 7 | Summary of consultations and stakeholder engagement plan |
| ☒ | Annex 8 | Gender assessment and project/programme-level action plan **(template provided)** |
| ☒ | Annex 9 | Legal due diligence (regulation, taxation and insurance) |
| ☒ | Annex 10 | Procurement plan **(template provided)** |
| ☒ | Annex 11 | Monitoring and evaluation plan **(template provided)** |
| ☒ | Annex 12 | AE fee request **(template provided)** |
| ☒ | Annex 13 | Co-financing commitment letter, if applicable **(template provided)** |
| ☒ | Annex 14 | Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule |

### H.2. Other annexes as applicable

| | | |
|---|---|---|
| ☐ | Annex 15 | Evidence of internal approval **(template provided)** |
| ☒ Annex 16 | | Map(s) indicating the location of proposed interventions |
| ☒ Annex 17 | | Multi-country project/programme information **(template provided)** |
| ☐ Annex 18 | | Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project |
| ☐ Annex 19 | | Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity |
| ☒ Annex 20 | | First level AML/CFT (KYC) assessment |

| | | |
|---|---|---|
| ☒ | Annex 21 | Operations manual (Operations and maintenance) |
| ☒ | Annex 22 | Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)[88] |
| ☒ | Annex 23 | Technical description of agrometeorological and rainfall stations |
| ☒ | Annex 24 | Climate risk and vulnerability assessment |
| ☒ | Annex 25 | Country resilience and fragility note for Guinea, Senegal, and Togo |

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

***Appendix 1***

Summary Table of Countries -specific Risks and Risks Mitigation Measures

| | | |
|---|---|---|
| ***Togo*** | | |
| *Risk 1: Land security measures are insufficient to attract the private sector and protect small farmers* | Moderate | Mitigation 1: The SESA has identified mitigation measures in the project, including the implementing decrees of the new Land Code and the establishment of single window for land tenure in Kara |
| *Risk 2: Climate change and harmful practices of the people could degrade natural resources (NR)* | Moderate | Mitigation 3: CRA practices and the development of alternative livelihoods should sustainably enhance the project effects |
| *Risk 3: Lack of interest by the private sector could limit investments planned in the Agro-park* | Moderate | Mitigation 4: (i) Meetings were held with the private sector (including Kalyan Group, key investor) and their needs taken into account; (ii) the Project will finance infrastructure for the development of agro-parks and production areas; (iii) support for the implementation of NP reforms is planned (land security, standardization and quality control, PPP, etc.); and (ii) technical assistance (including legal, technical, management, PPP etc.) will be made available to APRODAT |
| *Risk 5: Delays in project implementation due to poor knowledge of AfDB procedures by APRODAT* | High | Mitigation 5: Substantial institutional support (see action plans) for APRODAT is envisaged under the project |
| *Risk 5: The significant benefits expected from the project could be monopolised by a minority* | Moderate | Mitigation 6: The SESA (including RAP, and ESIMP, PDC), and Saemaul Foundation provide for the implementation actions (Population Resettlement Action Plan, ESIMP and PDC) that will have positive impacts on the people |
| Senegal | | |
| *Risk 1: Private Sector (PS) Lack of Interest Could Limit Investment* | High | Mitigation 1: (i) The PS will be consulted during the evaluation and thereafter;(ii) the PIU and the SCE will put in place the required facilities (infrastructure, one-off facilities, etc.), and;(iii) credit facilities will be put in place |
| *Risk 2: Climate change-CC and harmful population practices can degrade natural resources (RN)* | Moderate | Mitigation 2: Integrated NR management, increased agricultural productivity, and access to markets and agricultural insurance will need to mitigate the effects of CC and harmful practices |
| *Risk 3: The expected benefits could be grabbed by a minority (including for land)* | Moderate | Mitigation 3: The measures included in the EESS (including the GSWP), the ESMP / PAR, and the IEC program of the project should allow to mitigate it |
| *Risk 4: Delays in project implementation due to insufficient control of Bank procedures* | High | Mitigation 4: Institutional support and regular training will be provided to the CEP, AGEX, and the Borrower (Ministries in charge of Finance, Economy) |
| Guinea | | |

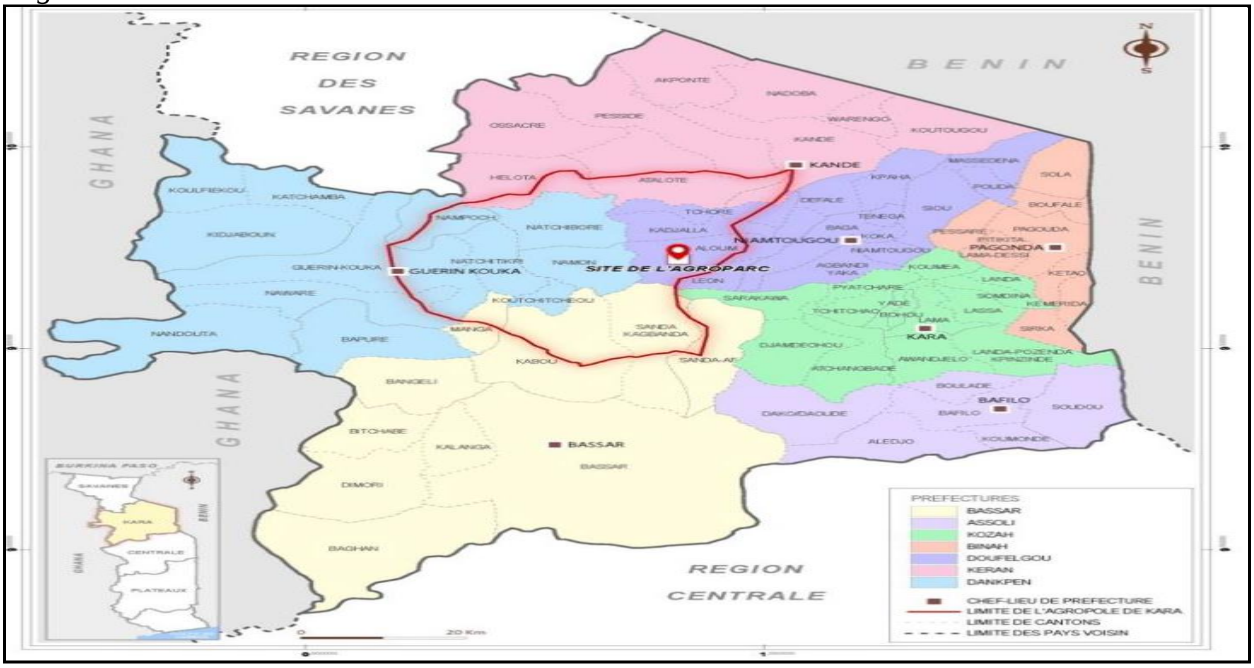
[88] Annex 22 is mandatory for mitigation and cross-cutting projects.

| | | |
|---|---|---|
| *Risk 1: Lack of commitment of oversight authorities to support the coordination team and slippage on programme implementation* | Moderate | The programme is one of central government's priorities and was designed on the basis of the systematic consultation of stakeholders including representatives of the administration and municipalities concerned |
| *Risk of weak private sector mobilisation* | Moderate | The programme was designed on the basis of a platform of exchange of information with the private sector whose concerns are taken into account in the programme's final formulation, a guarantee of the SME involvement in, and ownership of, the programme |
| *Slow acceptance by certain stakeholders could delay the programme's implementation* | Moderate | The programme was designed on the basis of systematic sensitisation and consultation to ensure its gradual ownership by all stakeholders |

***Appendix 2:***

*Maps of SCPZs Program Interventions Areas*

*Togo*

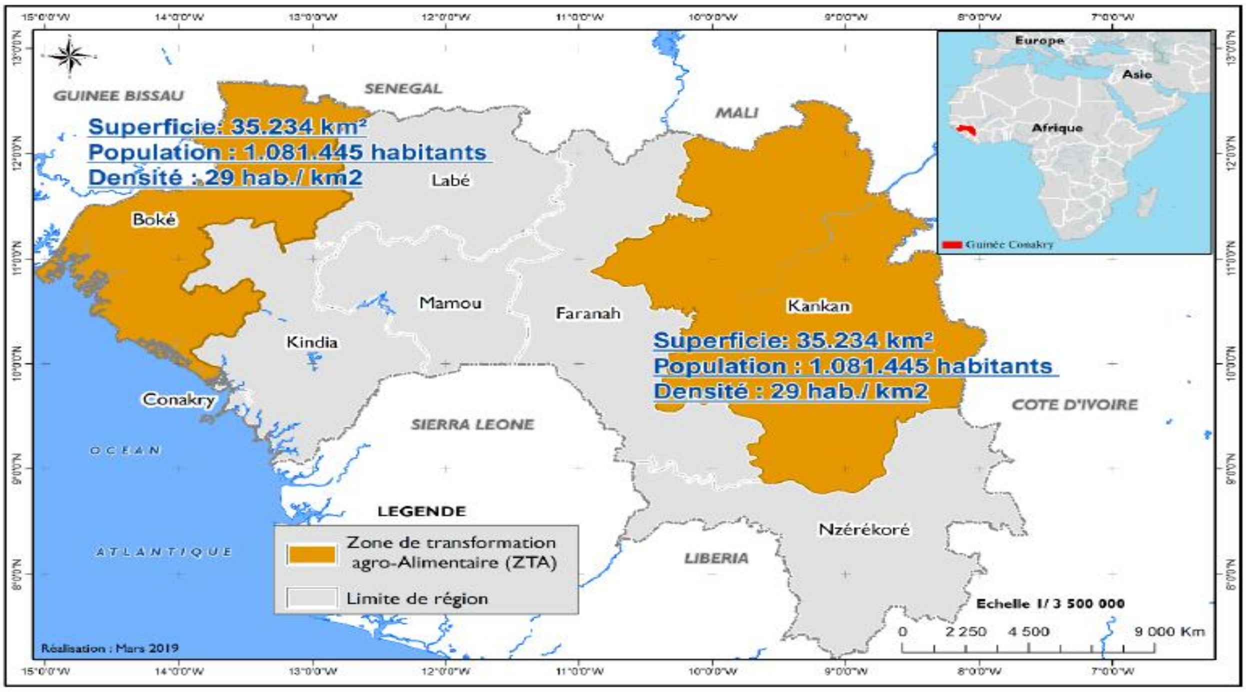



Map of PTA Intervention Area in the Kara Region

*Senegal*


Map of South Agropole in Senegal

*Guinea*




Map of PDZTA-BK SCPZs

# Appendix 3A
## Bank's Portfolio in Togo as at 31 December 2016

| Sector | Project Title | Status | Window | Approval Date | Disbursement Deadline | Disbursed Amount (UA) | Approved Amount (UA) | Disb. rate (%) |
|---|---|---|---|---|---|---|---|---|
| Water/ sanitation | Toilets for All in Sokodé and Recycling of Faecal Sludge | OnGo | AWF | 19/04/2013 | 31/12/2017 | 140,054 | 935,760 | 15.0 |
| | *Sub-Total Water & Sanitation* | | | | | *140,054* | *935,760* | *15.0* |
| Agric | PPF - Project for the Development of Agropoles in Togo | Apv | ADF | 29-Feb-16 | 30-Jun-17 | 0 | 995,000 | 0.0 |
| | *Sub-total agriculture* | | | | | *0* | *995,000* | *0.0* |
| Gov't | Resource Mobilization and Institutional Capacity Building Support Project (PAMOCI) | OnGo | TSF | 09-Oct-14 | 30-Jun-19 | 245,052 | 5,000,000 | 4.9 |
| | Fiscal Governance Support Project (PAGFI) | Apv | TSF | 17-Feb-16 | 31-Dec-20 | 0 | 3,080,000 | 0.0 |
| | | | TSF | 17-Feb-16 | 31-Dec-20 | 0 | 11,920,000 | 0.0 |
| | *Sub-Total: Governance* | | | | | *245,052* | *20,000,000* | *1.2* |
| Social | Kara and Lomé Markets Reconstruction and Traders Support Project | OnGo | ADF | 22/01/2014 | 31-Dec-18 | 0 | 1,930,000 | 0.0 |
| | | | ADF | 22/01/2014 | 31/12/2018 | 325,826 | 1,650,000 | 19.7 |
| | FAPA - Kara and Lomé Markets Reconstruction and Traders Support Project | OnGo | FAPA | 26-Jan-15 | 31/12/2018 | 0 | 588,061 | 0.0 |
| | Youth Training and Employment Project | Apv | ADF | 28-Oct-15 | 31-Dec-20 | 0 | 6,670,000 | 0.0 |
| | | | NTF | 28-Oct-15 | 31-Dec-20 | 0 | 6,500,000 | 0.0 |
| | | | TSF | 28-Oct-15 | 31-Dec-20 | 0 | 1,330,000 | 0.0 |
| | *Sub-Total – Social Sector* | | | | | *325,826* | *18,668,061* | *1.7* |
| | *National Total* | | | | | *710,933* | *40,598,821* | *1.8* |
| Multinational transport | Benin/Togo: Lomé-Cotonou Road Rehabilitation and Abidjan-Lagos Corridor Facilitation Project - Phase 1 | OnGo | ADF | 05/10/2011 | 31/12/2016 | 992,576 | 4,810,000 | 20.6 |
| | Togo/Burkina: Community Road (CU9) Rehabilitation and Transport Facilitation along the Lomé-Ouaga Corridor | OnGo | ADF | 27-Jun-12 | 31/12/2017 | 12,276,319 | 17,800,000 | 69.0 |
| | | | ADF | 27-Jun-12 | 31/12/2017 | 13,642,755 | 30,230,000 | 45.1 |
| | | | TSF | 27-Jun-12 | 31/12/2017 | 1,859,120 | 21,500,000 | 8.6 |
| | | | EU-AITF | 23-Feb-15 | 31/12/2017 | 467,267 | 923,915.19 | 50.6 |
| | *Sub-total for Transport* | | | | | *29,238,037* | *75,263,915* | *38.8* |
| | *Multinational Total* | | | | | 29,238,037 | 75,263,915 | 38.8 |
| | **Total - Public Sector** | | | | | 29,948,970 | 115,862,736 | 25.8 |
| | **GRAND TOTAL** | | | | | **29,948,970** | **115,862,736** | **25.8** |

# Appendix 3B

## Bank's Portfolio in Senegal as at 31 December 2016

| | Sector / Operation | | Approval Date | Approved Amount (MUA) | Disbursed Amount (MUA) | Disbursement Rate (%) | Closing Date |
|---|---|---|---|---|---|---|---|
| | **RURAL** | | | | | | |
| 1 | Community Roads Project in support of the National Local. Development Programme (PNDL) | - ADF | 17-July-13 | 15.00 | 9.13 | 60.9 | 31-Dec.18 |
| | | - OPEC | 11-Dec.-13 | 7.10 | 1.89 | 26.6 | 30-June19 |
| 2 | Lake Guiers Restoration Project (PREFELAG) | - ADF | 4-Sept.-13 | 15.00 | 6.48 | 43.2 | 31-Dec.18 |
| | | - GEF | 4-Sept.-13 | 0.93 | 0 | 0 | 31-Dec.18 |
| 3 | Louga-Matam-Kaffrine Food Security Support | - ADF | 26-Apr.-13 | 2.00 | 0.66 | 33.2 | 31-Dec.18 |
| | | - GAFSP | 26-Apr.-13 | 28.34 | 6.15 | 21.7 | 31-Dec.18 |
| | **Sub-Total/Average** | | | **68.37** | **24.31** | **35.6** | |
| | **INFRASTRUCTURE** | | | | | | |
| 4 | Dinguiraye-Nioro-Keur-Ayib Road Project - ADF | | 28-May-14 | 23.77 | 3.57 | 15.0 | 30-June18 |
| 5 | Project to Rehabilitate RN2 and Open up Access to Morphil Island - ADB | | 16-Dec.15 | 97.41 | 0 | 0 | 31-Dec.19 |
| 6 | Digital Technology Park Project - ADB | | 21-Oct-15 | 49.15 | 0 | 0 | 31-Dec.-20 |
| | **Sub-Total/Average** | | | **170.33** | **3.57** | **2.1** | |
| | **WATER AND SANITATION** | | | | | | |
| 7 | Project to Improve Management and Recovery of Faecal Sludge in Ziguinchor - AWF | | 23-Apr.-13 | 1.02 | 0.29 | 28.7 | 31-Dec.17 |
| 8 | Water and Sanitation Sector Project | - ADF | 23- Apr.-14 | 20.00 | 0.80 | 4.0 | 31-Dec.18 |
| | | - RWSSI | 23- Apr.-14 | 4.84 | 0.09 | 1.8 | 31-Dec.18 |
| | **Sub-total Average** | | | **25.86** | **1.18** | **4.5** | |
| | **SOCIAL** | | | | | | |
| 9 | Youth and Women's Employment Promotion Support Project - ADF | | 23-Oct.-13 | 21.19 | 1.95 | 9.2 | 30-June19 |
| 10 | Senegal Virtual University Support Project- ADF | | 18-Dec.-13 | 3.38 | 0.27 | 7.9 | 30 June17 |
| | **Sub-total /Average** | | | **24.57** | **2.22** | **9.0** | |
| | **GOVERNANCE** | | | | | | |
| 11 | Private Sector Promotion Support Project - ADF | | 10-Sept.-12 | 4.04 | 2.22 | 54.9 | 30-June16 |
| | **Sub-total/ Average** | | | **4.04** | **2.22** | **54.9** | |
| | **TOTAL** | | | **293.17** | **33.50** | **11.4 %** | |
| | **Of which Bank Group only (ADB, ADF)** | | | **250.94** | **25.08** | **10 %** | |

## Appendix 3C

## Bank's Portfolio in Guinea as at 31 December 2016

| Sector/Project | Approv. Date. | Signat. Date | Compl. Date. | Source of Financing. | Amount Approved (UA) | | % Disb. | Age (months) | Project Performance | Risk |
|---|---|---|---|---|---|---|---|---|---|---|
| **Agriculture** | | | | | | | | | | |
| NERICA Dissemination Project - Guinea | 26.09.03 | 13.02.04 | 30.12.11 | ADF Loan | 3,000,000 | 1,940,331.50 | 64.68 | 84.8 | 2.22 | Non PP |
| **Total Agriculture** | | | | | **3,000,000** | **1,940,331.50** | **64.68** | **84.8** | **2.22** | |
| **Infrastructure/Energy** | | | | | | | | | | |
| TOMBO-GBESSIA AIRPORT ROAD UPGRADING TO FOUR-LANES | 13.07.05 | 22.07.05 | 31.12.10 | ADF Grant | 8,250,000 | 2,889,754.43 | 35.03 | 60.11 | 1.88 | PPP |
| GUINEA – SUPPLEMENTARY GRANT | 29.04.09 | 13.05.09 | 31.12.12 | ADF Grant | 5,170,000 | 269,760.78 | 5.22 | 24.3 | - | Not rated |
| Rehabilitation of Electric Power Networks | 29.10.08 | 13.05.09 | 31.12.13 | ADF Grant | 12,000,000 | 1,513,210.75 | 12.61 | 24.8 | 2.07 | Non PP |
| RURAL ELECTRIFICATION PROJECT | 21.01.11 | 15.02.11 | 31.12.15 | ADF Grant | 14,960,000 | 0 | 0 | 5.0 | - | Not rated |
| **Total Infrastructure/Energy** | | | | | **40,380,000** | **4,672,725.96** | **11.57** | **28.55** | **1.97** | |
| **Social Sector** | | | | | | | | | | |
| Education IV | 13.07.05 | 22.07.05 | 31.12.11 | ADF Grant | 14,000,000 | 8,130,265.83 | 58.07 | 60.11 | 1.94 | PPP |
| UPPER AND MIDDLE GUINEA PDSD PHASE II | 09.02.11 | 15.02.11 | 31.12.13 | ADF Grant | 5,000,000 | 551,082.78 | 11.02 | 4.0 | 2.78 | Non PP |
| **Total Social Sector** | | | | | **19,000,000** | **8,681,348.61** | **45.69** | **32.05** | **2.36** | |
| **Multi- Sector** | | | | | | | | | | |
| CAPACITY BUILDING SUPPORT PROJECT. | 26.07.06 | 15.09.06 | 31.12.11 | ADF Grant | 2,500,000 | 1,196,032.18 | 47.84 | 48.11 | 2.06 | Non PP |
| PARCGEF | 31.01.11 | 15.02.11 | 31.12.14 | ADF Grant | 7,544,000 | 0 | 0 | 5.0 | - | Not rated |
| ECONOMIC REOFRM SUPPORT PROGRAMME | 18.05.11 | 10.06.11 | | ADF Grant | 20,000,000 | 0 | 0 | 1.0 | 2.70 | Non PP |
| **Total Multi Sector** | | | | | **30,044,000** | **1,196,032.18** | **47.84** | **18.03** | **2.38** | |
| **GRAND TOTAL – 10 On-going Projects** | | | | | **92,424,000** | **16,490,438.2** | **17.8** | **31.7** | **2.23** | **2 PPP/ 5 Non PP/ 3N. Rated** |

**No-objection letter issued by the national designated authority(ies) or focal point(s)**

MINISTERE DE L'ENVIRONNEMENT
ET DES RESSOURCES FORESTIERS
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SECRETARIAT GENERAL
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DIRECTION DE L'ENVIRONNEMENT
-------------

RT

REPUBLIQUE TOGOLAISE
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Travail – Liberté – Patrie

**To: The Green Climate Fund ("GCF")**

**Lomé , le** 0 6 JUIN 2018

Re: Funding proposal for the GCF by AFDB regarding "Kara Region Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains"

**Dear Madam, Sir,**

We refer to the project "Kara Region Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains" in Togo as included in the funding proposal submitted by AFDB to us on 05 june 2018

The undersigned is the duly authorized representative of AGRIGNAN Esso Sam Abdou Rassidou, the focal point of Togo.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Project as included in funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Togo has no-objection to the project as included in the funding Proposal;
(b) The project as included in the funding Proposal is in conformity with Togo's national priorities, strategies and plans; and
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding Proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We are also confirm that our no objection applies to all projects or activities to be implemented within the scope of the project.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Ministère de l'Environnement et des Ressources Forestières
Direction de l'Environnement
RT

**NAME: AGRIGNAN Esso Sam Abdou Rassidou**

**TITLE: National Focal Point of Togo**

*B.P. : 4825 – Tél : (228) 22 21 33 21 / 22 21 51 97 – Fax (228) 22 21 03 33 / 22 21 51 97*
*247, Rue des Nîmes – E-mail : ragrignan80@yahoo.fr / denv_togo@yahoo.fr*

**MINISTERE DE L'ENVIRONNEMENT,**

**DES EAUX ET FORETS**

..............................

***Point Focal du Fonds Vert pour le Climat***

....................................

**REPUBLIQUE DE GUINEE**

*Travail-Justice-Solidarité*

......................

**Conakry, le 24 janvier 2020**

**No.29 | FVC........./MEEF/PFFVC/2020**

*Le Point Focal*

Au Fonds Vert pour le Climat

(FVC)

**Re: Funding proposal for the GCF by African Development Bank regarding Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains.**

Dear Madam,

We refer to the programme: **Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains** in the Republic of Guinea as included in the funding proposal submitted by African Development Bank to us on 30th October 2019.

The undersigned is the duly authorized representative of the National Directorate of the Environment, the National Designated Authority of the Republic of Guinea.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to **the Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains** as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of the Republic of Guinea has no-objection to the **Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains** as included in the funding proposal;

(b) The **Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains** as included in the funding proposal is in conformity with the Republic of Guinea national priorities, strategies and plans;

(c) In accordance with the GCF's environmental and social safeguards, the project **Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains** as

**Lansébounyi – Coléah Commune de Matam BP : 3118 Conakry – Rép. De Guinée**

**Tel : (+224) 622 21 31 51 / 657 54 54 93 – Email : dml54@live.fr**

included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme **Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains** as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the **Staple Crops processing Zone (SCPZ): Promoting sustainable agricultural value chains.**

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards.

Fonds Vert Pour le Climat (FVC) Le Point Focal République de Guinée

**Mohamed Lamine DOUMBOUYA**

Lansébounyi – Coléah Commune de Matam BP : 3118 Conakry – Rép. De Guinée
Tel : (+224) 622 21 31 51 / 657 54 54 93 – Email : dml54@live.fr

Un Peuple - Un But - Une Foi

**MINISTERE DE L'ENVIRONNEMENT ET DU DEVELOPPEMENT DURABLE**

**DIRECTION DE L'ENVIRONNEMENT ET DES ETABLISSEMENTS CLASSES**

DEEC

**N°......10...........MEDD/DEEC/DCC**

**Dakar, le** 03 MARS 2020

To: The Green Climate Fund ("GCF")

**Mr. Yannick Glemarec**
Executive Director
G-Tower, 24-4 Songdo-dong, Yeonsu-gu
Incheon City, Republic of Korea

Re: Funding proposal for the GCF by "African Development Bank (AfDB)" on "***Staple Crops Processing Zone (SCPZ): Promoting sustainable agricultural value chains*** "

**Dear Sir,**

We refer to GCF-AFD co-financing programme " ***Staple Crops Processing Zone (SCPZ): Promoting sustainable agricultural value chains*** "in Senegal as included in the funding proposal submitted by African Development Bank to us on 30 th January 2020.

The undersigned is the duly authorized representative of the National Designated Authority of Senegal.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Programme "***Staple Crops Processing Zone (SCPZ): Promoting sustainable agricultural value chains*** "as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Senegal has no-objection to the Programme as included in the funding proposal;
(b) The Programme as included in the funding proposal is in conformity with Senegal's national priorities, strategies and plans;
(c) In accordance with the GCF's environmental and social safeguards, the Programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the Programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the Programme "***Staple Crops Processing Zone (SCPZ): Promoting sustainable agricultural value chains***".

We acknowledge that this letter will be made publicly available on the GCF website.

D.E.E.C
Direction de l'Environnement et des Etablissements Classés

Kind regards,

Madeleine Diouf SARR

AND Senegal

Direction de l'Environnement et des Etablissements Classés (DEEC)
Parc Forestier de Hann, route des Pères MaristesBP : 6557 Dakar Tél : +(221) 33 859 17 58
106, rue Carnot - tél : +(221) 33 821 63 49
Web:www.denv.gouv.sn e-mail : cellulecom.deec@gmail.com

**Environmental and social safeguards report form pursuant to para. 17 of the IDP**

| **Basic project or programme information** | |
|---|---|
| **Project or programme title** | Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains |
| **Existence of subproject(s)to be identified after GCF Board approval** | [No] |
| **Sector (public or private)** | Public |
| **Accredited entity** | African Development Bank |
| **Environmental and social safeguards (ESS) category** | Category A |
| **Location – specific location(s) of project or target country or location(s) of programme** | Guinea [Boke and Kankan]<br>Senegal [South Agropole]<br>Togo [Kara] |
| **Environmental and Social Impact Assessment (ESIA)(if applicable)** | |
| Date of disclosure on accredited entity's website | The ESIA aspects are included in the ESMF reports disclosed asper the section below |
| Language(s) of disclosure | [ English and French] |
| Explanation on language | [French is the official language of the 3 countries] |
| Link to disclosure | Please see links in ESMF section below |
| Other link(s) | Click here to enter a date. |
| Remarks | Confirmation that the ESIAs and the ESMFs are consistent with the requirement for a Category A programme is subject to further due diligence on the Funding Proposal |

| Environmental and Social Management Plan (ESMP) (if applicable) | |
|---|---|
| Date of disclosure on accredited entity's website | 22 June 2023 |
| Language(s) of disclosure | English and French |
| Explanation on language | Translation of French version disclosed on national systems |
| Link to disclosure (at Country level) | For Guinea:<br><br>https://www.adazguinee.com/en/_files/ugd/9edb0c_ccc0a34abb174776a6612149837cbff4.pdf<br><br>https://www.adazguinee.com/en/_files/ugd/9edb0c_310041339ea1424385d4a3e07d78f2b8.pdf<br><br>For Senegal:<br>EESS2023.pdf (agropole.sn)<br>ESMF2023.pdf (agropole.sn)<br><br>For Togo:<br><br>https://agriculture.gouv.tg/wp-content/uploads/2023/06/TOGO_ESMF_SCPZ_updated_Version_English_June_2023Submitted22June.pdf<br><br>https://agriculture.gouv.tg/wp-content/uploads/2023/06/TOGO_CGES_AGROPOLE-KARA_Version-actualisee_Francais_22-Juin-2023.pdf |
| Other link(s) Disclosure on AE's site | For Guinea:<br><br>En: https://www.afdb.org/en/documents/guinea-special-agro-industrial-processing-zones-development-programme-pdzta-bk-p-gn-aag-004<br><br>Fr : https://www.afdb.org/fr/documents/guinee-programme-de-developpement-de-zones-de-transformation-agro-industrielle-de-boke-et-kankan-pdzta-p-gn-aag-004<br><br>For Senegal:<br><br>En :https://www.afdb.org/en/documents/senegal-south-agro-industrial-processing-zone-project-pzta-sud-p-sn-aag-004 |

| | |
|---|---|
| | Fr : https://www.afdb.org/fr/documents/senegal-projet-de-zone-de-transformation-agro-industrielle-du-sud-pzta-sud-ou-agropole-sud-p-sn-aag-004-0<br><br>For Togo<br><br>En : https://www.afdb.org/en/documents/togo-agro-food-processing-zone-project-pta-togo-p-tg-aag-002<br><br>Fr : https://www.afdb.org/fr/documents/togo-projet-de-transformation-agro-alimentaire-du-togo-pta-togo-p-tg-aag-002 |
| Remarks | Confirmation that the ESIAs and the ESMFs are consistent with the requirement for a Category A programme is subject to further due diligence on the Funding Proposal |
| **Environmental and Social Management (ESMS) (if applicable)** | |
| Date of disclosure on accredited entity's website | Click here to enter a date. |
| Language(s) of disclosure | [__] |
| Explanation on language | [__] |
| Link to disclosure | [__] |
| Other link(s) | [__] |
| Remarks | [__] |
| **Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)** | |
| Description of report/disclosure on accredited entity's website | Not applicable |
| Language(s) of disclosure | [] |
| Explanation on language | [] |

| | |
|---|---|
| Link to disclosure | |
| Other link(s) | [__] |
| Remarks | [__] |
| **Disclosure in locations convenient to affected peoples (stakeholders)** | |
| Date | Thursday, June 22, 2023 |
| Place | National websites of the Executing Entities of the three countries and on the AE's website as per above. |
| **Date of Board meeting in which the FP is intended to be considered** | |
| Date of accredited entity's Board meeting | Tuesday, November 28, 2023 |
| Date of GCF's Board meeting | Tuesday, October 10, 2023 |

**Note: This form was prepared by the accredited entity stated above**

# Secretariat's assessment of FP219

| | |
|---|---|
| Proposal name: | Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains |
| Accredited entity: | African Development Bank |
| Country/(ies): | Guinea, Senegal and Togo |
| Project/programme size: | Large |

## I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

| Strengths | Points of caution |
|---|---|
| Significant impact potential with large amount of greenhouse gas (GHG) emissions reduction and substantial number of beneficiaries in the three target countries, in the context of increasing food insecurity. | Financial management capacity assessments (FMCA) for some Executing Entities (EEs) have been found to be less than satisfactory and recommendations from the FMCA will be a condition prior to first disbursement. The accredited entity (AE) will be obliged to directly support the capacities of EEs and undertake a periodic review of their financial management capacity. |
| Proposed end-to-end value chain approach that blends support to sustainable production with market access, value addition and livelihood diversification. | Full concessionality to the end beneficiaries is demonstrated but the financial exit strategy should be strengthened further during implementation to allow continuation and scaling up of programme activities. |
| Boosting agricultural transformation as part of the Feed Africa Initiative, with great potential for scale up and replication in other African countries, as well as technology driven paradigm shift potential. | |
| Significant sustainable development potential, reducing degradation on 40,000 ha of farmland and increasing soil fertility on 31,000 ha; expected 50% increase in yields and 40% increase in income; and enhanced food and water security. | |

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the respective term sheet and addendum XXI, titled "List of proposed conditions and recommendations."

## II. Summary of the Secretariat's assessment

## 2.1 Project background

3. In Guinea, Senegal and Togo, the agricultural and land use, land use change and forestry sectors are the two main drivers of GHG emissions. The two sectors account for more than 80 per cent of the GHG emissions in the three countries. These three countries have also been suffering from the impacts of climate change, including increasing temperature and rainfall variability, which are putting rain-fed agriculture at risk.

4. The program, Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains, aims to reduce carbon emissions and increase resilience within the agricultural value chains in these three countries in Africa. Staple crops processing zones (SCPZs) are agriculture-based spatial development initiatives to concentrate agro-processing activities and boost agricultural production in sustainable ways.

5. The project will contribute to a direct reduction in carbon emissions of 678,155 tonnes of carbon dioxide equivalent (t $CO_2$ eq) per year – resulting in a reduction of 16,953,882 t $CO_2$ eq over the 25-year project lifespan – through the deployment of low-emission water management schemes, biogas technologies, climate resilient agriculture (CRA) and agroforestry systems. The project will also bring important adaptation benefits by contributing significantly to the reduction in vulnerability and increase in adaptive capacity of about 1,104,728 direct beneficiaries (54 per cent of them women) and 5,612,416 indirect beneficiaries (54 per cent of them women), including the most vulnerable farmers in the countries.

6. The project area will cover agro-industrial parks in Guinea, Senegal and Togo:

(a) In Guinea, the programme includes two agro-industrial parks located in the Boké and Kankan Regions covering 110,000 ha;

(b) In Senegal, the programme targets the Casamance natural region, comprising the administrative regions of Ziguinchor, Sédhiou and Koldaqui, covering 105,000 ha; and

(c) In Togo, the programme targets the Kara Region, which consists of seven prefectures, namely Assoli, Bassar, Bimah, Dankpen, Doufelgou, Kéran, and Kozah, covering 165,000 ha.

7. The total project financing cost is USD 271,703,525, with a request for GCF grant financing of USD 75,791,156 and loans of USD 26,999,832. The African Development Bank (AfDB) is providing both grants and loans of USD 10,140,378 and USD 85,622,259, respectively. Additionally, the Korea Fund is contributing USD 5,000,000 in grants; the West African Development Bank (BOAD) is providing USD 17,600,000 in senior loans; the Islamic Development Bank (IsDB) is providing USD 31,063,890 in senior loans; and the Governments of the three countries will be contributing a total of USD 19,486,009.

8. The project is submitted under environmental and social safeguards (ESS) category A.

## 2.2 Component-by-component analysis

*Component 1: Strengthening the resilience of critical SCPZs value chain infrastructure (total cost: USD 141.9 million; GCF cost: USD 43.6 million, or 31 per cent)*

9. Component 1 will invest in improving the climate resilience of critical infrastructure in the agricultural value chains of the three target countries. GCF funding will be used twofold: (i) to improve access to drip irrigation distribution systems powered by solar pumps to support CRA practices; and (ii) to increase access to renewable energy technology for the processing of staple food crops and for clean cooking. GCF funding will complement the baseline investment, financed by AfDB and the co-financiers, that will create agroprocessing hubs and agricultural

transformation centres that will in turn establish various infrastructure and facilities to boost agricultural production in the target countries.

10. The component presents clear justification for GCF additionality in bringing adaptation and mitigation results. The solar-powered irrigation systems are estimated to reduce emissions over a productive area covering 39,179 ha, while benefitting 475,068 smallholder farmers by improving access to water. The off-grid solar-powered agricultural equipment for processing and harvesting is also expected to reduce the existing high crop losses and build resilience through diversified livelihoods. In addition, the promotion of biogas digesters will reduce the overdependence on charcoal and firewood as energy sources for rural households. The proposed activities are assessed to be adequate and reasonable, and are aligned with GCF objectives and results areas.

11. The component requests a combination of loans and grants from GCF in the ratio of 25 per cent and 75 per cent, respectively. Low concessional loans will be absorbed by the three sovereign states and only the grants will be provided to the end beneficiaries, who are the grassroot smallholder farmers. While the justification for higher grants is acknowledged, it is advisable to develop business models for service providers, especially with energy service companies (ESCOs) and farmers, to ensure financial profitability in the long term as an exit strategy.

*Component 2: Promotion of climate resilient agricultural practices and technologies adoption among smallholder farmers (total cost: USD 92.9 million; GCF cost: USD 59.2 million, or 64 per cent)*

12. Component 2 will focus on promoting CRA practices and support their widescale adoption. In this component, the GCF funding will be focused on responding to the identified climate hazards, exposure and vulnerability of smallholder agriculture in the targeted areas. The broad range of proposed CRA practices and technologies will respond to specific local conditions and include drought-tolerant varieties, crop rotation, minimum tillage techniques, mixed cropping, agroforestry, and integrated nutrient management, which are all well-known measures recognized for their adaptation benefits in the African context. These CRA solutions are well aligned with the GCF approach to building the resilience of the agricultural sector, particularly with the first paradigm-shifting pathway indicated in the sectoral guide for agriculture and food security: promoting resilient agroecology.

13. The uptake of CRA by local farmers would require appropriate training and incentives to support them in realizing tangible economic benefits from resilient agriculture. As such, these activities will be supplemented with awareness raising, capacity-building and institution-building activities for value chain actors. In addition, as in component 1, the programme will support the development of business models to strengthen financial sustainability in the long term and solidify the adoption of CRA solutions. Finally, the programme will also support the availability of better climate services for the agriculture sector in the targeted areas by closing gaps in the climate information and early warning systems in the three countries. These interventions are aligned with the GCF paradigm-shifting pathway on facilitating climate informed advisory and risk management services, as contained in the agriculture and food security sectoral guide.

14. Similar to component 1, the component requests a combination of loans and grants from GCF in the ratio of 25 per cent and 75 per cent, respectively. It is estimated that over 70 per cent of the grants will be dedicated to covering the incremental costs of mitigation and adaptation activities, especially the proposed CRA approaches. The expected impacts are considered significant and entail substantial adaptation benefits as well as support for smallholder farmers' livelihoods. The proposed activities are assessed to be adequate and reasonable to support local communities while contributing to reducing emissions from agricultural and forestry activities.

# III. Assessment of performance against investment criteria

## 3.1 Impact potential *Scale: High*

15. The project has significant impact potential as a mitigation project, with 0.7 Mt $CO_2$ eq in annual emission reduction and 16.9 Mt $CO_2$ eq over the 25-year project lifespan. These reductions will be mainly achieved through the adoption of solar photovoltaics and biogas in the agricultural value chains, as well as carbon sequestration from CRA and improved agroforestry practices. The programme's baseline investment involves road construction and rehabilitation, which may generate emissions. Such emissions have been estimated by the AE and were found to be minor. The programme should carefully monitor and report all possible GHG emissions to be generated from the baseline programme activities, and take appropriate measures to minimize negative environmental externalities.

16. The project's avoided emissions and carbon sequestration were calculated using the clean development mechanism methodology for solar installations as well as a combination of existing literature and country-specific data from the Food and Agriculture Organization of the United Nations and the Intergovernmental Panel on Climate Change regarding biogas digesters, climate-resilient crops and others. Activity data and emissions factors were found to be conservative (e.g. using an emission factor of 0.64 t $CO_2$ eq/MWh instead of the standard 0.8 t $CO_2$ eq/MWh).

17. The project will have substantial adaptation and livelihood improvement benefits for 1,104,728 direct beneficiaries (54 per cent of them women) and 5,612,416 indirect beneficiaries (54 per cent of them women), who will be supported through the end-to-end, farm to fork programme interventions. Livelihood resilience will be enhanced through an integrated approach that includes access to climate services and capacity development, better climate-responsive practices and technologies, improved energy and water access, as well as diversification and value addition.

18. The AE asserts that, compared to similar projects, conservative estimates were used to calculate the direct and indirect beneficiaries. The calculation of indirect beneficiaries used 5 to 15 per cent of each country's total population, depending on the size and number of agro-industrial parks in each country. As access to markets is key to achieving the objectives of the programme, the programme should ensure that all target beneficiaries – such as smallholder farmers, agricultural cooperative societies, farm-based associations, and agribusiness enterprises – will have full access to the markets and yield benefits from their produce.

## 3.2 Paradigm shift potential *Scale: High*

19. The programme was designed as an integrated initiative to build the resilience of staple crop value chains from farm to fork. Through this approach, the programme will not only build the resilience of the production part of these food systems, but also promote market access, value addition, processing and reduction of food losses as mechanisms to generate a long-term, sustainable transition of agriculture in the targeted areas.

20. Potential for scaling up and replication: The programme was designed to boost agricultural transformation as part of the Feed Africa Initiative, with great potential for scale up and replication in other African countries. The first phase of the Initiative is currently being piloted in Guinea, Senegal and Togo, with the expectation of scaling up to and replication in other AfDB regional member countries (RMCs). To date, preliminary field visits have been conducted in Côte d'Ivoire, Nigeria, Senegal, the United Republic of Tanzania, and Zambia, at the request of the respective Governments, for the implementation of phase II of the SCPZ

programme in these countries. Hence, the success of phase 1 is highly crucial for the scaling up and replication of the programme in other RMCs.

21. A number of strategies were built into phase I of the programme with respect to scaling up and replication in other RMCs of AfDB. Firstly, the programme was designed to train a "critical mass" of project expects (i.e. training of trainers) across all programme interventions, especially for the GCF-financed components that can easily be scaled up and replicated in other regions and countries. This critical mass of experts will be drawn upon to provide hands-on experience as international consultants during the implementation of phase II. Furthermore, for each activity financed by GCF under phase I, a comprehensive training manual will be prepared and developed to facilitate the replication of similar activities especially in phase II of the programme. Finally, several training workshops have been factored into the programme to facilitate knowledge sharing among value chain actors.

22. Contribution to the creation of an enabling environment: The programme will bring knowledge resources and capacity-building to the private sector, particularly potential agribusiness investors. This will help in creating an enabling environment for crowding in additional private sector financing into agriculture. Equally, the project will strengthen the technical capacity of key government agencies, universities, local non-governmental organizations and other key stakeholders on climate-resilient practices and low carbon emission technologies, which will deliver both adaptation and mitigation benefits. This will provide the necessary critical mass of technical personnel that will enable the sustainability of the technology adoption even after the completion of the project. Also, the access of women to finance to invest in CRA practices has been identified as a major impediment to the growth of women-led agribusiness ventures in the target countries. With affordable access to finance to invest in CRA practices, women will be empowered to act as key change agents for climate change adaptation in these vulnerable communities.

## 3.3 Sustainable development potential *Scale: High*

23. Environmental co-benefits: In addition to the above-mentioned reduction of 16.9 Mt $CO_2$ eq of GHG emissions during the programme lifespan, the programme will support the reduction of degradation of 40,000 ha of farmland and increase in soil fertility of 31,000 ha. Other environmental benefits will include: (i) soil conservation and reduction of erosion; (b) improved tree cover in farmlands, catchment areas and other components of the cascade ecosystems (such as silt barriers, salt/mineral balancing areas, etc.) that will have several interlinked environmental benefits such as improved microclimate, improved soil structure, bioremediation for irrigation and drinking water quality, and increased biodiversity; (iii) restoration of ecosystem integrity, goods, and services; and (iv) preservation of biodiversity in farmlands and agroforests.

24. Economic and social co-benefits: The programme is well developed to generate a wide range of economic and social co-benefits, including improved incomes and reduced poverty, improved food security and nutrition, employment generation, opportunities for women empowerment, as well as general livelihood improvements from energy access and clean cooking solutions.

## 3.4 Needs of the recipient *Scale: High*

25. The vulnerability of the countries and the target beneficiaries and their adaptation needs are generally explained throughout the proposal, but without detailed descriptions of the scale and intensity of the countries' exposure to climate risks. The impacts of climate change, particularly incidents resulting from heat stress and rainfall variability, on agricultural

production and livelihoods are presented. Moreover, the vulnerability of smallholder farmers, including women, especially with respect to their food insecurity and undernourishment is acknowledged in the funding proposal.

26. All three target countries are least developed countries with high levels of indebtedness and budget constraints in covering and sustaining climate change adaptation. The proposal highlights the lack of public infrastructure and social services, including roads, markets, education and so on, as the major vulnerability factors in the countries. The proposal also notes the lack of access to finance as the constraint on building the capacity of producers along the agricultural value chains.

27. For each country, the proposal explains the needs of the recipients based on the national fiscal situation and investment gaps. Although such justification is acknowledged, it could have been strengthened by providing a wider investment strategy, including a financial estimation of the cost of implementing climate change adaptation and mitigation measures in agriculture at the national level, and the associated shortfalls in the national budget. Such an exercise would have provided a clear road map or pathway for climate resilience building in the sector from a financial perspective, and would clarify the level of contribution from GCF.

## 3.5 Country ownership *Scale: High*

28. The overall assessment of this programme's country ownership is high.

29. The proposal demonstrates that the project contributes to the countries' nationally determined contributions and its various national development and climate change strategies. The proposal highlighted in a high-level manner how and to what extent the programme will contribute to the target countries' resilience building in the agriculture sector and in line with the nationally determined contributions, yet more specific linkages could have been made to allow for better tracking. The project was designed on the basis of rigorous consultations with relevant stakeholders, including the GCF national designated authorities in the participating countries. A stakeholder engagement plan was submitted as an annex to the funding proposal. Finally, the countries' interest in this programme has been maintained and reconfirmed, even when accounting for the long programme development timeline.

## 3.6 Efficiency and effectiveness *Scale: Medium to high*

30. The overall assessment of this programme's efficiency and effectiveness is medium-high.

31. The funding proposal package presents a financial analysis for two streams of project activities: (i) low carbon energy-related activities (solar-powered irrigation and electricity from biogas) and (ii) agroforestry activities. The analysis contains an estimation of the major financial values from GCF-financed activities. However, it does not provide a complete economic analysis with all the costs and benefits of the programme presented in the funding proposal, particularly on activities financed by co-financiers.

32. The analysis includes a sensitivity analysis, which calculates financial results using three different discount rates: (i) the low concessional loan pricing of GCF with 0.75 per cent interest rate; (ii) 12 per cent social discount rate; and (iii) the private sector ESCOs rate of 20 per cent. The results show all positive net present values (NPVs) and internal rates of return (IRRs) regardless of the discount rates used for agroforestry, and mixed results for energy components. The AE explained that the 12 per cent social discount rate was the standard rate it used for its renewable energy projects in Africa, and is considered to be an appropriate value in capital-scarce countries in the African region.

33. While the financial model presents a cash flow analysis with the major GCF-financed activities, the model could have been improved in several areas. These areas concern underlying data and assumptions used, including the lack of country-specific field data per type of intervention for estimating costs. Also, the valuation of anticipated economic benefits could help in estimating the impacts of the programme, including those financed by co-financiers. In addition, comprehensive sources could strengthen the model's validity and justification, especially with detailed references for all the underlying data.

34. The financial adequacy and appropriateness of concessionality is assessed to be medium. The programme requests high concessional loans from GCF, although the financial model results show positive returns with low concessional loan terms for the analysed activities. The request for high concessionality is rather justified given the insights of the qualitative assessment carried out during the appraisal stage of the programme, which considered the high indebtedness of the three target countries as well as the impacts of COVID-19 on these economies. The high concessional loan from GCF is also sovereign-backed as it would leverage private sector investment and stimulate low-carbon and CRA value chains. Therefore, the need for concessionality is justified.

35. The cost-effectiveness is assessed to be high. On the mitigation side, GCF funding would cover USD 2.65 per t $CO_2$ eq over the 25-year programme lifetime, considering the mitigation portion (44 per cent) of the GCF funding. This is significantly lower than the price range of USD 40–80/t $CO_2$ eq to meet the Paris Agreement target. Adaptation cost efficiency is also high, with USD 8.5 per beneficiary, considering the adaptation portion (56 per cent) of the GCF funding. This cost per beneficiary is in the low range of the GCF portfolio in the agricultural sector in Africa.

36. The co-financing ratio is 1:1.64, which is slightly lower than the target in the GCF updated Strategic Plan 2020-2023. However, a significant amount of co-financing has been included in the programme from various co-financiers in different financial instruments. The programme shows high potential for mobilizing additional public and private resources through stimulating and greening the agricultural value chains for attractive commodities. The programme should ensure that the high concessionality from GCF does not crowd out private sector financing, including from the ESCOs in the target regions.

# IV. Assessment of consistency with GCF safeguards and policies

## 4.1 Environmental and social safeguards

37. **Programme background.** The programme is a multi-country programme that aims to reduce climate change vulnerability and GHG emissions within the agricultural value chains in the target countries. It aims to strengthen the value chains infrastructure and governance systems, and promote CRA practices and infrastructure as well as the adoption of low-carbon energy technologies such as biodigesters and solar photovoltaic systems. The expected environmental co-benefits of the programme include increased biodiversity due to reduced farmland degradation from tree planting activities; improved soil fertility as a result of implementing soil conservation and erosion reduction approaches; and increased groundwater quality and quantity as a result of improved percolation and reduced use of agrochemicals. The social co-benefits include increased access to electricity from renewable energy sources, thereby improving the health conditions of beneficiaries, and improved food security and nutrition as a result of the availability of staple crops year-round.

38. **Environmental and social risk category and safeguard instruments.** The AE has classified the programme as category 1 (which is equivalent to GCF category A) due to its magnitude and scale as well as proximity of some of the sites to environmentally and socially

sensitive ecosystems and settlements. The programme would also have diverse impacts that would trigger the operational safeguards (OS) of the AE. The AE prepared a country-level strategic environmental and social assessment as well as environmental and social management frameworks (ESMFs) as the sites, technologies and designs of some component subprojects have not been fully identified yet.

39. **Compliance with GCF ESS standards.** The paragraphs below summarize the programme's compliance with GCF ESS standards and requirements.

40. **ESS 1: Assessment and Management of Environmental Risks and Impacts.** The ESMFs presented the planned interventions under the programme and their general locations, and identified their likely significant positive and negative impacts. Strategic and general environmental and social management measures have also been identified. Key country policies, laws, and regulations relevant to the programme, along with the operational safeguards of AfDB and the ESS standards of GCF, were also presented. The ESMFs indicated that each subproject under the programme will be subjected to a selection/screening process and the required environmental and social assessments based on the environmental impact assessment system of the host country. They also outlined the detailed steps involved from screening to evaluation of impacts. Additionally, the ESMFs identified the general management/mitigation measures for significant impacts and risks; outlined the process for assessing and managing subproject/activity-specific impacts and risks; and provided for the institutional arrangements and capacity-building for the implementation of safeguard measures. The component activities of the programme will also be subjected to the countries' environmental impact assessment requirements.

41. **ESS 2: Labour and Working Conditions**. The programme will involve significant construction activities in the selected agro-processing zones, including the support infrastructure (i.e. power, roads, etc.). The operations of the processing plants and associated facilities will also require staff and workers. Although labour and working conditions is covered under Operational Safeguard OS5 (Labour conditions, health and safety) of the AfDB, there is a need to ensure that the risk of violation of workers' rights and other labour management issues is addressed. The environmental guidelines for contractors and environmental clauses in contracts should include provisions for compliance not only on occupational health and safety but also on labour management standards. There is also a need to ensure that the assessments of subprojects/component activities will cover risks of violation of worker rights in the facilities, including gender-based discrimination, exploitation and endangerment of children, among others. Considering the potential for labour issues (e.g. child labour/forced labour) in the solar photovoltaic supply chain, the AE should ensure that these issues are avoided and/or mitigated in the implementation of the programme and that measures are incorporated in the management plans for each of the target countries. The approaches of the AE should include the development and implementation of a workforce management plan; the inclusion of environmental and social clauses, including health and safety standards at work and a code of good conduct to be signed by all workers, in the contracts with companies in charge of the works; and ensuring that there are no child/forced labour issues throughout programme implementation.

42. **ESS 3: Resource Efficiency and Pollution Prevention**. The requirements under ESS 3 are covered under Operational Safeguard OS4 (Pollution Prevention and Control, Hazardous Materials and Resource Efficiency) of the AE. The ESMFs provided brief background information on the pollution control laws and regulations and the prevailing pollution and resource efficiency issues in the programme countries. Existing pollution issues in the regions covered by the programme include salinization of groundwater due to saltwater intrusion (Senegal and Guinea); sedimentation/silting of streams and valleys due to natural (water and wind) erosion, erosive farming practices (slash and burn) and from mining; water pollution from industrial activities and mining (Guinea); inadequate waste management (Guinea) and lack of sanitation

facilities; air pollution in urban areas due to vehicular and industrial emissions and burning of wastes; and threat to soil and water from use of fertilizers and pesticides. The potential negative impacts relating to ESS 3 are mainly from the construction and operation of infrastructure such as roads, solar and biogas power generation, irrigation, buildings, and agricultural produce handling/processing facilities as well as the establishment and maintenance of agricultural farms, agroforestry and silvicultural farms, waste management systems, and community forests.

43. In terms of resource efficiency, the programme is generally seen to have positive impacts, particularly on the use of water and energy. Efficient use of water will be maximized with the promotion of drip irrigation technology among smallholder farms. The programme will also utilize renewable energy (solar and biogas) sources. Although the programme will include the development of underground water boreholes, which are sensitive to drought, it will also involve the development of small dams and reservoirs, which could increase water supply and improve infiltration into aquifers.

44. Potential water pollution will be in the form of sedimentation from earth moving activities during construction as well as contamination of surface and groundwater by agrochemicals from agricultural runoff and liquid discharges of crop/food processing facilities and biodigesters. Potential air pollution will be in the form of dusts during construction of infrastructure and from air emissions from crop/food processing facilities in the SCPZs. Construction activities could result in short-term impacts on the air and water quality due to generation of dusts during dry days and sedimentation of water channels during rainy days. This could also be experienced in the quarry sites. In the long term, the expansion of agricultural areas may likely worsen current problems of soil erosion. The management plans should ensure that measures to control dusts and sedimentation from construction activities will be adequately implemented. The operations of processing facilities will likely generate wastes (e.g. rejected damaged fruits, trimmings, husks, washings, animal wastes and effluents coming from biogas digesters). If not properly handled and/or treated, these wastes may contaminate surface and groundwater and could generate objectionable odour nuisance to nearby communities. This should likewise be appropriately mitigated during implementation. It is very likely that the use of pesticides will increase under the programme given the prevalence of pests and diseases on some of the crops and trees promoted by the programme. Hence, the programme should ensure that an integrated pest management approach will be implemented in the control of pests and diseases and should train growers on proper handling and storage of pesticides.

45. The ESMFs provided measures on potential pollution impacts. However, it is understood that each subproject under the programme will have to be screened and assessed in terms of these issues. It is therefore important that the screening and assessment of subprojects under the programme include issues on resource efficiency and pollution.

46. **ESS 4: Community Health, Safety and Security**. Construction related safety issues, including increased risk of traffic accidents, potential gender-based violence (GBV), and spread of sexually transmitted diseases, could also occur and have been addressed by the programme's ESMF. The risk of outbreak of vector-borne diseases was also identified as a risk as the programme areas are likely to experience influx of labour and economic migrants. Some of the measures envisioned include combating waterborne diseases through intensive information, education and communication campaigns, sensitization of populations on malaria prevention measures (i.e. treated mosquito nets), and awareness raising activities. The ESMFs also identified the possibility of social conflicts as one of the potential risks in the target countries. The AE has indicated that measures – such as developing training modules that consider the prevention and management of various conflicts (including land conflicts between farmers and livestock breeders), and conducting a more detailed resilience and fragility assessment during the implementation stage – will be implemented. These will be conducted as part of the environmental and social assessment of the subprojects.

47. **ESS 5: Land Acquisition and Involuntary Resettlement**. The construction of various infrastructure, such as processing facility complexes as well as the support infrastructure, would likely require land acquisition. The ESIA assessment identified "expropriation and involuntary displacement of populations" as one of the major potential impacts/risks of the programme. The impacts of land acquisitions for the programme were identified in the ESMFs and may include economic displacements due to loss of land or loss of access to land or access to land-based resources such as forests and grazing areas, leading to loss of traditional means of livelihood. Physical displacements (i.e. loss of dwellings) are deemed less likely especially in Guinea due to the sparsely populated programme areas. The ESMFs indicated that individual subprojects under the programme will be screened for involuntary land acquisition and shall be required to develop resettlement action plans should there be land acquisition and involuntary resettlement concerns.

48. **ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources**. The programme safeguard documents noted the significance of some regions (e.g. the Casamance region) in terms of forest and natural habitat for wildlife. They also noted that the fragile environments of these regions were constantly under pressure from both natural and anthropogenic factors, including deforestation due to clearing for crops and orchards, logging and fuelwood gathering. The construction of processing facilities and support infrastructure and the development of community forests could also result in a loss of natural vegetation cover. In the long run, the programme may encourage the expansion of orchards and cultivated areas, which may put further pressure on the remaining forests and natural habitats. Some of the mitigating measures identified include reforestation to compensate for lost trees, promotion of intensive organic farming, awareness campaigns, and avoidance of wildlife habitats during construction and rehabilitation of temporary quarries. The ESMFs have identified the following as mitigation measures: (i) establishment and management of community forests; (ii) compensatory reforestation and reclamation; (iii) restoration and protection of the natural habitat around development and other infrastructure; and (iv) protection of sensitive areas. A further detailed environmental and social assessment should be conducted and appropriate mitigation measures should be put in place to address potential risks on land degradation and habitat loss that may result from project implementation.

49. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples**. Indigenous Peoples may be negatively affected by the project and, accordingly, the AE has included country-level Indigenous Peoples Planning Frameworks (IPPFs) or its equivalent in the ESMFs. While the addition of IPPFs to the ESMFs meets the basic requirements of the Indigenous Peoples Policy and guides the preparation of Indigenous Peoples Plans (IPP), the following recommendations are provided to cover remaining gaps. As per the Indigenous Peoples Policy, an IPP should be prepared when there are impacts to Indigenous Peoples and not just where there is acquisition of land from Indigenous populations/local communities. In annex 25 of the funding proposal, titled “Country Resilience and Fragility Note for Guinea, Senegal, And Togo”, the high-level analysis raises several flags and reinforces the ESMF conclusion that Indigenous and vulnerable groups have limited rights and opportunities. This should be addressed in resulting environmental and social management plans and IPPs. With no analysis at the ESMF and IPPF levels on intercommunity conflict, the resultant IPPs should include robust conflict sensitivity analyses, particularly between pastoralists and farmers, with accompanying mitigation measures and recommendations. The AE is recommended to ensure that consultations are undertaken, including with pastoralist and other Indigenous and vulnerable groups in the countries, to ensure their inclusion in the locations where the programme will be implemented. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE and the EEs.

50. **ESS 8: Cultural Heritage.** Cultural heritage is covered under Operational Safeguard OS1 (Environmental and social assessment) of the AfDB. The safeguard documents allow for identification of any cultural heritage sites in the programme areas that are at risk of being

adversely affected by the programme during construction as well as provide for mitigation measures should this be encountered. Such measures include the stipulation in agreements for contractors to immediately notify the relevant authority in the country in case of discovery of sites or assets of cultural significance. Where the proposed subproject location is in an area where tangible cultural heritage is likely to be found, chance find procedures will be incorporated into the management plan.

51. **Implementation arrangements**. Various arrangements have been established to manage the implementation of mitigation measures in the target countries. At the country level, the Program Implementing Unit, which will be established under each EE, will be responsible for the establishment of the environmental and social management functions of the programme. The Program Implementing Unit will recruit environmental and social safeguard experts who will ensure the implementation of the management plans, coordinate and monitor the environmental and social aspects of the programme, and interface with other stakeholders involved in the safeguards such as the country's environment agencies (e.g. the Guinean Office of Audit and Environmental Compliance Agency in Guinea, the Environmental and Classified Establishments Division in Senegal, and the National Agency for Environment Management in Togo), contractors and private investors. The execution of specific environmental and social management measures will be the responsibility of the contractors and private investors, who will be required to appoint their own environmental and social safeguards focal persons.

52. **Stakeholder engagement and grievance redress mechanism**. The programme underwent a series of consultations during the preparation of the proposal. The public consultation involved various stakeholder groups and organizations. While the safeguard documents provide for stakeholder consultations, the AE also indicated the development of a stakeholder engagement plan during programme implementation as part of the individual subproject environmental and social assessment process. In addition to the existing grievance redress mechanism of the AE, the programme also provides for the establishment of a complaints management mechanism and outlines procedures for handling complaints. It also outlines the key features of the mechanism, which include accessibility, handling and processing of complaints, responding to the complainants, and monitoring and evaluation. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim is made.

53. **On sexual exploitation, abuse and harassment (SEAH)**. The revised GCF Environmental and Social Policy, adopted by decision B.BM-2021/18, requires safeguarding from SEAH in GCF-financed activities. The AE included SEAH safeguards in the three ESMFs for Guinea, Senegal and Togo, and also reflected some of the GBV/SEAH risks and mitigation measures in the gender assessment and gender action plan (GAP), respectively. The Guinea ESMFs recognize the risk of GBV/SEAH due to the presence of foreign workers on the sites, which may lead to risks of SEAH in relation to the beneficiary communities. Considerations will also be made to ensure that decision-making and operations are effectively managing ESS risks, which include SEAH, and ensure the protection/prevention of the same to all persons that will be impacted by the programme. In Togo, the programme will facilitate the development and delivery of SEAH trainings, and conduct information/awareness sessions. In Senegal, the South Agropole Project will set up a system for reporting, receiving, recording and processing complaints relating to GBV/SEAH, and SEAH cases will be managed by competent SEAH and/or gender experts. The Secretariat recommends that the AE ensure that SEAH complaints, are handled confidentially as reported in the submitted ESMF. The grievance redress mechanisms should be as gender responsive and survivor centred as possible, and the monitoring and reporting mechanisms should be strengthened. Overall, the SEAH safeguarding framework of the AE aligns with the GCF Revised Environmental and Social Policy SEAH provisions and requirements.

## 4.2 Gender policy

54. The AE has provided a gender assessment and GAP, and therefore the proposal complies with the requirements of the GCF Gender Policy.

55. The gender assessment is based on a literature review of available information and consultations on the programme in the beneficiary countries on ESS matters from as early as 2017. It is not clear whether gender issues were discussed in the consultations or how stakeholder engagement undertaken during the preparation of the programme was used to identify the needs and priorities of both men and women.

56. The gender assessment examines the overall gender context within Guinea, Senegal and Togo, where the programme will be implemented. The legal, policy and regulatory framework for promoting gender equality is presented for all beneficiary countries. All countries have established institutional mechanisms, including national ministries for women's affairs in the case of Guinea and Togo, with mandates for supporting the implementation of policy frameworks for the advancement of gender equality and women's empowerment. Senegal has two directorates with similar mandates – one for equity and gender equality and another for family affairs with a division for women.

57. The impacts of climate change have detrimental effects on agriculture, including limiting the capacity to produce food from staple crops. Traditions and cultural barriers negatively affect women by limiting their access to resources, such as land and water, and participation in decision-making. For instance, in Togo, female-headed households have smaller plots than those headed by men. They also have less access to water for irrigation, which limits their capacity to provide food for their families, thereby increasing their vulnerability to climate change impacts. The programme intends to increase the resilience of beneficiary communities to climate change impacts by providing agricultural inputs and improving access to markets and modern processing equipment to reduce losses and enhance production and incomes. Women are often responsible for gathering water and fuel in traditional agrarian societies, and these tasks are becoming more laborious, challenging and time consuming because of longer distances they have to travel due to the impacts of climate change. Most of the populations in the beneficiary countries depend on rain-fed agriculture for daily subsistence.

58. The programme-level GAP contains activities aligned with the programme's components, including indicators with baseline figures and targets, and responsibilities. Sex-disaggregated targets for smallholder women farmers and female-headed households are included as well. The GAP responds to some of the issues identified in the gender assessment and includes activities that will provide access to programme benefits – such as improved access to water for irrigation, access to finance, access to solar energy for processing agricultural produce, and training in CRA – for both women and men, including female-headed smallholder farming households.

59. Implementation arrangements include having a gender expert in the project implementation unit of each target country. The gender experts are expected to monitor, evaluate and report on the implementation of the GAP by programme partners. Financial resources from the overall programme budget have been allocated for the implementation of the GAP.

60. The AE defines women-led enterprises as businesses that are majority owned by women, led by women, or have a significant portion of women on their board. The AE is recommended to establish the percentage of women that own or sit on the board of an enterprise, and this will be used as the criteria to identify that an enterprise is women-led.

61. The GAP could be improved by including additional activities that can be implemented by the programme. The AE is recommended to revise the action plan accordingly. For example, improvement in incomes of smallholder farmers, agricultural cooperative societies/farmer-

based associations, and women-led agribusiness enterprises could be achieved through improved access to markets for agricultural produce (with corresponding indicators and targets to measure actual improvements); access to biogas energy technologies; decrease in losses of agricultural produce owing to improved post-harvest processing; and training activities. Women empowerment in agricultural value chain development – through awareness raising, training courses and capacity-building exercises – should be included to ensure that both women and men in managerial positions are fully enabled to carry out their roles and responsibilities in the agro-industrial parks.

62. The AE is recommended to undertake further analyses as well as consultations on gender-related matters with relevant stakeholders, including women. The consultations should be used to identify the needs and priorities of both men and women in the regions where the programme will be implemented. This task should be undertaken prior to commencing implementation. The analyses and findings should then be used to prepare project-level GAPs, and the programme-level GAP should also be updated accordingly and submitted to the Secretariat.

## 4.3 Risks

### 4.3.1. Overall programme assessment (medium risk)

63. The programme will provide concessional financing to agriculture activities that will contribute to reducing emissions and improving resilience of ecosystems and beneficiaries in Guinea, Senegal and Togo. GCF is requested to provide a senior loan of USD 27.0 million to the Governments of the target countries and a grant of USD 75.8 million. The AE shall co-finance a senior loan of USD 85.6 million and a grant of USD 10.1 million. Other co-financers – including the Korea Fund, West African Development Bank, Islamic Development Bank, and the governments of the target countries – are providing senior loans and grants. Total co-financing, including from the AE, will add up to USD 168.9 million.

64. All countries in the programme are least developed countries. The ratio of GCF grant (~75 per cent) and loan (~25 per cent) to be provided will be similar in all target countries and was proposed by the AE based on consultations with the countries. The AE indicated that over 70 per cent of the grants will finance activities with high incremental costs but with significant climate impacts, such as the CRA activities, agroforestry and biogas generation.

### 4.3.2. Accredited entity/executing entity’s capability to execute the current programme (medium-high risk)

65. The AfDB, the AE, has a track record of financing multiple projects in different sectors of the three programme countries. The net commitment the AE has made in each country ranges from UA (African Development Bank Units of Account) 233 million to UA 1.3 billion (~USD 326 million to USD 1.82 billion). The AfDB is rated as “Aaa” by Moody’s.

66. A designated ministry or government agency in charge of trade or industrial development is an EE in each country. A detailed FMCA was carried out by the AE. Apart from the EE in Senegal, the other EEs have challenges due to gaps in full operationalization (Togo) and absence of personnel and processes (Guinea). The assessment of the EE recommended filling these gaps within specified deadlines as condition precedents to the first disbursement from AfDB. The same condition is applied to the first GCF disbursement in the term sheet.

### 4.3.3. Project-specific execution risks (medium risk)

67. ESCO selection: The private sector/ESCOs will be selected by the EE of each country through a competitive bidding process. The private sector providers will add their own equity and debt resources to design, construct, operate and maintain the assets to provide the expected services for agreed user fees. However, the current selection criteria do not include any minimum amount of contribution. In addition, given that the procurement capacity of the EEs is key to selecting the ESCOs, the capacity gaps of EEs may delay the process.

68. Carbon credits: The current term sheet does not include any restriction on the sale of carbon credits. Thus, should there be any, the climate impact achieved by the GCF intervention may be reduced by offset credits monetization or double counting financing.

69. Impact risk: This is a cross-cutting programme with a mitigation impact of 16.9 million t $CO_2$ eq over the programme lifetime of 25 years. However, the final target in the logical framework is 8.5 million t $CO_2$ eq (over a project implementation period of five years), which accounts for about 50 per cent of the total impact. Thus, GCF will not be reported on the full ex-post impact as presented in funding proposal. The Secretariat recommends that the AE extend the impact monitoring and reporting period to cover the impact as presented in the funding proposal.

70. Credit risk: Under the programme, GCF will provide sovereign loans to below investment grade rated entities – Senegal (Ba3), Togo (B3), and Guinea (not rated). The credit profiles of these countries remain weak due to increased liquidity risks and external vulnerability as a result of pandemic-related economic and fiscal shocks. However, the highly concessional loan from GCF with a long-term repayment period reduces the financial burden for the countries.

### 4.3.4. Project viability and concessionality

71. A financial and economic analysis was carried out for various types of activities. As for interventions related to energy, the analysis indicates that, with GCF financing, the negative NPV becomes positive with an IRR ranging from 0.1 to 9 per cent by replacing grid-connected power with biodigesters and solar photovoltaic cells. Some agriculture interventions show high economic internal rates of return (68–99 per cent) over 25 years, considering increased yields from crops (e.g. coffee, mango and cashew with drip irrigation technology) as benefits. The AE considers that concessional financing is required as the project interventions are provided to the most marginalized groups in the three countries who lack access to financial resources to invest in CRA activities.

### 4.3.5. GCF portfolio concentration risk (low risk)

72. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results areas and single proposal is not material.

### 4.3.6. Compliance risk (medium risk)

73. Guinea, Senegal, and Togo are not subject to United Nations Security Council (UNSC) resolutions. The AfDB, as a GCF AE, advised that it is not aware of any individual, executing entity, beneficiary and/or implementing entity that is subject to UNSC sanctions in Guinea, Senegal or Togo.

74. The AfDB confirmed that an integrity assessment – covering anti-money-laundering and countering the financing of terrorism (AML/CFT) and other integrity-related issues – is undertaken as a standard operational practice for all funding operations. This is also the case with fiduciary capacity assessments and credit risks assessments. These assessments have been carried out for the project and the AfDB has incorporated relevant information in the funding proposal and the operational manual.

75. The AfDB has also undertaken a risk assessment of the fiduciary risks and capacities of the beneficiary countries. This assessment is a key document for dialogue with the countries regarding risk ratings and agreed mitigation measures to address noted fiduciary risks, including corruption, AML/CFT and prohibited practices issues. The AfDB has also undertaken a review of UNSC sanctions, security and exchange commissions and financial oversight authorities, as well as a review of politically exposed persons (PEPs) and other high-risk relationships of the technical responsible parties.

76. As a multilateral development bank (MDB), the AfDB has sound internal policies and joint agreements with other MDBs, such as the Agreement for Mutual Enforcement of Debarment Decisions; General Principles and Guidelines for Sanctions; MDB Harmonised Principles on Treatment of Corporate Groups; and Uniform Framework for Preventing and Combating Fraud and Corruption.

77. The AfDB has due diligence guidelines and screening procedures for AML/CFT and provides guidance on mainstreaming AML/CFT measures in its internal operations and activities. The AfDB is applying the same procedures for assessing AML/CFT risks to the SCPZ programme. Screening tools used include the International Transfer Readiness (ITR), Politically Exposed Persons Check Tool, Accuity, LexisNexis, Google Advance Search and other open-source tools.

78. The AfDB confirmed that procurement planning for the SCPZ programme has been undertaken and included an assessment of national procurement frameworks and the capacity of the EEs to ensure efficient and effective procurement of goods and services. The bank readiness review process also ensures that procurement, financial management and disbursement arrangements for the programme are related to the issues and risks identified.

79. The AfDB advised that it will undertake periodic reviews of adequacy for procurement and financial management arrangements specified in the FMCA during on-site implementation missions. This will include a review of procurement and an assessment of its compliance with AfDB procedures.

80. The AfDB confirmed that there is no intention to distribute or disburse cash, vouchers, commodities or other items of value, either directly or indirectly, to beneficiaries. Beneficiaries have been identified and detailed information in this regard is provided in the funding proposal. Controls for fraud and corrupt and prohibited practices will follow the internal policies and procedures the AfDB uses for its operations.

81. The AfDB advised that its Guidelines for Preventing and Combating Corruption and Fraud in Bank Group Operations establishes basic principles, rules and procedures for preventing and combating corruption as well as for ensuring ethical conduct of staff and relevant individuals and institutions involved in funding operations such as the SCPZ project.

82. The AfDB has a Procurement Review Committee, an Oversight Committee on Corruption and a Fraud, and an Ethics Committee that ensure activities are compliant with AfDB rules and policies. Moreover, whistle-blower protection ensures that measures are in place to protect anyone who reveals fraud, corruption or any wrongdoing in AfDB funding operations and to ensure they are safe from reprisals.

83. There are also activities geared towards strengthening internal capacity for countering fraud, corruption and prohibited practices, including increased on-site supervision missions and support to RMCs to enhance their capacity to monitor, report and handle incidents of fraud, corruption and prohibited practices.

84. The Secretariat has reviewed the project in accordance with relevant GCF Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a compliance risk rating of "medium" and has no objection to this proposal proceeding.

85. The ORMC/Compliance Team would like to remind the AfDB, as the AE, of its continuing obligations and responsibilities with regard to monitoring and reporting any risks for money-laundering, terrorist financing, or prohibited practices among the intended counterparties, EEs, beneficiaries, persons involved, or any of the proposed activities.

86. Summary risk assessment and recommendation:

| **Summary risk assessment** | | **Risk assessment** |
|---|---|---|
| Overall project/programme | Medium | The AE assessment of the EEs reveals that two of the three EEs have capacity gaps that need to be improved before first disbursement. The selection of ESCOs will be carried out by the EEs, and the capacity issues of the EEs may delay the process depending on the progress made in each country in terms of filling the capacity gaps of the respective EE. |
| AE/EE capability to implement the project/programme | Medium-high | |
| Project-specific execution | Medium | |
| GCF portfolio concentration | Low | |
| Compliance | Medium | |

## 4.4 Fiduciary

87. The AfDB is the AE for this programme. The EEs of the programme are the Government of Togo, acting through the Agency for Promotion and Development of Agropoles in Togo, for Togo; the Government of Senegal, acting through the Ministry of Industrial Development and Small and Medium Industries, for Senegal; and the Government of Guinea, acting through the Authority for the Development and Administration of Special Economic and Industrial Zones, for Guinea.

88. During implementation, the AE will perform periodic review of the adequacy of the procurement and financial management arrangements specified in the FMCA during on-site implementation missions to assess, among others, the integrity of the accounting system, the efficiency of internal controls, the application of controls, and the smooth delivery of planned financial resources to the SCPZ programme.

89. The AE has conducted a capacity assessment of each EE. The financial management systems for the project at the country level are not fully satisfactory and the overall fiduciary risks are substantial because of the absence of critical staff, proper manuals, and relevant software. Recommendations for each programme country based on the results of the FMCA were stated in sections B.4 and G.3 of the funding proposal.

90. In relation to accounting management, legal agreements will require the pilot SCPZ countries to: (i) maintain accounts and books as well as prepare financial statements for the programmes and projects financed by the resources and administered resources of the AE; and (ii) implement accounting, administrative and financial procedures documents based on the policies of the AE. These accounts and records should provide justification for the use of funds disbursed to the programme and be prepared in accordance with applicable accounting standards.

91. As part of the fiduciary duties of the AE, it should review the performance of external auditors once a year upon receipt of the annual audit report to ensure that the work completed complies with the terms of reference, and that the audit was performed based on the agreed auditing standards.

92. The general conditions applicable to loan and grant agreements require that all programme and subproject accounts be audited each year in accordance with the relevant generally accepted standards. Each country must have the accounts of the project or programme audited by qualified and independent accountants that meet the requirements of the AE.

93. The audited financial statements, the auditor's report, and the management letter must be received by the AE no later than six months after the end of the financial year to which the documents relate. Non-compliance with this requirement will result in the immediate suspension of further replenishments to the special account.

## 4.5 Results monitoring and reporting

94. As a cross-cutting project, the intervention is foreseen to result in an overall reduction in GHG emissions amounting to 16.9 Mt $CO_2$ eq over the project lifetime of 25 years, while benefiting 1,104,728 direct beneficiaries and 5,612,416 indirect beneficiaries in Guinea, Senegal and Togo as per the GCF core indicators.

95. The theory of change adequately explains how change is understood to come about. The "if, then, because" logic is appropriately formulated in the diagram provided in the funding proposal. Moreover, the stated results, outcomes and interlinkages are defined in a manner that is adequately supportive of meeting the ultimate project goal. One economic and social co-benefit is identified by the project and will be monitored in its logical framework, however it is not captured well visually in the theory of change diagram.

96. Overall, the logical framework was found to meet the GCF results management framework and performance measurement framework requirements. The final version of the logical framework was designed with relevant details, including reporting on the appropriate impact and outcome indicators for mitigation as well as the social and economic co-benefit indicators. The outputs are adequately aligned with the fund level impacts and outcomes. The logical framework is thus expected to provide a sufficient basis for measuring the desired results.

97. The implementation timetable was provided in a format that would enable progress assessment during the implementation period, with the milestones and deliverables properly integrated to determine implementation performance in concrete terms.

98. The monitoring and evaluation plan in annex 11 was revised as per the Secretariat guidance. Logical framework indicators were included, and a monitoring budget was allocated. The total monitoring and evaluation budget (excluding the costs for interim and final evaluations) is USD 3.5 million. However, it is not clear how this amount is incorporated in annex 4 (project budget) of the funding proposal.

## 4.6 Legal assessment

99. The Accreditation Master Agreement was signed with the Accredited Entity on 8 November 2017 (the "**AMA**"), and it became effective on 7 November 2019.

100. The AE has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme.

101. The proposed programme will be implemented in the Togolese Republic, the Republic of Senegal and the Republic of Guinea, countries in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in these countries, which are risks that need to be further assessed.

102. The status of the bilateral negotiations with each of the countries of implementation is as follows:

(a) **Togolese Republic:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Togolese Republic on 22 September 2016. The latest communication was sent by the GCF Secretariat to the Government of the Togolese Republic on 29 August 2017.

(b) **Republic of Senegal:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Republic of Senegal on 30 September 2015. The latest communication was sent by the GCF Secretariat to the Government of the Republic of Senegal on 29 August 2017.

(c) **Republic of Guinea:** The GCF Secretariat has not yet dispatched a draft agreement on privileges and immunities and a background note to the Government of the Republic of Guinea, hence, negotiations have not yet started.

103. The Heads of the GCF Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF be made only after GCF has obtained satisfactory protection against litigation and expropriation in the countries, or has been provided with appropriate privileges and immunities.

## 4.7 List of proposed conditions (including legal)

104. To address the matters raised in the previous section, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Submission by the AE to the Fund of a certificate or legal opinion, in a form and substance satisfactory to the GCF Secretariat, within 180 days of the Board approval, confirming that the AE has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme;

(b) Signature of the funded activity agreement, in a form and substance satisfactory to the GCF Secretariat, within 180 days from the date of Board approval or the date the AE provides a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and

(c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

# Independent Technical Advisory Panel's assessment of FP219

| | |
|---|---|
| Proposal name: | Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains |
| Accredited entity: | African Development Bank |
| Country/(ies): | Guinea, Senegal and Togo |
| Project/programme size: | Large |

# I. Assessment of the independent Technical Advisory Panel

## 1.1 Impact potential *Scale: Medium to high*

1. Guinea, Senegal and Togo which are situated in the Horn of Africa and are part of the Sahel and Sudanese agroecological zones of Africa, are among the countries that are already being impacted by the effects of climate change as a result of historical warming and variable precipitation trends. Their vulnerabilities to climate extremes are high, due to the strong dependence of their economies on rain-fed agriculture which is largely dominated by smallholder farmers (about 80 per cent) for daily subsistence. Reliance on rain-fed farming and pastoralism for income and subsistence mean that livelihoods and food security in these regions and countries are strongly affected by climate variability and change. On top of this challenge, farmers in these countries have no or inadequate access to climate information to guide their farm management decision-making processes.

2. The proposed project supports a more sustainable low-carbon and climate-resilient development pathway for the climate-sensitive agriculture value chains, and for vulnerable smallholder farmers and rural communities. The project seeks to achieve its aims through strategic investments and interventions in the entire agricultural value chain, from the production stage to processing, via the two major components: component 1 on "Strengthening of critical SCPZs value chain infrastructure and management governance"; and component 2 on "Promotion of climate resilient agricultural practices and technologies adoption among smallholder farmers".

3. The accredited entity (AE) submitted the original proposal for B29 in 2021, and the current proposal has been revised to take into account feedback from that review stage. While the overall goal in the current funding proposal remains the same, there are some key changes in the present submission compared with the previous submission, as follows:

(a) Only three countries – Guinea, Senegal and Togo– are covered. Ethiopia is no longer included;

(b) The theory of change for the resubmitted funding proposal[1] has been revised compared with that of the previous funding proposal;[2] and

(c) Output 2.2 of the resubmitted funding proposal now focuses on "Increased community resilience through capacity-building, awareness and institutional strengthening on climate information early warning systems (CIEWS) for risks preparedness and readiness", which calls for capacity-building of value chains' agents/communities and development of agricultural processing support infrastructure (activity 2.2.1), and the establishment of climate information and early warning systems in staple crops processing zones (SCPZs) (activity 2.2.2).[3]

4. Table 1 shows a comparison of the funding proposal submission for the twenty-ninth meeting of the Board (B.29) and the current submission (for B.37).

[1] "IF the required infrastructure, technical and financial support as well as climate-based agro-advisory services for staple crops processing zones (SCPZs) are provided, THEN the sustainability of agriculture value chains in three countries will be ensured BECAUSE critical SCPZ value chain infrastructure are strengthened and made more climate resilient, and adoption of sustainable climate resilient practices, technologies and innovations among smallholder farmers are promoted, thus leading to improved agricultural productivity and value addition, better forest and land use management practices, access to renewable energy and generation contributing to avoid over 10.87 million t $CO_2$ eq in GHG emission of the countries' NDCs targets by 2030."

[2] "IF both the required infrastructure, technical and financial assistance support for staple crops processing zones (SCPZs) are provided, THEN the agriculture value chains in 4 countries will adopt a low emission and climate resilient path BECAUSE critical SCPZ value chain infrastructure are strengthened and made more climate resilient and climate resilient practices, technologies and innovations adoption among smallholder farmers are promoted, thus leading to improve agricultural productivity and value addition, better forest and land use management practices, access to renewable energy and generation contributing to avoid 21.2 million t $CO_2$ eq in GHG emission of the countries' INDCs targets by 2030."

[3] The previous funding proposal had "Improved access to low-carbon energy technologies for staple food crops processing to reduce post-harvest losses" as output 2.

Table 1: Comparison of the original and revised funding proposals

| FP Submitted for B29 | | | FP Submitted for B37 | | |
|---|---|---|---|---|---|
| **Component** | **Output** | **Activity** | **Component** | **Output** | **Activity** |
| **Component 1:** Strengthening of critical SCPZs values chain infrastructure | **Output 1.1:** Improved access to drip irrigation distribution systems (DIDS) powered by solar pumps to support climate resilient agricultural (CRA) practices | **Activity 1.1.1:** Support access to small-scale agricultural water management (AWM) infrastructure | **Component 1:** Strengthening the resilience of critical SCPZs values chain infrastructure | **Output 1.1:** Improved access to drip irrigation distribution systems (DIDS) powered by solar pumps to support climate resilient agricultural (CRA) practices | **Activity 1.1.1:** Support access to small-scale agricultural water management (AWM) infrastructure |
| | **Output 1.2:** Enhanced access to solar energy technology for staple food crops processing to reduce post harvest losses | **Activity 1.2.1:**Improving agricultural cooperatives, FBAs value chain infrastructure for food processing | | **Output 1.2:** Enhanced resilience of SCPZs energy supply through increased access to renewable energy | **Activity 1.2.1:**Improving agricultural cooperatives, FBAs value chain infrastructure for food processing |
| | | | | | **Activity 1.2.2:**Deploying low-carbon energy technologies for enhanced resilience |
| | **Output 1.3:** Increased agricultural productivity through enhanced access to Agro-Industrial Parks (AIPs) support facilities and access infrastructure | **Activity 1.3.1:** Development of SCPZs Access Infrastructure & Support for Agro-Park Management Governance | | **Output 1.3:** Increased agricultural productivity through enhanced access to Agro-Industrial Parks (AIPs) support facilities and access infrastructure | Activity 1.3.1: Development of SCPZs Access Infrastructure & Support for Agro-Park Management Governance |
| **Component 2:** Promotion of climate resilient agricultural practices and technologies adoption among smallholder farmers | **Output 2.1:** Adoption of CRA and agro-forestry practices among smallholder farmers | **Activity 2.1.1:** Promoting CRA among smallholder farmers | **Component 2:** Building the Resilience of SCPZs Agricultural Products Supply Value Chain | **Output 2.1:** Adoption of CRA and agro-forestry practices among smallholder farmers | **Activity 2.1.1:** Promoting CRA among smallholder farmers |
| | | **Activity 2.1.2:** Promoting of Agro-forestry practices among smallholder farmers | | | **Activity 2.1.2:** Promoting of Agro-forestry practices among smallholder farmers |
| | **Output 2.2:** Improved access to low-carbon energy techologies for staple food crops processing to reduce post harvest losses | **Activity 2.2.1:** Deploying low-carbon energy technologies | | **Output 2.2:** Increased community resilience through capacity building, awareness and institutional strengthening on climate information and early warning systems (CIEWS) for risks preparedness and readiness. | **Activity 2.2.1:** Capacity Building of Value Chains Agents/Communities' & Development of Agricultural Processing Support Infrastructure |
| | | | | | **Activity 2.2.2:** Establishment of climate information and early warning systems in SCPZs |
| | **Output 2.3:** Awareness raising, capacity building, community and institutional strengthening on agricultural value chains management. | **Activity 2.3.1:** Capacity Building of Value Chains Agents/Communities' & Development of Agricultural Processing Support Infrastructure | | | |
| **Component 3:** Project Management Component | **Output 3.1:** Project management and coordination | **Activity 3.1.1:** Project management, evaluation and coordination | **Component 3:** Project Management Component | **Output 3.1:** Project management and coordination | **Activity 3.1.1:** Project management, evaluation and coordination |

### 1.1.1. Adaptation impacts

5. Under the previous submission for B.29, the independent Technical Advisory Panel (iTAP) assessed that the funding proposal was not ready for endorsement, particularly because the overall climate rationale of the project was found to be weak. The issues raised were as follows:[4]

(a) The individual feasibility studies per country (annex 2 of the B.29 submission) did not contain any assessments, studies or discussions on climate; and climate assessments were only presented in the funding proposal (section 5.1.1) and the consolidated feasibility study;

(b) No climate risk and vulnerability assessment (CRVA) had been done to establish the vulnerability context of the targeted farmer beneficiaries and agroprocessing actors, critical resources (i.e. water resources, forests, land, soil), agriculture-related built

[4] See B.29 iTAP assessment of the proposal "Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains". Available upon request.

infrastructure (e.g. irrigation, roads, power, water facilities) and investments (e.g. in crops, livestock, equipment, agroprocessing and market facilities);

(c) No water resource assessments had been done, and the funding proposal relied on the availability of water to irrigate staple crops for its success. Moreover, the accredited entity (AE) did not clearly address the impact of the changing climate on water resources;

(d) The funding proposal had a weak basis for selecting climate adaptation measures because of the absence of a CRVA based on good science. The iTAP then found that the method for choosing appropriate climate adaptation and resilience measures to address the identified context of vulnerabilities could not have been applied without the potential risk of maladaptation (e.g. water over-extraction, forest clearing due to agricultural land expansions) and wasteful use of GCF resources. There was no document showing that the selected climate-resilient agriculture (CRA) and agroforestry management (AFM) measures had been discussed with the direct farmer beneficiaries in the project area. Specific CRA and AFM activities to be conducted per project area per country were not clearly presented, especially in the budget and procurement plans;

(e) There was no clear funding allocation for capacity-building on CRA and AFM, especially for the smallholder farmers;

6. Overall, the funding proposal had a weak consolidated feasibility study and country feasibility study.

7. Under the present resubmission for B.37, the above critical issues have been addressed as explained below. On the basis of the new information the iTAP assesses the adaptation impact potential as medium to high. The resubmitted funding proposal, together with the revised annex 2 (feasibility study) and a new annex 24 (CRVA for Staple Crops Processing Zone) have been prepared for Guinea, Senegal and Togo, and and these address the issues identified in the original iTAP review (para. 5 (a)–(e)).

8. In particular, the target farmer beneficiaries are now part of the category “value chain agents/communities” under activity 2.2.1, as requested by the iTAP review (para. 5 (e)) and as discussed in the AE’s response to the original iTAP review, dated 13 July 2021.

9. The iTAP considers that the additional information provided in the resubmission has gone a long way to support its assessment of the adaptation impact potential of this funding proposal.

### 1.1.2. Mitigation impacts

10. The funding proposal seeks to achieve greenhouse gas (GHG) mitigation via (a) integration of low-carbon technologies in the agricultural value chains, from crop production to agroprocessing; and (b) implementation of carbon sequestration via agroforestry interventions.

11. The iTAP estimates that, if implemented successfully beginning in 2024, the proposed project for the three countries would lead to an annual reduction of 6,193,845 tonnes of carbon dioxide equivalent (t $CO_2$ eq) by 2029 (year 6), 14,586,726 t $CO_2$ eq by 2030 (year 7) and 126,075,936 t $CO_2$ eq by year 25 (see annex 3A). The GHG emissions will mainly be the result of GHG abatement from agroforestry (93 per cent) and the energy and waste sectors (7 per cent).[5] On the basis of these projections, the SCPZ project has the potential to contribute up to 17 per cent of the combined nationally determined contribution targets of the countries by 2030.

12. The project’s basic approach to delivering renewable energy services through private sector providers (i.e. energy service corporations (ESCOs)) has great potential in improving the

economics and performance of the agricultural supply chains in sub-Saharan Africa. The proposed low-carbon technologies for crop production (via solar-powered drip irrigation distribution systems (DIDS); and agroprocessing facilities (via solar-powered lights, cold-chains and biogas for heat, power and steam for other processing stages) are all powerful technological advances that will, if introduced successfully, reduce GHG emissions and contribute to lower processing costs by reducing reliance on diesel.

13. Under the previous submission, the AE had not demonstrated, other than perhaps for the solar elements, that the proposed renewable energy solutions were realistic and implementable. No market studies were provided that would have shown, for instance, the willingness of farmers to pay for these services and there was no assessment of the chosen delivery channel (i.e. ESCOs) in terms of viability, barriers and resolution of contractual challenges.

14. The previous iTAP assessment noted that the proposed GCF investments in CRA and AFM were focused on assets (“equipment”). The funding proposal was also unclear regarding whether the targeted farmers were on board, incentivized and willing to participate in and deliver the key outputs under component 2 that would result to high carbon sequestration potential. The CRA and AFM activities were not clearly identified and defined in each country, and the targeted farmer beneficiaries were yet to be identified and perhaps also yet to be consulted.

15. Moreover, the original funding proposal (i.e. project narratives, monitoring and evaluation, project implementation plans) mentioned capacity-building activities, but the budget and procurement plans (annexes 4 and 10, respectively) did not include or itemize, for each project component, the capacity-building, training and workshops per country. In addition, the key activities that would enable the transfer of skills and knowledge to cause behavioural changes among the targeted farmer beneficiaries had not been properly presented.

16. In the revised funding proposal, the revised annex 2 (Feasibility study section 6, appendix 3) and the new annex 7 (Stakeholder consultation and engagement plan), the “willingness to pay” issue has been addressed. Stakeholder consultations show that the target farmer beneficiaries along the business value chains have expressed willingness to pay for the ESCO services. The new annex 24 (CRVA for SCPZ), annex 7 (Stakeholder consultation and engagement plan) and annex 3 (Economic & Financial Analysis (EFA) for irrigation and agroforestry) clarified and demonstrated the process of how the CRA and AFM activities were selected from an array of adaptation measures identified during the CRVA.

17. On the basis of the above revisions, the iTAP assesses the mitigation impact as medium to high.

## 1.2 Paradigm shift potential *Scale: High*

18. The holistic and integrative approach of the funding proposal, if properly implemented, shows a clear paradigm shift potential; a shift that will transform the targeted agricultural areas into modern agriculture centres in Guinea, Senegal and Togo . The project focuses its interventions on the production, post-production, processing and market access stages of staple crops such as rice, maize, cassava, banana, sesame, mango, cashew, coffee and vegetables, as well as horticulture and woodlot.

### 1.2.1. Transformation of traditional agriculture into climate-smart agriculture

19. The project focuses on improving farmers’ skills for the production stages through the introduction of selected CRA and AFM practices (component 2, output 2.1). The interventions

include the introduction and use of CIEWS (component 2, output 2.2) and the integration of climate-smart technologies (i.e. use of DIDS) (component 1, output 1.1).

20. For the post-production and processing stages, the project focuses on enhancing access to reliable and sustainable climate-resilient energy facilities (component 1, outputs 1.2 and 1.3), logistics infrastructure such as water irrigation, farm-to-market roads, ICT, storage and support service facilities in SCPZs (component 1, outputs 1.3). These interventions will be achieved through the development of strategically located agroprocessing zones for staple crops in the regions. This holistic approach will trigger the shift "from implementing agriculture in the normal 'business-as-usual' way to one that is an integrated value chain from the production to processing stages via low-emission, climate-resilient development pathways across the agricultural value chains in Guinea, Senegal and Togo".[6]

21. The project interventions on the mitigation side are technically well considered and they are aligned to deliver a comprehensive paradigm shift, if implemented correctly and with continuing support from the governments of the participating countries. Their scalability would enable agricultural processing hubs to become centres of energy supply, driving other productive and beneficial uses of clean energy, for example for health or education, thereby driving economic growth and development. Providing these solutions through private sector entities that would have a strong motive and technical capability to expand supply to other uses is also a great opportunity for clean development and private sector involvement.

### 1.2.2. Replicability and monitoring, reporting and verification for knowledge dissemination

22. The project activities are highly replicable, and capacity-building activities will be required to ensure knowledge and learning are achieved. The replicability of the project to other staple crop growing regions and/or other countries will require the robust documentation of the proposed project's implementation progress, results and impacts throughout implementation.

23. The iTAP strongly recommends that, in order to achieve the potential replication and scalability in the countries and the wider region, a robust monitoring, reporting and verification protocol should be implemented and ensured through regular checks by the Secretariat and through encouraging the dissemination of knowledge.

24. Behavioural changes among the target smallholder farmers along the agriculture value chains will be key in sustaining the paradigm shift. Thus, the paradigm shift potential will be high if the targeted farmer beneficiaries and agrobusiness value chain stakeholders (a) adopt relevant mitigation skills (i.e. use of low-carbon technologies in agri- and agro-processing chains and provision of data and information in support of monitoring, reporting and verification plans); and (b) sustainably practice adaptation (e.g. CRA, AFM, CIEWS) measures.

25. The funding proposal describes how these specific adaptation measures were determined, beginning with the conduct of the CRVA and the identification of appropriate adaptation measures that will lead to increased transformative capacities and not just increased absorptive and adaptive capacities.

26. **Risks to paradigm shift.** There are substantial risks to achieving the paradigm shift. However, the revised funding proposal has addressed these issues by including appropriate market studies and stakeholder consultations among farmers, validation of the market for clean energy ESCOs, and documentation outlining the proposed handling of contracts and potential contractual issues. Although the coronavirus disease 2019 (COVID-19) pandemic had already subsided at the time of the revisions, the AE also addressed the risk regarding how the COVID-

[6] Funding proposal, p. 45, para. 4.

19 pandemic would impact the sustainability of the present and future agriculture business value chains, and this has been integrated into the relevant sections of the funding proposal.

27. In view of the above discussions and balancing the overall impact potential and the risks, the iTAP assesses paradigm shift potential as high.

## 1.3 Sustainable development potential *Scale: High*

28. **Economic co-benefits.**[7] Overall, the implementation of the SCPZ is expected to usher in new economic opportunities benefiting 634,472 people via (a) new and/or improved livelihoods and income generating activities, (b) reduced losses and damage from more resilient agriculture value chains and the introduction of CIEWS; (c) improved access to and efficient use of critical agricultural resources as well as infrastructure and facilities; (d) transfer of new climate-smart agricultural and AFM skills and knowledge. For the GCF funded components, an estimated total of 281,728 people will benefit from these interventions (see annex 22).

29. **Environmental co-benefits.** Interventions on CRA and AFM in general in the SCPZ will lead to improved soil and local environments due to a reduction in or avoidance of inputs from fertilizers and pesticides. The use of DIDS and the utilization of organic agri-waste for biodigesters within the SCPZ will reduce municipal solid waste and contribute to cleaner air, especially with the displacement of diesel.

30. **Social co-benefits.** The provision of power from renewables in the SCPZ could expand over time due to the scalability of renewable energy power solutions at a time when their cost is also reducing. This would facilitate power use for other social infrastructure (e.g. health centres, schools and childcare) and other important community uses.

31. **Gender action plans.** The gender action plans (for each country participating) target women smallholder farmers, women-led agribusiness enterprises and women-led civil society organizations, as described in annex 8 of the funding proposal.

32. **Sustainable Development Goals.** The funding proposal can be linked to several of the Sustainable Development Goals (SDGs), including strong gender equality (SDG 5), affordable and clean energy (SDG 7), decent work and economic growth (SDG 8), industry, innovation and infrastructure (SDG 9), reduced inequalities (SDG 10), sustainable cities and communities (SDG 11), responsible consumption and production (SDG 12), partnerships for the goals (SDG 17), life on land (SDG 15) and zero hunger (SDG 2).

33. If the project is implemented properly the sustainable development potential of the project could potentially be high.

## 1.4 Needs of the recipient *Scale: Medium*

34. **Needs of the governments.** Guinea, Senegal and Togo which are situated in the Horn of Africa and the Sahel, are among the major climate hotspots of the continent. As countries in the climate-sensitive agroecozones of sub-Saharan African, Guinea, Senegal and Togo are recognized as vulnerable countries that have urgent and immediate needs to stimulate productivity and add value, increase competitiveness, generate employment and increase incomes of the most vulnerable people and communities in the agriculture sector. The project targets the critical bottlenecks in the agribusiness value chains affecting these countries. Component 1 aims to strengthen crop post-production and processing value chains and improve access to markets via the introduction of DIDS, renewable energy power supplies ( solar and biogas) and agriprocessing value chain infrastructure needs (e.g. cold storage,

[7] See funding proposal, section D.3 "Sustainable development".

warehouses, ICT facilities, farm-to-market roads and irrigation water supply). Component 2 addresses the capacity needs of “value chain agents” (i.e. farmers, but also the staff of project executing agencies, project implementing units and agro-industrial parks) via integration of CRA, AFM and SCPZ CIEWS at the crop production stages.

35. **Needs of the farmer beneficiaries and agro-crop processing stakeholders.** It is estimated that 80 per cent of the smallholder farmers in the SCPZ rely on rain-fed agricuture for daily subsistence; thus it is clear that these farmers are the most vulnerable to adverse climate change impacts. Other actors along the post-production to market stages of the agribusiness value chains are also vulnerable, especially those who have limited access to sustainable energy sources, and also rely on inefficient distribution channels and limited markets for their produce. The activities under components 1 and 2 address these needs. The project is expected to improve the direct and indirect beneficiaries’ incomes and livelihoods, food security and access to resources (i.e. knowledge, skills and training, climate-friendly technologies, CRA and AFM inputs and processing facilities, and the market) if implemented well. The funding proposal highlights that the direct beneficiaries of the project amount to 6.9 per cent of the total population for Togo, 1.4 per cent for Senegal and 3.8 per cent for Guinea, with indirect beneficiaries amounting to 12.17 per cent, 19.5 per cent and 14.4 per cent for Togo, Senegal and Guinea, respectively. The iTAP considers that the funding proposal could be improved if the eligbility criteria for the smallholder farmers to participate in these activities were greatly enhanced, as follows:

36. Although the proposal mentions smallholder farmers in general, the eligibility criteria for selecting smallholder farmers are not fully inclusive, as can be seen from the eligibility criteria for the main activities; namely, under activity 1.1.1, activities 1.2.1 and 1.2.2, activity 1.3.1, activities 2.1.1 and 2.1.2, and activities 2.2.1 and 2.2.2. In summary, as currently configured, the eligibility criteria target and prefer the least vulnerable of the cohort of vulnerable farmers, or those that are the most bankable/creditworthy. For example, the eligibility criteria states a “Preference for agricultural cooperation societies, farmer-based organizations, or smallholder farm households that are willing to belong to a farmer water user group, or farmer water user community to promote water harvesting through on-farm catchment (small ponds and reservoirs), for irrigational use, including necessary O&M”, or farmers that can commit to contributing to the necessary operation and maintenance (O&M) of on-farm water catchment ponds and reservoirs. Similarly, activity-related criteria are not fully inclusive: “Preferences for ACSs, FBAs, WABEs, and SHFHs practising any form of water-coping strategy in agriculture such as water harvesting, use of boreholes, small reservoirs and dams” (activity 1.1.1); “Capacity to contribute funds either by having own funds or taking loans” (activities 1.2.1 and 1.2.2); “Proximity to local and international markets, transport networks and telecommunications”, “A formally registered SHFH with a cooperative society, FBA or WABE”, “Currently practising any form of climate-resilient activities in its daily operations such as use of drought resistance and improved yield seeds, among others”, “Has a strong business plan to develop a trading agribusiness with potential downstream value chain linkages” (activity 1.3.1); and “A formally registered SHFH with an ACS, or FBA for more than 1 year”, “Preference for beneficiaries that are currently practising any form of climate-resilient activities in daily farming operations such as water harvesting, mixed agro-forestry practice, livestock rearing, use of drought resistance and improved yield seeds, among others” (activities 2.1.1 and 2.1.2);

37. Such preferences and/or eligibility criteria create a potential risk of segregating and excluding the neediest vulnerable informal smallholder farmer households from the above-mentioned main activities. These traditional farmers need as much or even more assistance than the morebankable cohort of farmers to enable them to transform from their old ineffective farming methods to a climate-smart agriculture approach; and

(a) Only activity 2.2.1 includes other smallholder farmer households located within the SCPZ areas that will not benefit from the main activities, although that activity also excludes those who are outside the perimeter of the SCPZ.

38. Considering the above, the iTAP rates the needs of the recipient as medium with the possibility of high rating should the AE enhance the eligibility criteria for the targeted smallholder farmers.

## 1.5 Country ownership *Scale: High*

### 1.5.1. Alignment with national climate strategy and policies

39. The funding proposal provides evidence of how the project is aligned with the 2015 nationally designated contributions (NDCs) of each country (table B.1.1), which stress the importance of adaptation and resilience in the agriculture, land use, forestry and energy sectors; the countries' national adaptation programmes of action (annex 6) and the Special Agro-Industrial Processing Zones Development Programmes (section 3.2 and annex 2); as well as with in-country strategy papers, and the African Development Bank (AfDB) Strategy for the Industrialization of Africa (2016–2025).

### 1.5.2. Capacity of accredited entities or executing entities to deliver

40. The AE for the project is AfDB and the executing entities are the Development and Administration of Special Economic and Industrial Zones of Guinea, the Ministry of Industrial Development and Small and Medium Industries of Senegal and the Agency for Promotion and Development of Agropoles of Togo. The capacity-building to be provided by AfDB under component 2.3 will enhance the project management capacities of these entities to oversee the projects.

41. AfDB has relevant experience and expertise in the sector, as well as in the countries, on the implementation of projects related to clean energy supply, CRA and food security. AfDB has been involved in various SCPZ projects in these countries, and chief among these is the AfDB flagship programme "Feed Africa initiative 2016–2025" where SCPZ plays a major role.

### 1.5.3. Engagement with Relevant public and private stakeholders

42. For agriculture-based economies such as Guinea, Senegal and Togo, there is evidence of a strong sense of ownership and serious engagement among the SCPZ-targeted stakeholders, as shown in the new annex 7 (Stakeholder consultation and engagement plan), which records the interactions with the relevant local and national government entities for each country (e.g. ministries of energy, agriculture, fisheries, industry, national hydromet agencies, mayors), and the various civil society organizations, private sector representatives from the energy, agriculture and finance sectors, women's groups, and centres of excellence (universities).

### 1.5.4. Risks to country ownership

43. The previous submission lacked a fully documented stakeholder consultation and, from the previous iTAP's perspective, this posed a considerable risk to country ownership at the implementation level.[8] However, the revised funding proposal includes a stakeholder consultation and engagement plan (annex 7).[9] This has enabled the current iTAP to clearly

[8] See B.29 iTAP Assessment of AfDB SCPZ.
[9] In the previous submission, the Annex 7 Stakeholder plan was missing though it was checked as part of the funding proposal annexes The AE stated during the AE–iTAP–Secretariat Zoom meeting on 11 May 2021 that they had indeed submitted said annex to the Secretariat.

understand the relevance and alignment of the proposed interventions from the ground up. The stakeholder consultations do show the need and strong support for the project interventions to address the many cross-cutting climate-related issues and challenges that are affecting the targeted beneficiary farmers and agroprocessing stakeholders and enterprises in the business value chains.

44. The iTAP assess country ownership for the project as high.

## 1.6 Efficiency and effectiveness *Scale: Medium*

45. The total project cost of the resubmitted proposal is USD 271,703,525, with the GCF allocation of USD 102,790,988 (38 per cent of the total project costs) in the form of grants amounting to USD 75,791,156 (74 per cent) and loans of USD 26,999,832 (26 per cent). AfDB and other co-financiers will contribute USD 149,426,527.95 (55 per cent of total costs); while government in-kind contributions will amount to USD 19,486,009.0 (7 per cent of total costs).

46. Overall, the GCF finance leverage a total of USD 168,912 537, giving a leverage ratio of 1:1.64, made up of USD 160,883,989 for grants (leverage ratio 1:1.57), and USD 91,333, 527 for loans (leverage ratio 1:0.9).

47. In terms of GCF country fund allocations, 37.8 per cent of GCF funds are allocated to Togo amounting to USD 38,898,617.38 (74 per cent in grants, 26 per cent in loans); 32.4 per cent is for Senegal, amounting to USD 33,333,722.33 (67 per cent in grants, 33 per cent in loans); and 29.7 per cent is for Guinea, amounting to USD 30 517,648.04 (80 per cent in grants, 20 per cent in loans).

48. In terms of use of GCF proceeds, a detailed breakdown of fund use is presented in table 2on GCF financed activities disaggregated by countries.

| GCF Financed Activities Disaggregated By Countries | | | | |
|---|---|---|---|---|
| Component | | | | |
| Togo | Description | Budget | Loan | Grant |
| Biodigester Volume in m3 | 9,447.1 | $2,834,118.00 | $696,788.61 | $2,137,329.39 |
| Solar PV -Pumping (Total Watts Capacity Needed) | 1,018,248.0 | $1,527,372.00 | $375,515.56 | $1,151,856.44 |
| Solar PV-Lighting (Total Watts Capacity Needed) | 7,166,527.4 | $7,166,527.37 | $1,802,170.45 | $5,364,356.92 |
| DIDS (Land Area in Ha) | 15,428.0 | $6,942,600.00 | $1,706,888.92 | $5,235,711.08 |
| CRA (Land Area in Ha) | 15,428.0 | $11,278,617.50 | $2,732,703.15 | $8,545,914.36 |
| Agroforestry (Land Area in Ha) | 10,000.0 | $5,000,000.00 | $1,229,286.52 | $3,770,713.48 |
| CIEWS (Equipment) | | $4,149,382.50 | $1,456,633.96 | $2,692,748.54 |
| *Sub-total forTogo* | | *$38,898,617.37* | *$9,999,987.18* | *$28,898,630.19* |
| Senegal | Description | Budget | Loan | Grant |
| Biodigester Volume in m3 | 5,218.6 | *$1,565,577.00* | *$554,097.39* | *$1,011,479.61* |
| Solar PV -Pumping (Total Watts Capacity Needed) | 788,040.0 | *$1,182,060.00* | *$418,331.03* | *$763,728.97* |
| Solar PV-Lighting (Total Watts Capacity Needed) | 3,273,085.2 | *$3,273,085.23* | *$1,144,587.21* | *$2,128,498.02* |
| DIDS (Land Area in Ha) | 11,940.0 | *$5,373,000.00* | *$1,901,504.70* | *$3,471,495.30* |
| CRA (Land Area in Ha) | 11,940.0 | *$7,790,617.50* | *$2,770,817.50* | *$5,019,800.01* |
| Agroforestry (Land Area in Ha) | 20,000.0 | *$10,000,000.00* | *$3,539,000.00* | *$6,461,000.00* |
| CIEWS (Equipment) | | *$4,149,382.50* | *$671,610.76* | *$3,477,771.74* |
| *Sub-total for Senegal* | | *$33,333,722.23* | *$10,999,948.59* | *$22,333,773.64* |
| Guinea | Description | Budget | Loan | Grant |
| Biodigester Volume in m3 | 9,911.0 | $2,973,302.58 | $608,722.08 | $2,364,580.92 |
| Solar PV -Pumping (Total Watts Capacity Needed) | 779,460.0 | $1,169,190.00 | $239,333.19 | $929,856.81 |
| Solar PV-Lighting (Total Watts Capacity Needed) | 4,250,655.0 | $4,250,655.04 | $879,976.66 | $3,370,678.38 |
| DIDS (Land Area in Ha) | 11,810.0 | $5,314,500.00 | $1,087,878.15 | $4,226,621.85 |
| CRA (Land Area in Ha) | 11,810.0 | $3,511,235.42 | $708,795.27 | $2,802,440.15 |
| Agroforestry (Land Area in Ha) | 10,000.0 | $5,000,000.00 | $1,023,500.00 | $3,976,500.00 |
| CIEWS (Equipment) | | $8,298,765.00 | $1,451,690.85 | $6,847,074.15 |
| *Sub-total for Guinea* | | *$30,517,648.04* | *$5,999,896.21* | *$24,517,752.25* |
| Costs of ESS translation | | $41,000.00 | | $41,000.00 |
| | | *$41,000.00* | | *$41,000.00* |
| *Grand Total all Countries* | | *$102,790,987.64* | *$26,999,831.98* | *$75,791,156.08* |

49. It is worth noting that the GCF funds will primarily be benefiting the vulnerable smallholder farmers engaged in climate-smart agriculture and agroforestry value chains. GCF funds are matched with the riskier activities of the project (i.e. the project components where it is harder to ensure they are sustainable in case of poor implementation and O&M, or weak farmer stakeholder participation and commitment).

50. The iTAP also notes that there is zero co-financing for the mitigation project components from the AE or beneficiaries (i.e. a ratio of 1:0). The project proposes a budget of USD 43.5 million for mitigation activities in energy supply (solar and biodigesters), of which about USD 32.1 million (i.e. 74 per cent of the total) is requested from GCF as grants and the balance of USD 11.4 million is requested from GCF as loans.

51. The cost estimations for the solar component are realistic, even though it is not clear why three quarters of the funding should be provided as grants in a market where commercial finance is available in at least some locations. For the biodigesters, however, the cost estimations appear high and, as they are based on 6 cubic-metre digesters, they do not take into account economies of scale that occur with the installation of larger biodigesters.[10]

[10] International Energy Agency. Accessed in May 2021. "Average costs of biogas production technologies per unit of energy produced (excluding feedstock)", 2018.

52. **Cost per direct beneficiary.** Given the project's target of 1.1 million direct beneficiaries, the estimated cost per beneficiary is USD 245.95, while the GCF cost per beneficiary is USD 93.05.

53. **Cost of carbon mitigated at the programme level.** On the basis of the emission calculations worksheet (annex 3A SCPZ EFA), total emission reductions over the project's 25-year duration amount to 126,075,936 t $CO_2$ eq. The total cost of carbon mitigated is USD 2.16 per t $CO_2$ eq , and the GCF's cost of carbon is USD 0.82 per t $CO_2$ eq. Assuming the project starts delivering mitigation results by 2024, the estimated total emissions mitigated by 2030 (year 6) will amount to 14,586,725 t $CO_2$ eq; and the estimated total project cost per t $CO_2$ eq is USD 18.63, while GCF cost per t $CO_2$ eq at year 6 is USD 7.05.

54. **Economic analysis.** The funding proposal identifies the following streams of potential benefits from the energy mitigation project:

(a) Reduced use of kerosene;

(b) Reduced use of diesel;

(c) Reduced use of wood charcoal;

(d) Replacement of urea fertilizer with higher-performance bio slurry; and

(e) Social benefits of avoided carbon emissions amounting to USD 31 per t $CO_2$ eq.

55. The cost–benefit analysis primarily considered the fuel replacement at different discount rates: social (12 per cent) and private sector (20 per cent). The analysis demonstrated the need for GCF finance only by using the GCF interest rate as an implied discount rate of 0.75 per cent, which would make the net present value become positive.[11]

56. In view of the above considerations, the effectiveness and efficiency of the renewable mitigation component of the project is rated as medium.

57. Overall, the effectiveness and efficiency is medium.

# II. Overall remarks from the independent Technical Advisory Panel

58. The development of a project focusing on strengthened agricultural production and supply chains and the utilization of renewable energy on both the production side and in staple crop processing is welcome, given that these are both critically important areas of potential climate mitigation and adaptation action, with a high potential for positive economic and social impacts.

59. The iTAP assessment concluded that this funding proposal has addressed the key issues raised in the previous submission by:

(a) Strengthening the climate rationale for targeted project areas;

(b) Submitting a CRVA;

(c) Clarifying climate adaptation and resilience measures per project area/country;

(d) Clarifying the potential risk of maladaptation (e.g. over-extraction of water resources);

(e) Updating and improving the feasibility study, due to the lack of updated assessments on the impacts of the COVID-19 pandemic on the crop production and processing chains;

(f) Submitting the annex on stakeholder consultation and management plans;

[11] See Annex 3A SCPZ EFA program level digester analysis.

(g) Addressing the availability of draft last-mile agreements on access to renewable energy services with the target beneficiaries; and

(h) Adding relevant data and information on the viability of ESCOs selling power to target markets (farmers using solar for DIDS, agroprocessing zones and biogas).

60. Given the above changes implemented by the AE for this resubmission, the funding proposal is endorsed by iTAP.

61. However, to strengthen further the resubmitted funding proposal and ensure clarity of documentation as well as the sustainability of proposed interventions, the iTAP also strongly recommends that the AE:

(a) Seriously address the issue of having a fair and inclusive criteria for all target smallholder household farmer beneficiaries that reasonably stand to benefit from the project interventions;

(b) Integrate a robust monitoring, reporting and verification protocol over the project lifetime and ensure such implementation through regular checks by the Secretariat and encouraging the dissemination of knowledge; and

(c) Submit a complete and final set of documents that are satisfactory to the Secretariat and iTAP prior to the signing of the funded activity agreement.

62. The iTAP notes that the funding proposal has several documents that remain inconsistent with the content of the proposal or are in draft forms, some are in French (e.g. the willingness to pay survey document), and some contain amounts/figures that are inaccurate. As such, the following documents need to be properly updated, cleaned completely, aligned for consistency with the revised project design, and submitted in final versions:

(a) Funding proposal sections B2 and B3 (e.g. theory of change diagram and outcome statements);

(b) Annex 3A, SCPZ EFA (e.g. emission reduction calculations; country calculations on GHG emissions still include crop abatement; energy demand for Guinea);

(c) Annex 4, SCPZ Budget plan revised (e.g. budget notes, assumptions and figures do not match information in the funding proposal; some financiers, for example, for Ethiopia are still included);

(d) Annex 14 SCPZ Term sheet; and

(e) Annex 17 Multi-country project information.

63. The iTAP also recommends that the AE submit copies of last-mile agreements with the target beneficiaries showing alignment of funding proposal outputs and indicators, scope of activities per project component, and duties and responsibilities, to ensure delivery of funding proposal outputs/deliverables. This will ensure that the project interventions on CRA (activity 2.1.1), AFM (activity 2.1.2) and CIEWS (activity 2.2.2) among the target farmer beneficiaries will bring about behavioural transformation. Based on the stakeholder consultations and engagement plan (annex 7), these last-mile agreements have already been requested but remain pending, however, these last-mile agreements pertain to energy services only.

# Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP219)

| | |
|---|---|
| Proposal name: | Staple Crops Processing Zone (SCPZ): Promoting Sustainable Agricultural Value Chains |
| Accredited entity: | African Development Bank |
| Country/(ies): | Togo, Senegal, and Guinea |
| Project/programme size: | Large |

| **Impact potential** |
|---|
| Thank you for the compliments. As you rightly mentioned your comments in the previous iTAP round contributed to the improvement of the FP package.<br><br>Kindly note that Guinea, Togo and Senegal were mistakenly mentioned to be in Horn of Africa instead of "West Africa". Thank you |
| **Paradigm shift potential** |
| Thank you for your recommendation. Through Activity 3.1.1 (Project management, evaluation, and coordination) the project will ensure robust monitoring, evaluation, reporting and learning to facilitate replicability and scaleup in other regions of the country and in other countries. |
| **Sustainable development potential** |
| Thank you for highlighting the alignment of the programme with some key SDGs elements and several benefits that the intervention promises to provide. |
| **Needs of the recipient** |
| Thank you for your very insightful comments. We will take account of your recommendations and refine the selection criteria.<br><br>Kindly note that Guinea, Togo and Senegal were mistakenly mentioned to be in Horn of Africa instead of "West Africa". Thank you |
| **Country ownership** |
| Thank you for the compliments. |

| **Efficiency and effectiveness** |
| --- |
| Thank you for the compliments |
| **Overall remarks from the independent Technical Advisory Panel:**<br>We would like to sincerely thank the iTAP for their time taken to review and provide insightful and valuable comments which led to the improvement of the FP package. Your recommendations are well noted. Based on these recommendations:<br>a. We will refine the FP accordingly to ensure a fair and inclusive criterion for all target smallholder household farmer beneficiaries that could reasonably benefit from the project interventions<br>b. A robust monitoring, reporting and verification protocol will be put in place (under component 3) throughout the project lifetime and the effective dissemination of knowledge will be encouraged<br>c. Noted. We will submit all clean updated copies of the FP package<br><br>Please we like to draw your attention to some instances in the iTAP Assessment report, where Guinea, Togo and Senegal were mistakenly mentioned to be in Horn of Africa instead of “West Africa”. |

# STAPLE CROPS PROCESSING ZONEs (SCPZs): Promoting Sustainable Agricultural Value Chains.




## GENDER ANALYSIS/ASSESSMENT GUIDE AND ACTION PLAN (GAP):

African Development Bank

April 2023

# A. GENDER PLAN OF ACTION FOR TOGO'

## A.1 GENDER ANALYSIS/ASSESSMENT

**Context and Background**

The Togolese population is estimated at 7,552,318 inhabitants in 2015, of which 51.4% are women. Togo has made progress on gender equality in the area of legislative reform. However, the country ranks 134th out of 149 countries in the 2018 Global Gender Gap Report. This low ranking is the result of a number of adverse socio-economic indicators. Thus, the incidence of poverty is higher for households headed by women than for those headed by men (57.5% against 54.6%) and increased for the former while it is decreasing for women. the latter, between 2011 and 2015 (PND 2015).

Legal and Policy Framework and Gender Institutional Mechanism: Togo is a party to all international conventions on women's rights and gender equality. At the national level, the gender legal framework does not contain any particular discrimination. The Constitution of the Republic of 1992 stipulates equality before the law of all citizens without distinction (Article 2). In addition, it recognizes the right to equality and dignity (art.11), the right to development (art.12) and property (article 27). Following the adoption of the National Policy on Equity and Gender Equality (PNEEG), significant progress has been made at the legislative level, including: (i) mainstreaming gender in the electoral code; 2013; (ii) The revision of the Code of Persons and the Family providing the Togolese woman with a set of rights allowing better protection; (iii) the adoption in 2015 of a new penal code with the prohibition of violence and discrimination against women, (iv) the adoption in June 2018 of a new land and land code which guarantees the access to land for women as well as men. However, many challenges remain. It is in this perspective that the government, through the Ministry of Social Action, the Promotion of Women and Literacy, adopted a new National Strategy for Equity and Gender Equality 2019-2028 ( SNEEG) with the support of the African Development Bank (ADB).

Promoting gender and women's empowerment is one of the areas of intervention of the Ministry of Social Action, Advancement of Women and Literacy. It is led by the Directorate General of Gender and the Advancement of Women (DGGPF). The Government has also set up in each sectoral ministry, a cell composed of three persons of high decision-making level, as gender focal point. As for the women's organizations of the civil society, they have regrouped in a Coordination of Women's Organizations of Togo since 2016 to work synergistically in areas such as gender-based violence, women's economic empowerment and women's participation in decision-making.

**Governance:** the 2012 Electoral Code revised by the 2013 Law, has incorporated exceptional measures to improve the participation of women. The guarantee of female candidates for the legislative elections is reduced by half. The principle of parity of the lists of candidates for parliamentary elections is enshrined in Article 220 paragraph 5 to concretise the political will of the Head of State expressed in his speech to the nation in December 2012.

It is also noted a significant advance of the presence of women in decision-making spheres. The number of women in the National Assembly rose from 9 out of 81 deputies in 2007 and to 16 out of 91 deputies in 2013. In percentage terms, the proportion of women deputies increased from 11.1% in 2013 to 18.7 % in 2015 and 2018.On the government, 5 women were appointed ministers in 2010 and 2012, 06 in 2013 and 05 in 2016. In 2018, women ministers represent 23.07% against 76.93% of male ministers. In small numbers at the level of the traditional chieftaincy, they are more and more represented as notable. According to the staffing analysis report of the public administration, in 2017, women civil servants make up 19.30% against 80.7 male civil servants.

**Education:** the abolition of preschool and primary school fees, as well as the reduction of school fees for girls at secondary level (70% compared to 100% for boys), have greatly contributed to increase the school enrolment rate of girls and achieving parity at the primary level. For example, in 2017, the primary school attendance rate was 88.7% for both boys and girls. There is also a clear increase in women's literacy rate from 50% to 63.4% between 2015 and 2018.

**Health:** The maternal and infant mortality rates remain, respectively, 401 deaths per 100,000 live births and 49%, in 2014 (EDS-III). According to the EDST survey (2013-2014), nearly 29% of women have suffered physical violence since the age of 15. The rate of early marriages is about 32% and the regions of Savanes (Tamongue and of Nadjoundi) and the Central Region (Cantons of Lama-Tessi and Kri-Kri) have a prevalence of early marriages of more than 35%. The prevalence rate of chronic malnutrition is almost identical among female and male children (27% versus 28%). The prevalence of anaemia among women aged 15 to 49 is 48% (64% of pregnant women), whereas it is only 20% for men (FAO, IEH, 2016: 47-48).

**Employment and entrepreneurship:** 74.6% of women of working age (15-64 years) are economically active compared to 79.1% for men. They are the majority in the informal sector and play an important role in agriculture (51.1% of the agricultural population), services and trade, where they account for 24.2%. women are also present at all stages of production and generally cultivate two different plots: their "personal" plots that they obtain after marriage, usually small areas whose income is used to feed the family and "common" farming. or family on which men enjoy full enjoyment. In addition to domestic work and work in their own fields, women have the obligation to work in the family fields. Women also play an important role in the processing and marketing of agricultural products from the family fields, but management is the responsibility of men. Women use traditional tools because the processing units set up remain insufficient and the material and financial resources required to access to research-based technologies is not easily accessible by most of them. 9.4% of villages have corn huskers, 5.9% have rice huskers, and 6.2% have huskers for coffee. In the implementation of PNIASA, certain technologies have been successfully introduced, including rice steamers, cane planters, a smoked fish processing platform (APRM, PNIASA, 2015).

Rural women benefit from actions carried out with a view to getting them better organized in associations, groups, cooperative societies, in order to enable them to benefit from technical support. In 2012, there were 6,010 farmer organizations in activity (DSID , 2012), of which 66% of mixed type and 14% of female type receiving support at the organizational level for the marketing of cereals and several actions to make them solvent and credit recipients. However, the inadequacy of female agents in the agricultural sector, especially at the grassroots level, is a limiting factor. The Institute of Technical Support Advice (ICAT) has only 5.4% of female agents in the field (ICAT, 2014). The number of NGOs working in the sector is not high and very few women work in these NGOs because of their small numbers in the sector, precarious conditions and socio-cultural constraints. Due to this numerical insufficiency, the supervision of women's groups and individual women often comes back to the male agents whose methodologies of accompanying these agents often exclude women, because it is the chiefs of exploitation generally male which shelter the units of demonstration. As a result, because of their limited contact with extension and advisory services, women do not get their fair share of advice from extension and other services (seeds and fertilizers ...) delivered by these services. In addition, 86.1% of households headed by men against only 13.9% of households headed by women benefit from irrigation. At the National Agricultural Training Institute of Togo no specific module on the taking into account of the kind in the agricultural sector is developed within this institute. Outside, statistics show that only 14% of women work in the Ministry of Agriculture, Livestock and Water Resources.

**Access to land ownership:** only 20% of women between the ages of 45 and 49 own land. The 2013 National Census of Agriculture shows that female heads of households have smaller plots than their male counterparts. Household concentration Depending on the size of the farms, the size of farms is between 1 to 3 hectares for men and, for women, between 0.5 and 2 hectares. As a result, women are confined to food crops (maize, cowpeas, groundnuts, rice, cassava, sweet potatoes, vegetables, etc.), which generally have more immediate, less profitable and more household consumption (GIZ, INADES, ProDRA, 2015: 19-20 and GIZ, GFA, ProSeCal, 2016: 07).

The project area is one of the poorest, with a fairly visible impact on women and children. Women are involved in most of the product lines: rice, soybean, sesame, maize. Land insecurity, access to affordable financing to invest in climate resilient agricultural practices, lack of capacity in climate resilient agricultural practices and technology adoption, poor access to small water infrastructure for market gardening and horticulture, lack of access to post-harvest handling equipment to reduce losses, and poor outlets for the marketing of agricultural products are some of the major problems for women in the project area.

The following measures and actions will be taken to boost the role of women and maximize their contribution to the project: (i) grouping of women farmers by food product line (rice, maize, soybean and sesame); (ii) allocation of at least 30% of developed land to groups of women farmers; (iii) allocation of a 30% quota, at least, for women industrial promoters in the CTA; (iv) facilitation of women's access to modern agricultural production and processing equipment: inputs (seeds, fertilizers and phytosanitary products), processing materials and equipment (rice parboiling, maize husking); (v) building of the entrepreneurial capacities of women's groups in the production, storage, processing, packaging and marketing of agricultural products; (vi) building of women's capacities in business plan preparation, marketing and networking for the market and access to financing; (vii) facilitation of access to basic services, such as water and electricity; and (viii) facilitation of the process of obtaining civil status documents, such as identification and nationality cards. The budget to be allocated for specific gender promotion and women's empowerment activities is UA 7.2 million. The project is classified under category 2, according Gender Marker System.

## A.2 GENDER ACTION PLAN: TOGO

| Activities | Indicators and Targets | Timeline | Responsibilities |
|---|---|---|---|
| ***Impact Statement**: Strengthened climate resilience of the agricultural value chain, to help stimulate productivity, value addition, competitiveness and income generation for the most vulnerable communities, including women and girls.*<br><br>***Outcome Statement**: Improved Access to Climate Resilient Livelihood Options for at least 201,500 Smallholder Women Farmers including Women-led Agribusiness Enterprises (Farm-Based Associations and Cooperative Societies).* | | | |
| ***Output(s) Statement 1: Critical SAPZ value chain infrastructure strengthened and made more climate resilient*** | | | |
| ***(i)** Smallholder women farmers (SHWF) provided with access to finance to invest in drip irrigation technology adoption* | • At least 20% SHWF in project areas | By 20XX +5 | Executing Agency |
| *(ii) Increase in women-led agricultural cooperative societies and farm-based associations, adopting low carbon technologies to reduce post-harvest losses of staple food crops in programme areas* | • At least 50% of women-led agric business enterprises in project area | By 20XX+5 | Executing Agency |
| *(iii) SHWFs and women-led agribusiness enterprise (WABEs), provided with access to low carbon technologies for renewable energy generation* | • At least 50% of women-led agric business enterprises in project area | By 20XX+5 | Executing Agency |
| ***Output(s) Statement 2: Climate resilient agricultural practices and technologies adoption among smallholder farmers promoted*** | | | |
| ***(i)** SHWFs provided with access to finance to invest in climate resilient agriculture and agroforestry practices* | • At least 50% of SHWF in project areas | By 20XX+5 | Executing Agency |
| *(ii) SHWFs and WABEs trained in the adoption of climate resilient livelihood options and low carbon technologies* | • At least 60% of women-led agric business enterprises in project area | By 20XX+5 | Executing Agency |
| *(iii) SHWFs and WABEs trained in climate smart agro advisory services and digital technology adoption* | • At least 75% of SHWFs and WABEs in project area | By 20XX+5 | Executing Agency |
| ***(iv)** SHWFs and WABEs trained in the maintenance of installed hydromet equipment* | • At least 20% of SHWFs and WABEs in project area | By 20XX+5 | Executing Agency |
| ***Output(s) Statement 3: Enabling environment for the adoption of sound agribusiness policies within the SCPZs enhanced*** | | | |
| (i)*Women-led civil society organisations (WCSOs) empowered on land Reforms* | • 30% WCSOs in project areas | By 20XX+5 | Executing Agency |
| *(ii) More female-headed households (FHH) empowered with managerial responsibilities at SCPZs* | • 50% of managerial staff at SCPZs | By 20XX+5 | Executing Agency |

# B. GENDER PLAN OF ACTION FOR SENEGAL

## B.1 GENDER ANALYSIS/ASSESSMENT

***Context and background***

Senegal's population is predominantly rural (54.8%) and has a young demographic structure (50% of the population is under 18 years of age). Women represent 50.1% of the population. The country has made significant progress in gender mainstreaming. However, the World Economic Forum ranks Senegal 94th out of 149 countries in the Gender Gap Report in 2018, reflecting the persistence of large social disparities between men and women in the country. Several constraints related to the patriarchal nature of society hinder the achievement of gender equality in Senegal, including insufficient enforcement of laws in favor of women, the persistence of social and cultural constraints.

It is established that economic constraints and cultural norms prevent women from having optimal access to natural resources and formal jobs, which implies that their livelihoods or the improvement of family livelihoods (nutrition, health, education, food, energy, income, etc.) depend on initiatives in climate-sensitive sectors such as agriculture, forestry, water collection, energy, etc. Indeed, women have distinct vulnerability, exposure to risk, inadequate coping capacity, and inability to recover from climate change impacts (Masika 2002[1]). Women's empowerment strategies are directly affected by the upheavals of climate change: they walk longer distances to find food, fuel and water, which are becoming increasingly scarce, and to care for family members exposed to the health risks associated with climate change. They spend less time on education, income-generating activities or participation in decision-making processes in the community, which further accentuates gender inequalities. In addition, gender inequalities in the distribution of assets and opportunities (e.g. difficulties in access to land, credit and permits to use forest resources), make women's choices even more limited in the face of climate change.

It is recognized that economic constraints and cultural norms prevent women from having optimal access to natural resources and formal employment, which implies that improving family livelihoods (nutrition, health, education, food, energy, income, etc.) depends on initiatives in climate-sensitive sectors such as agriculture, forestry, water harvesting, energy, etc. Women are clearly vulnerable, at risk, lack adaptive capacity and are unable to recover from the effects of climate change (Masika 2002). Their empowerment strategies are directly affected by the upheavals of climate change: they walk longer distances to find food, fuel and water, which are becoming increasingly scarce, and to care for family members exposed to the health risks associated with climate change. They spend less time in education, income-generating activities or participation in community decision-making processes, further exacerbating gender inequalities. In addition, gender inequalities in the distribution of assets and opportunities (e.g., difficulties in access to land, credit and permits to use forest resources) further limit women's choices in the face of climate change.

Gender inequalities in the region covered by the project heighten women’s vulnerability, which is required to better mainstream their needs into the project and to focus on the integration of women’s empowerment initiatives. Women are still few from the climate change and natural resource-related decision-making processes at all levels. Equal participation in community-based decision-making remains a complex and difficult goal to achieve, especially in the contexts of highly unequal gender relations. Despite significant effort, in general, women’s participation has low social visibility, and this has repercussions on social development strategies implemented to support or assist vulnerable households. Resilience is inconceivable without rural women

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***Legal and policy framework and Institutional mechanism***

In 2001, Senegal adopted a constitution that reaffirms the equality of women and men. Senegal has set important milestones in terms of gender mainstreaming in public policies. These include the law on absolute parity between men and women in elective bodies, the Prime Minister's circular letter on gender budgeting through the Medium-Term Sectoral Expenditure Frameworks in 2008 The country also has a National

[1] Masika, Rachel. 2002. “Gender and Climate Change.” Genderand Development Journal 10 (2): 2–9.

Strategy for Gender Equality and Equity (SNEEG) 2016-2026 structured around four issues: (i) the enhancement of women's social position and the strengthening of their potential; (ii) the economic advancement of women in rural and urban areas; (iii) the promotion of the equitable exercise of the rights and duties of women and men and the strengthening of women's access and position in decision-making spheres; and (iv) the improvement of the impact of interventions in favor of gender equality and equity.

The National Gender Mechanism for the Advancement of Women is responsible for ensuring the political and operational implementation of the public authorities' commitment to equality between women and men and the advancement of women. The Directorate of Equity and Gender Equality (DEEG) was created by decree n°2008-1045 and the Family Directorate has a division in charge of Women. In addition, gender units have been set up at the level of sectoral ministries. The Ministry of Agriculture and Rural Development has an operational gender unit that coordinates capacity building activities on gender equity and equality in farmers' organizations and leadership of members of women's farmers' organizations.

***Governance***

Senegal is cited as one of the top 7 African countries in terms of women's representation in politics: there were about 41.7% women in local authorities and 41% in the National Assembly in January 2019. However, women are still under-represented in political life both as voters, elected officials and in public administration. For example, there are only about ten female mayors and few of them are heads of a large municipality.

***Poverty and migration***

The incidence of poverty at the national level is on a downward trend, falling from 46.7% in 2011 to 34.6% in 2015. It is estimated at 59.9% for men against 42% for women in rural areas. The project area has a poverty rate higher than the national average, which particularly affects women given their lack of access to productive resources.

International migration is 6.1 in the region and remains the most important of the Casamance regions. The migration phenomenon mainly concerns the class composed of young people between 15 and 24 years of age and that of men than women (11,533 male migrants compared to 10,916 female migrants). Migration of household members who do not send remittances is likely to increase household food insecurity and spouses of international migrants are worse-off in several domains of empowerment, including decision-making on productive activities and agricultural income, and access to information.

***Socio-economic indicators***

In terms of health, life expectancy at birth in 2018 was estimated at 67.5 years, above the average for sub-Saharan Africa (60.7 years). Women live longer on average than men (69.7 years compared to 65 years, respectively).

Maternal mortality remains high overall at 434 per 100,000 live births. There are 459 deaths in rural areas compared to 398 deaths in urban areas per 100,000 live births. In 2017, the infant and child mortality rate reached 56% in 2017.The fertility rate is 4.6 children per woman. In rural areas, a woman has an average of 5.9 children compared to 3.5 for a woman living in urban areas (DHS 2016). The proportion of births attended by skilled personnel was 68% in 2017. Modern contraceptive use has improved significantly: the contraceptive prevalence rate was 28% in 2017. Female genital cutting remains a strong cultural practice (24% of women aged 15 to 49 years excised in 2017) and 27% of women have been victims of gender-based violence (GBV).

Regarding food and nutrition security, according to the Rural Food Security and Nutrition Survey (ERASAN 2014), rural households, which are food insecure, represent 30% of households (about 153,728 households), of which 12% are in severe and 18% in moderate situations. The food insecurity rate is higher among female-headed households, where it is 40.4% compared to 29.4% for male-headed households. The regions of Ziguinchor (63%), Sédhiou (52%) and Kolda (52%) have the highest rates of food insecure households.

Nearly one in two Senegalese children aged ten or over (45.4%) can read and write in any language (53.7% for men and 37.7% for women). At the pre-school level, girls dominate with 51.2% while boys represent 48.8%. The rural environment has the lowest enrolment rate at only 35.4%, while the urban environment polarizes 64.6%. For primary education, the rural environment dominates with 51.2% of learners. Girls represent about 51% in rural areas. In the field of secondary education, girls are more in the minority in rural areas (38.7%) than in urban areas (45.4%). Finally, for higher education, men represent 60.5% of the workforce compared to 39.5% for women. This difference is the result of the low representation of women (27.0%) in rural areas. For vocational training, women are generally more disadvantaged than men: 92.5% of women do not have access to it compared to 86.3% for men.

**Employment / Unemployment**

The National Employment Survey of Senegal (ENES), conducted in 2015, shows that the labour force participation rate is 65.2% at the national level. It is more important in men (72.1%) than among women (52.3%). At national level, in terms of employment rates, there is a clear predominance of young men (55.2%) over young women (34.6%). The distribution of jobs by age group shows the same disparities to the disadvantage of women. The unemployment rate for people aged 15 and over was estimated at 15.7% in the 4th quarter of 2017. Women are more affected (22.1%) than men (9.6%). According to the ENES 2015, the underemployment is 27.7%. This indicator reveals a significant disparity by gender. Indeed, the level of women underemployment linked to women's working time is very high, compared to that of men (40.3% and 20.9% respectively).

In the project area ;

- Sédhiou region, unemployment in the region is quite high with a rate of 21.7%, of which 16.5% for men and 12.1% for women, and represents a fairly frequent phenomenon;
- Kolda region: the employment rate is 55.6% for men and 60.0% for women. According to residence, it is 42.5% in urban areas and 46.5% in rural areas. The unemployment rate observed is 38.8% overall in the Kolda region. It is 30.6% for men and 53.7% for women;
- Ziguinchor region: the employment rate is 52.9% for men and 32.0% for women. The unemployment rate is 26.5%, of which 18.4% for men and 42.1% for women.

***Agriculture***

Women are very active in the processing and marketing of food products, livestock and fisheries. Generally, women work more on family land and in subsistence activities. In the informal sector, they are widely represented in small trade (40%) and in services. Women have little access to training and research services, and illiteracy affects 69.3% of them. They have difficult living conditions, particularly in terms of access to drinking water, sanitation, health, transport and communications.

Some of them are involved in export crops, particularly horticultural products (fruit and vegetables) and handicrafts, even if they face significant difficulties related to access to promising markets, information, training and services for marketing, packaging, labeling (e.g. labeled products, organic certifications), and promising outlets. They face unfair competition from manufacturers and exporters.

Women are scarcely present in cash crops, except as unskilled labour. The production and marketing of non-timber forest products is an activity mainly practiced by women. They use these products as nutritional supplements, for family pharmacopoeia but also to generate income.

At the level of agricultural value chains, women are less integrated and play the least skilled roles than men due to their low levels of literacy and education, their low income and limited access to factors of production (capital, land) and financing. They are often mere employees, while men occupy managerial positions. They also dominate activities generating higher added value thanks to their strong purchasing power enabling them to acquire means of production. For the crop production sub-sector, at the level of the various links in agricultural value chains, women are more involved in sowing or transplanting and harvesting, in the processing, conservation and marketing of agricultural products, while men are more involved in the supply of inputs and production (phyto treatment, watering, soil preparation, manure spreading)

Despite their poor access to factors of production, women occupy a significant share of the rural labour force and play an important role in the various links in the agricultural value chain. Thus, according to the annual rainfed crop survey, 80.7% of farms in 2015 were run by men compared to 19.3% by women, while in 2014, 16.4% of farms were run by women compared to 83.6% by men. This shows an improvement in the situation of women

Generally, women generally have little access to land, due to their dependence on men's land management and other productive resources (seeds, fertilizers, loans). In this regards, women farmers organizations often appears as a means of economic emancipation that allows them access to means of production and resources. Thus, men control 93.6% of the cultivated areas compared to 6.4% for women, with the exception of rainfed rice cultivation where women farmers cultivate 62.7% of the plots. In general, women access land indirectly or even through family farm managers, village chiefs and landowners. This access is done through verbal agreements without written records. Their access is therefore not exempt from customary law. Since women are expected to marry and leave the family, the traditional system of land tenure considers that giving them access rights to land could lead to a dislocation of the family's land assets.

Regarding access to seeds, women face difficulties in obtaining them due to both their male head of household status and their low income :70.1% of men were able to build up personal seed reserves against 53.7% of women; 30.8% of men bought seeds without subsidies against 20% of women.

Because of their status as inferior, women have less access to the labour force than men in the exploitation of their plots. On the other hand, men, because of their status as heads of households, can mobilize family labour. This explains why most of them cultivate small areas of about 0.4 ha, while men cultivate an average of 1.3 ha. In addition, women are unable to maintain their plots properly, due to not only the overload of domestic work and lack of sufficient manpower for tillage, but also because of their poor access to agricultural inputs such as plant protection products (herbicides, etc.) and agricultural equipment.

The project runs in a social environment where gender inequalities are still alive and accentuated by the migration of men. This situation reinforces the triple role of women with its corollaries of work overload and poverty. Thus, the project will take into account the specific needs of women and men as well as young people of both sexes. The aim is to reduce the inequalities of access for women and men of all social categories to the opportunities offered by the Agropole. Thus, the actions to be carried out are: (i) equity in the implementation of activities in the project (selection of projects, training, etc.), (ii) taking into account the specific needs of women and girls in the choice of infrastructures to be financed, (iii) support to women's and youth groups, (iv) recruitment of an expert in strengthening gender-specific organizations in the PIU. Thus, the project would be classified in category 2 according to the Bank's gender marker system.

## B.2 GENDER ACTION PLAN: SENEGAL

| Activities | Indicators and Targets | Timeline | Responsibilities |
|---|---|---|---|
| ***Impact Statement**: Strengthened climate resilience of the agricultural value chain, to help stimulate productivity, value addition, competitiveness and income generation for the most vulnerable communities, including women and girls.*<br><br>***Outcome Statement**: Improved Access to Climate Resilient Livelihood Options for at least 100,000 Smallholder Women Farmers including Women-led Agribusiness Enterprises (Farm-Based Associations and Cooperative Societies).* | | | |
| ***Output(s) Statement 1: Critical SAPZ value chain infrastructure strengthened and made more climate resilient*** | | | |
| ***(i)** Smallholder women farmers (SHWF) provided with access to finance to invest in drip irrigation technology adoption* | • At least 20% SHWF in project areas | By 20XX +5 | Executing Agency |
| *(ii) Increase in women-led agricultural cooperative societies and farm-based associations, adopting low carbon technologies to reduce post-harvest losses of staple food crops in programme areas* | • At least 50% of women-led agric business enterprises in project area | By 20XX +5 | Executing Agency |
| *(iii) SHWFs and women-led agribusiness enterprise (WABEs), provided with access to low carbon technologies for renewable energy generation* | • At least 50% of women-led agric business enterprises in project area | By 20XX +5 | Executing Agency |
| ***Output(s) Statement 2: Climate resilient agricultural practices and technologies adoption among smallholder farmers promoted*** | | | |
| ***(i)** SHWFs provided with access to finance to invest in climate resilient agriculture and agroforestry practices* | • At least 50% of SHWF in project areas | By 20XX +5 | Executing Agency |
| *(ii) SHWFs and WABEs trained in the adoption of climate resilient livelihood options and low carbon technologies* | • At least 60% of women-led agric business enterprises in project area | By 20XX +5 | Executing Agency |
| *(iii) SHWFs and WABEs trained in climate smart agro advisory services and digital technology adoption* | • At least 75% of SHWFs and WABEs in project area | By 20XX +5 | Executing Agency |
| ***(iv)** SHWFs and WABEs trained in the maintenance of installed hydromet equipment* | • At least 20% of SHWFs and WABEs in project area | By 20XX +5 | Executing Agency |
| ***Output(s) Statement 3: Enabling environment for the adoption of sound agribusiness policies within the SCPZs enhanced*** | | | |
| (i)*Women-led civil society organisations (WCSOs) empowered on land Reforms* | • 30% WCSOs in project areas | By 20XX+5 | Executing Agency |
| *(ii) More female-headed households (FHH) empowered with managerial responsibilities at SCPZs* | • 50% of managerial staff at SCPZ | By 20XX+5 | Executing Agency |

# C. GENDER PLAN OF ACTION FOR GUINEA

## C.1 GENDER ANALYSIS/ASSESSMENT

***Context and Background***

***Demographics***

With 51.7% women, Guinea's current population is estimated at 11,883,516[2], including 1.9 million for the city of Conakry. The rate of urbanization is low: only 33% of the population lives in urban areas and more than 50% of the urban population resides in Conakry. The population growth rate remains high, although it has stalled in recent years, from 3.1% in 1983-96 to 2.2% for 1996-2014 (RGPH3). It is the result of this strong growth, an extreme youth of the population. More than half of the population (51.3%) is under the age of 18 and more than three-quarters of the population is under 35 (77.4%). Life expectancy at birth is estimated in 2018 at 61.7 years for women, compared to 58.8 years for men. In 2014, more than four in ten women (47.3%) were of childbearing age, or 24.5% of the total population.

**Socio-cultural data**

The current Guinean population is the result of a deep mix between indigenous populations and migrants who have come in successive waves (RGPH 2014). It is made up of several ethnic groups that live in harmony. Several dialects or languages are spoken in the country, the main ones being Soussou, Malinké, Peul, Kissi, Toma and Guerzé. Religiously, there is a high representation of Muslims (89.1%); Christians account for 6.8%, animists 1.6% and people without religion 2.4% (RGPH 2014). The variety of religious customs and beliefs induces different behaviors regarding certain practices such as marriage and dowry. The dowry, sum of money or property given by the fiancé (or his family) to the father of the bride (Binet; 1958) is a very old tradition that modern Guinean legislation has enshrined (Article 249 of the Civil Code of February 2016) by providing that the future spouses may be symbolically suitable for the determination of a dowry, in kind or in cash, to be given to the bride-to-be. In fact, rites and practices vary and women are endowed according to rules specific to ethnic groups. In some Guinean ethnic groups, the high value, but especially the social perceptions and representations of the dowry make it an instrument of domination of the man over the woman, since interpreted as a support of the title of property granted to the latter on his woman who is now one of her property. Polygamy is widespread, with a prevalence of 48% (MICS 2016). The same is true of free unions. Some ethnic groups practice levirate and sororat.

The country is marked by a socio-cultural context that is unfavorable to women's rights, with the persistence of harmful traditional practices such as forced and early marriage and female genital mutilation/cutting (96.8%, MICS, 2016). Guinean customs limit women's power to the administration of the household, to the domestic aspects of family and community management. In this regard, their activities are agricultural, domestic work and the education of children[3]. Unsurprisingly, the AfDB's Gender Equality Index ranked Guinea 48th out of 52 countries assessed in 2015[4], with customary rules and practices devoting the distribution of roles and tasks between men and women (revised PNG, 2017). With a gender index of 0.439 (OECD SIGI Index), Guinea is among the 8 countries (78th out of 86) with the largest disparities between women and men in the non-OECD area and among one of the ten African countries with the highest gender disparities.

***Legal Framework and Gender Promotion Policy***

**Legal and regulatory framework for promoting gender equality in Guinea**

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2 http://www.stat-guinee.org/

3 Institut National de la Statistique – RGPH 2014 - Rapport d'analyse des données - thème : Situation des femmes - Analystes : LENO Émilie Bernadette & KABA Ibrahima ; 2017

4 Banque africaine de développement ; Autonomiser les femmes africaines : Plan d'action - Indice de l'égalité du genre en Afrique ; 2015

In its preamble, Guinea's Constitution proclaims its adherence to the ideals and principles, rights and duties established in the African Charter of Human and Peoples' Rights and its additional protocols on women's rights. Article 8 states that men and women have the same rights. The country has signed and ratified all international conventions on women's rights and gender equality, including the Convention on the Elimination of All Forms of Discrimination against Women (CEDEF). The country has a legal arsenal that guarantees the principle of equality between men and women. However, some notable discriminatory provisions relate to the legal age of marriage (s. 280) which is 17 years for women and 18 for men, family authority (s. 324), since "the husband is the head of the family" and, as a result, the choice of the residence is his responsibility (s. 247 and 331) and that he may object to his wife practicing the profession of his choice (s. 328). In the event of divorce, the woman will only be able to obtain custody of the children until the age of 7 (s. 359). Adultery is considered grounds for divorce if committed by the wife. For the husband, it will only be considered a ground for divorce if the husband has "maintained his concubine in the matrimonial home" (s. 341 and 342). In addition to the juxtaposition of legal orders (religious, customary and modern), there is a lack of application of the laws, in a context where these texts and the mechanisms of redress are unknown to the population in particular women (due to illiteracy and socio-cultural burdens).
On the other hand, the existing texts enshrine the principle of non-discrimination in access to land ownership. Article 1 of the Guinean Land Code allows any individual or legal person to hold the right of ownership on the land and the buildings he wears. But the practices are mainly based on the preponderance of customary law, which discriminates more against women, both for access to land and for the right to inheritance. Currently, reflections are under way on land reforms. However, the general statements on land held in 2016 were not an opportunity to invite gender issues into these debates.

***Policy framework for gender promotion***

The Government at the end of May 2018 has only 4 women out of 33 portfolios, compared to 7 in the previous government. In 2010, the Electoral Code provided for a quota of 30% representation for women in Article 129. But these quotas have not been met. This provision was abolished in the Electoral Code promulgated on 27 July 2017. The main political parties in the 2014 municipal and legislative elections had 22.85% in the Union of Democratic Forces of Guinea - UFDG (out of 6,521 candidates, there are only 1,490 women) and 26% in the Union of Democratic Forces - UFR (3642 candidates, 951 of whom are female candidates). In the 2015 presidential election, out of eight candidates, there was only one woman, a candidate of the Guinean Green Party. According to statistics provided by the Independent National Electoral Commission (CENI), out of a total of 29,669 councilors for the 342 constituencies in the 2014 municipal and legislative elections, there were only 7070 women, or 23.83% of compared to 77% of men. Women MPs currently account for only 23% of the seats in parliament, or 26 out of 114. Women often hold junior positions in the public sphere. There are no spaces for preparing young girls for positions of responsibility. Illiteracy and the lack of financial resources of women are also constraints to their political participation.

Guinea has joined the Beijing Action Programme and the 2030 Sustainable Development Programme based on the achievement of the Sustainable Development Goals (SDGs). The National Gender Policy adopted in 2011 and revised in 2017 is a contribution to addressing the inequalities that exist between women and men in Guinea. It proposes a systemic approach involving all key sectors and players in the country's development. Several other policies, national strategies, programs and projects contribute to gender equality.

One can quote without being exhaustive: the National Policy of Social Protection; National Family Policy; National Policy for the Protection of Children; National Health Policy; Agricultural Development Policy; National Environment Policy; The National Strategy to Combat Gender-Based Violence; Energy Policy Statement; The National Strategy to Combat the Feminization of HIV/AIDS; The Strategic Plan for the Implementation of Recommendations 1325 and 1820 on Women, Peace and Security; The Strategic Plan to Implement the African Women's Decade in Guinea; National Health Development Program; National Reproductive Health Program; Women and Youth Development Fund; Socioeconomic Reintegration Project for Youth and Women as part of the Peace Consolidation Fund; Social nets; Basic Education for All Program; Village Community Support Program; Project Supporting The Promotion of Gender; Rehabilitation Program of Women's Self-Promotion Support Centers.

However, this policy and policy framework would be more effective if a few other reforms were accelerated, including those related to gender-sensitive budgeting and gender cell operationalization within departments. The Ministry of Planning and Cooperation undertook in 2010 with the support of the United Nations Population Fund an assessment of the level of integration of population, reproductive health and gender issues into Guinea[5]. Moreover, the extension of the national gender policy as revised in 2017 remains insufficient; its effective operationalization is slow to take place, not only because of budget shortfalls, but also in the absence of a multi-sector plan to implement this instrument, with a coherent and realistic budgetary framework.

***Institutional mechanisms for gender promotion***

The Ministry of Social Affairs, Women's Promotion and Children's Affairs is central to the development, development and coordination of the Government's policy in the areas of its jurisdiction. The main mission of the National Directorate of Women's Promotion and Gender is to promote women's rights and protect them from all forms of violence and discrimination, with central and decentralized structures. In addition, several MECHANISM mechanisms can be noted: (i) gender cells in all ministries, (ii) gender-specific thematic group which provides a strategic interface between the government, the United Nations system and civil society organizations, (iii) a network of women ministers and MPs, (iv) regional anti-VBG committees, v) national observatory on VBG, vi) National Committee for the Abandonment of Female Genital Mutilation/Cutting, vii) Monitoring Committee on CEDEF Recommendations, and viii) National Steering Committee for the Implementation of Recommendations on Women, Peace and Security. Among the public institutions is the Directorate General for women's empowerment and promotion (CAP/WOMEN).
While these structures are making an intensive effort to promote gender in its many facets, it should be noted that the staff capacity is very inadequate, with most of the organic framework positions not being filled, mainly in decentralized structures. The capacity to formulate projects and programmes on the one hand and to research and mobilize funding, on the other hand, should be strengthened to enable them to better structure the responses to the expectations of their targets, to equip themselves with resources logistics, human, financial and appropriate technical and methodological tools and improve their performance. The same is true of communication, archiving, reporting and capitalization capabilities. Women's entrepreneurial skills development skills deserve to be strengthened within the Women's Empowerment and Promotion Branch.

In 2015, the Government, by Order No. 2015/1257 of the Minister in charge of the Public Service, established a Gender and Equity Service within each ministerial department with the mission of implementing and monitoring the government's policy in the gender and equity. Within departmental departments, the operationalization of gender services is done at variable speeds; less than half have dedicated offices, less than a quarter have office furniture and less than 10% have computer equipment and an operational internet connection. In order to make these structures more efficient, they need to be provided with the right logistical, human, financial and technical and methodological tools, particularly for monitoring gender consideration in sectoral projects and programmes, production and dissemination of an annual follow-up report.

Civil society organizations working to promote gender equality and empower women in Guinea play a role of citizen watch. They are close to communities and populations and are key partners of the state in the formulation and implementation of public action. In addition, they are key players in information, awareness, training and advocacy for the ownership of political and strategic instruments to promote and protect the rights of men and women. A list of the main organizations is included in the appendix. Although very active, these structures face challenges, the most important of which are the lack of expertise in project formulation on issues of equality, research and mobilization of resources, the lack of resources to drive endogenous processes. They need support in archiving, funding and disseminating their results. They have difficulty relaying at the level of national authorities the learnings resulting from their participation in regional and international workshops and forums. A study of training, structuring and logistics needs could lead to an interesting and useful project to build the capacity of CSO networks to promote its kind in Guinea.

5 Guinée ; évaluation du niveau d'intégration des questions de population, de sante de la reproduction et de genre dans les politiques et programmes de développement ;
2010 - web2.unfpa.org/public/about/oversight/evaluations/docDownload.unfpa?docId=51

***Gender and Cross-Cutting Issues***

**Activity and employment**

The Statistical Yearbook 2016 and RGPH3 (2014) show that the proportion of the employed labour force aged 6 and over was 52.8% in 2014, of which 61.5% for men compared to 44.5% for women, with large disparities according to the environment, 38.9% for urban and 60.7% for rural people. The net participation rate of this group is 55% (64.8% for men versus 46.1% for women; 43.3% in urban areas compared to 61.7 in rural areas). For the 15-64 age group, it is 63.8%, higher among men (78.4%) women (51.4%). For the 15-19 age group, it is 44.4%, with disparities by living environment (29.1% in urban compared to 56.4% in rural areas). It is 51.7% for people who are literate compared to 67.3% for non-literate people.
The informal sector occupies 95.2% of the population. The majority of informal jobs (77.5%) rural development in the broadest sense (agriculture, hunting, forestry and fishing). The formal sector occupies 4.8% of the labour force, mainly in public administration. The employment rate is relatively low for women (41.7% versus 72.1% for men). The labour force unemployment rate is 5.2%, and is higher for men (6.3%) women (3.9%). People with a high degree are relatively at risk of unemployment. Data from the statistics service METFP-ET, 2017 taken by the revised DPNG document, show that Guinean women make up only 9.7% of the formal sector labour force, very low in terms of their demographic weight (51.7%).

**Entrepreneurship**

On women's entrepreneurship, The Global Gender Gap Report 2017 cites Guinea as one of the countries that have made substantial progress in covering gender gaps in women's participation and economic opportunities. With a score of 0.813 in this area, Guinea ranks 10th in the world out of 144 countries. Women work mainly in agriculture (over 80%). But they are present in the foodservice, crafts, mining, local product processing and small business subsectors, but they are often far from providing leadership. In the artisanal mining sector, for example, there are more than 100,000 farmers, with at least 800,000 dependents. It is estimated that 40% of these people are women. The average age of these women is 29 years for those in diamond mining and 26 years in gold. 81% of women working in the diamond sector are illiterate, compared to 91% in the gold sector. Almost all work on behalf of men (90%). For 25% of these women, incomes are only used to meet the primary needs of their families. They work in very precarious conditions: 7 days a week and an average of 10 to 12 hours of work per day; Informal access to operating sites; - inadequate infrastructure: promiscuity, housing offering no comfort.

In general, women continue to face difficulties, the most important of which are their low access and control of productive resources, their time/budget workload and their submission status that limit their access to certain opportunities and markets, limited entrepreneurial, managerial and organizational skills, difficult relationships with tax authorities and infrastructure shortfalls, energy and transport deficiencies notably. Guinean women's entrepreneurship needs a new lease of life to play a leading role in women's empowerment.

**Education**

MICS 2016 establishes that only just under two in five women (39%) know how to read and write. Literacy of young older women is negatively correlated with age; those aged 15-19 (41.6%) are more literate than those aged 20-24 (36%). Among children of primary school age (age 6) in Guinea, only 43% are admitted in the first year (42% among girls and 45% among boys). The gross primary school enrolment rate (GST) has changed positively over the decade, from 78% in 2010 to 85% in 2016. The TBS of primary school girls is still about 7 to 9 points below average (INS, 2016). The dropout rate is still a concern, 9% for the whole, 10% for girls and 8.2% for boys. Unfortunately, the number of children dropping out during the cycle increases as children progress through this cycle, as evidenced by the drop-out rate in the fifth grade, which was 27.99% in 2015.

In 2015-2016, general secondary education recorded a gender parity index of 0.74 and a Gross Enrolment

Rate (GST) of 38% for all, of which 28% for girls and 49% for boys. The TBS of girls, which was 75.6% at the primary level, drops in secondary school to 28%. The reasons for this decline lie in the weight of traditional values and early marriage of young girls. The situation of girls in the technical sectors is of greater concern. Of the 18,182 applicants, including 6,191 girls, in the 2017 session exit exams, only 3,138 were declared admitted. Regarding gender parity in ETFP institutions in 2017, out of a total of 35,199, there were 17,487 girls (49.7%) compared to 17,712 boys (50.3%). In order to promote girls' access to technical courses, certain challenges must be met, including: (i) the low attendance of technical and technological sectors by girls, women; (ii) the lack of qualified trainers in technical and vocational education institutions; (iii) inadequate training in relation to employment needs; girls' low awareness and motivation in technical occupations.

In higher education, the parity index has changed little, from 0.35 in 2011 and 2012 to 0.40 in 2013 and 2014 (INS, Yearbook 2016). At the 2018 session of the Single Baccalaureate, there were 99,030 applicants, 37,106 of whom were girls, or 37.5% (Ministry of National Education). A biometric student census is underway. Its completion will provide more reliable and up-to-date data. There is an under-representation of girls/women in higher education institutions (IES) in general and in particular in Science, Technology, Engineering and Mathematics (Stem). To achieve the sustainable development goals associated with education, strong literacy and second-chance schools for girls and women need to be established, and awareness-raising efforts need to be strengthened. on girls' enrolment in school and the fight against dropping out of secondary school in particular. In higher education, incentives should be developed to encourage girls to complete secondary school and prosper in higher education.

**Information and Communication Technologies**

According to MICS 2016 only 8.9% of the women surveyed read a newspaper or magazine, 40.2% listen to the radio and 41.3% watch television at least once a week. Overall, 42.3% are not regularly exposed to any of the three media, while 57.5% are exposed to at least one and 5.6% to all three media types each week. From an age perspective, women under the age of 25 are more exposed to all three types of media than older women. More than six women aged 15-19 (63.2%) 20-24 years (65.3%) out of ten are exposed to at least one of the three media outlets, at least once a week. This proportion decreases to 56.3% for women aged 25-29 and 45.6% for those aged 45-49. 8% of women aged 15-24 have used a computer, 5.7% have used it in the past year and 4% have used it at least once a week in the month prior to the survey.

Overall, 19.4% of women aged 15-24 have used the internet, while 17% have used it the previous year. The proportion of young women who used the internet more frequently, at least once a week in the month before the survey, is lower at 15.4%, while the percentage of young women aged 15-24 who used a computer during the last 12 months before the survey is 5.7%. Women's access to information remains limited. The information comes to them through traditional and local communication channels such as meetings, training, or the market and sometimes radios. Women's access to ICT remains very limited despite the efforts already made. This is mainly due to the low level of education of women. The mobile phone is by far the most common property in households (84% of households own a telephone at least). However, the data disaggregated by gender of the telephone owner in the household are lacking. These figures support ICT training programmes for girls and women to facilitate their connection to opportunities, as well as the establishment of community radio stations, to improve their exposure to information.

**HIV, maternal and child health**

According to the PDNES, HIV/AIDS remains a public health problem with a prevalence of 1.7% in 2015 while the Government's target was less than 1.5%. This rate is almost twice as high among women (2.1%) men (1.2%). The prevalence of the virus in pregnant women is higher than that of the general population. This is mainly due to the fact that HIV testing is mandatory during antenatal consultations. The rate of access to ARVs for pregnant women increased from 17% in 2011 to 62% in 2014 (PDNES), marking an improvement in limiting the spread of the AIDS virus from mother to child. The vast majority of women aged 15-49 say they have heard of AIDS, 71% (84% in urban areas compared to 61% in rural areas). Nearly 47% say they use condoms as the main means of preventing HIV transmission. The percentage of women aged 15-49 who know that HIV can be transmitted during pregnancy is 45%, 46% during childbirth and 46% during breastfeeding. The percentage of those who know that HIV can be transmitted from mother to child and that this risk can be reduced if the mother takes special medications is 35%. These figures appeal to public and private actors and call on them to intensify programmes aimed at behavioral change, prevention campaigns and sexuality education for both girls and women and men.

MICS 2016 reveals that the level of infant mortality for the most recent period (0-4 years prior to the survey) is estimated at 44 per 1000 live births. Health risk factors for women remain poverty, inaccessibility of health centers and low levels of maternal education. Despite efforts, access to health services is still marked by a lack of infrastructure and health personnel at almost every level of the urban and rural shelter pyramid. Regarding reproductive health, MICS 2016 found that 31% of women aged 15 to 19 started their fertile lives (26% having already had a live birth and 5% being pregnant with their first child at the time of the survey). In addition, a significant proportion of women aged 15 to 19 had a live birth before the age of 15 (6% for the whole, 10% in urban areas compared to 4% in rural areas). The total fertility rate for women aged 15-49 is 4.8. The overall fertility pattern by age indicates that procreation begins early in Guinea (MICS 2016). Early pregnancies, expressed by the percentage of women aged 20-24 who had at least one live birth before the age of 18, is 36.9%.

The contraceptive prevalence rate (percentage of women aged 15-49 who are currently married or in a common-law relationship who use (or whose partner uses) a contraceptive method (modern or traditional) is very low, about 8.7%. Adolescent girls are much less likely to use contraception than older women. Only 6% of women aged 15-19 who are married or in union currently use a contraceptive method compared to 8% of women aged 20-24, while contraceptive use among older women is 11% for each age group aged 30-34 and 35-39 years old. These figures call for an intensification and better targeting of family planning and reproductive health programs.

**Climate change**

The main climatic disturbances identified are the decline in rainfall, violent and recurrent droughts since the 1970s, early and frequent floods (Kankan- 2001, Boké - 2003, Gaoual- 2005, etc.), disturbances to the rainfall regime. They are responsible for the drying up of rivers, the drying up of soils, the destruction of vegetation cover, the decline in agricultural production, the resurgence of waterborne diseases, especially in the northern part of the country. These changes have undermined agropastoral populations; herds have been decimated by lack of water and pasture. The devastation on food crops has led to food insecurity problems. Traditional roles for women exacerbate their vulnerability and precariousness in the face of the negative effects of climate change.

According to Guinea's National Climate Change Adaptation Action Plan, more than 80% of the population lives and works in rural areas. Their livelihoods are increasingly degraded. Thus, all socio-economic groups, which depend on ecosystems and their resources to meet their subsistence needs, are vulnerable. While a plurality of actions are planned in the PANA or in the National Determined Planned Contribution (NDPC) under the United Nations Convention on Climate Change (UNFCCC), none of the documents say how women and women people are affected by climate change in Guinea, nor how they are affected by adaptation actions/measures, even if they mention the low resilience to natural disasters to which the country is exposed (floods, floods, drought). Although the country has national policy documents and programmes on climate change, they do not take gender into account in their orientation. There are the lack of baseline studies to measure the impact of climate change on men and women and to assess adaptation strategies based on the specific types of activities in agriculture, health, and water, for example, where it is postulated that the effects of climate change are observed by a decline in land productivity, the drying up of watercourses, the upheaval of economic activities, health problems in the scarcity and distance from biomass (firewood). The risks of annihilation of sources of income and livelihoods related to agriculture, livestock and fishing are greater in these different areas/sectors and can aggravate the already complex nutritional situation of the country.

Water resources are thus threatened by the activity of men and women due to deforestation, bushfires, pastoral and agricultural nomadism, the manufacture of baked bricks, carbonization and gold riverbank gullying, the destruction of forest galleries, the silting of the riverbed, the reduction of flow, pollution and the loss of biodiversity. Building the gender capacity of departmental executives in the development of climate change policy instruments and a national study on gender and climate change, including a review of endogenous strategies adaptations developed by men and women would bring significant added value.

**Water and sanitation**
According to MICS 2016 data, only one in four households (25%) exists in a hand-washing place. Moreover, among households where a place for hand washing exists, only 13% had both water and soap (or another cleaning product) at the specific site. This proportion is higher in urban areas (21%) rural (8%). In 10% of Guinean households, there was no soap, water or other hand detergents. It shows that in Guinea, the percentage of households that have soap or other cleaning products anywhere in the dwelling is 38%. For example, hospital-acquired infections are common due to practices that do not meet hygiene standards such as clean hands, instruments, maintenance of premises and management of biomedical waste in some health care stations. The same conditions for not respecting basic hygiene rules are available to all types of households, but there is a need to educate men and women about the issues of compliance with basic hygiene rules.

MICS 2016 shows that access to an improved source of drinking water is still a challenge in Guinea. Overall, 82.1% of the population use an improved source of drinking water for drinking, with large disparities: 98% in urban areas compared to 72.2% in rural areas; 99% in large cities compared to 96.5% in secondary cities; the situation in the Kankan region (88.4%) and Forest Guinea (87.6%) contrast with the Mamou region (60%) and that of Middle Guinea (67%). The use of drinking water from improved sources increases with the level of education of the head of household (95% in households whose head has secondary education or higher, 82.2% for the primary level and 77.4% for those who are said to be without level). MICS also reports that in rural areas, 21% of the population does not use sanitation facilities. People who use them most often use pit latrines with slabs (24%) and slabless pit latrines (41%). The bush and fields remain commonly used places of comfort (21%). The use of a sanitary facility shared or shared by the population is common not only among households using improved toilets (24%) but also among households that use unimproved toilets (13%). Members of urban households (49%) are more likely than those of rural households (29%) to share their sanitary facilities with members of other households.

This situation raises important gender issues given the importance of toilets and places of comfort in women's hygiene. Indeed, it is more than humans, vulnerable to the health risks associated with poor quality water. Its level of requirement on the quality of toilets is higher: cleanliness, drinking water giving the possibility of intimate toilet without risk of infection, guarantee of total discretion and safety, changing rooms where it can change, changing a baby, ... bins where she can eventually throw out diapers or worn towels. Toilets and places of comfort that are not comfortable for women are a hindrance to women's participation in economic and social activities and call for adequate government action. It is suggested to develop and implement a gender, water and sanitation program, with an important component on menstrual hygiene management.

**Gender-Based Violence (VBG)**
Guinea's context is one of a high prevalence of gender-based violence in a variety of forms. The percentage of women aged 15-49 who believe it is justified for a husband to beat or beat his wife under certain conditions is 70.1%. Early and/or forced marriages are a major societal problem. 54.6% of women aged 20-49 married or were in a relationship before the age of 18. More seriously, 21.1% of women aged 15-49 married or were in a relationship before the age of 15. The percentage of women aged 15-49 who report having undergone some form of FGM/C is 96.8% (INS/MICS, 2016). The practice seems to be more common in rural areas (98%). The prevalence of FGM/C among girls aged 0-14 remains high (45.3%). Similarly, girls from the poorest households are more likely to have experienced this practice (50%) than their counterparts in the households of the richest quintiles (41%).

The decline in the prevalence of female genital mutilation/cutting (FGM/C) among women must remain a major political objective. The Government, in collaboration with its partners, has made considerable efforts in the fight against VBG, including the adoption of legislation and regulations prohibiting and severely punishing FGM/E54 and the implementation of implementation of various initiatives aimed at training, raising awareness and raising awareness among stakeholders. At the police level, there is an Office for the Protection of Gender, Children and Moeurs (OPROGEM), with a mandate to combat violations of women's and children's rights and to protect morals. However, understaffing, lack of training for existing staff and failure to take into account violence in remote areas of the country prevent the effective functioning of the office. The Gendarmerie has a "Child and Gender Protection Division" in the Directorate of Judicial Investigations; but it is located only in the commune of Matam (Conakry), and this service lacks sufficient and adequately trained staff.

Despite this legal and institutional mobilization, the practice of FGM/C has increased in recent years (Report on Human Rights and the Practice of Female Genital Mutilation/Cutting in Guinea, 201655). In fact, the percentage of women aged 15-49 who said that the practice of FGM/C should continue is 67.2% (MICS, 2016), reflecting some approval for female genital mutilation/cutting. Support for prosecution is higher among out-of-school women (73%) those with primary and secondary education (68% and 50%)56. Other data on VBGs are included in Appendix 6. abnormally high; (ii) on the other hand, the results of those investigations may be reduced by modesty; indeed, the prevalence of certain practices such as domestic/domestic violence is underestimated because the phenomenon is killed by victims when they consider it a private matter and fall within the secret garden. There are also, as in any retrospective survey, underestimates related to memory defects, particularly among people with relatively distant exposure periods.

***Support for the development of agricultural and industrial value chains***

Guinean agriculture accounts for an average of 17.5% to 23% of GDP over the last five years, and accounts for 52.6% of the working population, 75.4% of which is in rural areas (RGPH3, 2014). It is largely dominated by family-owned farms. The main agricultural export sectors are coffee, cocoa, cotton, cashew, palm oil, rubber, horticultural products. Vegetable production is the main source of income for 57% of rural people, while 30% of rural people derive their income from livestock. Rice is the main food crop with 80% of farms, 67% of the area being enclosed, and occupies 23% of primary GDP, 11% of imports and 6% of national GDP. The country has nearly 6.2 million hectares of arable land, of which only 25% is cultivated annually. Livestock, forestry and fishing account for 17%, 14% and 4% of primary GDP, respectively. The animal sectors concern the rearing of cattle, sheep, goats, pigs, poultry, milk and egg production37. Industrial and artisanal fishing activities are carried out, but most of the activities of the subsector are carried out around fish farming activities carried out by hand. Fishing generates nearly 10,000 direct jobs and provides 40% of the animal protein consumed in the country.

According to the 2013 PNIASA report, agricultural activities are largely carried out by women. There is an average of 144 women per 100 men in the national agricultural population, or 87% of the female labour force. In general, women are present throughout the agricultural production process, from land preparation to post-harvest activities. They devote 80% of their working time to agricultural activities. Despite this reality, they control only a few resources in the sector they devote to family maintenance and child-rearing. Men, on the other hand, despite their relatively low participation in agricultural work, are masters of decisions about land-sharing and the choice of areas to be developed.

The Report on Women's Access to Land and Technology Resources in the Republic of Guinea (CEA, 1998) states that across Guinea, traditional systems provide women with land-use rights, depriving them of the rights to control and control transfer. These rights of use are based on women's relationships with men in the family. For example, in the event of divorce, separation or widowhood, women may lose their rights to the land they exploit. Advocacy initiatives for gender-friendly land reform have not yet produced significant results. Thus, for example, despite the strong involvement of women in the rice sector, the management of rice fields remains a reality mainly reserved for men; almost 91% of rice acreage is managed by men compared to only 9% of areas managed by women (ANASA, General Report of the Agricultural Survey 2014-2015, 2015)

Women's illiteracy limits their access to agricultural training and extension services, climate information, market information and market opportunities. Their time/budget workload and submission status hinder their access to certain opportunities, especially when they take them away from their place of residence (PNG; 2017). Inadequate road infrastructure, low sensitivity to the gender of commercial equipment and weak cross-cutting gender considerations in agricultural sector policies, strategies, plans and programs are all challenges for the full development of men and women in the agricultural sectors. Indeed, the ANASA report (2016) indicates that almost all of the agricultural population (80.54%) ages 7 and over, has not been to school; Only 1% of the population, ranging in age from 20 to 79, report having received literacy training at some point. RGPH3 (2014) establishes that 62.8% of people without education end up in agriculture.

***Conclusion and Recommendations***

While the legal framework for protecting the rights of men and women in Guinea is satisfactory, there are shortcomings in the application of texts due to socio-cultural factors as well as the juxtaposition of legal orders (religious, customary and modern), in a context where these texts and mechanisms of redress are unknown to the population and in particular to women (due to illiteracy). Thus, women's rights, although guaranteed by the current texts, remain largely unprotected in practice, especially with regard to inheritance, access to family property, choice of family residence, choice of activity women's economic scans. On these matters, the decision is usually up to the man or is made in his favor by the community actors.

The policy and policy framework for gender promotion, as well as institutional mechanisms for gender promotion, are consistent and effective. However, ownership and mastery of the gender concept is not certain among policy, technical and administrative decision-makers. The integration of gender into policy instruments at the national, sectoral or local level is not yet sufficient to generate significant impacts on reducing gender inequalities. Budget and gender skills gaps are factors that limit the coverage rate and penetration rate of most women's empowerment initiatives.

Women and men are affected differently by the challenges of accessing modern energy sources. In fact, the well-being of men and women is affected differently by the quality of access to energy. This is linked, for example, to the difficulty of household chores in the absence of electricity, but also to the difficulties of maintaining a quality technical plateau in health centers, particularly in the area of reproductive health. In this regard, the integration of gender into energy sector policies, plans, strategies and reforms is of the utmost importance. The same is true of gender integration at all stages of the energy sector infrastructure project and program cycle. Possible solutions are better management of the social risks associated with the massive influx of foreign workers, the promotion of women's employment and entrepreneurship in and around projects, strengthening women's leadership in local development decisions. Similarly, in agricultural and industrial value chains, the mastery of data on the roles and responsibilities of men and women, on their positioning in the sectors, but also on their opportunities and constraints, on the extent of gaps in access and control of critical resources, their specific needs and perspectives, is the first step for relevant intervention.

In many programmes and projects, gender integration remains marginal and non-cross-cutting, with participation quotas or specific activities, but without inclusion in the outcome frameworks of indicators that can measure effects and impacts of action on men and women; there is, for example, little ownership of the "Gender Response to Regional and National Agricultural Investment Plans to Meet the Zero Hunger Challenge in the ESAO region." The aim of this initiative is to analyze gender inequalities in access to important resources, knowledge, opportunities and markets, which contribute to low agricultural productivity as well as food and food insecurity and Nutritional. Reference studies and recent gender data on men and women in agricultural activities and agri-food in Guinea are non-existent. Thus, there is no reliable and up-to-date data to measure the extent of the challenges faced by women such as: poor access and control of productive resources (crop land, credits, inputs, techniques, technologies and means of work, including production, storage, conservation and processing equipment, low market access, poor organization/structuring of women's groups; weak entrepreneurial skills and low participation in the sectors, particularly in the juicy processing, labeled and marketing segments of agricultural products.

Finally, in most development sectors, actions to establish, establish and sustain gender equality are producing insufficient results. It is therefore imperative that a gender-sensitive culture be developed at all levels of Guinean society so that appropriate solutions are made to the causes of gender inequalities. This requires awareness, awareness and gender training at all levels aimed at increasing the capacity to perceive inequalities so that as few Guineans as possible continue to remain indifferent to the discrimination and inequality.

## C.2 GENDER ACTION PLAN: SENEGAL

| Activities | Indicators and Targets | Timeline | Responsibilities |
|---|---|---|---|
| ***Impact Statement**: Strengthened climate resilience of the agricultural value chain, to help stimulate productivity, value addition, competitiveness and income generation for the most vulnerable communities, including women and girls.*<br><br>***Outcome Statement**: Improved Access to Climate Resilient Livelihood Options for at least 160,000 Smallholder Women Farmers including Women-led Agribusiness Enterprises (Farm-Based Associations and Cooperative Societies).* | | | |
| ***Output(s) Statement 1: Critical SAPZ value chain infrastructure strengthened and made more climate resilient*** | | | |
| ***(i)** Smallholder women farmers (SHWF) provided with access to finance to invest in drip irrigation technology adoption* | • At least 20% SHWF in project areas | By 20XX +5 | Executing Agency |
| *(ii) Increase in women-led agricultural cooperative societies and farm-based associations, adopting low carbon technologies to reduce post-harvest losses of staple food crops in programme areas* | • At least 50% of women-led agric business enterprises in project area | By 20XX +5 | Executing Agency |
| *(iii) SHWFs and women-led agribusiness enterprise (WABEs), provided with access to low carbon technologies for renewable energy generation* | • At least 50% of women-led agric business enterprises in project area | By 20XX +5 | Executing Agency |
| ***Output(s) Statement 2: Climate resilient agricultural practices and technologies adoption among smallholder farmers promoted*** | | | |
| ***(i)** SHWFs provided with access to finance to invest in climate resilient agriculture and agroforestry practices* | • At least 50% of SHWF in project areas | By 20XX +5 | Executing Agency |
| *(ii) SHWFs and WABEs trained in the adoption of climate resilient livelihood options and low carbon technologies* | • At least 60% of women-led agric business enterprises in project area | By 20XX +5 | Executing Agency |
| *(iii) SHWFs and WABEs trained in climate smart agro advisory services and digital technology adoption* | • At least 75% of SHWFs and WABEs in project area | By 20XX +5 | Executing Agency |
| ***(iv)** SHWFs and WABEs trained in the maintenance of installed hydromet equipment* | • At least 20% of SHWFs and WABEs in project area | By 20XX +5 | Executing Agency |
| ***Output(s) Statement 3: Enabling environment for the adoption of sound agribusiness policies within the SCPZs enhanced*** | | | |
| (i)*Women-led civil society organisations (WCSOs) empowered on land Reforms* | • 30% WCSOs in project areas | By 20XX+5 | Executing Agency |
| *(ii) More female-headed households (FHH) empowered with managerial responsibilities at S**CPZ**s* | • 50% of managerial staff at SCPZ | By 20XX+5 | Executing Agency |